

Basics of Analog Flow: A Design-Oriented Approach

Rapid Adoption Kit (RAK)

Product Version: Virtuoso ICADVM20.1 ISR30, Spectre 21.1 ISR16, PEGASUS 22.20,
Quantus 22.1.0-p089
June 2024

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Contents

Purpose.....	4
Audience	4
Terms.....	4
Overview.....	5
Module 1: Initial Setup	6
Lab 1: Generating a cds.lib File for Virtuoso.....	6
Lab 2: Creating a New Library and Cell.....	13
Module 2: Schematic Design.....	17
Lab 3: Schematic Creation Using Virtuoso Schematic Editor	17
Lab 4: Symbol Creation	24
Module 3: Schematic Testbench	29
Lab 5: Testbench Creation.....	29
Lab 6: Modifying DUT Parameters.....	41
Module 4: Pre-Layout Simulation	45
Lab 7: Simulation Setup Using ADE Explorer and Assembler	45
Lab 8: Waveform Results Analysis	60
Lab 9: Generating Expressions.....	74
Lab 10: Design Tuning Using Parameters.....	82
Lab 11: Backannotation of Tuned Parameters	93
Module 5: Layout	95
Lab 12: Layout XL	95
Lab 13: Transistor Abutment.....	100
Lab 14: Placement	106
Lab 15: Routing.....	115
Lab 16: DRC Tests.....	128
Lab 17: LVS (Layout Versus Schematic).....	135
Module 6: Post-Layout Simulation.....	141
Lab 18: Using Quantus to Create a DSPF File.....	141
Lab 19: Using Quantus to Create a Smart View.....	143
Lab 20: Creating a Config View Using Virtuoso Hierarchy Editor	146
Lab 21: Post-Layout Simulations on ADE Assembler	151
Support	157
Feedback	157

Purpose

This RAK introduces the steps of creating an analog design from initial requirements to the sign-off stage.

Audience

This RAK is intended for designers who are new to Virtuoso or those who would like an overview of the complete custom flow using the latest Cadence tools.

For this RAK, you do not require prior knowledge of Cadence tools.

Terms

ADE	Analog Design Environment
LMB	Left Mouse Button
CIW	Command Interpreter Window

Overview

This workshop highlights the main steps of the analog design flow, focusing on improved designer productivity by using the latest features available in ICADVM20.1.

For this RAK, you must have ICADVM20.1 (or IC6.1.8), Spectre21.1, PEGASUS 22.20, and Quantus 22.1 in your path and the necessary licenses for them.

Module 1: Initial Setup

You need a PDK to create the design. Cadence Generic Process Design Kits (GPDK) and standard cell reference libraries are provided with Virtuoso and Innovus products. They are intended to be a representative of the actual semiconductor processes.

For this RAK, you will use the latest version of the 45nm GPDK. You can find this PDK at <https://support.cadence.com>. You do not require to download it for this RAK because it is provided in the database.

Action 1: Unpack the RAK database and cd into the **BAF_RAK** directory.

```
$ tar -xzf BAF_RAK.tar.gz
```

Action 2: Enter the main directory.

```
$ cd BAF_RAK
```

The following two directories will be shown:

- **gpdk045_v_6_0:** This contains GPDK files.
- **solutions:** This contains the files representing the completed modules and labs.

To use any of the results from the `solutions` directory, copy all files inside the desired Lab and paste them into the `BAF_RAK` directory by replacing any existing files. For example, if you want to do Lab 9, copy all files from `BAF_RAK/solutions/Module4/Lab9` and paste them into the `BAF_RAK` directory, and you will have all the necessary setup to start the ninth Lab.

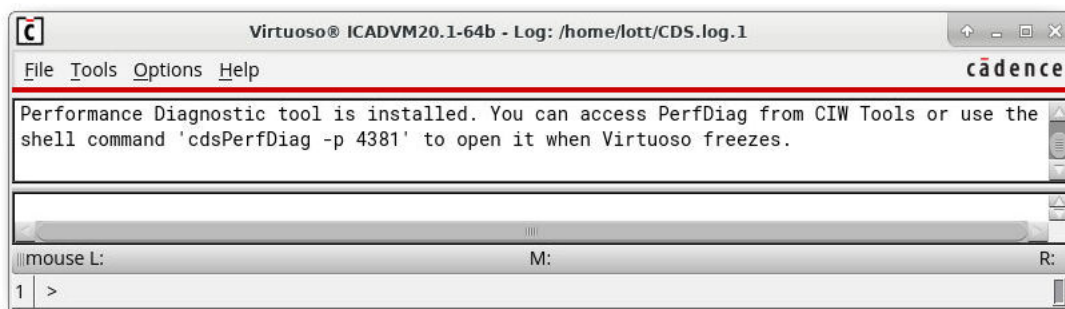
Lab 1: Generating a cds.lib File for Virtuoso

The first step in the analog flow is to write a file referencing the libraries required for your design.

Action 3: Invoke Virtuoso in your xterm.

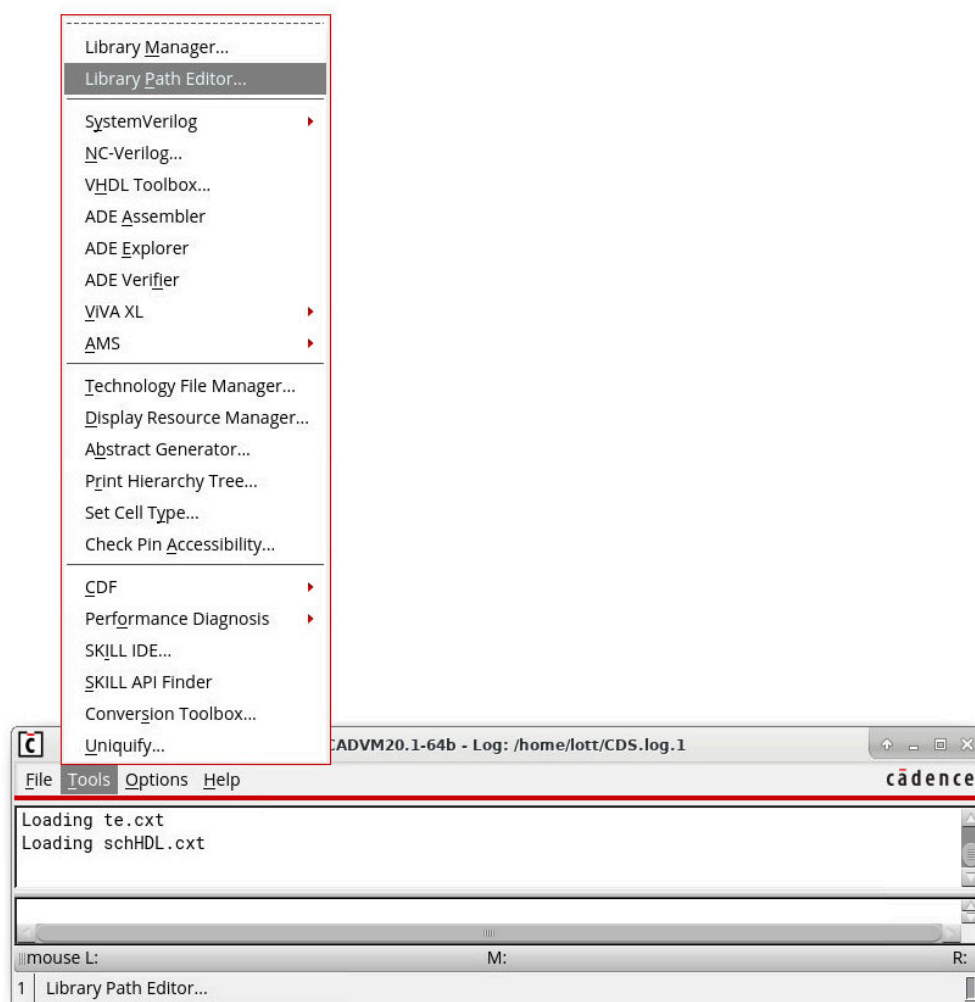
```
$ virtuoso &
```

Basics of Analog Flow: RAK

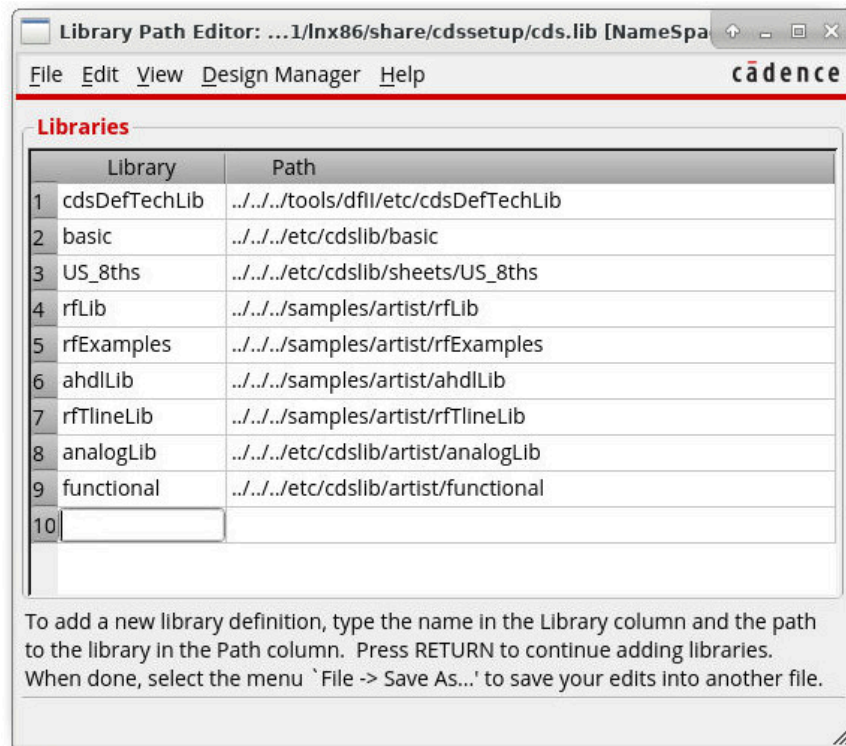


The CIW will open.

Action 4: Go to **Tools > Library Path Editor** from the CIW.

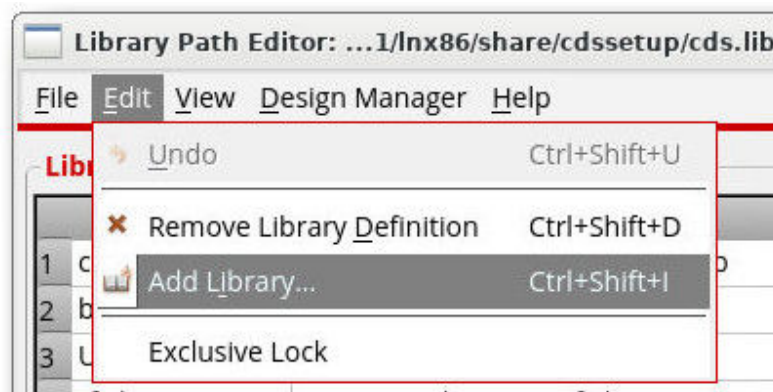


The **Library Path Editor** opens.

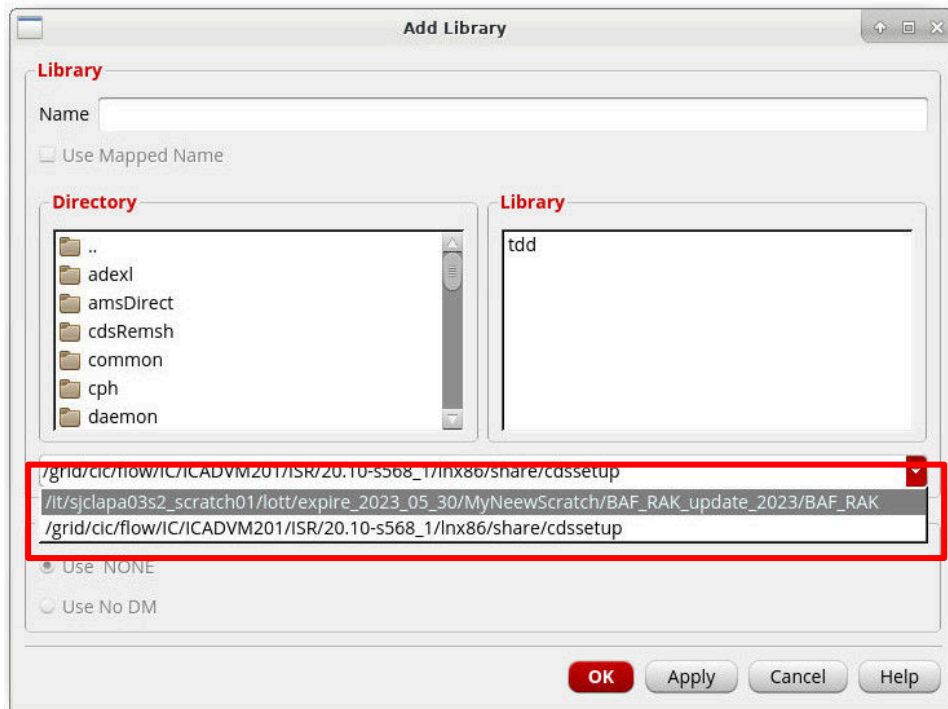


You will see several libraries already defined. These are the default libraries provided by Cadence and include generic components for the testbench creation. For the actual design, you need to include the GPDK with the process-specific (45nm) components.

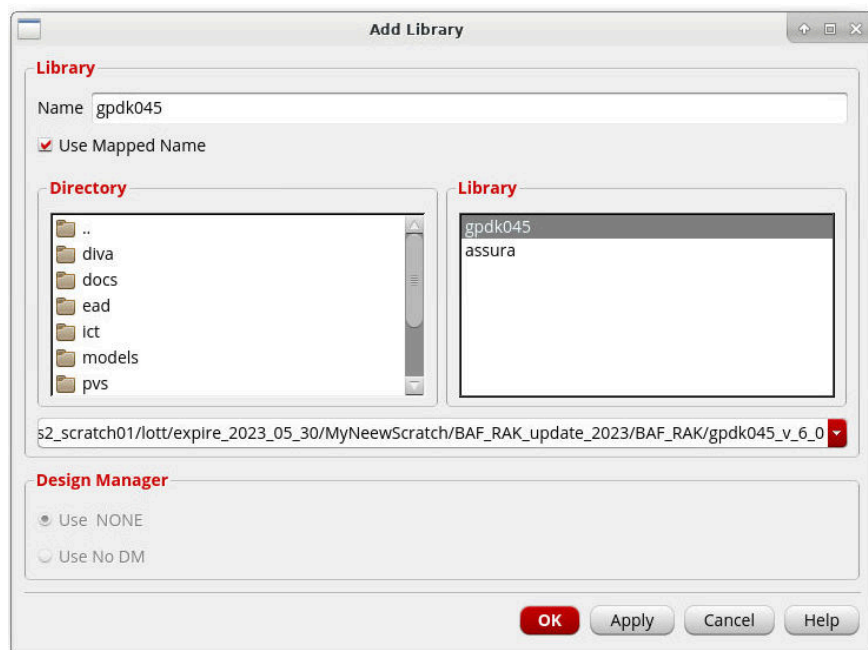
Action 5: Go to **Edit > Add Library**.



Action 6: In the **Add Library** form that opens, click on the pull-down menu and select the path where you extracted the GPDK.

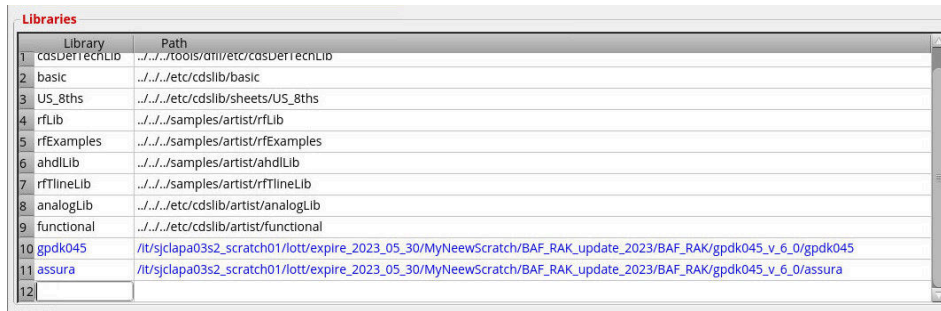


Action 7: In the **Directory** field, double-click on **gpdk045_v_6_0**, select the **gpdk045** library, and click **Apply**. Repeat this for the **assura** library. Click **OK** to close this window.

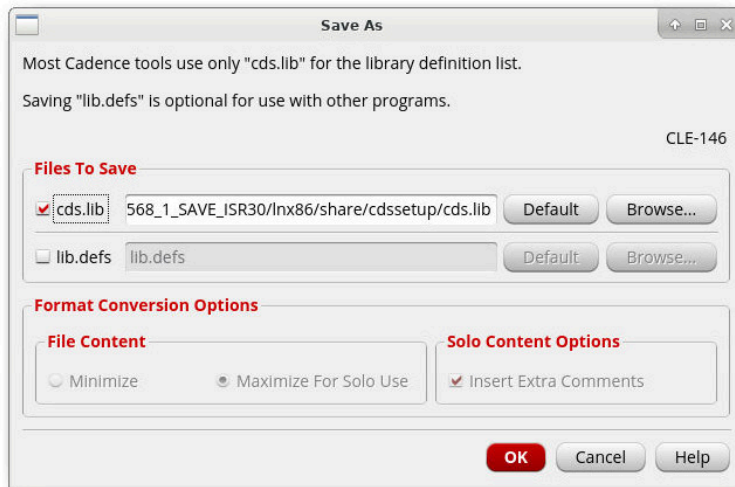


This step will add these libraries to your **Path Editor** form, as shown below.

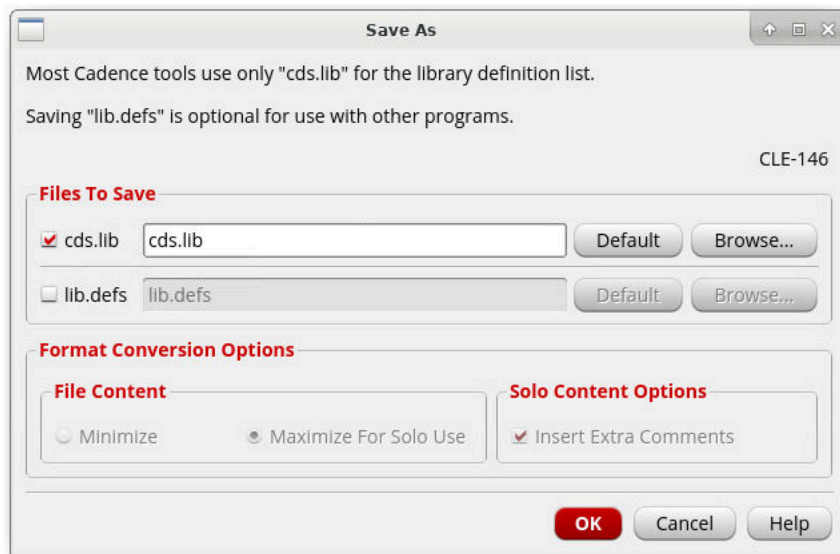
Basics of Analog Flow: RAK



Action 8: Select **File > Save As**. The **Save As** form will open.



Action 9: In the **cds.lib** field, erase all data from the field and write “cds.lib”. Then, click **OK** and close the **Library Path Editor**.



This will create a new `cds.lib` file in your project directory (where you invoked Virtuoso).

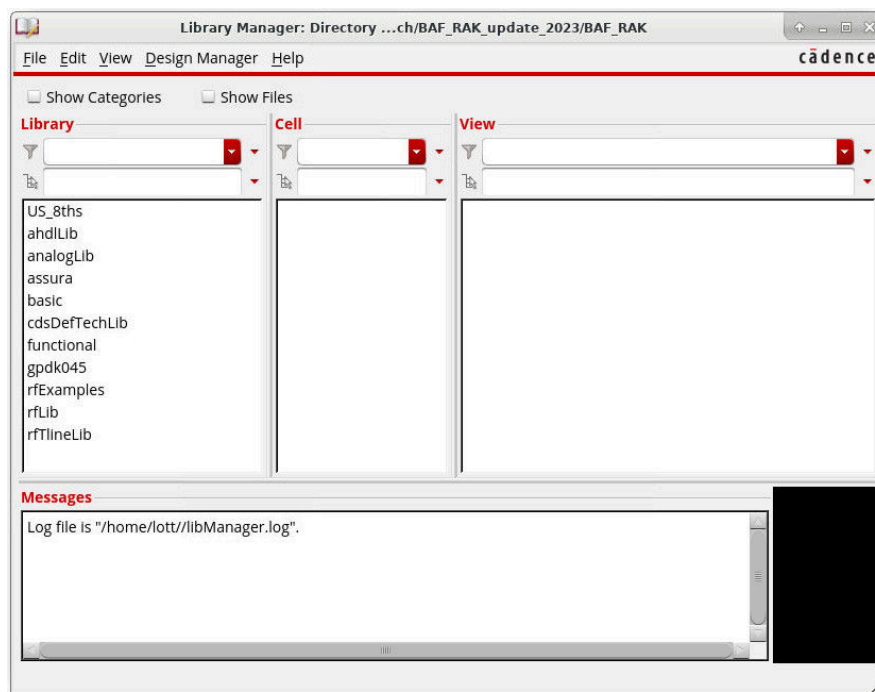
Action 10: Open the new `cds.lib` file in a text editor.

```
# File Created by John Doe
# assisted by CdsLibEditor
SOFTINCLUDE /grid/cic/flow/IC/ICADVM201/ISR/20.10-s568_1/lrx86/share/cdssetup/dfII/cds.lib
SOFTINCLUDE /grid/cic/flow/IC/ICADVM201/ISR/20.10-s568_1/lrx86/share/cdssetup/hdl/cds.lib
SOFTINCLUDE /grid/cic/flow/IC/ICADVM201/ISR/20.10-s568_1/lrx86/share/cdssetup/pic/cds.lib
SOFTINCLUDE /grid/cic/flow/IC/ICADVM201/ISR/20.10-s568_1/lrx86/share/cdssetup/sg/cds.lib
DEFINE gpd045 gpd045_v_6_0/gpd045
DEFINE assura gpd045_v_6_0/assura
```

As you can see, the `cds.lib` file is filled with the libraries added in Action 7. It also includes the default libraries.

Action 11: Close the file without making any changes. You can manually create/edit the `cds.lib` file in any editor.

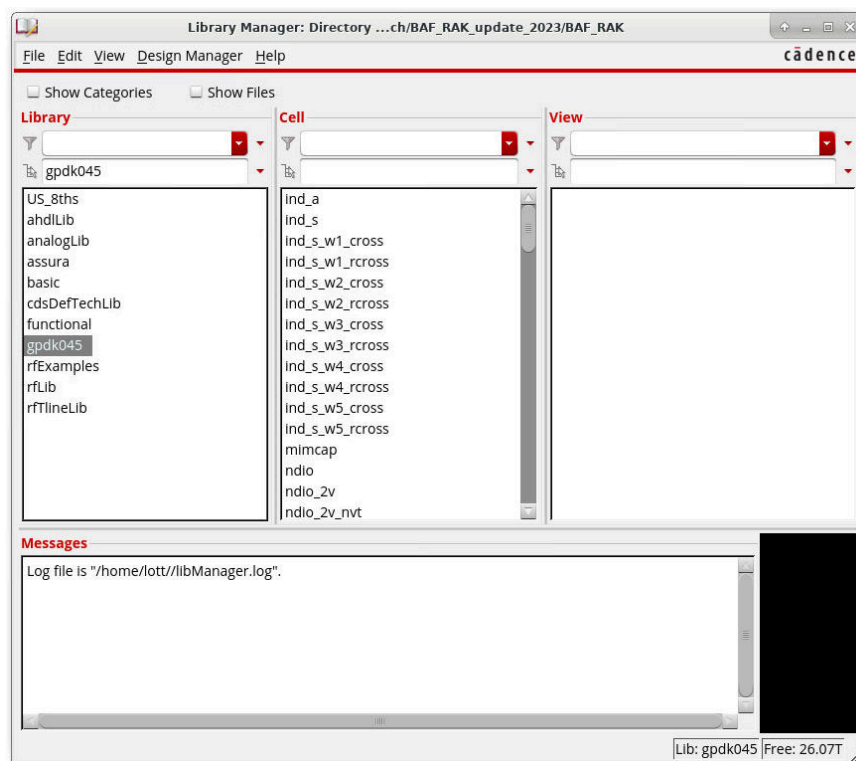
Action 12: In the CIW, go to **Tools > Library Manager**.



Observe that all the libraries you need for your project are available.

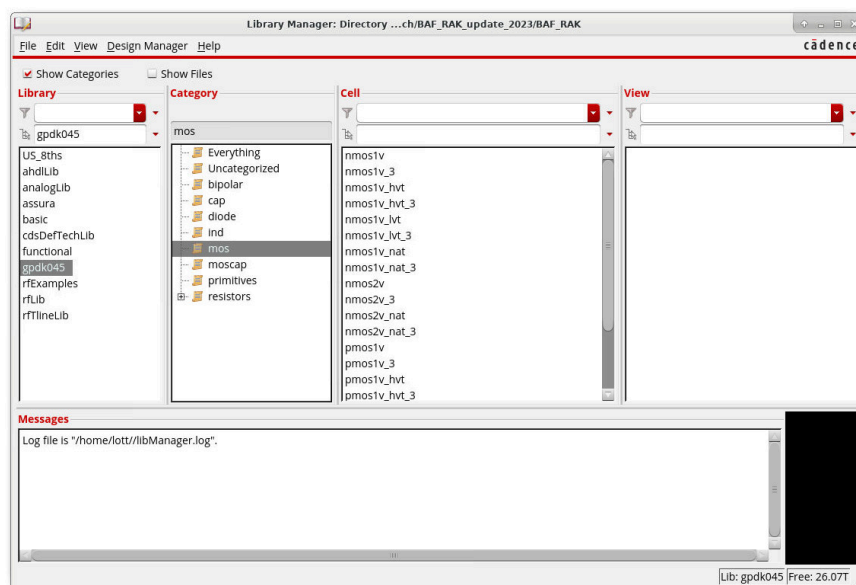
Action 13: In the **Library Manager**, click on the **gpd045** library.

Basics of Analog Flow: RAK



These are the basic components you might need for your design. PDK also includes process constraints, Spectre models, DRC/LVS rules, and QRC extraction decks.

Action 14: Click on the **Show Categories** checkbox and select the **mos** category.



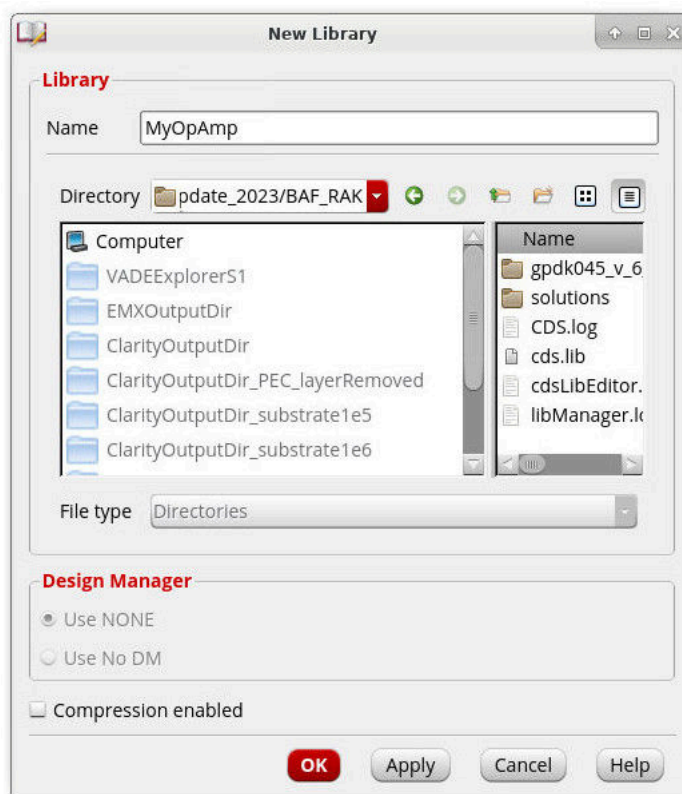
The **Category** column will filter the display of components to show all **mos** devices.

Lab 2: Creating a New Library and Cell

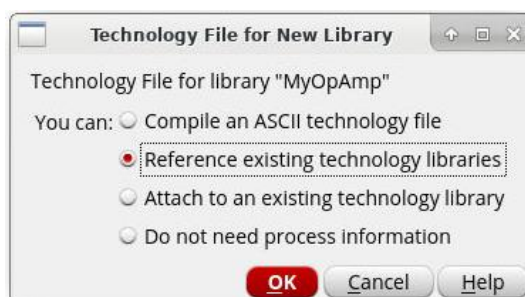
The next stage is to create a library for your DUT and testbench.

Action 15: In the **Library Manager**, go to **File > New > Library**. The **New Library** form will open.

Action 16: Select the path of your project directory and enter **MyOpAmp** in the **Name** field. Then, click **OK**.

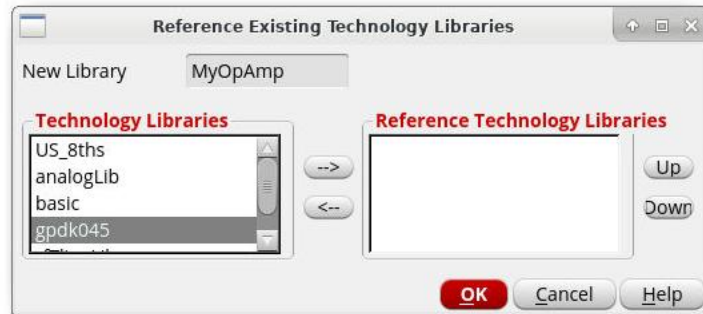


The **Technology File for New Library** form will open, as shown below.

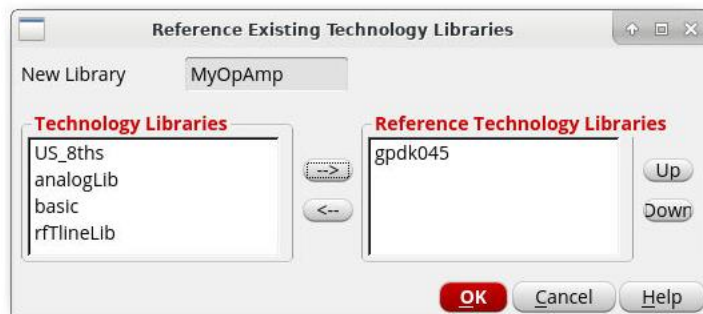


Action 17: In the **Technology File for New Library** form, select the **Reference existing technology libraries** option and click **OK**.

The **Reference Existing Technology Libraries** form will open.

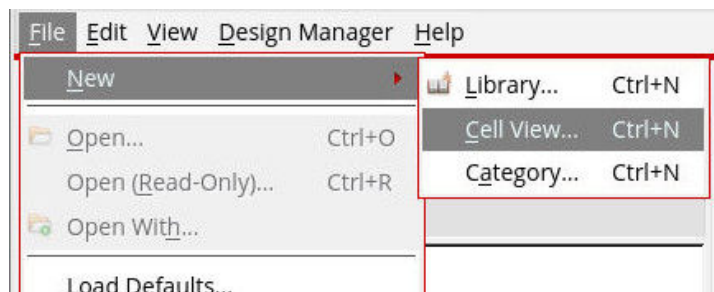


Action 18: Select the **gpdk045** library and click the right arrow icon to include it in the **Reference Technology Libraries** list box. Click **OK**.



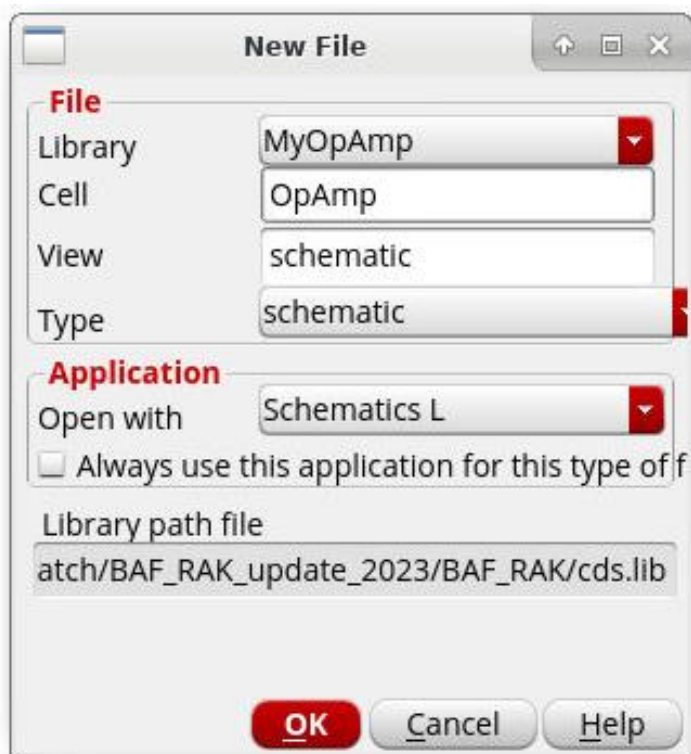
This ensures that the newly created library uses the technology of the GPDK. The new library will then appear in the **Library Manager**.

Action 19: Click on **MyOpAmp** in the **Library Manager**. Then, go to **File > New > Cell View**.



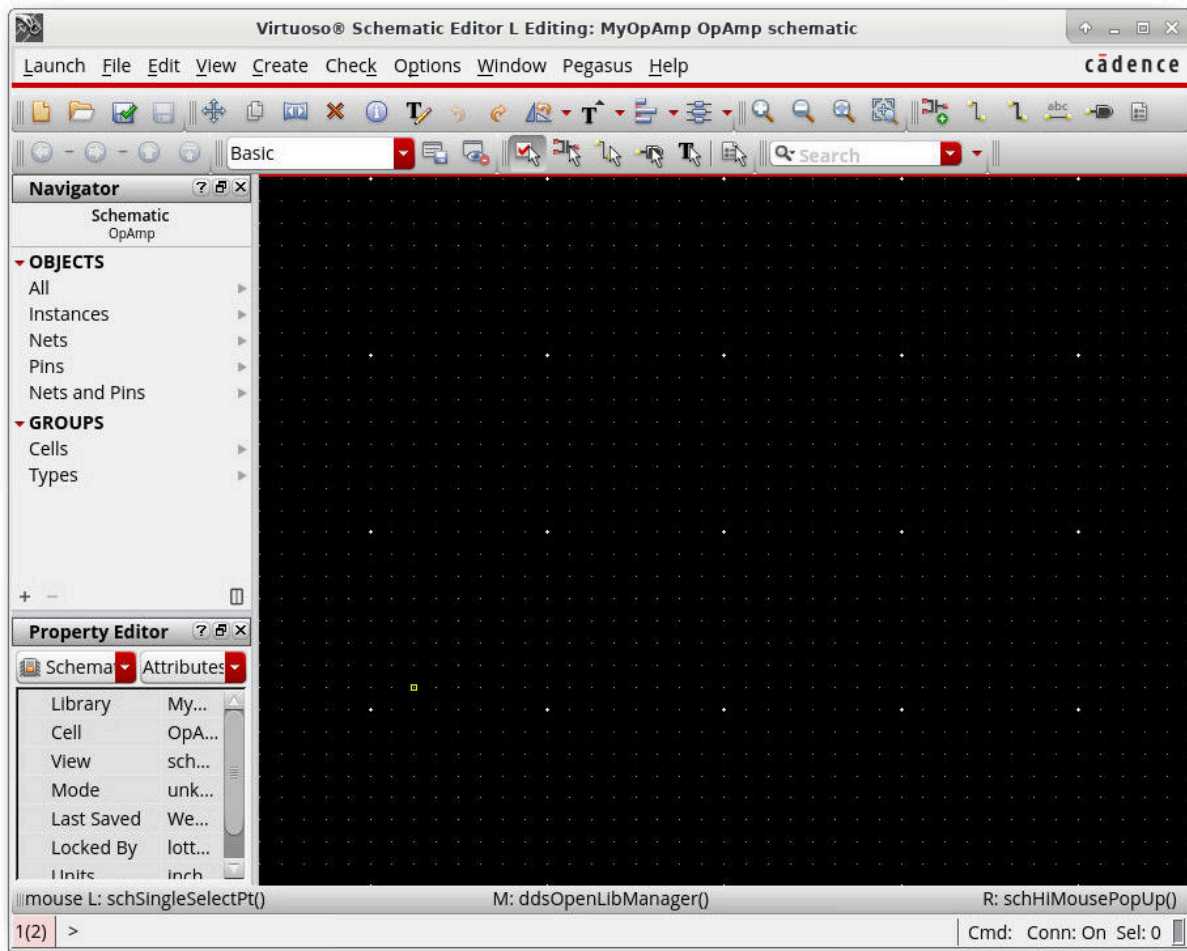
This will open the **New File** form.

Action 20: In this form, enter **OpAmp** as the **Cell** and leave the other entries as-is. Click **OK**.



This will create a new cell named “OpAmp” in the **MyOpAmp** library and a new view of that cell named **schematic**. The schematic cellview is automatically opened.

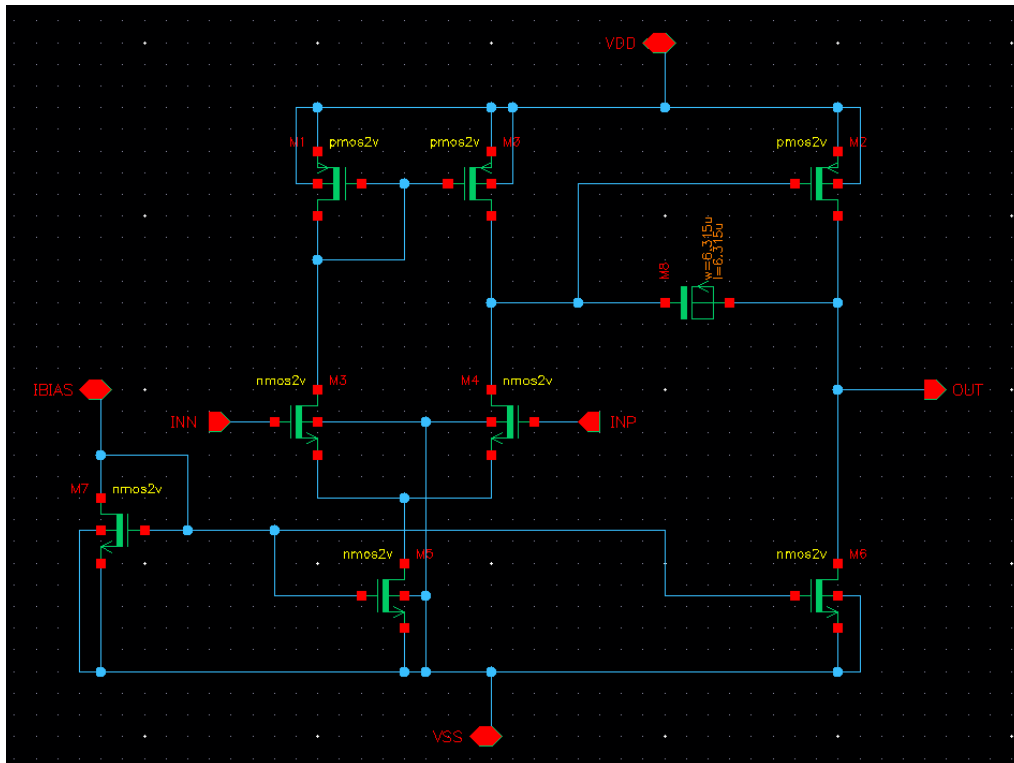
Basics of Analog Flow: RAK



Action 21: Leave this view open and proceed to the next module.

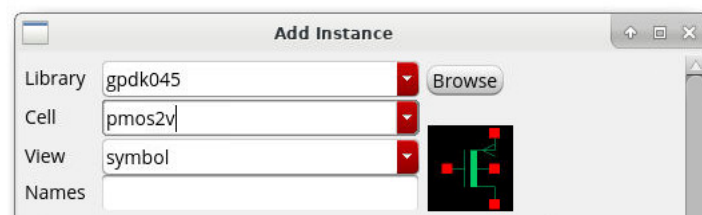
Module 2: Schematic Design

In this module, you will create a schematic view of your design. The topology of your design will be a Two-Stage Operational Amplifier. The final schematic will look like the one shown below.



Lab 3: Schematic Creation Using Virtuoso Schematic Editor

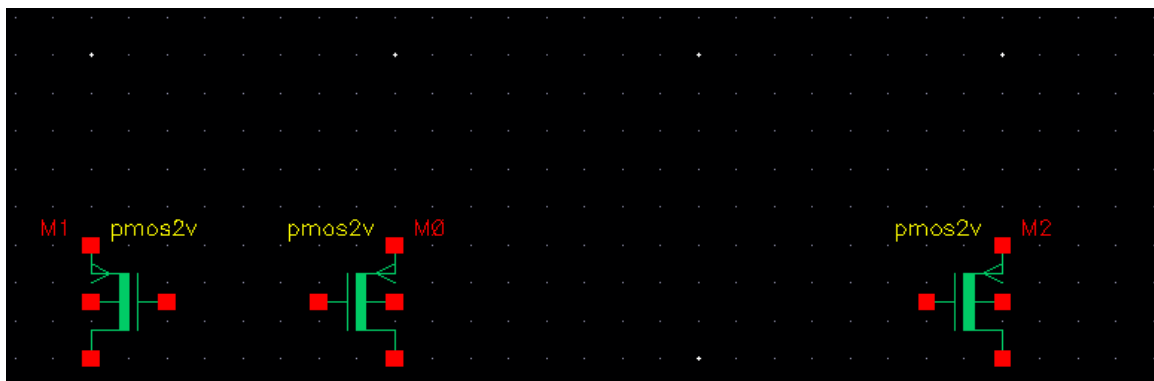
Action 22: On the schematic canvas, press **i** to open the **Add Instance** form. Fill in the form, as shown below, to place a four-terminal pmos device from the **gpdK045** library. You can type into these fields and use the autocomplete feature.



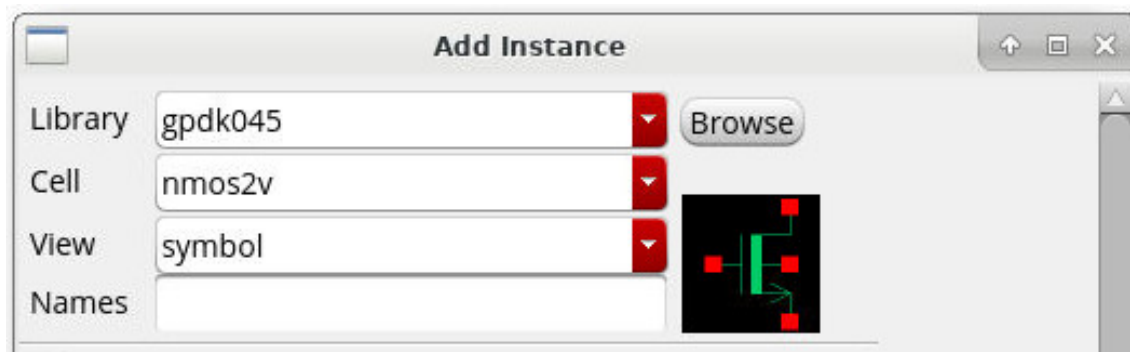
In the lower part of the form, you will see the default parameters for this component.

Action 23: Press **Hide** on the form. Then, place three pmos transistors, as shown below. Pay attention to the ordering so that you place the middle, then the left, and then the right-hand components. Press **Shift+R** to mirror the left-hand instance. Press **Esc** after you finish.

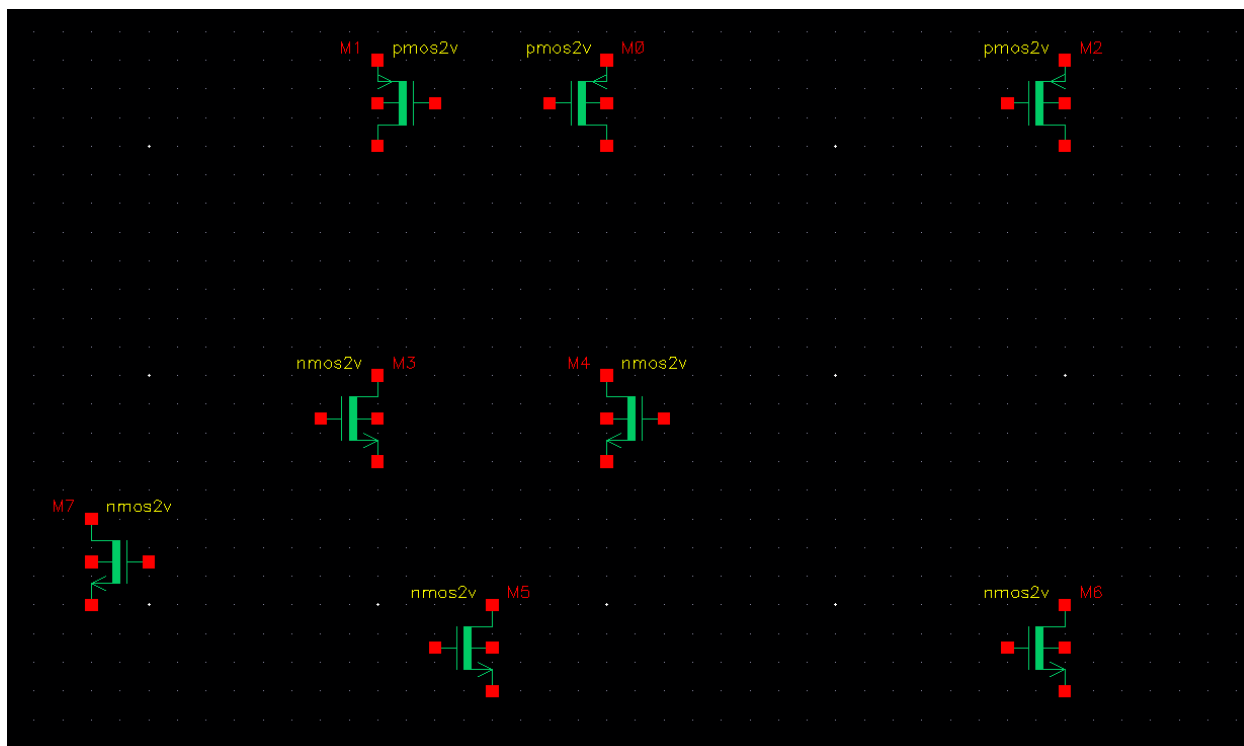
Note: The placement of components should match the image shown below.



Action 24: On the schematic canvas, press **i** to open the **Add Instance** form. Fill in the form to place a four-terminal nmos device from the **gpdk045** library, as shown below.



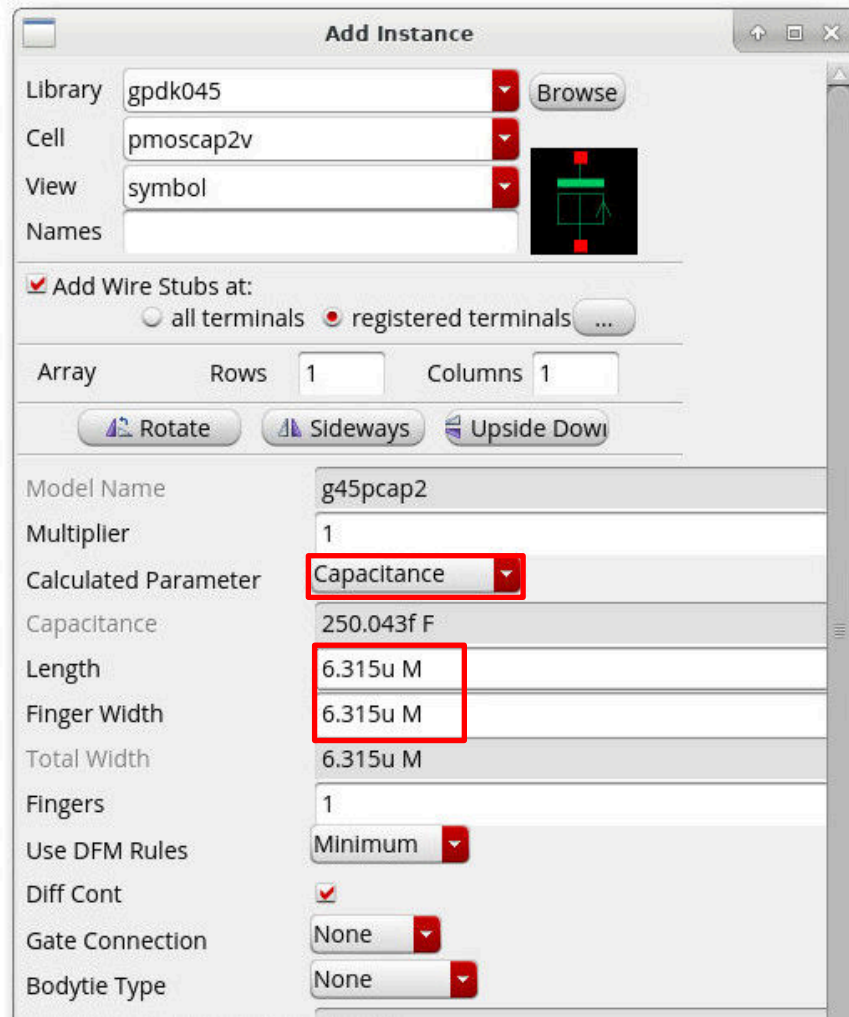
Action 25: Press **Hide** on the form. Then, place the five nmos transistors. Once again, pay attention to the order in which you place them so that the names match the image shown below. Press **Shift+R** to mirror as needed. Press **Esc** after you finish.



Now, you have inserted all MOSFET devices. You will now add the capacitor.

Action 26: On the schematic view, press **i** to open the **Add Instance** form. Select the **pmoscap2v** device from the **gpdsk045** library.

In the **Parameters** section, change the **Calculated Parameter** field to **Capacitance** and change both the **Length** and **Finger Width** parameters to **6.315u**, as shown below.

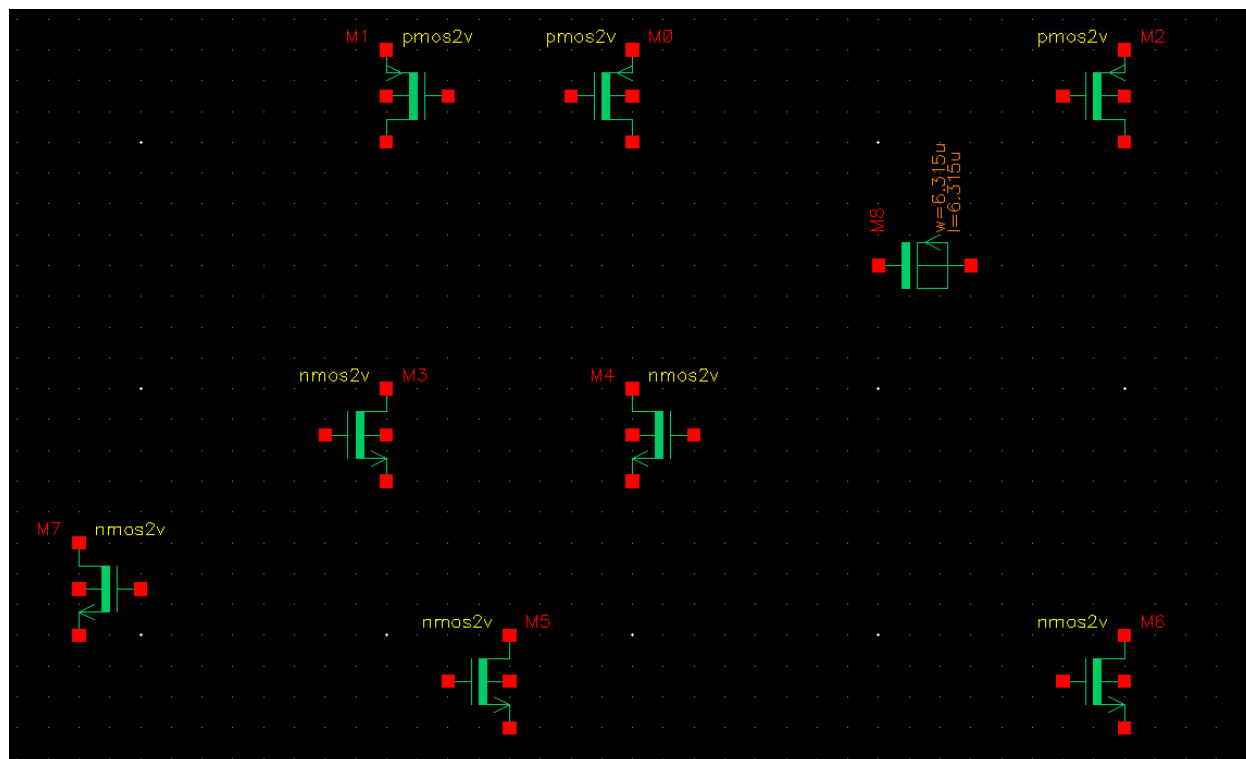


The 'Add Instance' dialog box is shown with the following settings:

- Library: gpd045
- Cell: pmscap2v
- View: symbol
- Names: (empty)
- ☒ Add Wire Stubs at:
 - ☐ all terminals
 - ☒ registered terminals
- Array: Rows 1, Columns 1
- Buttons: Rotate, Sideways, Upside Down
- Model Name: g45pcap2
- Multiplier: 1
- Calculated Parameter: Capacitance
- Capacitance: 250.043f F
- Length: 6.315u M
- Finger Width: 6.315u M
- Total Width: 6.315u M
- Fingers: 1
- Use DFM Rules: Minimum
- Diff Cont: ☒
- Gate Connection: None
- Bodytie Type: None

By choosing **Calculated Parameter** as **Capacitance**, you can edit the capacitor dimensions and see the resultant capacitance value.

Action 27: After you fill in the form, press **Hide**. Then, place the capacitor, as shown below. You can use the **Rotate** icon '  ' on the form as you place it. Then, press **Esc**.

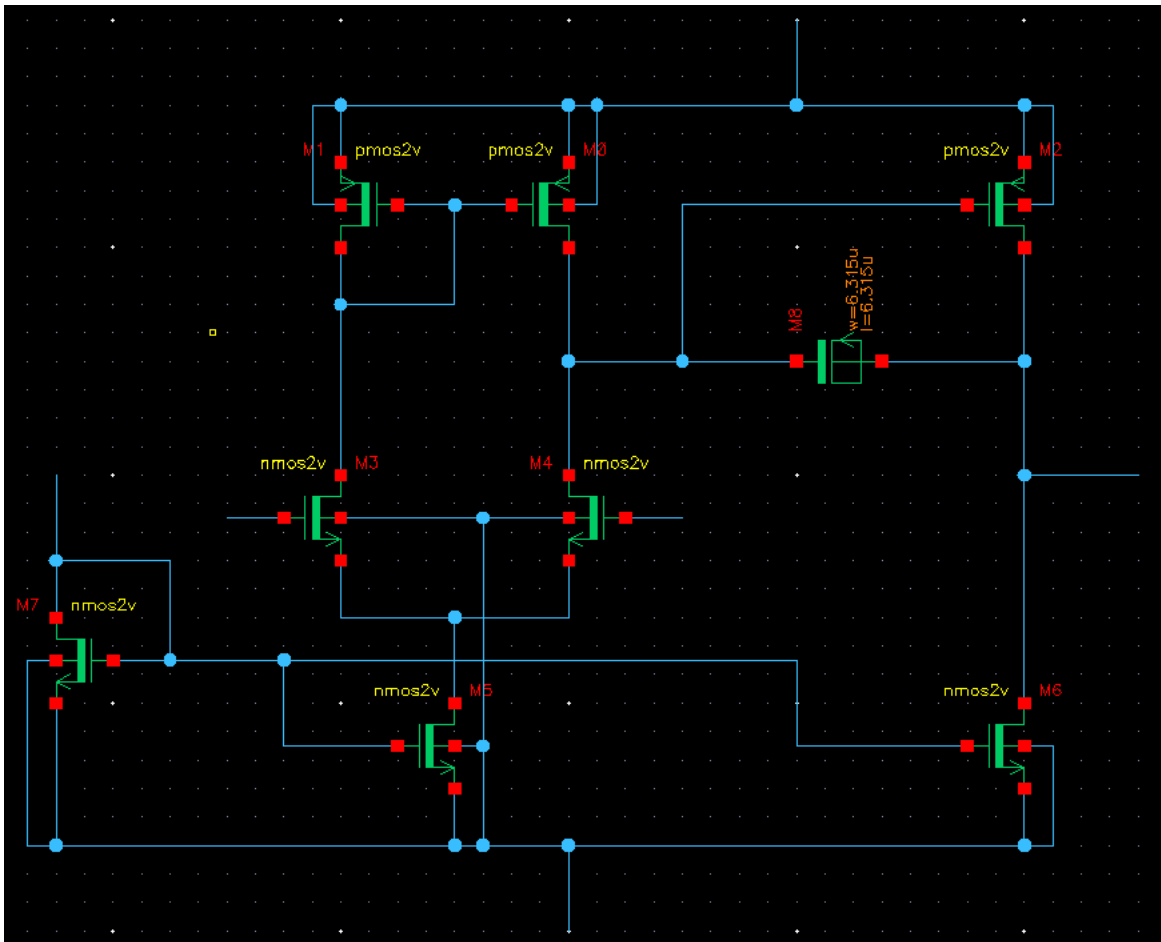


Action 28: Press **W** or select **Create > Wire** to invoke the ‘wiring’ command.

To create connections, click where you want the wire to start. Now, click again in free space to create an intermediate point OR on a device terminal OR another wire to end the command. Double-click or press **Enter** to complete the connection that ends in free space. You can also click+drag from one terminal and release it on the terminal where the connection ends.

You can also use the **S** bindkey to snap to the nearest pin or wire endpoint without moving the mouse. A yellow diamond will indicate the nearest routing target.

Action 29: Complete the connections as shown below.



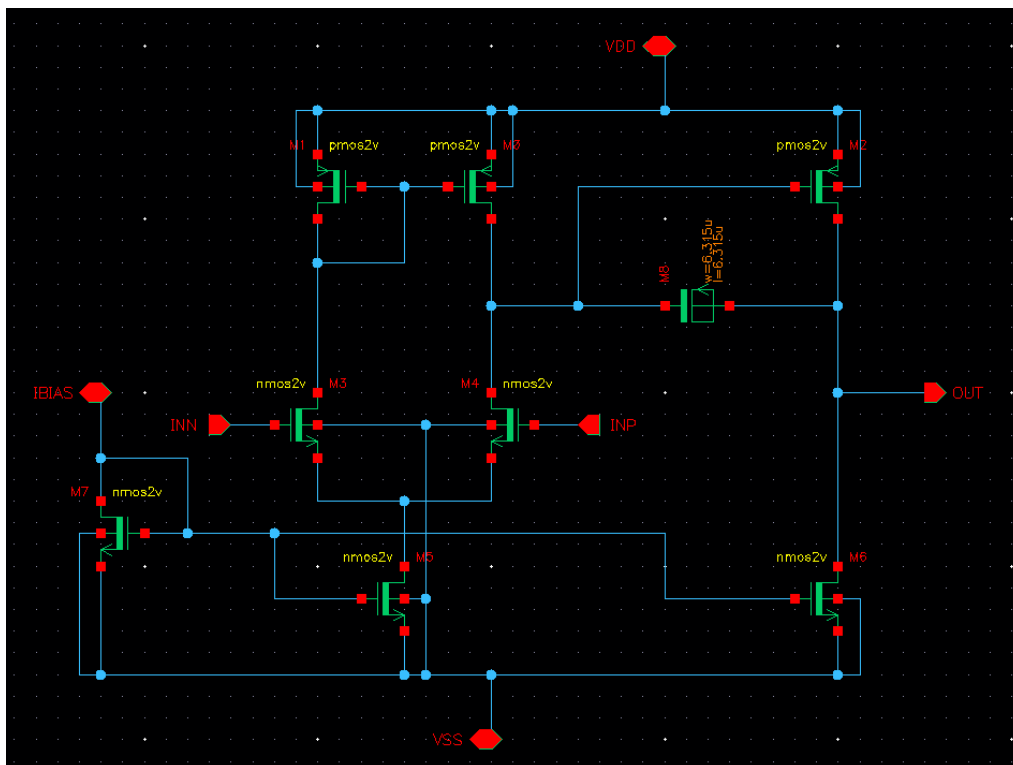
To delete a net or any other component, press **Delete** and select the component or use the '✕' icon.

Action 30: After you finish routing, press **Esc**. Then, press **P** to add pins. Fill in the **Names** field, as shown below, to create pins for this design.



Action 31: Change the value of the **Direction** field to **input** and place the **INN** and **INP** pins. Switch **Direction** to **R** to place the **OUT** pin. Place the last three pins as **inputOutput**. Use the **Rotation** icons on the dialog box to help with the placement.

Press **Esc** after you finish. The correct pin placement is shown below in the schematic.

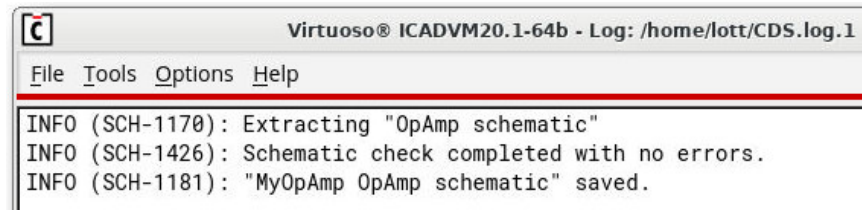


You can drag the name of a pin to make the text more visible if needed.

The schematic is now complete.

Action 32: Select the **Check and Save** icon . This checks the design against a number of built-in rules, floating nets, overlapping instances, shorts, and so on. **Check > Rules Setup** will show you the complete list of rules.

The following information should appear on your CIW to inform you that everything is correct:



If there are any issues, a **Schematic Check** window will open. Markers will appear on the canvas alerting you of any issues.

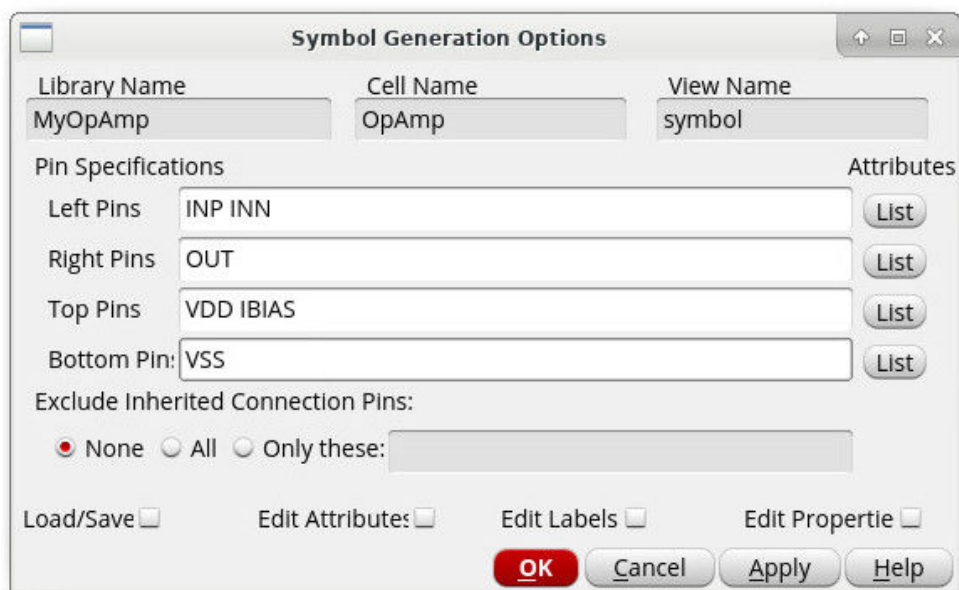
Lab 4: Symbol Creation

The last step in this module is to create a symbol for your design. Symbols are used to construct a hierarchy in designs.

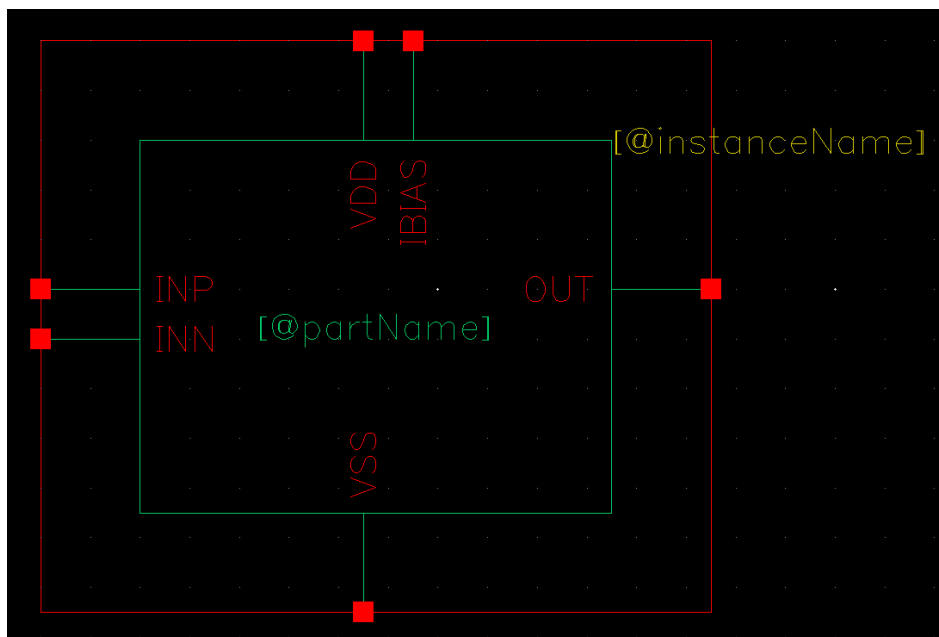
Action 33: In the Schematic Editor, select **Create > Cellview > From Cellview**. Leave the form as-is and click **OK**.

The **Symbol Generation Options** form will appear. This allows you to specify a location for placing each pin.

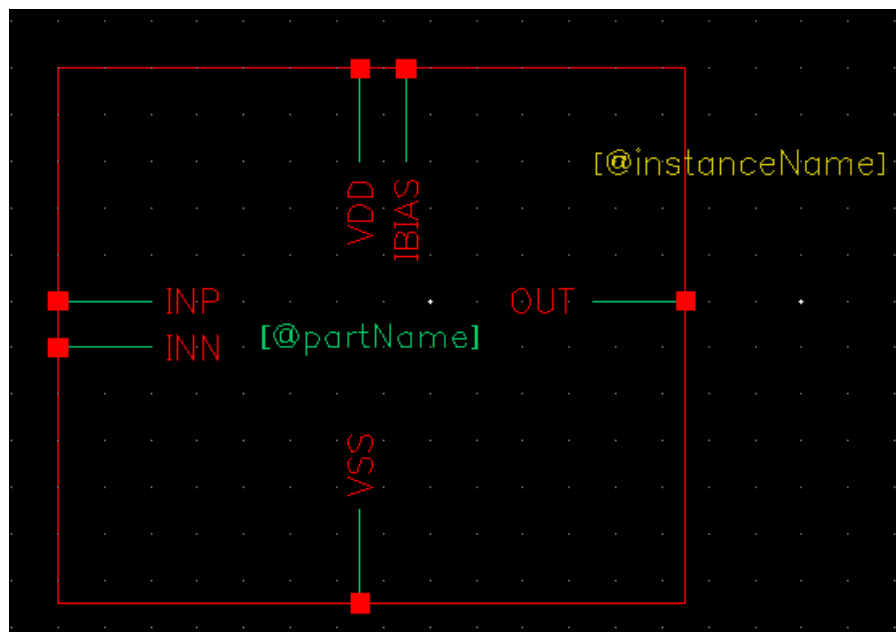
Action 34: Fill in the form as shown below and then click **OK**.



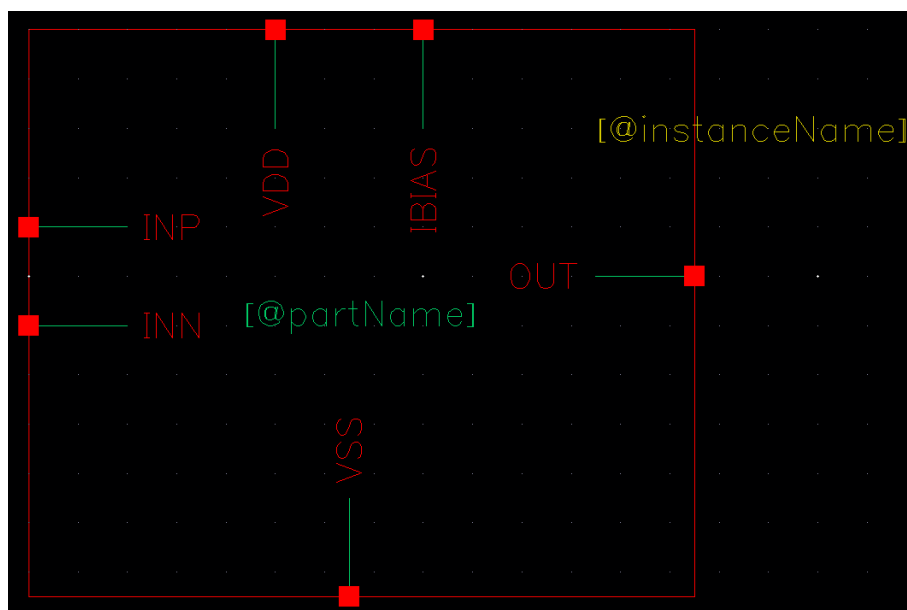
The symbol is generated with a rectangular shape. To make it resemble a traditional op-amp block, you need to edit the shape.



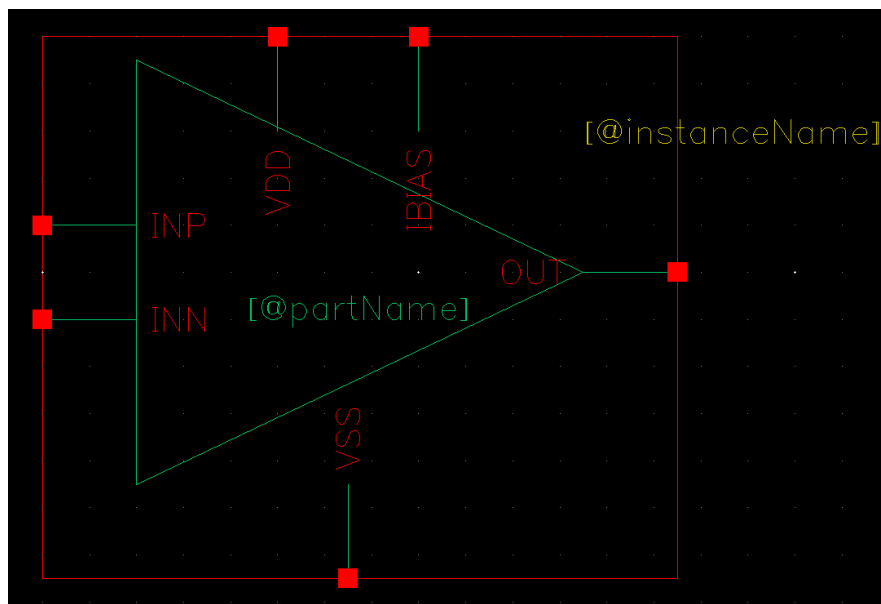
Action 35: Press  and select the green rectangle shape for deleting it. Then, press **Esc**.



Action 36: Drag the nets and pins apart by selecting and dragging them to create more space between them.

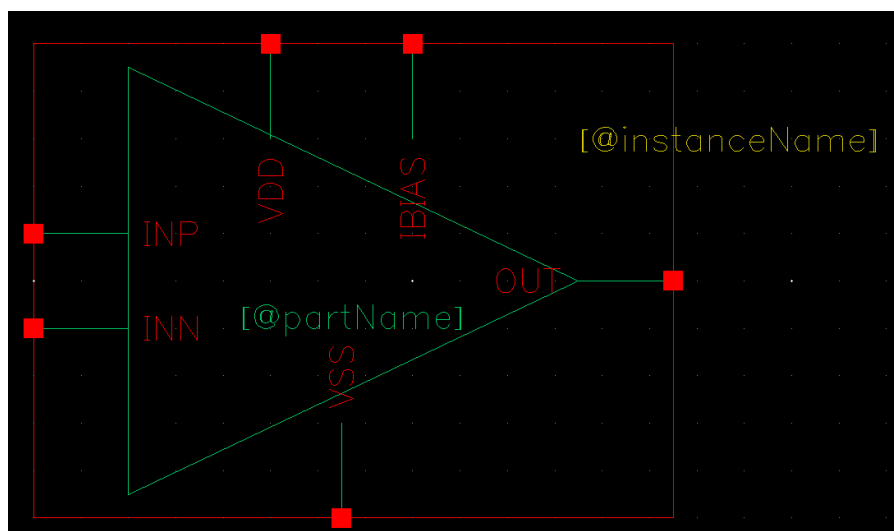


Action 37: Click the **Create Polygon** icon . Starting on the tip of the **INP** stub (the green line), draw a triangle like the one usually seen for an op-amp. Click twice when done.



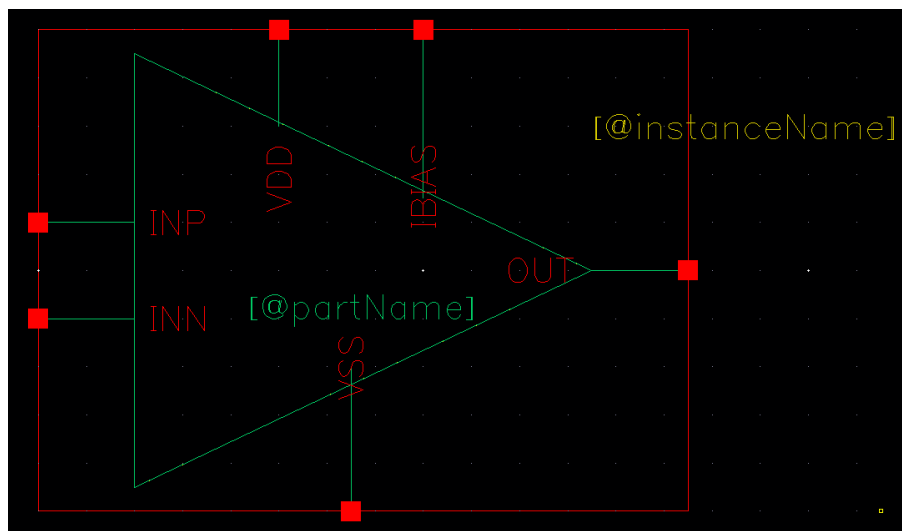
Now, you need to stretch the bottom part of the red rectangle upwards to make the op-amp symbol selection box more symmetrical.

Action 38: Select the bottom side of the red rectangle together with the **VSS** stub and pin and drag them up, as shown below.

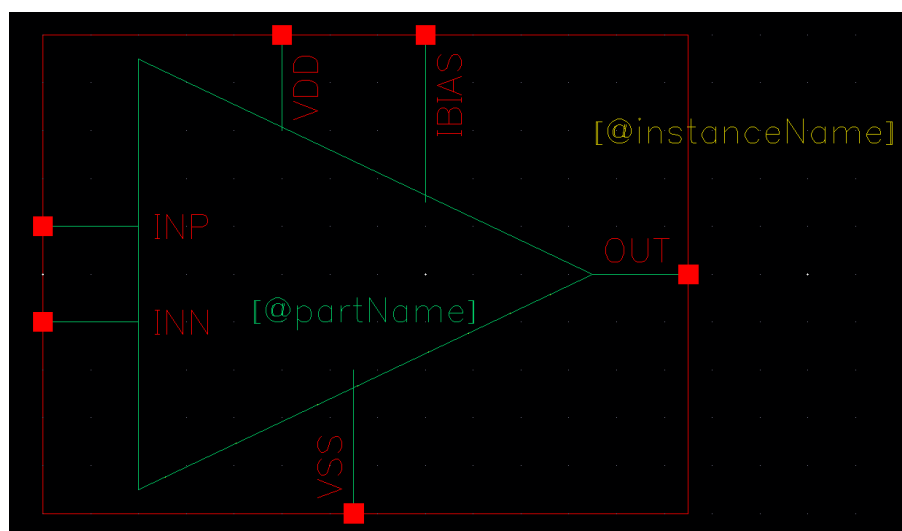


The only step left is to extend the stubs for **VSS** and **IBIAS** so that they connect to the triangle. This is an optional step to improve the look of the op-amp symbol.

Action 39: Select the tip of the **IBIAS** stub and drag it until it touches the triangle. Repeat the same for the **VSS** stub.



Action 40: Adjust the pin-label positions (red text) to make them more legible. To do this, select and drag them individually.



Action 41: Once the editing is complete, click on the **Check and Save** icon. If no other window pops up, the symbol is created correctly.

Action 42: Close the symbol and the OpAmp schematic views and proceed to the next module.

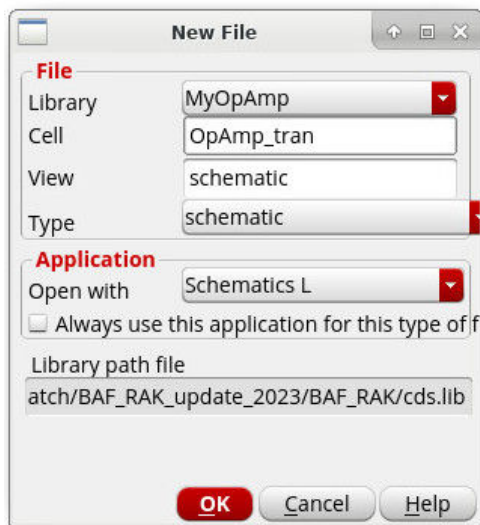
Module 3: Schematic Testbench

The next step is to create a schematic view that will be used for testing the op-amp design.

Lab 5: Testbench Creation

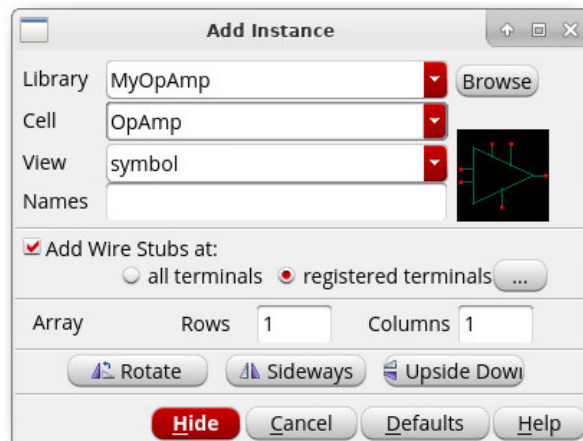
Action 43: Go to the **Library Manager**. Select **MyOpAmp** as the **Library** and go to **File > New > Cell View**.

Action 44: On the **New File** form, enter **OpAmp_tran** in the **Cell** field and click **OK**.

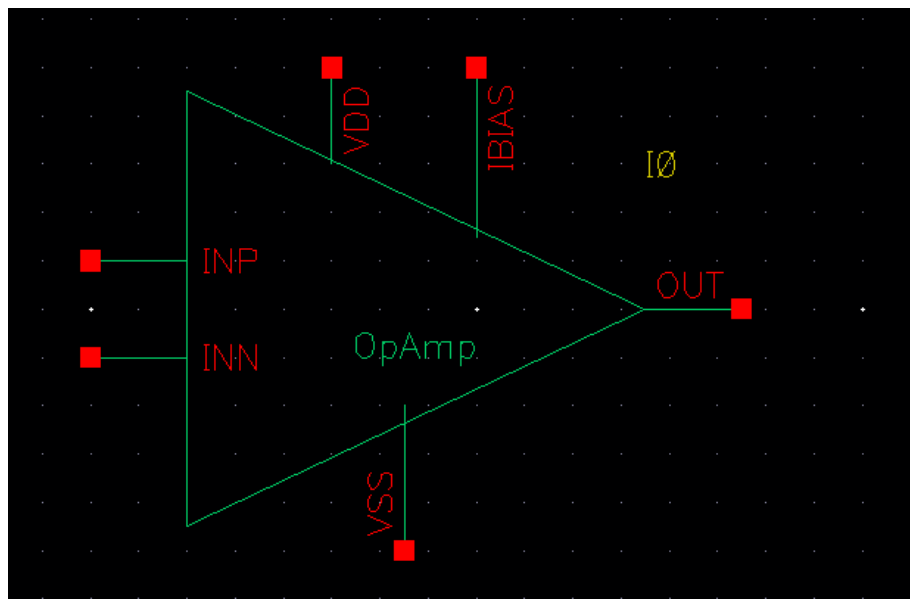


This will generate and open a new schematic view.

Action 45: Press **i** to invoke the **Add Instance** form and fill it in, as shown below.



Action 46: Press **Hide** and place the instance on the schematic. Then, press **Esc**.



The next step is to add the necessary DC sources.

Action 47: Press **i** and populate the **Add Instance** form, as shown below in the image on the left. Place the source in the schematic. Repeat twice more, using the other two images as a guide. Press **Esc** after it is finished.

It is not necessary to fill in the names of the instances. If the **Add Instance** form is hidden behind a window, you can press **F3** to show this form.

Basics of Analog Flow: RAK

The image displays three screenshots of the 'Add Instance' dialog box in Cadence, showing configurations for different analog components.

Top Left Screenshot (VDD_source):

- Library: analogLib
- Cell: vdc
- View: symbol
- Names: VDD_source
- ☒ Add Wire Stubs at:
 - ☐ all terminals
 - ☒ registered terminals
- Array: Rows 1, Columns 1
- Buttons: Rotate, Sideways, Upside Down
- Noise file name: (empty)
- Number of noise/freq pair: 0
- DC voltage: vdd V
- AC magnitude: (empty)
- AC phase: (empty)
- XF magnitude: (empty)
- PAC magnitude: (empty)
- PAC phase: (empty)
- Temperature coefficient 1: (empty)
- Temperature coefficient 2: (empty)
- Nominal temperature: (empty)
- Buttons: Hide, Cancel, Defaults, Help

Top Right Screenshot (VSS_source):

- Library: analogLib
- Cell: vdc
- View: symbol
- Names: VSS_source
- ☒ Add Wire Stubs at:
 - ☐ all terminals
 - ☒ registered terminals
- Array: Rows 1, Columns 1
- Buttons: Rotate, Sideways, Upside Down
- Noise file name: (empty)
- Number of noise/freq pair: 0
- DC voltage: vss V
- AC magnitude: (empty)
- AC phase: (empty)
- XF magnitude: (empty)
- PAC magnitude: (empty)
- PAC phase: (empty)
- Temperature coefficient 1: (empty)
- Temperature coefficient 2: (empty)
- Nominal temperature: (empty)
- Buttons: Hide, Cancel, Defaults, Help

Bottom Screenshot (Ibias_source):

- Library: analogLib
- Cell: idc
- View: symbol
- Names: Ibias_source
- ☒ Add Wire Stubs at:
 - ☐ all terminals
 - ☒ registered terminals
- Array: Rows 1, Columns 1
- Buttons: Rotate, Sideways, Upside Down
- Noise file name: (empty)
- Number of noise/freq pair: 0
- DC current: Ibias A
- AC magnitude: (empty)
- AC phase: (empty)
- Multiplier: (empty)
- Temperature coefficient 1: (empty)
- Temperature coefficient 2: (empty)
- Nominal temperature: (empty)
- XF magnitude: (empty)
- PAC magnitude: (empty)
- PAC phase: (empty)
- Buttons: Hide, Cancel, Defaults, Help

The following three DC sources were added:

- One **vdc** source named **VDD_source** with **DC voltage** set to **vdd**
- One **vdc** source named **VSS_source** with **DC voltage** set to **vss**
- One **idc** source named **Ibias_source** with **DC current** set to **ibias**

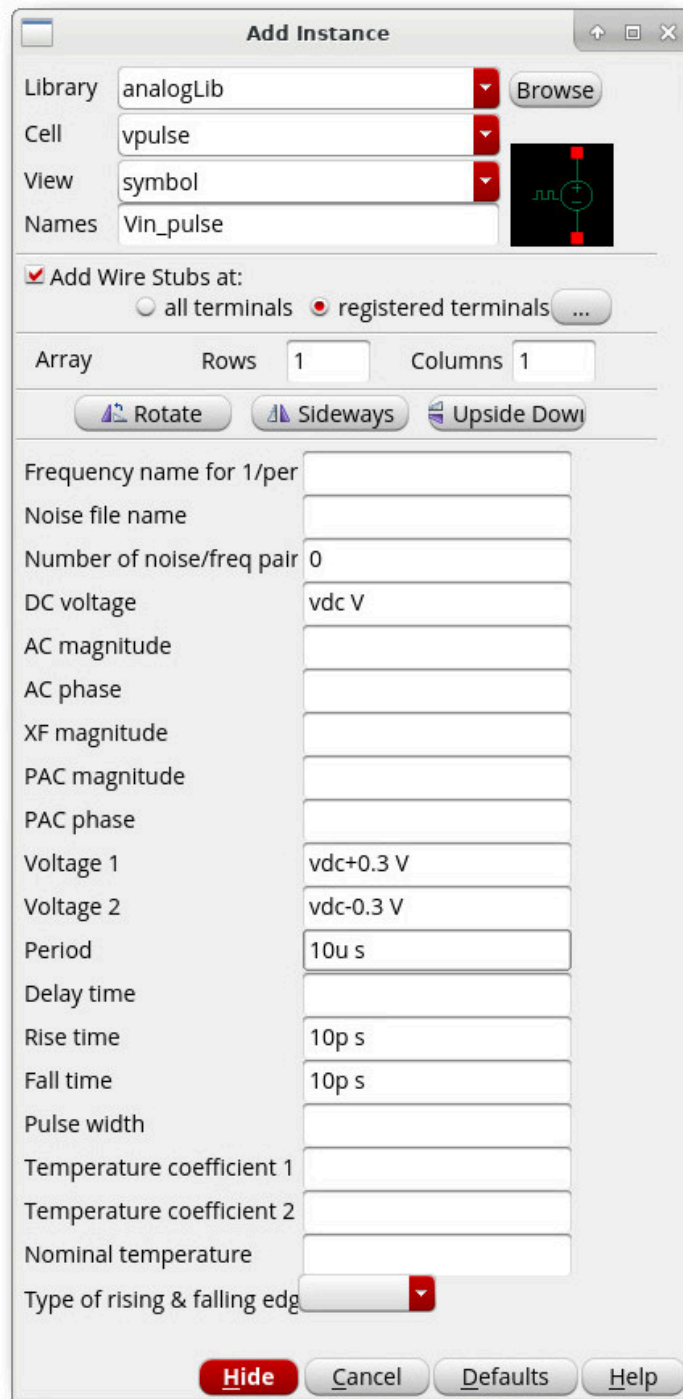
The **DC voltage** and **DC current** field entries are variables. These values will be set later for the simulation.

The schematic should look as follows:



To test this op-amp, you need to create a simple buffer configuration and use a square wave for the input. For that, you will need a pulse source.

Action 48: Press **i** and fill in the form, as shown below. Then, place the instance on the schematic close to the **INP** pin.



Add Instance

Library: analogLib Browse

Cell: vpulse

View: symbol

Names: Vin_pulse

☒ Add Wire Stubs at:
☐ all terminals ☒ registered terminals ...

Array: Rows 1 Columns 1

Rotate Sideways Upside Down

Frequency name for 1/per:

Noise file name:

Number of noise/freq pair: 0

DC voltage: vdc V

AC magnitude:

AC phase:

XF magnitude:

PAC magnitude:

PAC phase:

Voltage 1: vdc+0.3 V

Voltage 2: vdc-0.3 V

Period: 10u s

Delay time:

Rise time: 10p s

Fall time: 10p s

Pulse width:

Temperature coefficient 1:

Temperature coefficient 2:

Nominal temperature:

Type of rising & falling edge: ▼

Hide Cancel Defaults Help

This defines a square-wave signal with an amplitude of 700mV to 1.3V.

A load for this testbench is the next step. For example, a capacitor should be a good load to obtain realistic results of **Slew Rate**.

Action 49: Fill in the **Add Instance** form (as shown below) and place the cap close to the **OUT** pin.

Add Instance

Library: analogLib Browse

Cell: cap

View: symbol

Names:

☒ Add Wire Stubs at:
☐ all terminals ☒ registered terminals ...

Array: Rows: 1 Columns: 1

Rotate Sideways Upside Down

Model name:

Capacitance: CL F

Width:

Length:

Multiplier:

Scale factor:

Temp rise from ambient:

Initial condition:

Temperature coefficient 1:

Temperature coefficient 2:

Number of Polynomial Coefficients: 0

Capacitor Area:

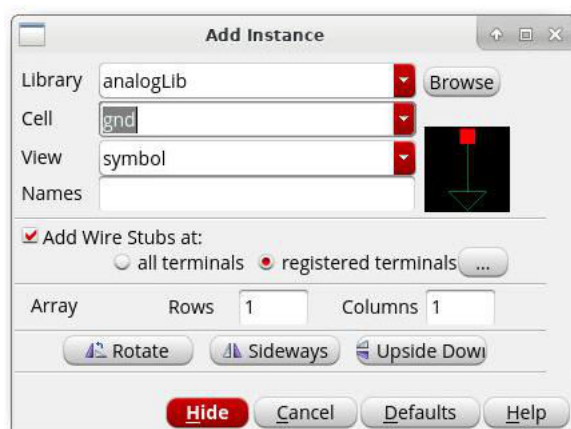
Capacitor Perimeter:

Hide Cancel Defaults Help

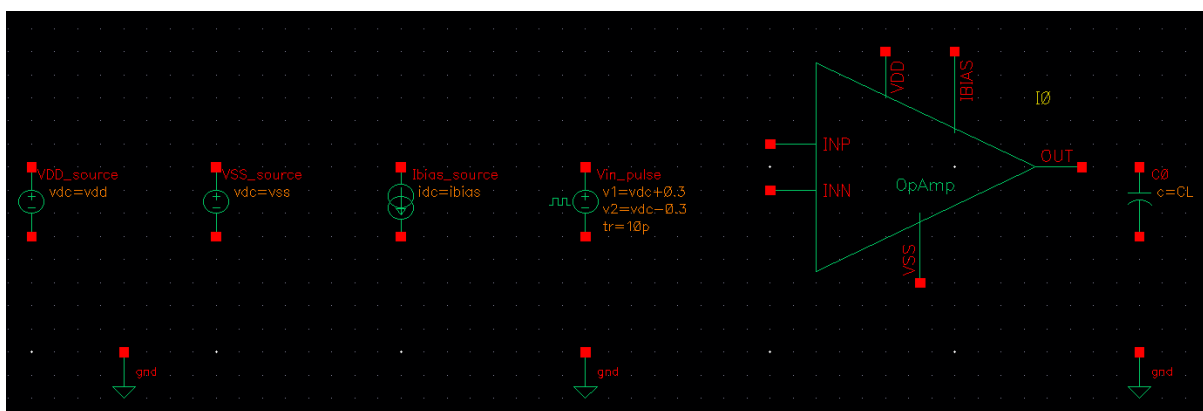
The last component to insert is the voltage reference **gnd**.

Action 50: Fill in the **Add Instance** form (as shown below) and place the component thrice: the first below the DC voltage sources, the second below the **Vin_pulse** source, and the third below the **Cap_Load**. Then, exit the **Add Instance** form by pressing **Esc**.

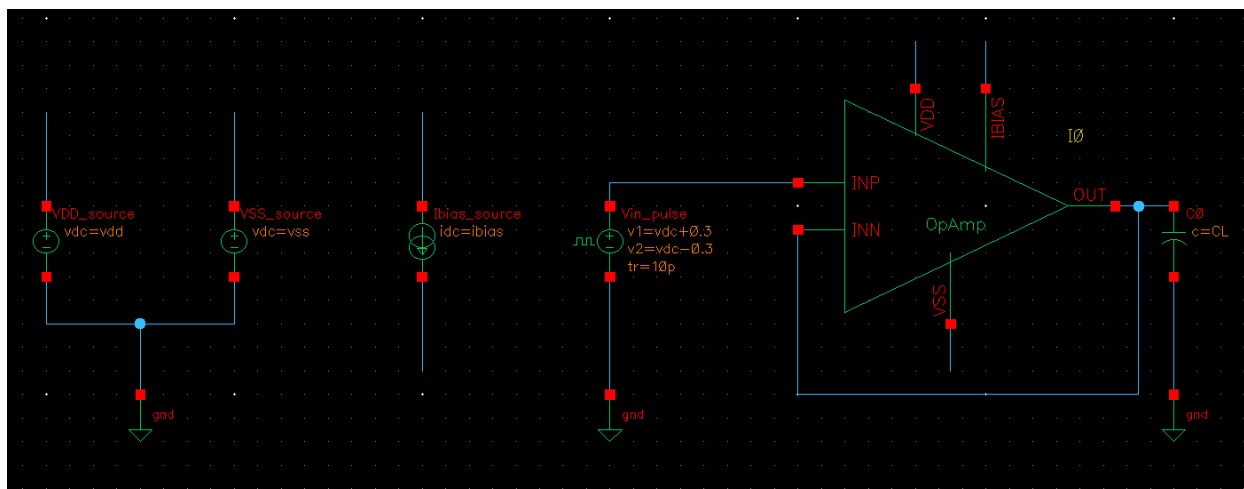
Basics of Analog Flow: RAK



The schematic testbench should look as follows:

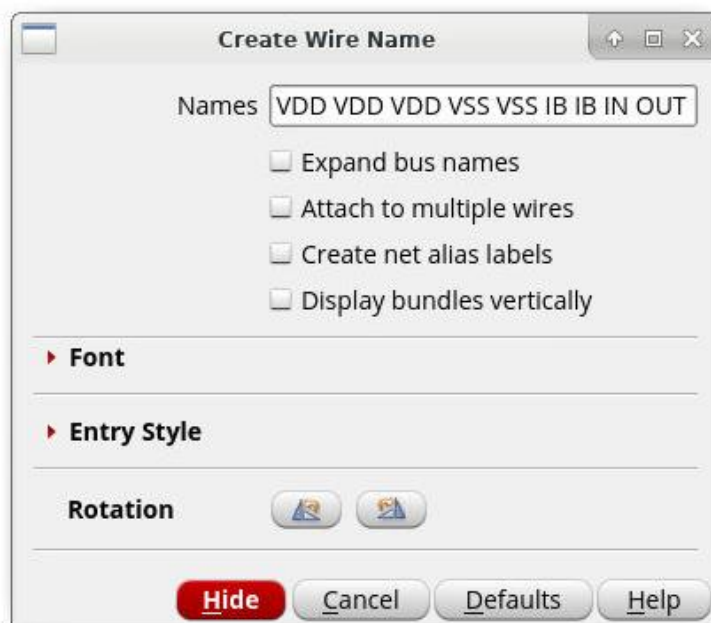


Action 51: Press **w** to invoke the 'wiring' command and route the testbench, as shown below. Press **Esc** after you finish.

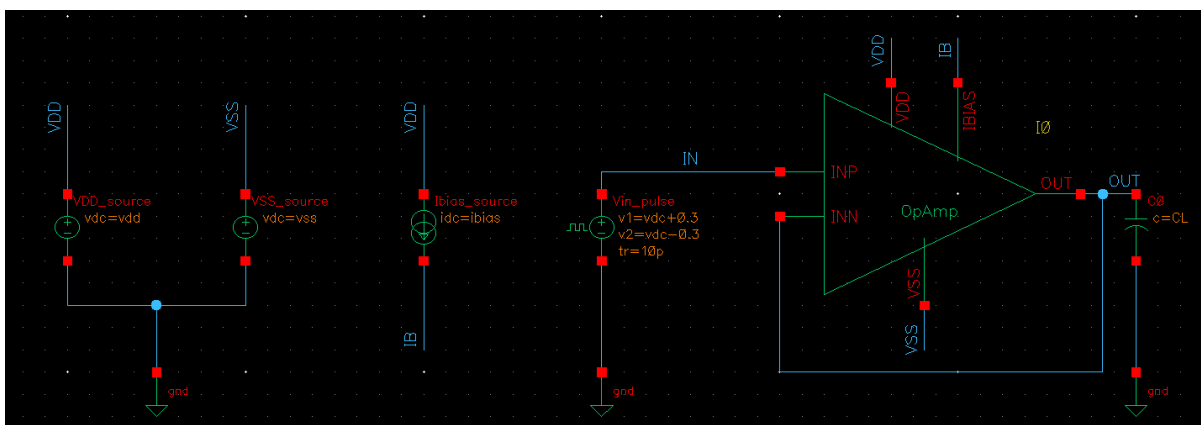


Note that to avoid crossing wires, sources are not wired to the DUT. Alternatively, you can connect by name.

Action 52: Click on the **Create Wire Name** icon  at the top of the toolbar OR press the **I** bindkey to invoke this form. Fill it in, as shown below.



Action 53: Press **Hide** and click on the wires to place each label. Press **Esc** after you finish. This schematic testbench is now complete.

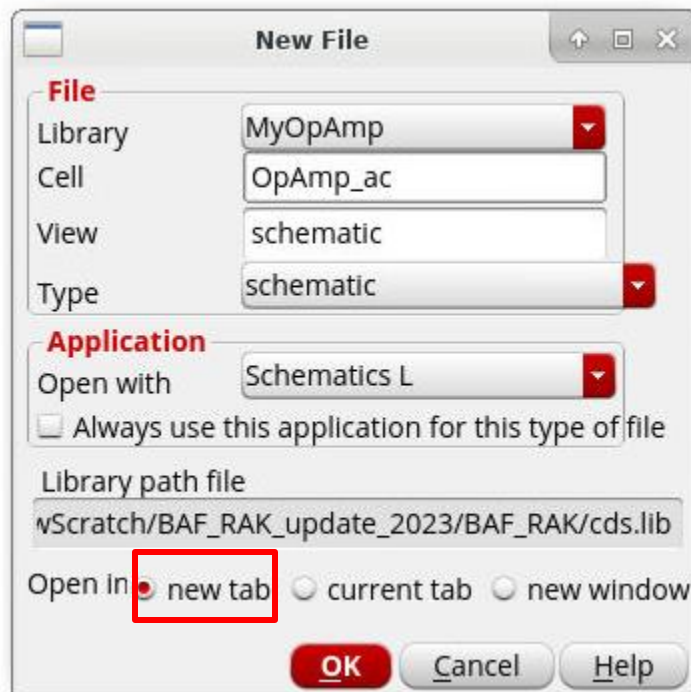


Action 54: Press the **Check and Save** icon . The following information should appear on your CIW to inform you that everything is correct:

```
INFO (SCH-1170): Extracting "OpAmp_tran schematic"  
INFO (SCH-1426): Schematic check completed with no errors.  
INFO (SCH-1181): "MyOpAmp OpAmp_tran schematic" saved.
```

A testbench is required to analyze the open-loop gain and bandwidth. You must create a second cellview for that.

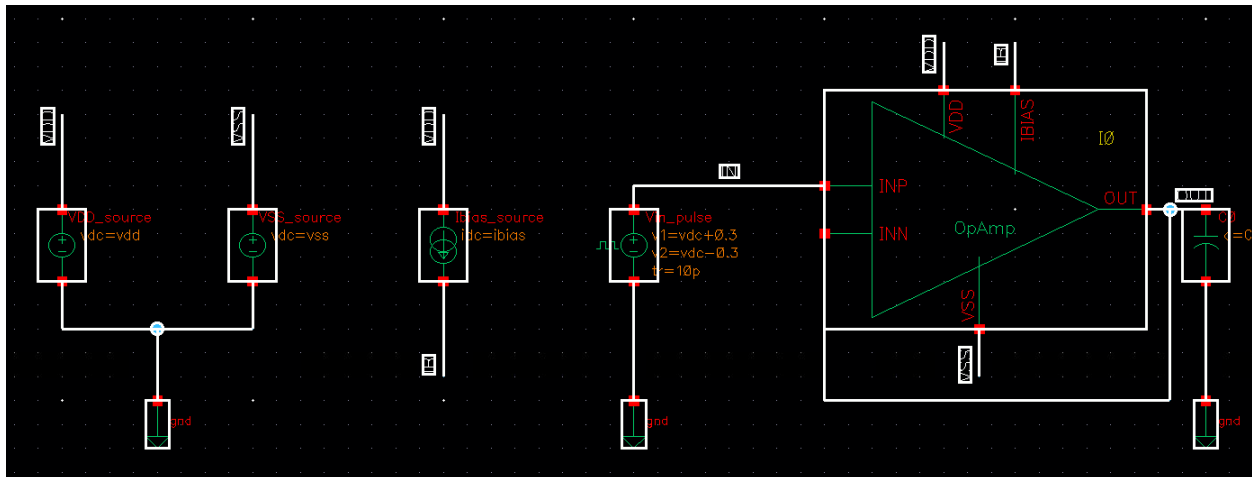
Action 55: Go to **File > New** in the **Schematic Editor Window** and complete the form, as shown below. Ensure that the **new tab** option is selected for **Open in**. Press **OK** when finished.



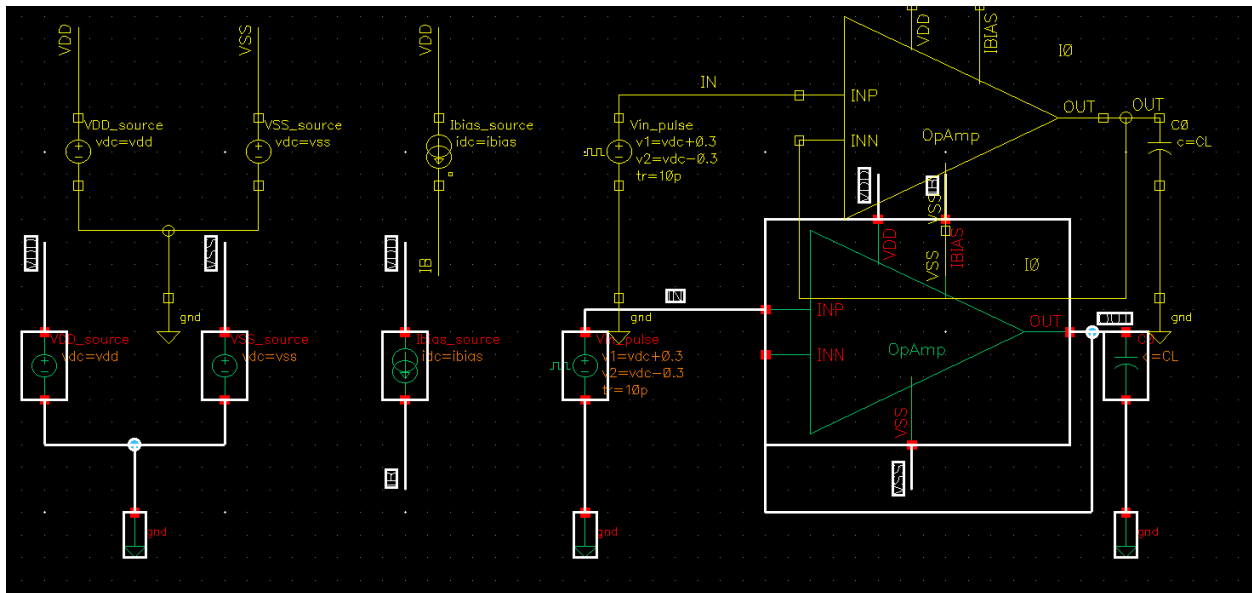
The new view will open in a new tab.

You can easily create the new testbench by copying the **OpAmp_tran** components and making suitable modifications.

Action 56: Go back to the **OpAmp_tran** tab. Press **Ctrl+A** to select the entire design, and then press **C** and click somewhere in the blank space to copy the design.



If you move the mouse around after you click, you will see ghosted images of all objects:



Action 57: Go to the new tab and paste the design by clicking in the free space.

Action 58: Select the `Vin_pulse` source and press `q`. Alter the properties to match the following screenshot:

Edit Object Properties

Apply To: ☐ only current ☒ instance

Show: ☐ system ☒ user ☒ CDF

Browse Reset Instance Labels Display

Property	Value	Display
Library Name	analogLib	off
Cell Name	vsource	off
View Name	symbol	off
Instance Name	Vin_ac	off

Add Delete Modify

User Property	Master Value	Local Value	Display
lvsignore	TRUE		off

parameters Expand Expand

CDF Parameter

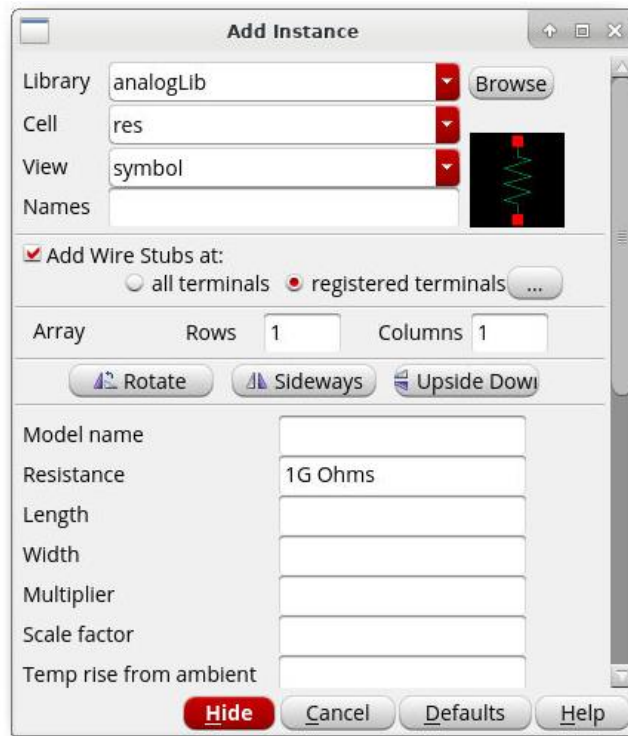
CDF Parameter	Value	Display
DC voltage	vdc V	off
Source type	dc	off
Display small signal param	☒	off
PAC Magnitude (Vpk)		off
PAC phase		off
AC Magnitude (Vpk)	vac V	off
AC phase		off
XF Magnitude (Vpk)		off
Display temperature param	☐	off
Display noise parameters	☐	off
Multiplier		off

OK Cancel Apply Defaults Previous Next Help

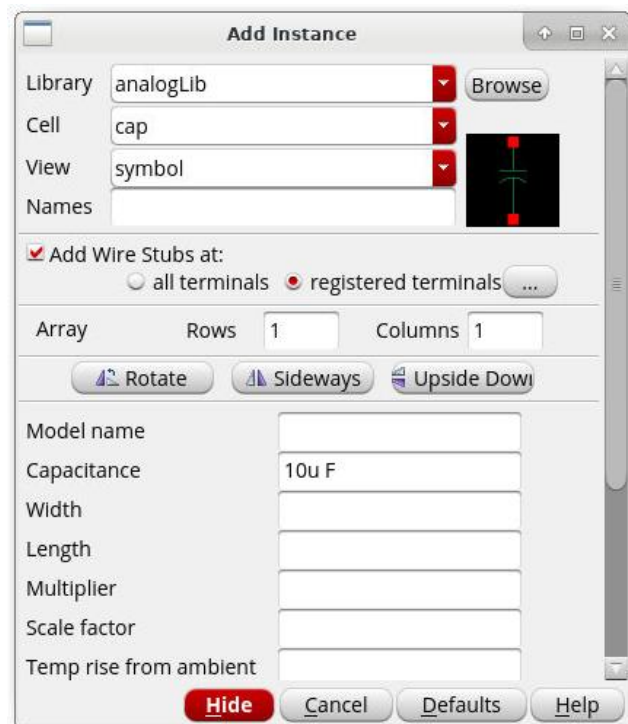
Action 59: Press **OK** when done.

The **vsource** component is a generic source that can be set to several different sources such as dc, sine, pulse, and so on, by changing the **Source type** option.

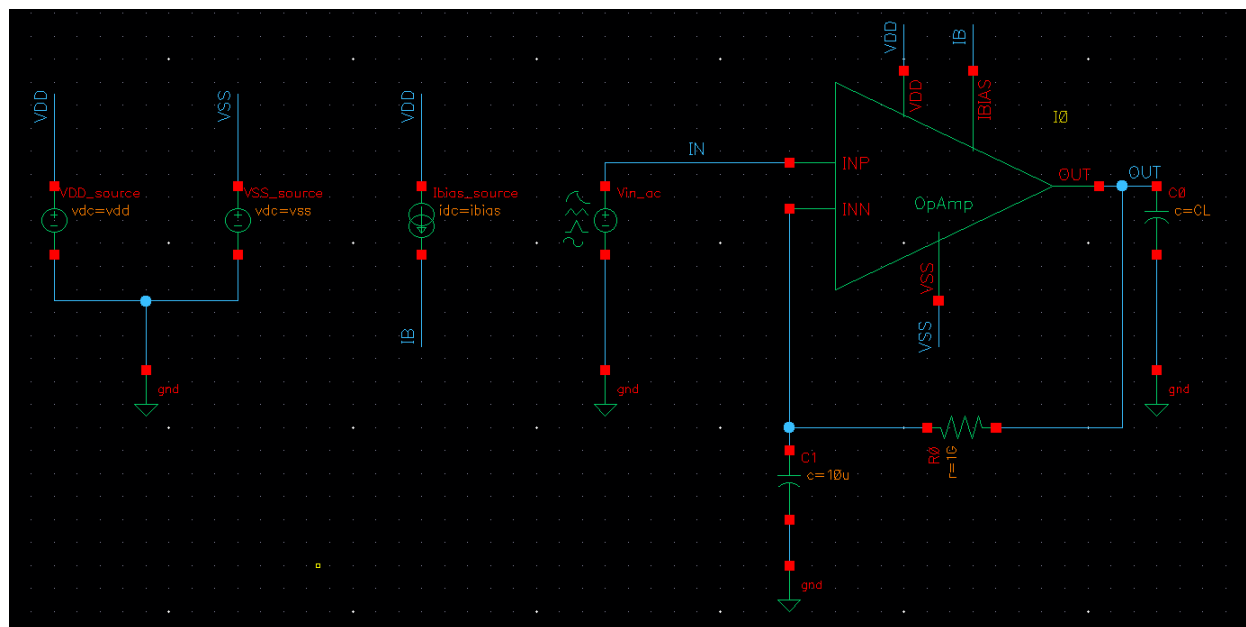
Action 60: Delete the feedback net connecting **INN** to **OUT** and place a resistor in its place with the settings shown below. As you place it, you can rotate it using the icons on the form.



Action 61: Place a capacitor from the **INN** pin to the ground. Now, add a ground component for the capacitor.



The second testbench should look as follows:



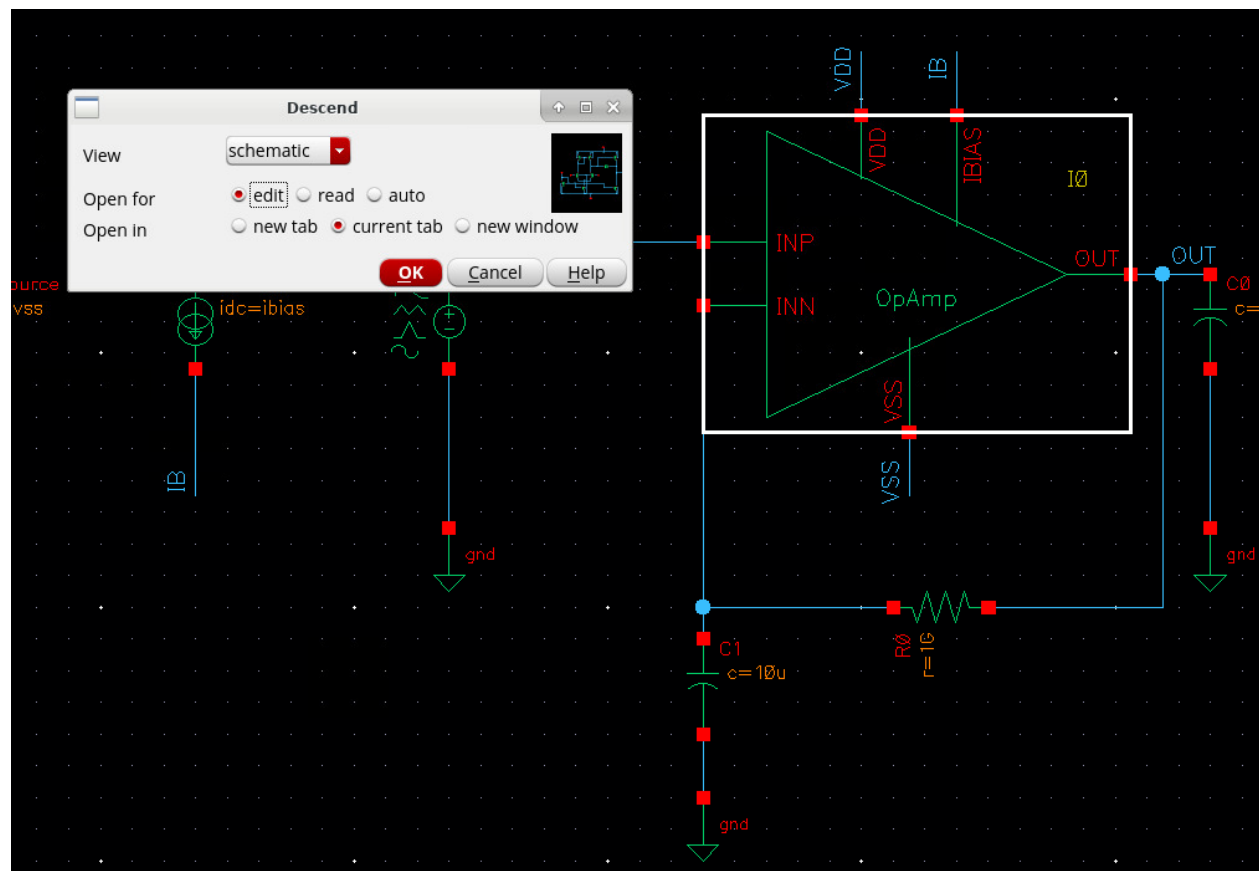
Action 62: Check and Save the schematic testbench. No warnings or errors should appear.

Lab 6: Modifying DUT Parameters

Observe that you have not yet set the width (W) and length (L) values for the NMOS and PMOS transistors of your op-amp, so they all are set to their default values.

In this section, you will use the **Navigator** and the **Property Editor** assistants to set these values.

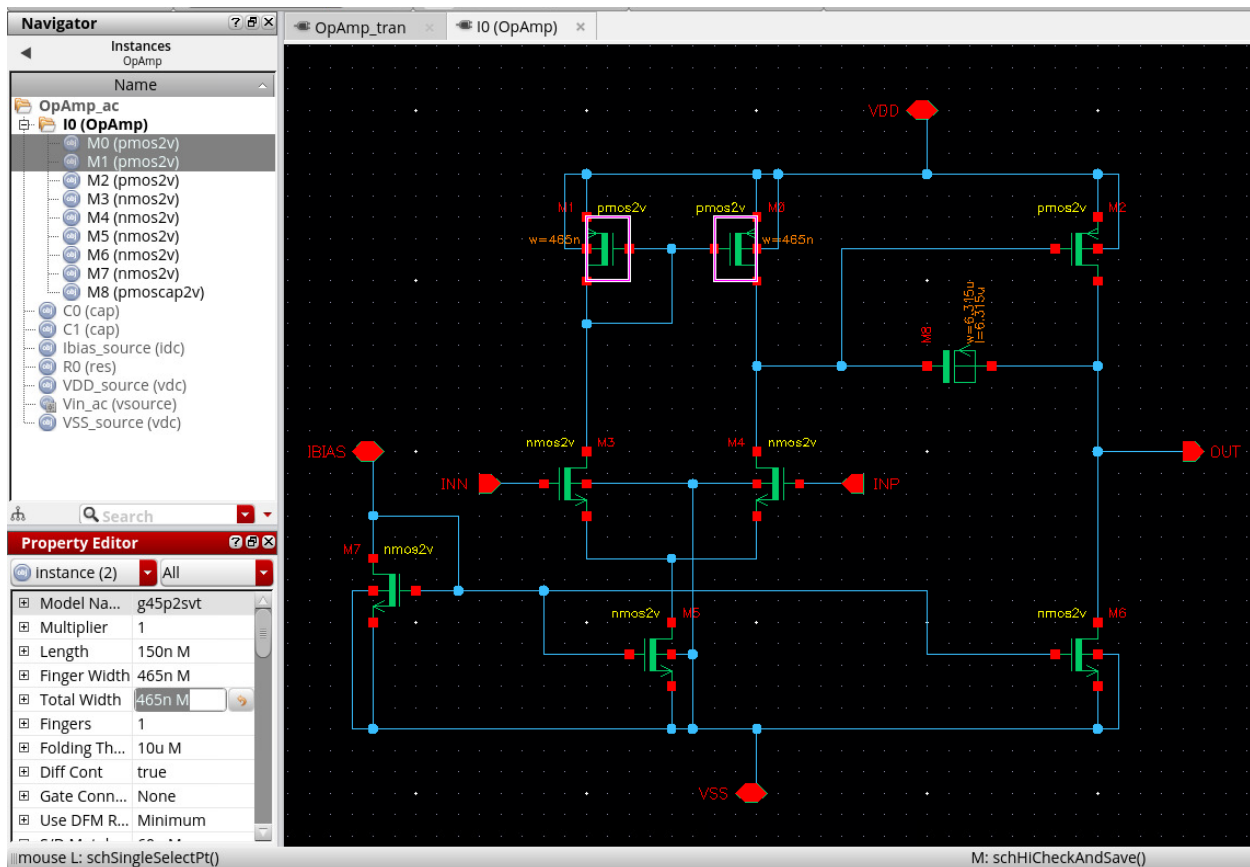
Action 63: In the **OpAmp_ac** schematic testbench, double-click the DUT symbol. In the **Descend** dialog box, make sure **schematic** is selected for **View**. Choose the **edit** option in **Open for** and click **OK**.



The OpAmp schematic is opened. In the left pane, you should see that both the **Navigator** and the **Property Editor** assistants are displayed. If not, invoke them individually by selecting **Window > Assistants** from the banner menu.

You will select devices using the **Navigator** assistant and edit parameters with the **Property Editor** assistant. The Navigator also displays nets and pins, making it a convenient way to locate items in a schematic.

Action 64: In the **Navigator** assistant, select the **Instances** category (**Object**). Select **M0** and **M1** using the **Ctrl** key for multiple selections. Search for **Total Width** on the **Property Editor** and change the value to **465n**. Press **Enter**.



Note: Do not write “m” for meters because it is automatically set after you press **Enter**.

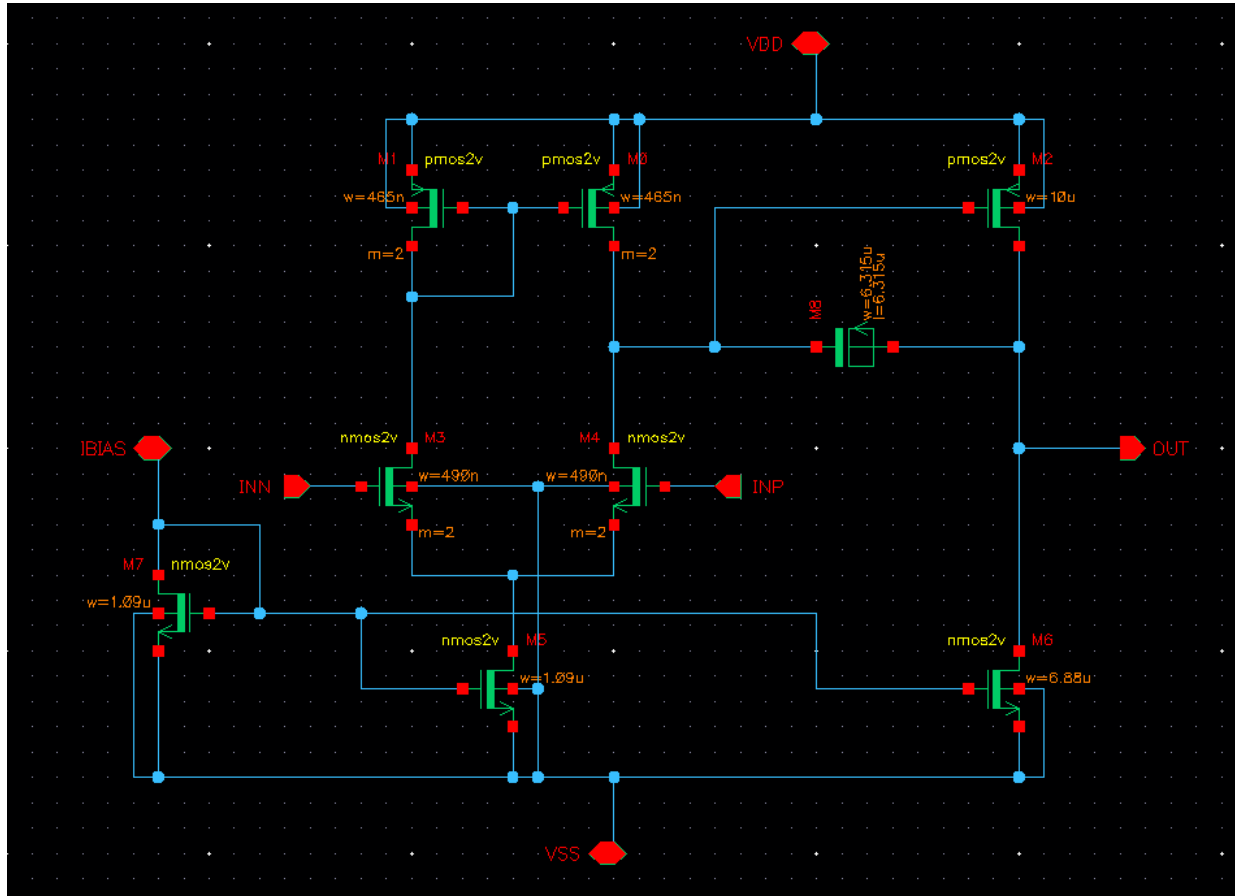
Since both transistors (M0 and M1) were selected, width values will be changed for both at once. Repeat this process for the other MOSFET elements, as shown below:

- M2 Total Width: 10u
- M3 Total Width: 490n
- M4 Total Width: 490n
- M5 Total Width: 1.09u
- M7 Total Width: 1.09u
- M6 Total Width: 6.88u

Note: For all transistors, the value of the **Fingers** field should be **1**.

Action 65: Select M0, M1, M3, and M4 and change their **Multiplier** value to 2.

After you finish, your schematic should look as follows:



Your schematic may differ in the labels that are displayed. Just ensure that the **w** and **m** values are correct.

Action 66: Click on **Check and Save** to save your design. No warnings should be displayed.

Action 67: Click on the up arrow  on the toolbar in the top-left corner to return to the top-level testbench. Alternatively, you can right-click on an empty space of the DUT schematic and select **Return** (ensure nothing is selected).

The AC schematic testbench view will appear again.

Action 68: Close this view by selecting the 'x' icon in the tab. You should still have the **OpAmp_tran/schematic** view displayed.

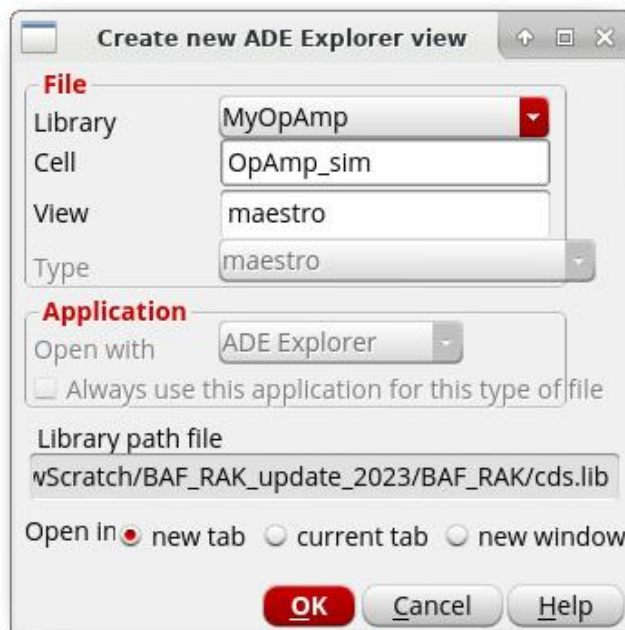
Module 4: Pre-Layout Simulation

Once the schematic testbenches are completed, you can evaluate the design and its basic functionality. You will calculate the slew rate, open-loop gain, gain bandwidth, DC offset, and settling time. The specifications for each of those output measurements are given below:

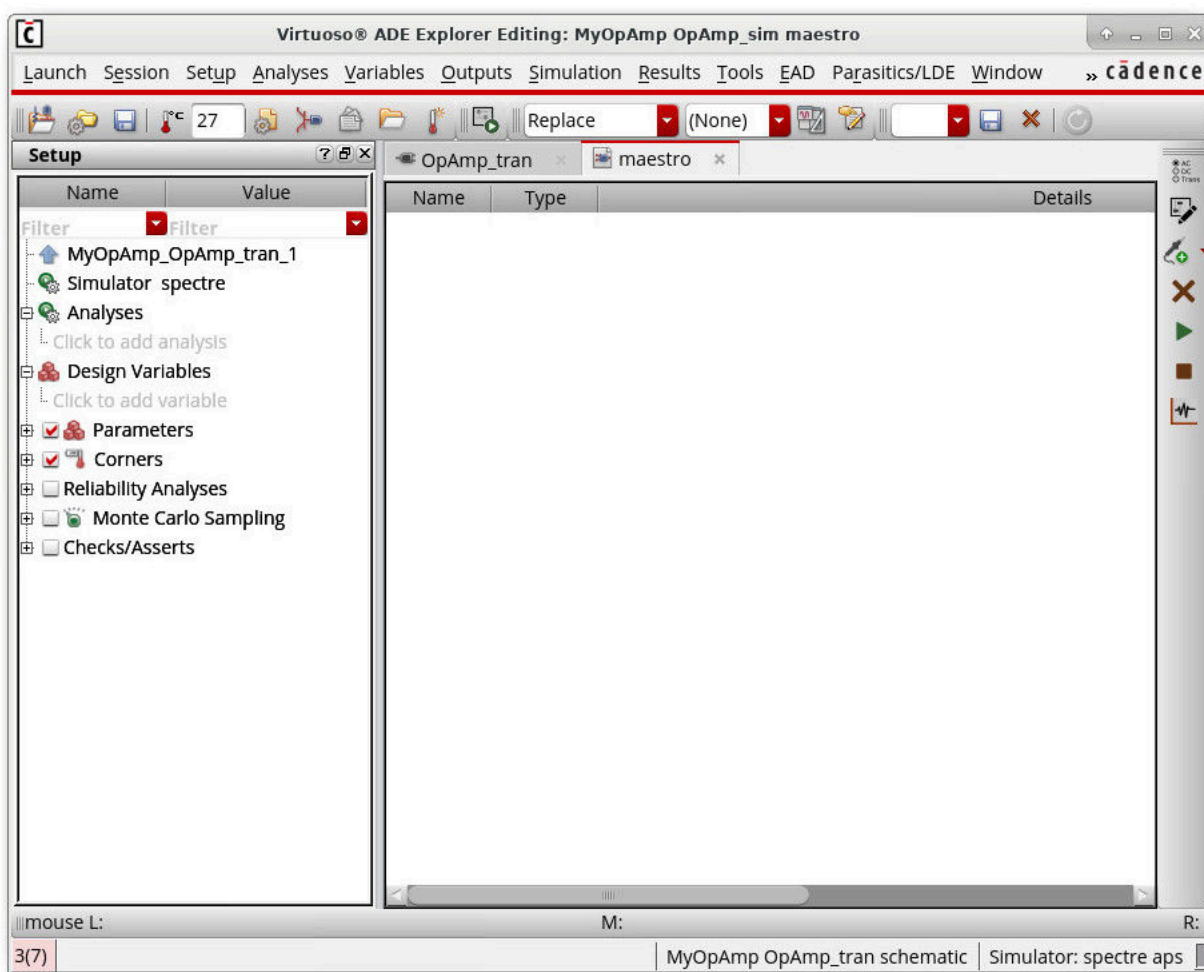
- Slew Rate ≥ 50 MV/s
- DC Open-Loop Gain ≥ 60 dB (1000V/V)
- Unity-Gain Bandwidth ≥ 50 MHz
- Output Offset $\leq \pm 10$ mV
- Settling Time ≤ 50 ns

Lab 7: Simulation Setup Using ADE Explorer and Assembler

Action 69: In the **OpAmp_tran** schematic, select **Launch > ADE Explorer**. Select **Create New View** and click **OK**. Change the **Cell** name in **Create new ADE Explorer view** to **OpAmp_sim** and click **OK** again.

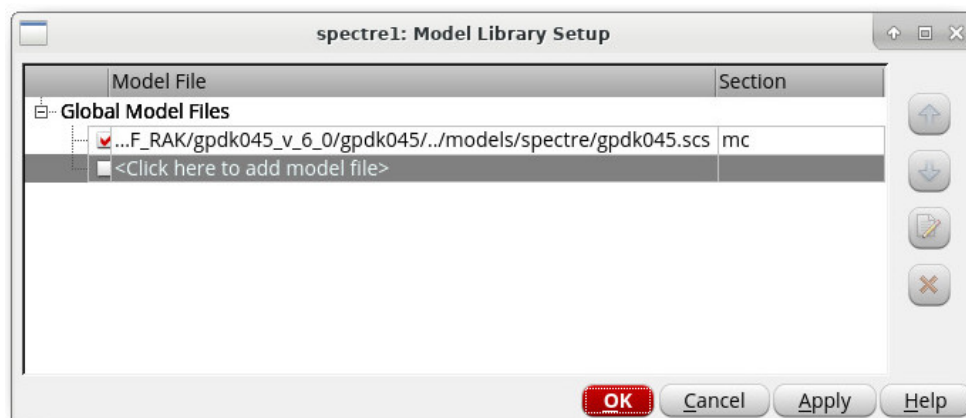


This will create and open a new **maestro** view, which is used by the **ADE Explorer** and **Assembler** simulation environments.



You will first set up PDK model library paths.

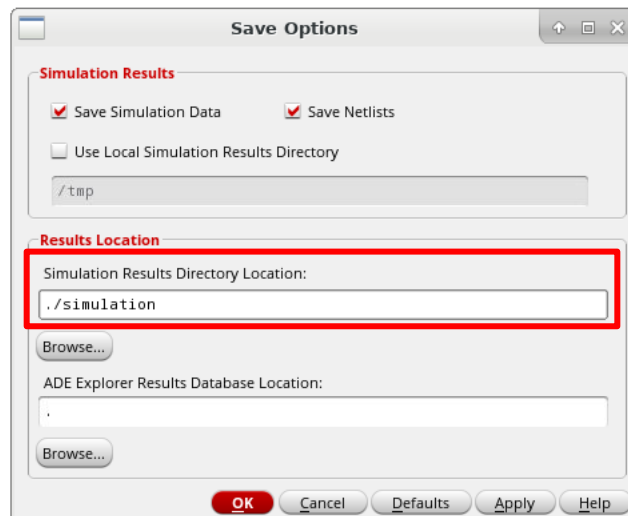
Action 70: Go to **Setup > Model Libraries**. Verify that a path is set like as follows:



Note: Your exact path will depend on the location where your RAK database is installed.

Action 71: Change **mc** to **tt** in the **Section** column and click **OK**.

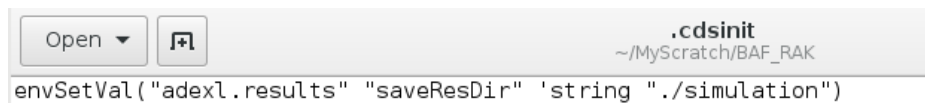
Action 72: Go to **Setup > Save Options** and change **Simulations Results Directory Location** from **~/simulation** to **./simulation**, as shown below.



This will save all your simulation data on a directory called “simulation”, created inside your project directory instead of your home directory. You can set this and other properties as default by using a .cdsinit file.

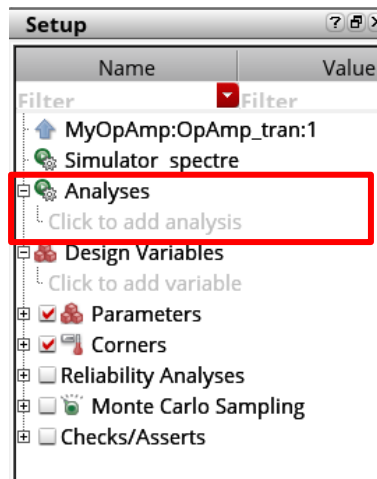
Action 73: In your project directory, create a file called “.cdsinit” and insert the text shown below. Then, save and close the file.

```
envSetVal("adexl.results" "saveResDir" 'string "./simulation")
```



Environment variable settings saved in the .cdsinit file get applied when you launch your Virtuoso session. This way, you will not have to modify this field whenever you invoke Virtuoso in this project directory.

Action 74: In the **Setup** assistant docked on the left side of the window, locate the **Analyses** section.

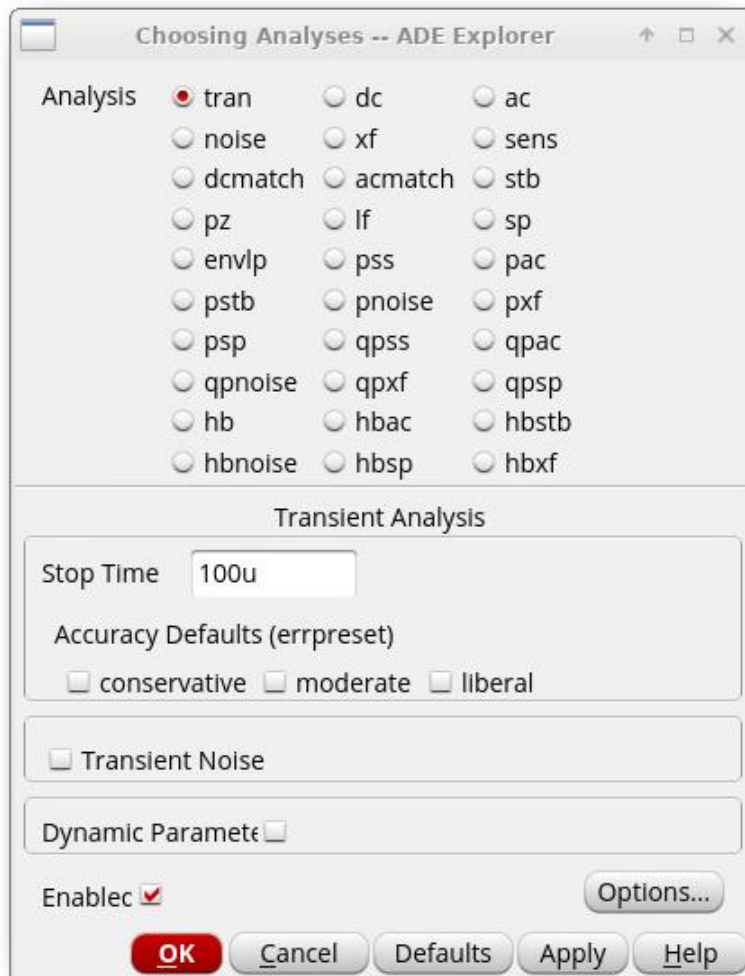


Select **Click to add analysis**. The following dialog box will appear:



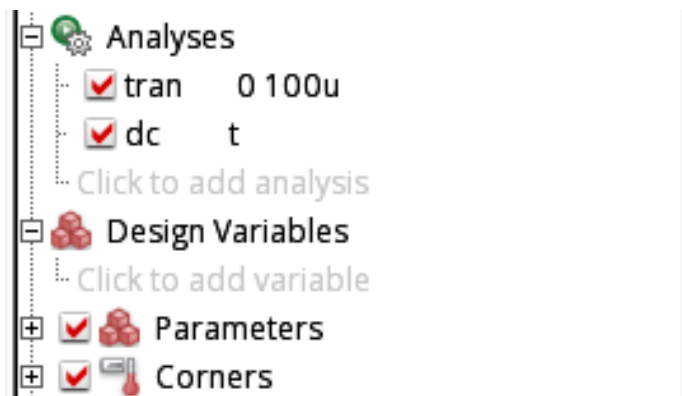
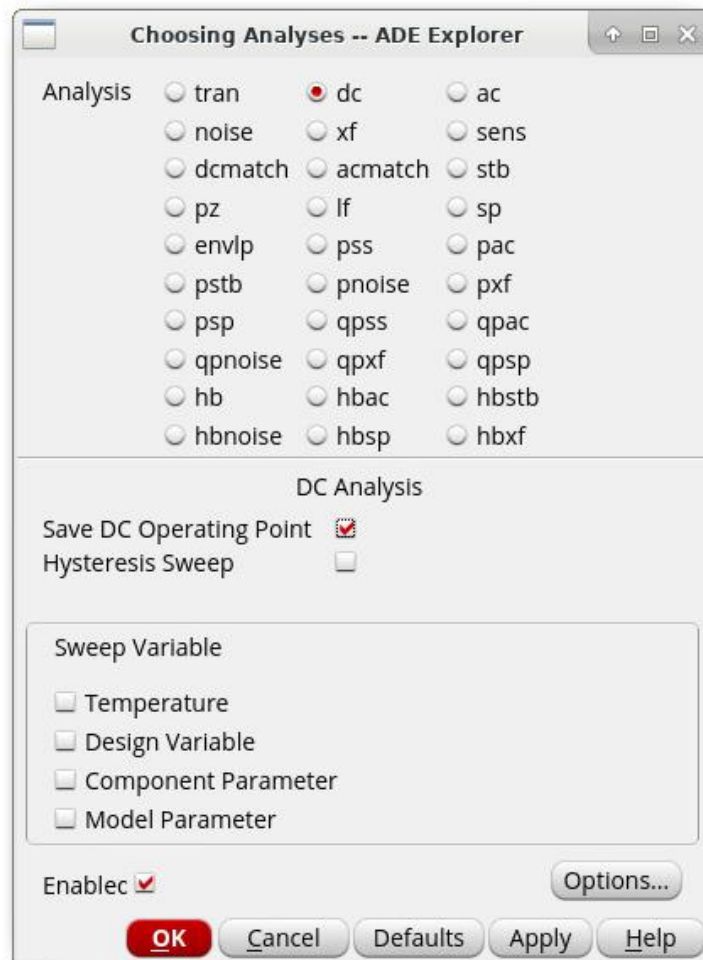
You will specify two analyses: **dc** and **tran**.

Action 75: Leaving **tran** as the analysis, set **Stop Time** as **100u** and click **Apply**.



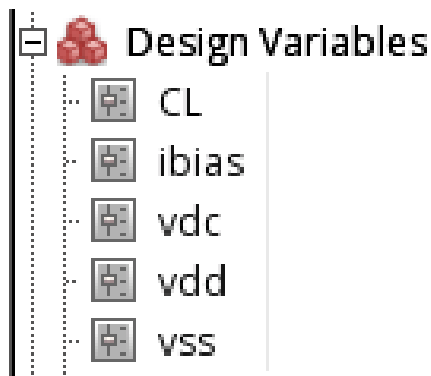
Now, add the dc analysis. The **tran** information will not be lost if you change the analysis type.

Action 76: Select **dc** from the list of **Analysis**, check the **Save DC Operating Point** checkbox, and click **OK**.



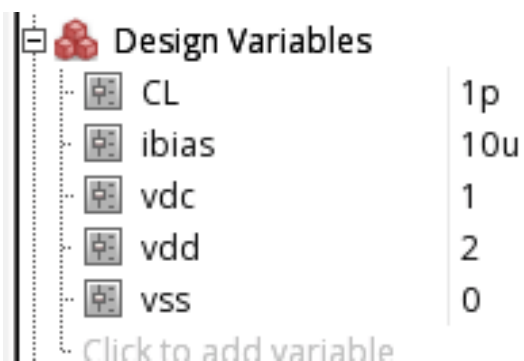
The **Setup** assistant shows the **Analyses**. The next step is to add **Design Variables**.

Action 77: Go to **Variables > Copy From Cellview**.



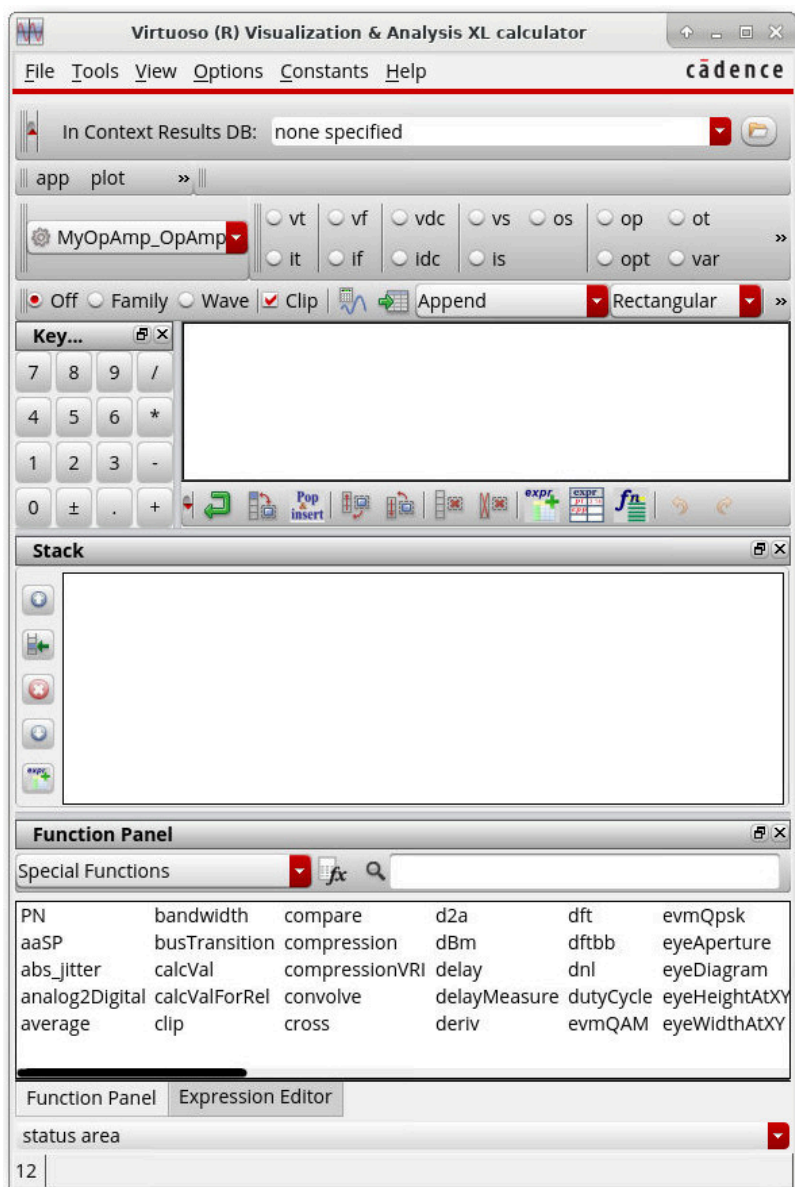
This scans the schematic view for string parameters and imports them into the **Explorer Setup** assistant.

Action 78: Click twice in the **Value** column of the **Design Variables** field to set each value, as shown below. Alternatively, use “Tab” shortcut to go to the next variable, after selecting the first field.

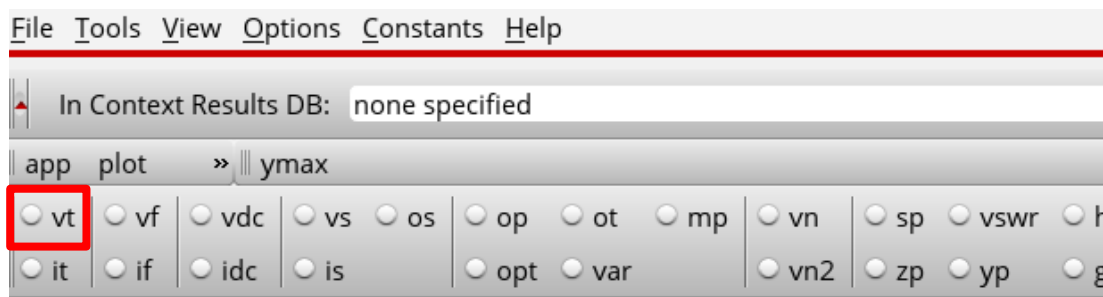


Next, you will specify the outputs for the simulation.

Action 79: Go to **Tools > Calculator**.



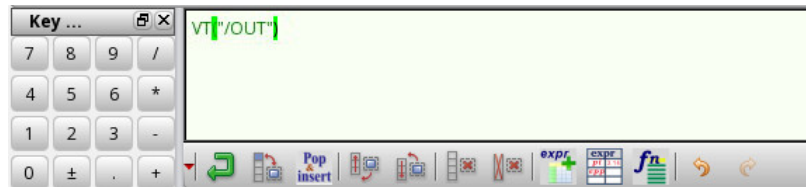
Action 80: Click on the **vt** boolean in the upper-left portion of the calculator.



This will open the schematic so that you can select a signal.

Action 81: Select the output net (wire) **OUT** in the schematic. Be careful not to select the terminals connecting to the wire.

The following expression will appear in the buffer of the calculator:



Action 82: Click the  icon to send this expression to the **ADE Outputs** pane. Move the calculator to a side and verify that you see the output added to ADE:

maestro x OpAmp_tran		
Name	Type	
	expr	VT("/OUT")

Action 83: In the **Name** column, double-click and enter **Vout_tran**. Press **Enter**.

Name	Type	
Vout_tran	expr	VT("/OUT")

Action 84: In the **Calculator**, click again on **vt** and select the net **IN**. Click on the  icon again and name the expression **Vin_tran**.

Name	Type	
Vout_tran	expr	VT("/OUT")
Vin_tran	expr	VT("/IN")

Action 85: In the **Calculator**, click on **vdc**. Select the net **OUT** again. Send the expression to the **Outputs** pane and name it **Vout_dc**.

Name	Type	
Vout_tran	expr	VT("/OUT")
Vin_tran	expr	VT("/IN")
Vout_dc	expr	VDC("/OUT")

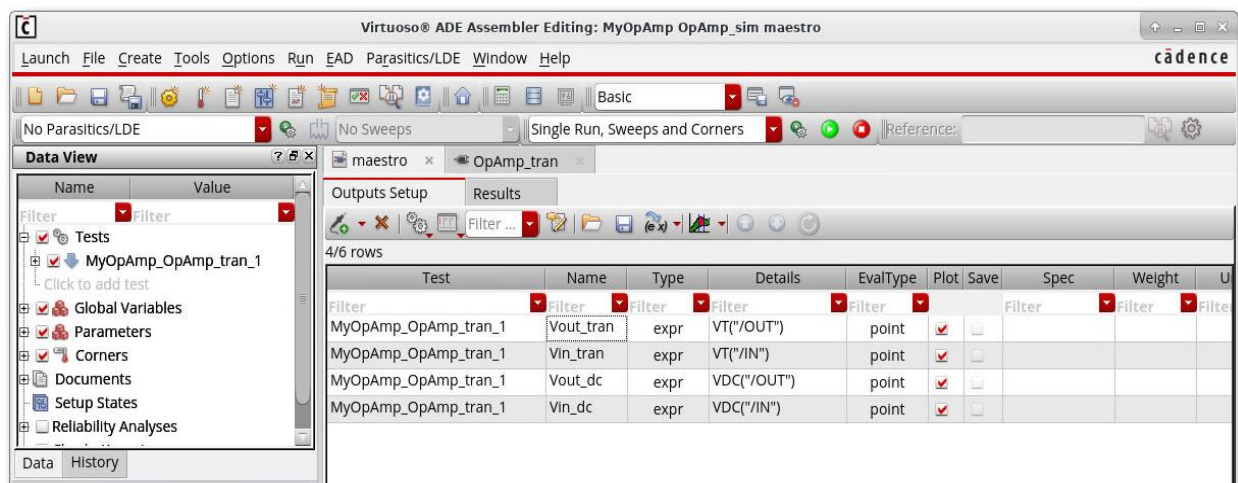
Action 86: In the **Calculator**, click on **vdc**. Select the net **IN**. Send the expression to the **Outputs** pane and name it **Vin_dc**.

Name	Type	
Vout_tran	expr	VT("/OUT")
Vin_tran	expr	VT("/IN")
Vout_dc	expr	VDC("/OUT")
Vin_dc	expr	VDC("/IN")

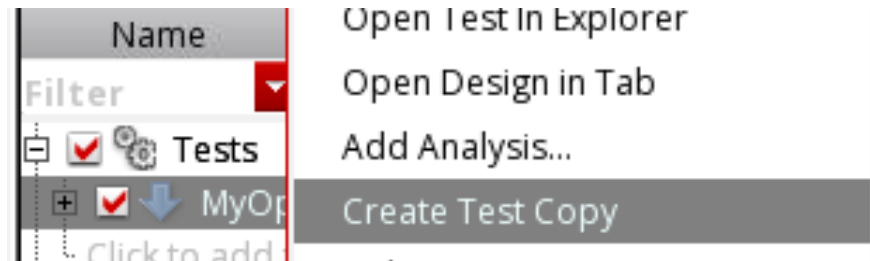
For open-loop gain and bandwidth, you will use the AC testbench view. You will add this second testbench to your setup by using **ADE Assembler**.

Action 87: Click on the up-arrow icon '↑' on the **Setup** assistant next to the testbench name.

This will invoke **ADE Assembler**, which allows multiple tests to be simulated on the same environment.

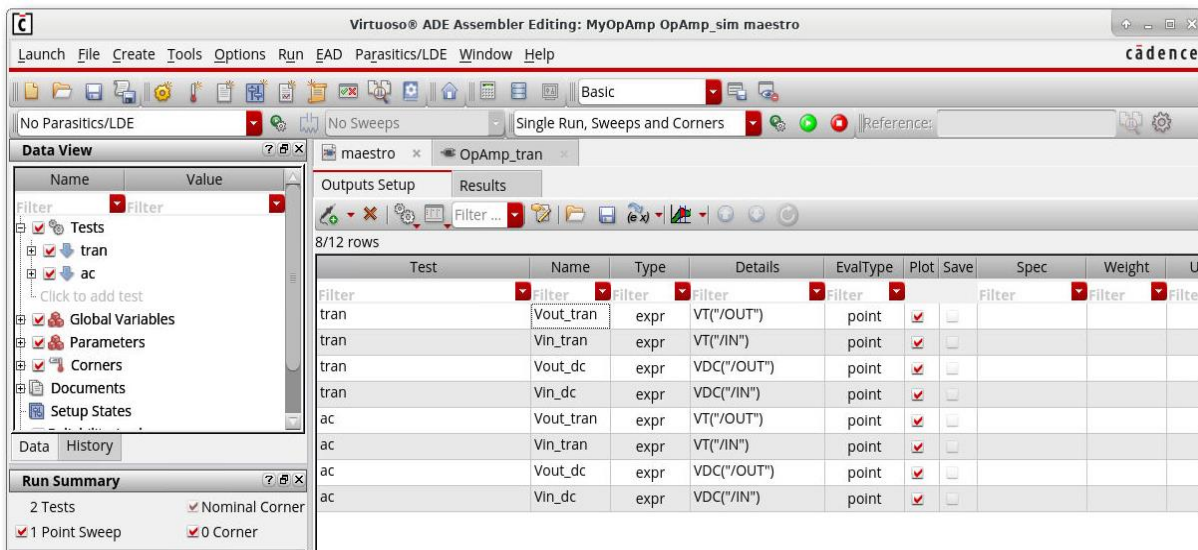


Action 88: Expand **Tests** on the **Data View** assistant. Right-click on the **MyOpAmp:OpAmp_sim:1** test and select **Create Test Copy**.



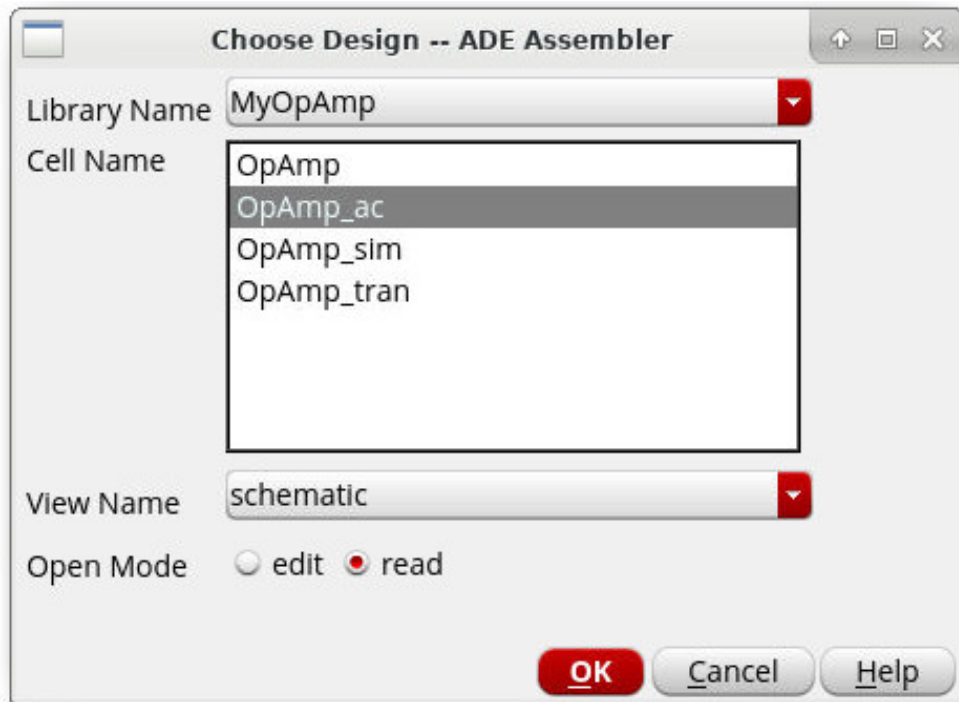
Action 89: Change the name of the first test to **tran** and of the second test to **ac**. To do this, select the test name text and then left-click on the name again. Type the new name in the inline editor.

Your setup should look as follows:



Action 90: Right-click on the **ac** test and select **Design**.

Action 91: In the **Choose Design** form, select **OpAmp_ac** and press **OK**.

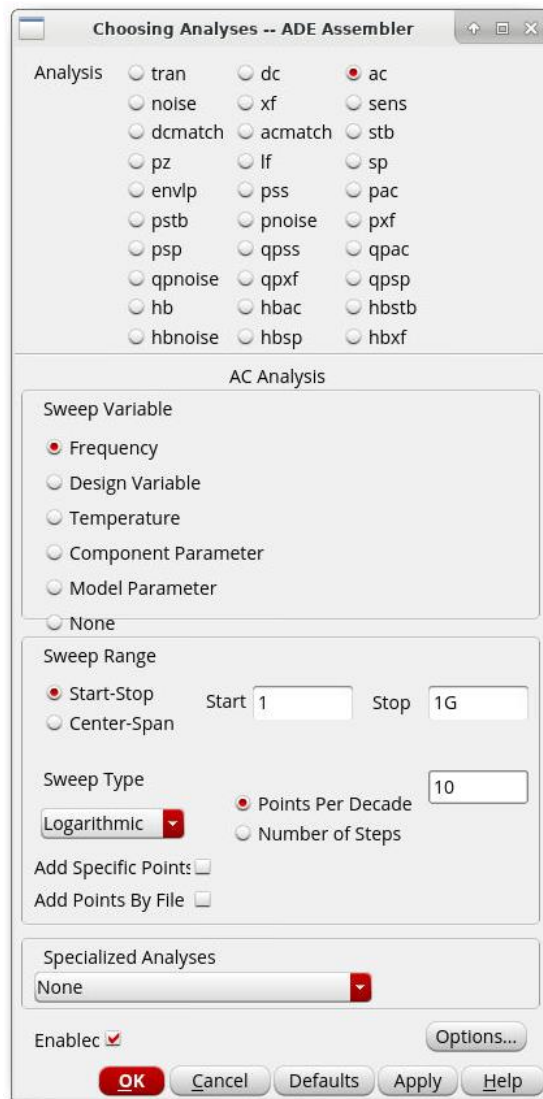


This changes the testbench to the OpAmp_ac schematic for the 'ac' test.

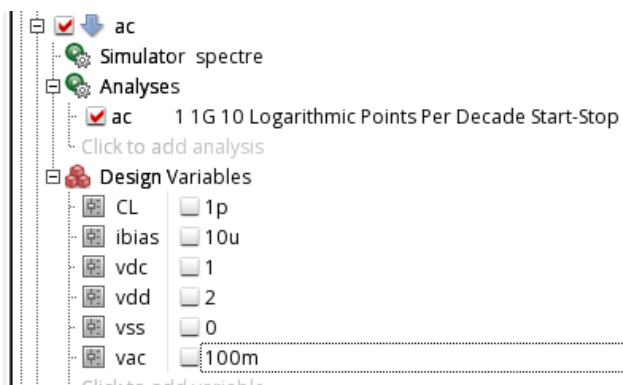
Action 92: Select all expressions in the **Outputs Setup** corresponding to the 'ac' test and delete them. Use the **Shift** key to select multiple items at once, right-click, and select **Delete Output**.

Action 93: Expand the **ac** test tree in the **Data View Assistant**, expand **Analyses**, and delete the two analyses that were copied from the 'tran' testbench.

Action 94: Add the following **Analysis** to the **ac** test:



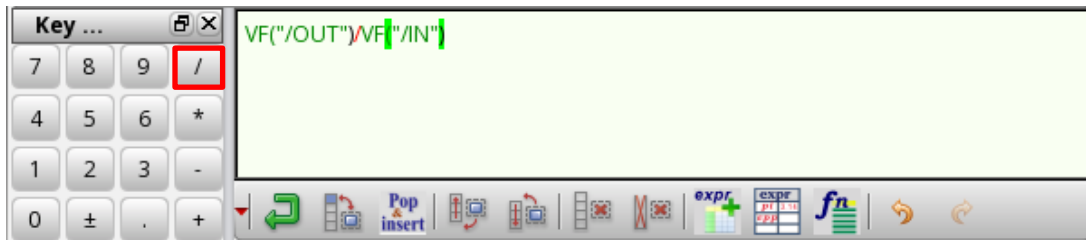
Action 95: Expand the **Design Variables** field and add the variable **vac** with a value of **100m**.



Action 96: In the **Calculator**, select **ac** as the test.



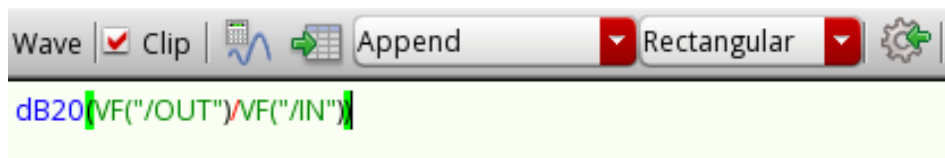
Action 100: Now, click the '/' button on the left, as shown in the following screenshot.



Action 101: Go to the **Function Panel** in the lower portion of the calculator. Select **All** from the drop-down box (if it is not already selected) and enter **db20** in the search field, as shown below.



Action 102: Select the **dB20** function.



This will calculate the gain across the ac frequency sweep, plotting the results in dB.

Action 103: Import this expression to the **Outputs** pane (as done in the previous section) and name it **Av_ac**.

Test	Name	Type	Details
Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")
tran	Vin_tran	expr	VT("/IN")
tran	Vout_dc	expr	VDC("/OUT")
tran	Vin_dc	expr	VDC("/IN")
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))

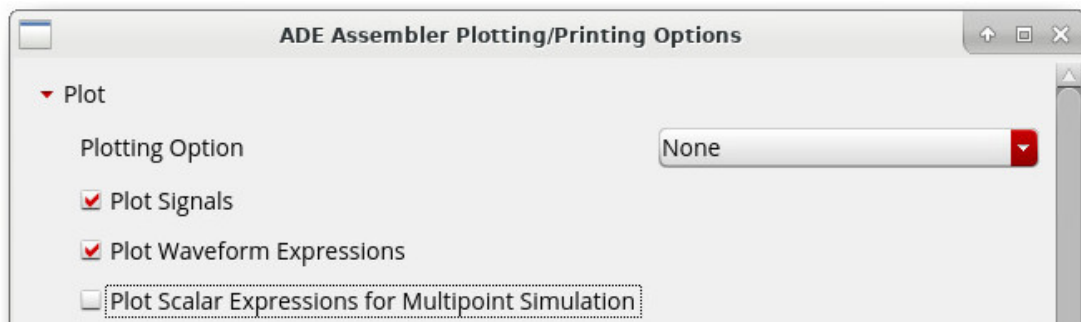
Action 104: Press the green icon '▶' on the top toolbar to start the simulation. You will see a UI update with a simulation status.

Lab 8: Waveform Results Analysis

After the simulation is finished, the **Results** tab will look like as shown below.

Test	Output	Nominal	
tran	Vout_tran	998.8m	
tran	Vin_tran	1	
tran	Vout_dc	998.8m	
tran	Vin_dc	1	
ac	Av_ac		

Action 105: Go to **Options > Plotting/Printing**. Deselect the **Plot Scalar Expressions for Multipoint Simulation** checkbox and click **OK** on the form.



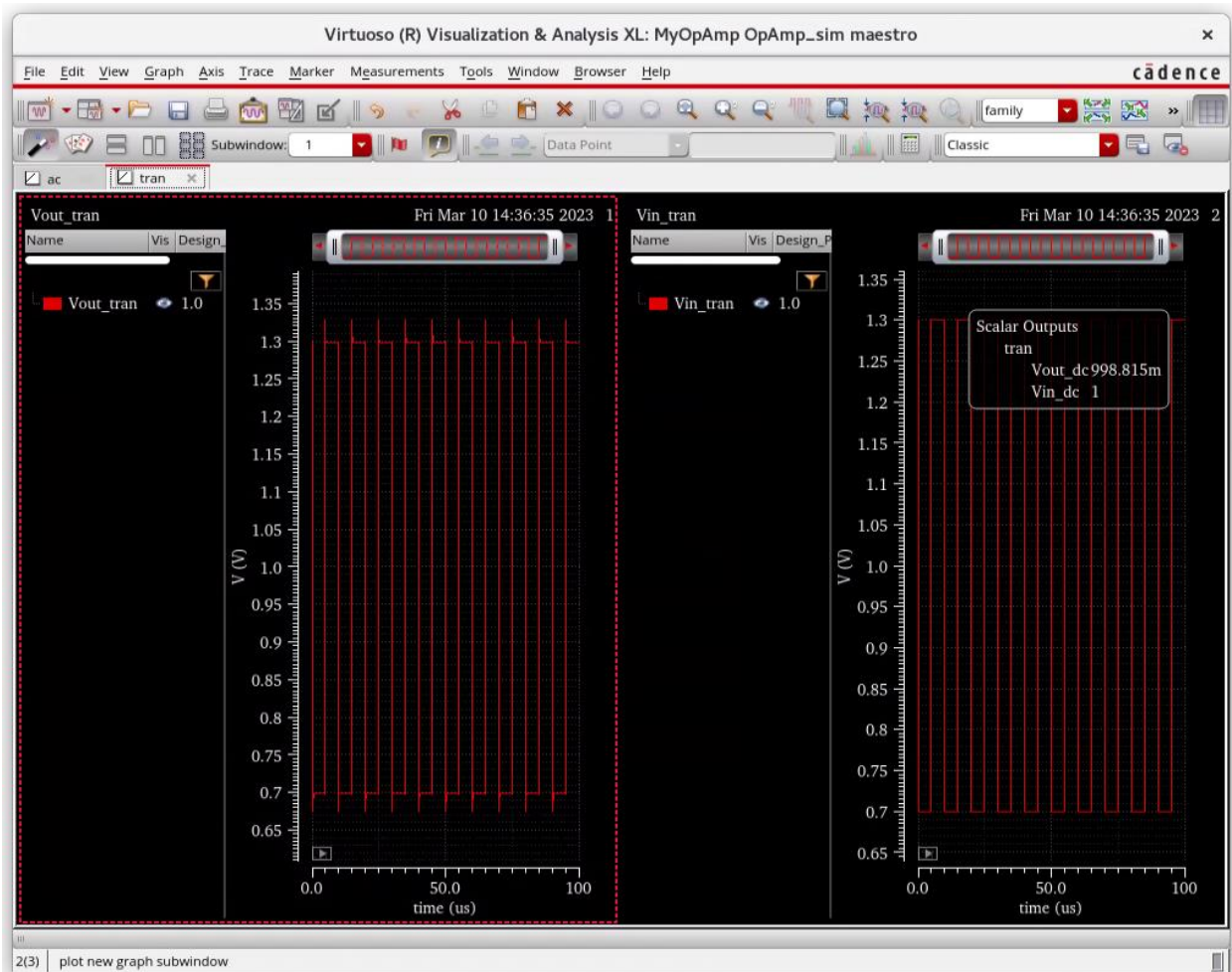
Note: If this option is checked and grayed out, change **Plotting Option** to **Auto**, uncheck the **Plot Scalar Expressions** option, and then change it back to **None**.

Action 106: To see the waveform results, click on the **Plot All** icon.



A ViVA XL (**Virtuoso Visualizations & Analysis XL**) window will appear with results as follows:

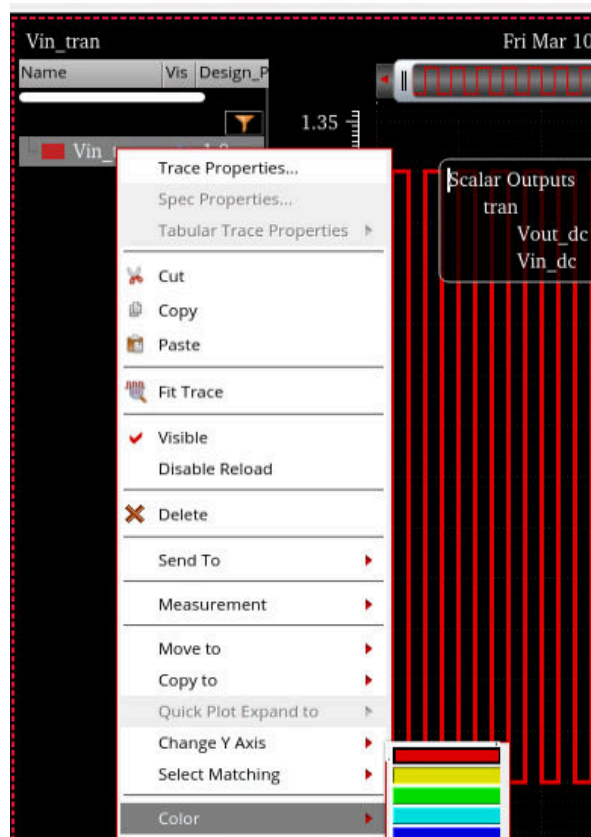
Basics of Analog Flow: RAK



If the ViVA window is docked to the **maestro** view, press the  icon.

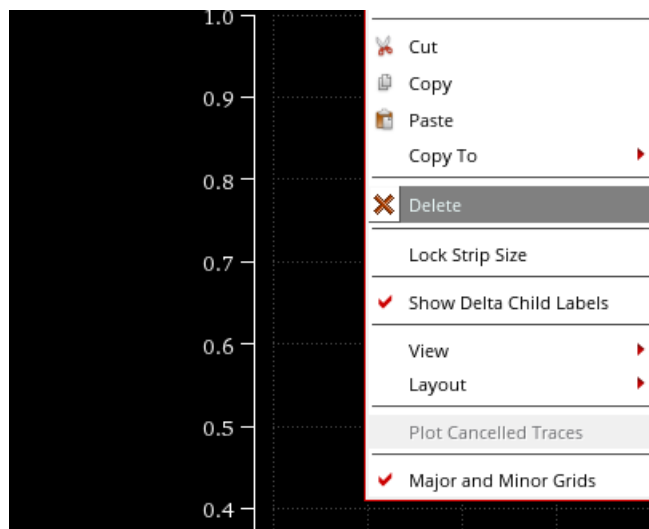
Two subwindows are displaying the **tran** test results. The scalar output values for **Vout_dc** and **Vin_dc** are also informed.

Action 107: Right-click on the **Vin_tran** label on the left side of the waveform and change the **Color** to yellow.

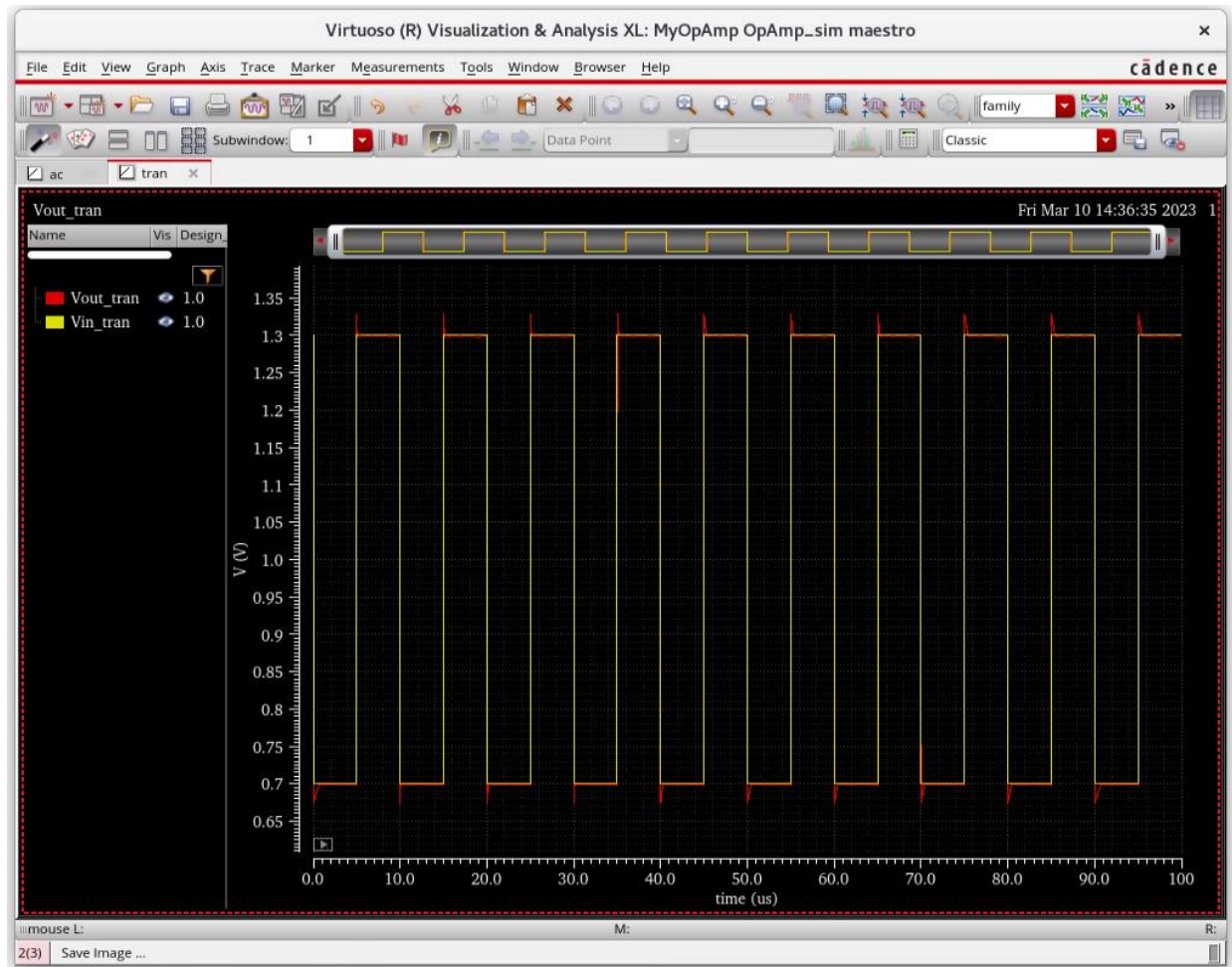


Action 108: Drag and drop **Vin_tran** in the other subwindow.

If the signals are still not overlaid, press the **overlay** icon  found in the upper-right toolbar. Delete the second subwindow by right-clicking on the empty graph and selecting **Delete**.



Basics of Analog Flow: RAK

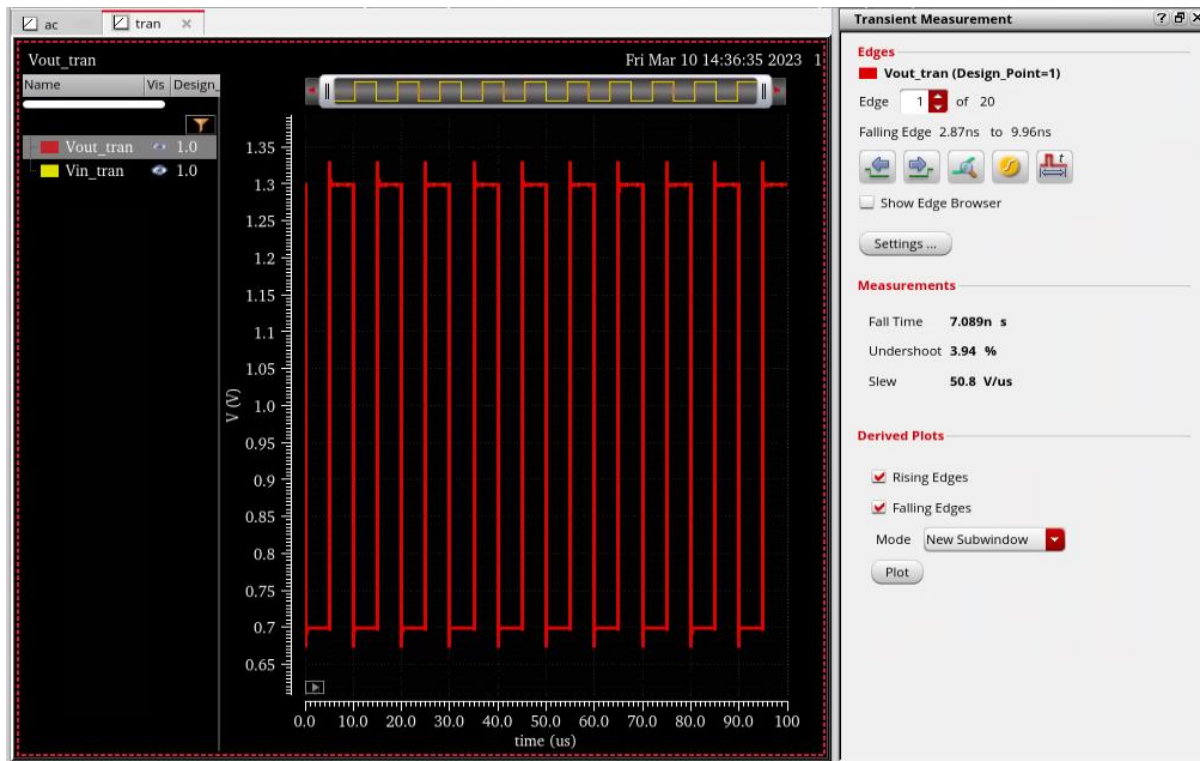


The op-amp works well as a buffer because the output and the input waves are almost equal.

In this section, you will use some manual methods for calculating measurements with signals. This is an introduction to assistants and markers. In a later section, you will create expressions for these measurements. If you wish to skip this section, move to Lab 9.

Action 109: Left-click on the **Vout_tran** legend text to select the signal and go to **Measurements > Transient Measurement**.

The **Transient Measurement** assistant opens, as shown below.



As you can see in the **Measurements** section, the first edge (falling) has a **Slew Rate** value of **50.8 V/us**. You need to find the slowest slew rate value over time.

Action 110: Press the **Edge** selector up arrow to scroll through the other edges. You will see that the smallest value is 50.8V/us. Thus, your worst-case **Slew Rate** is 50.8MV/s.

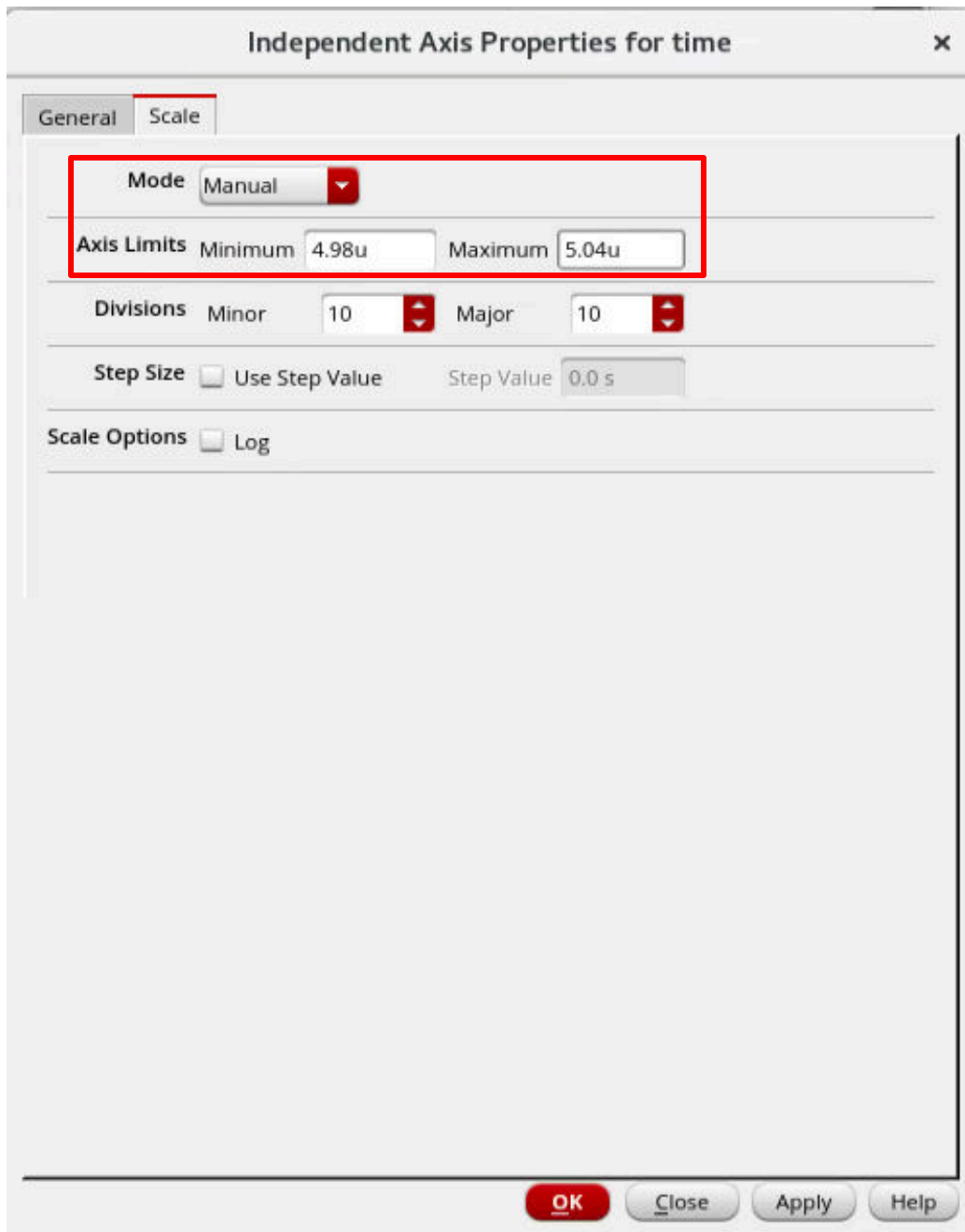
Action 111: Close the **Transient Measurement** assistant by selecting the 'x' icon in the toolbar.

To calculate the settling time, consider the time the wave reaches and stays between $\pm 1\%$ of its final value compared to the initial value. This is calculated as follows:

- Lower bound = $99\% \times (1.3V - 0.7V) + 0.7V = 1.294V$
- Upper bound = $101\% \times (1.3V - 0.7V) + 0.7V = 1.306V$

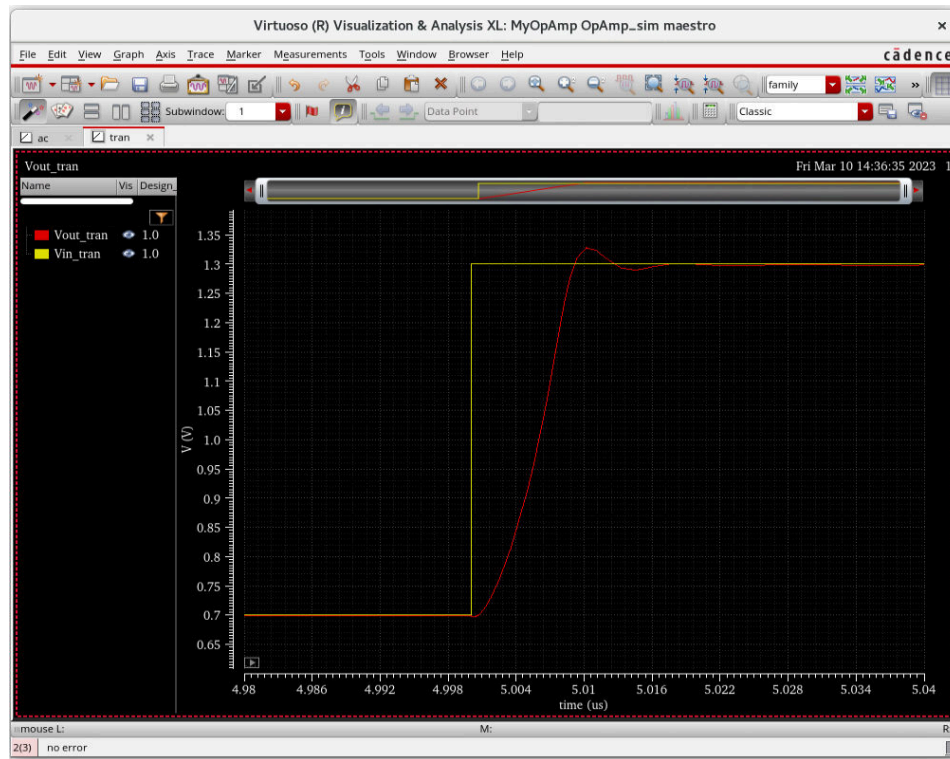
Action 112: Right-click on the x-axis and select **Axis Properties**.

Action 113: Go to the **Scale** tab, change **Mode** to **Manual**, and set **Axis Limits** between **4.98us** and **5.04us**. After that, click **OK**.



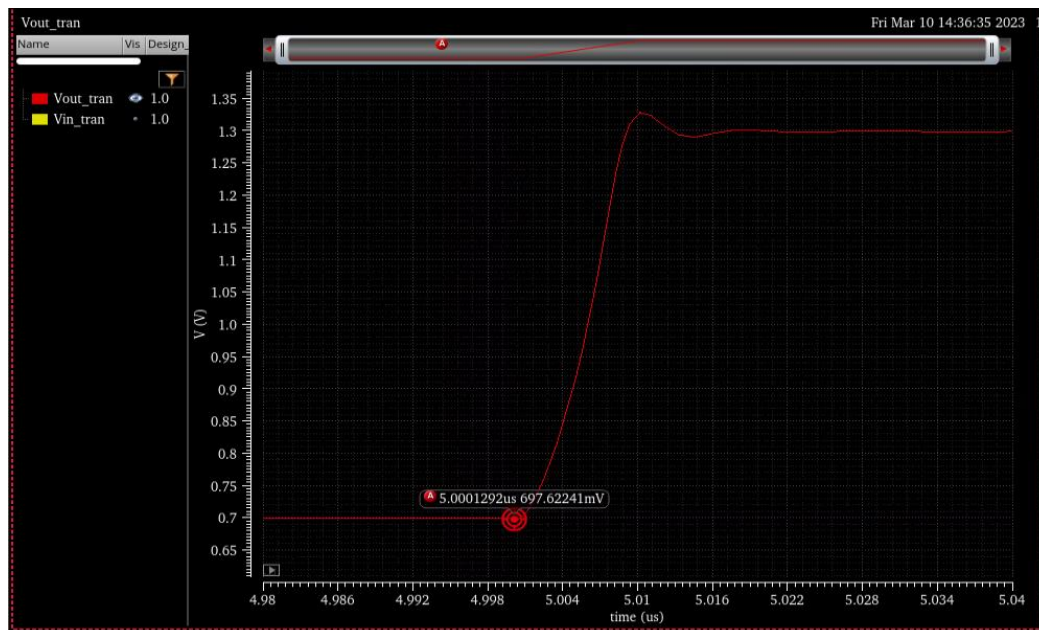
This will isolate the edge that you want to analyze.

Basics of Analog Flow: RAK



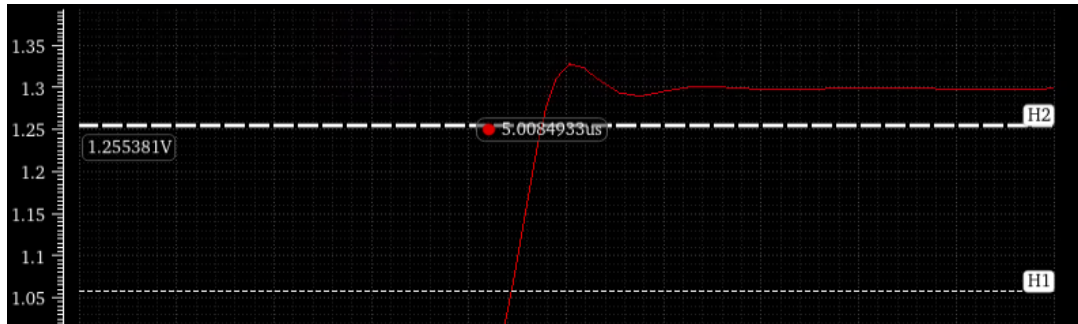
Action 114: Click on the **eye** icon  next to the **Vin_tran** signal to hide it.

Action 115: Hover your mouse close to the point 5us of **Vout_tran** and press **A** to mark the point, as shown below. You can manually adjust the position of the marker.

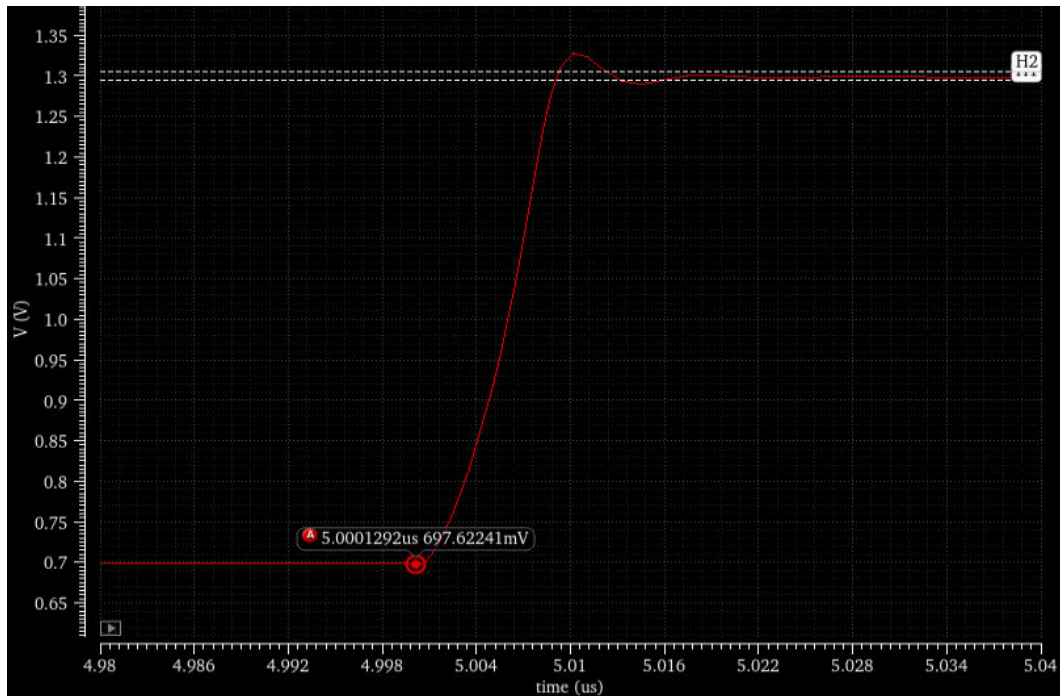


Place a second marker where the signal settles within the bounds calculated above. To make this placement easier, use horizontal markers for guidance.

Action 116: Press **H** close to the **1.055V** level to place a horizontal marker. Move your mouse up and place a second one.



Action 117: Double-click on the lower marker to open its properties. Set **Y Position** to **1.294V**. Repeat for the other marker, setting its **Y Position** to **1.306V**.

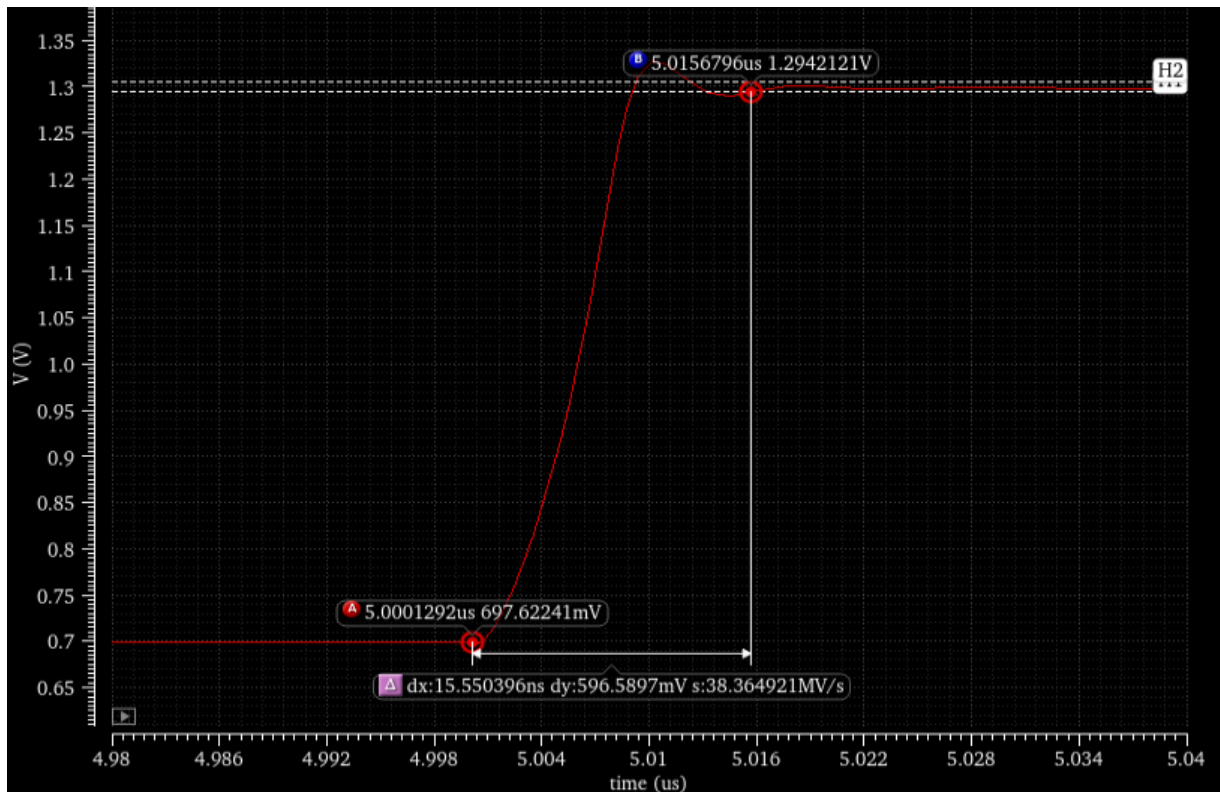


Action 118: Zoom into the region between the markers. You can do that by holding the **Ctrl** key and using the scroll wheel of your mouse.



Action 119: Hover your mouse over the **Vout_tran** signal where the voltage value enters and remains between the horizontal markers (“settling point”). Press **B** to place the second marker. Then, press **f** to see the full signal again.



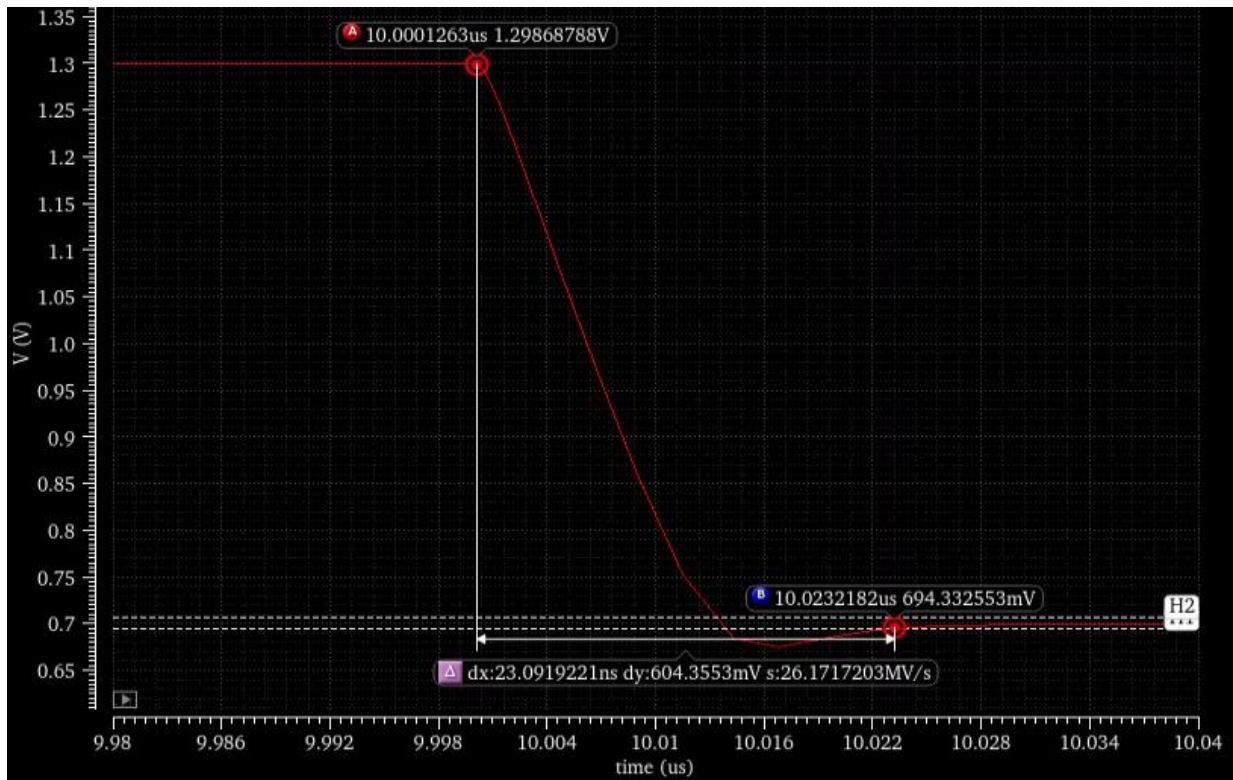


Do not worry if your numbers differ slightly from what is shown here. The **dx** value reports a **Settling Time** of about **15.6ns**.

Action 120: Go to **Marker > Delete All**.

Action 121: Repeat this process with the falling edge of the signal. This time, set the x-axis region between **9.98us** and **10.04us**. Place a marker at **10us**. Place horizontal markers at **694mV** and **706mV**.

Again, place a second marker when the signal enters and stays in the region defined by the horizontal markers.

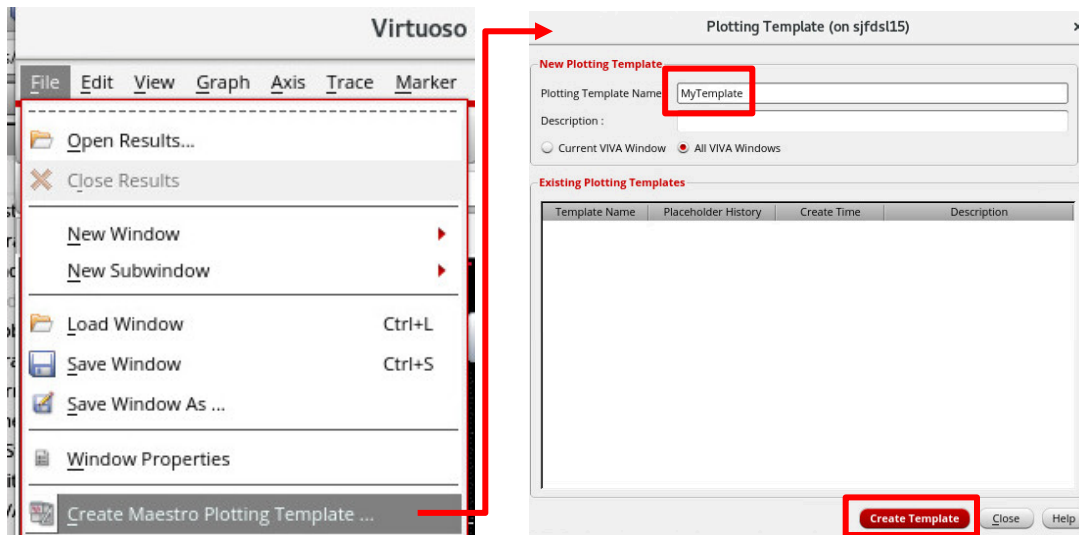


As you can see, the **dx** value is much larger for the falling transition. Therefore, considering the worst case, you can say that your final **Settling Time** is about **23.1ns**.

One interesting feature ViVA allows is to save plotting templates, in order to easily access them when using same (or similar) waveforms. Let's try it now.

Action 122: Go to **File -> Create Maestro Plotting Template ...**, write **MyTemplate** as template name and press **Create Template** button.

Basics of Analog Flow: RAK

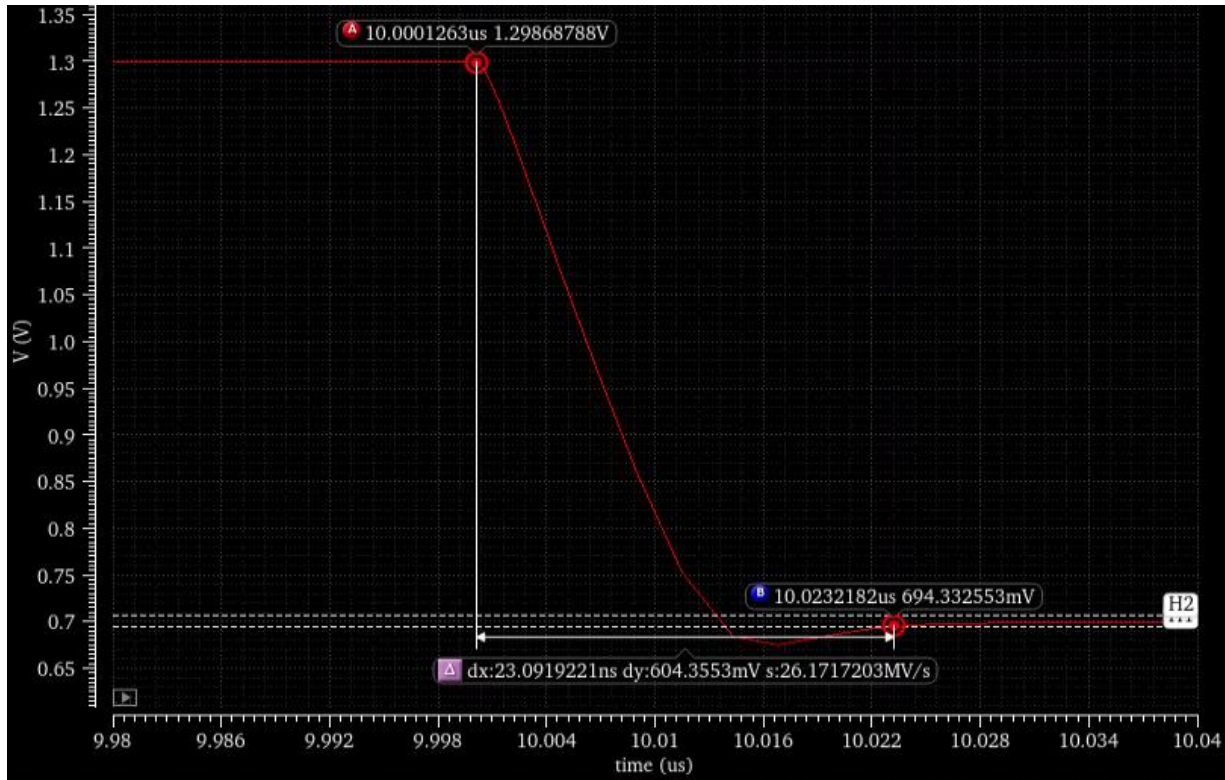


Action 123: Now close the **ViVA** window. In **maestro** tab, make sure “MyTemplate” is selected and press plot button again.

The image shows the 'Maestro' tab in the Virtuoso software. The top toolbar has a 'Plot' button (a line graph icon) highlighted with a red box. To its right, the 'Replace' dropdown menu is set to 'MyTemplate', which is also highlighted with a red box. Below the toolbar, a table displays test results. The table has 7 columns: Test, Output, Nominal, Spec, Weight, and Pass/Fail. The table contains 11 rows of data, with the first row being a header. The 'Pass/Fail' column shows 'pass' for most tests, with some tests marked as 'fail' (indicated by a red 'X' icon).

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
tran	Vout_tran				
tran	Vin_tran				
tran	Vout_dc	1.004			
tran	Vin_dc	1			
tran	Slew Rate	51.02M	> 50M		pass
tran	Settling Time	22.35n	< 50n		pass
tran	Offset	3.85m	range -10m 10m		pass
tran	Pdiss	184.9u	< 200u		pass
ac	Av_ac				
ac	UGB	121.1M	> 50M		pass
ac	Av0	60.36	> 60		pass

Same results will be plotted:



Now, move to the DC results.

Action 124: Go to the **maestro** tab. Do not close **ViVA**.

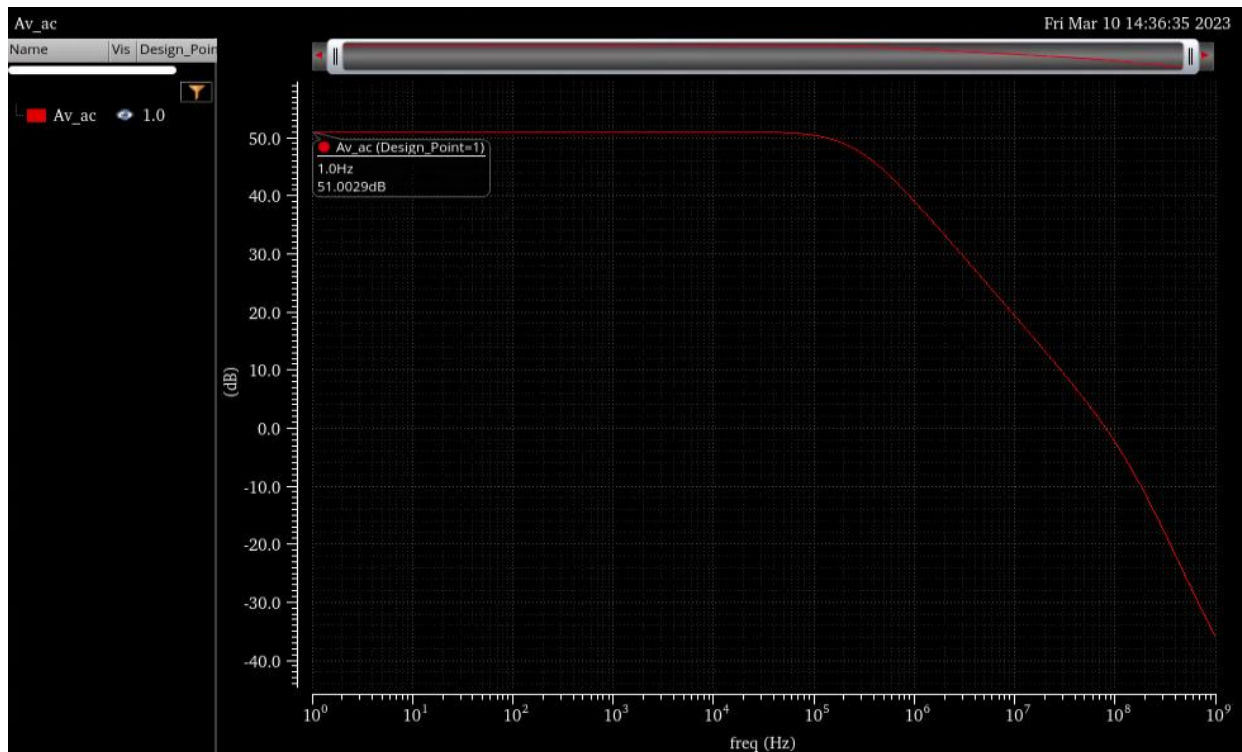
Test	Output	Nominal
Filter	Filter	Filter
tran	Vout_tran	
tran	Vin_tran	
tran	Vout_dc	998.8m
tran	Vin_dc	1
ac	Av_ac	

As you can see, the output DC value is **998.8mV**. Since the input DC value for this simulation is **1V**, the **DC Offset** is $(998.8 - 1000)\text{mV} = -1.2\text{mV}$.

Now, move to the AC results.

Action 125: Go to the **ac** tab on **ViVA**.

Action 126: Hover your cursor over the low-frequency portion of the signal.



Looking at this value, you can say that the **DC Open-Loop Gain** is about **51.0dB**.

Action 127: Move your cursor along the signal until you get to the point where the y value is around 0dB. Press **M** to put a simple marker on it.



From this marker, you can say that the **Unity-Gain Bandwidth** is about 79MHz or 80MHz.

As a review, the measurement specifications are as follows:

- Slew Rate $\geq 50\text{MV/s}$
- DC Open-Loop Gain $\geq 60\text{dB}$ (1000V/V)
- Unity-Gain Bandwidth $\geq 50\text{MHz}$
- Output Offset $\leq \pm 10\text{mV}$
- Settling Time $\leq 50\text{ns}$

You can see that all measurements meet the specifications, except for the **DC Open-Loop Gain**, which is 9dB below the required value. You need to tune your design to meet this specification.

Action 128: Close the **ViVA** window and the **Calculator**.

Lab 9: Generating Expressions

In the last section, you measured waveforms manually to derive expressions. This is useful, but you can increase productivity by creating reusable expressions.

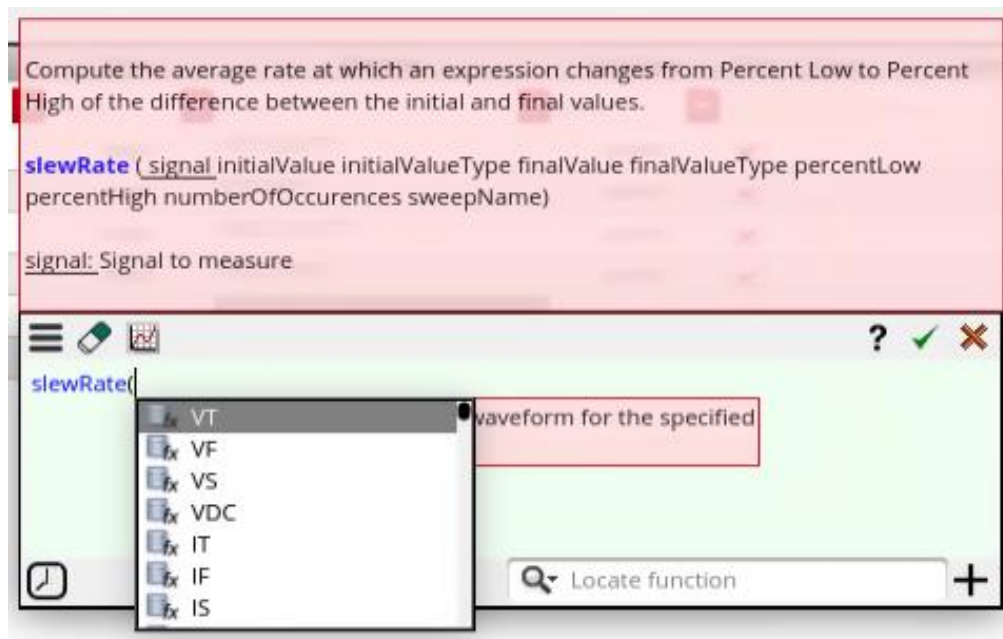
Action 129: Go to the **Outputs Setup** pane in **ADE Assembler** and click on the red down-facing arrow beside the **Add New Output** icon . Select **tran > Expression** to add an expression to the **tran** test.

Test	Name	Type	Details	EvalType
Filter	Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")	point
tran	Vin_tran	expr	VT("/IN")	point
tran	Vout_dc	expr	VDC("/OUT")	point
tran	Vin_dc	expr	VDC("/IN")	point
tran		expr		point
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))	point

Action 130: Double-click the **Details** column for the new expression and click on the **Open expression builder** icon  to open the **Expression Builder**.



Action 131: Type **slew**, and when the autocompleted string appears in the prompt, select **slewRate**.



As you can see, Expression Builder provides helpful prompts for building the measurement.

Action 132: For the **signal**, scroll down to the list of measurements and select **Vout_tran**. You can also start typing Vout_tran and take advantage of the autocomplete feature.

Action 133: For **initialValue**, enter **1.3**. Proceed to the next step by pressing the spacebar.

Action 134: For **initialValueType**, select **nil**.

Action 135: For **finalValue**, enter **0.7**.

Action 136: For **finalValueType**, select **nil**.

Action 137: For **percentLow**, enter **20**.

Action 138: For **percentHigh**, enter **80**.

Action 139: For **numberOfOccurrences**, select **nil**.

Action 140: For **sweepName**, select **time**.

Action 141: Press the closing parenthesis and click on the green icon '✓' (OR press **Enter**) to insert the expression.

Action 142: Name the expression as **Slew Rate**.

Test	Name	Type	Details
Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")
tran	Vin_tran	expr	VT("/IN")
tran	Vout_dc	expr	VDC("/OUT")
tran	Vin_dc	expr	VDC("/IN")
tran	Slew Rate	expr	slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time")
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))

Action 143: Repeat these steps to add a second expression. This time, you will build an expression for the settling time.

Action 144: Follow the process defined above to build the following expression:

```
settlingTime[Vout_tran 0.0 t 2e-06 t 1 nil "time"]
```

This expression uses **0us** as the initial time and **2us** as the final time, so it will measure the settling time of the first falling edge. Click on the green icon '✓' to insert the expression.

Action 145: Name the expression **Settling Time**.

Basics of Analog Flow: RAK

Test	Name	Type	Details
Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")
tran	Vin_tran	expr	VT("/IN")
tran	Vout_dc	expr	VDC("/OUT")
tran	Vin_dc	expr	VDC("/IN")
tran	Slew Rate	expr	slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time")
tran	Settling Time	expr	settlingTime(Vout_tran 0.0 t 2e-06 t 1 nil "time")
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))

Action 146: Add another expression to the 'tran' testbench. For that, click on the , and the last action performed will be repeated. Enter **Vout_dc - Vin_dc** in the **Details** column and click on the green icon  to insert the expression.

Action 147: Name the expression as **Offset**.

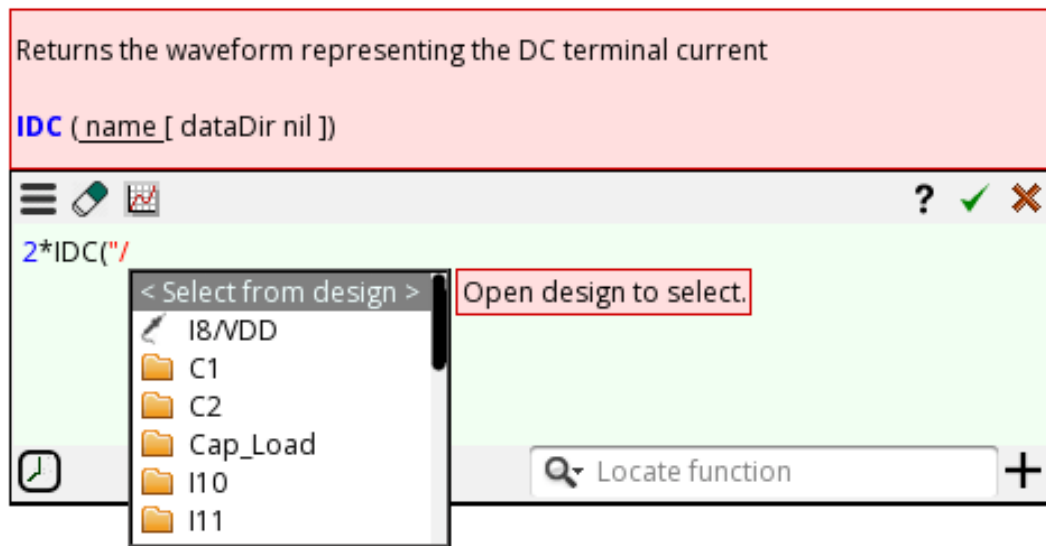
Test	Name	Type	Details
Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")
tran	Vin_tran	expr	VT("/IN")
tran	Vout_dc	expr	VDC("/OUT")
tran	Vin_dc	expr	VDC("/IN")
tran	Slew Rate	expr	slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time")
tran	Settling Time	expr	settlingTime(Vout_tran 0.0 t 2e-06 t 1 nil "time")
tran	Offset	expr	(Vout_dc - Vin_dc)
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))

You will also build an expression to measure dissipated power from the DC analysis.

Action 148: Add another new 'tran' testbench expression and bring up the **Expression Builder**, as done in the previous steps. Enter **2***.



Action 149: Enter **IDC** and click on **< Select from design >**.



Action 150: Select the top pin of the **VDD_source** component.

`2*IDC("/VDD_source/PLUS")`

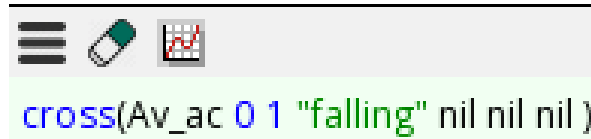
Action 151: Click on the green icon '✓' to insert the expression. Switch to the **maestro** tab and name the expression **Pdiss**.

Test	Name	Type	Details
Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")
tran	Vin_tran	expr	VT("/IN")
tran	Vout_dc	expr	VDC("/OUT")
tran	Vin_dc	expr	VDC("/IN")
tran	Slew Rate	expr	slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time")
tran	Settling Time	expr	settlingTime(Vout_tran 0.0 t 2e-06 t 1 nil "time")
tran	Pdiss	expr	(2 * IDC("/VDD_source/PLUS"))
tran	Offset	expr	(Vout_dc - Vin_dc)
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))

The value **2** is the total supply voltage range applied on the op-amp. Now, two expressions are left to add: dc gain and bandwidth. Those will be added to the **ac** test.

Action 152: Add an expression for the **ac** test (you will again need to use the downward-facing arrow '▼' next to the icon) and open the **Expression Builder**, as done in previous steps.

Action 153: Build the following expression:



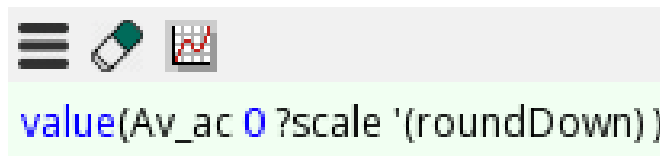
cross(Av_ac 0 1 "falling" nil nil nil)

Action 154: Insert the expression and name it as **UGB** (Unity-Gain Bandwidth).

Test	Name	Type	Details
Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")
tran	Vin_tran	expr	VT("/IN")
tran	Vout_dc	expr	VDC("/OUT")
tran	Vin_dc	expr	VDC("/IN")
tran	Slew Rate	expr	slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time")
tran	Settling Time	expr	settlingTime(Vout_tran 0.0 t 2e-06 t 1 nil "time")
tran	Pdiss	expr	(2 * IDC("/VDD_source/PLUS"))
tran	Offset	expr	(Vout_dc - Vin_dc)
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))
ac	UGB	expr	cross(Av_ac 0 1 "falling" nil nil nil)

Action 155: Enter another expression for 'ac' and type **value** in the **Expression Builder**.

Action 156: Build the following expression:



value(Av_ac 0 ?scale '(roundDown))

This will measure the **Av_ac** signal at the initial value (1Hz).

Action 157: Insert the expression and name it as **Av0**.

Test	Name	Type	Details
Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")
tran	Vin_tran	expr	VT("/IN")
tran	Vout_dc	expr	VDC("/OUT")
tran	Vin_dc	expr	VDC("/IN")
tran	Slew Rate	expr	slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time")
tran	Settling Time	expr	settlingTime(Vout_tran 0.0 t 2e-06 t 1 nil "time")
tran	Pdiss	expr	(2 * IDC("/VDD_source/PLUS"))
tran	Offset	expr	(Vout_dc - Vin_dc)
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))
ac	UGB	expr	cross(Av_ac 0 1 "falling" nil nil nil)
ac	Av0	expr	value(Av_ac 0 ?scale '(roundDown))

Action 158: Go back to the **Results** tab of the **maestro** view. Press  in the toolbar to reevaluate the output expressions. This feature allows an evaluation of expressions without rerunning the simulation if all of the required data is available.

The results are shown below.

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
tran	Vout_tran				
tran	Vin_tran				
tran	Vout_dc	998.8m			
tran	Vin_dc	1			
tran	Slew Rate	-50.82M			
tran	Settling Time	22.14n			
tran	Pdiss	-160.7u			
tran	Offset	-1.185m			
ac	Av_ac				
ac	UGB	81.01M			
ac	Av0	51			

As you can see, these results are similar to the ones you manually analyzed in the last section.

Action 159: Since you want absolute values for **Slew Rate** and **Pdiss**, insert the **abs()** function around both, as shown below. Evaluate again to see the change.

Basics of Analog Flow: RAK

Test	Name	Type	Details
Filter	Filter	Filter	Filter
tran	Vout_tran	expr	VT("/OUT")
tran	Vin_tran	expr	VT("/IN")
tran	Vout_dc	expr	VDC("/OUT")
tran	Vin_dc	expr	VDC("/IN")
tran	Slew Rate	expr	abs(slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time"))
tran	Settling Time	expr	settlingTime(Vout_tran 0.0 t 2e-06 t 1 nil "time")
tran	Offset	expr	(Vout_dc - Vin_dc)
tran	Pdiss	expr	abs((2 * IDC("/VDD_source/PLUS")))
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))
ac	UGB	expr	cross(Av_ac 0 1 "falling" nil nil nil)
ac	Av0	expr	value(Av_ac 0 ?scale '(roundDown))

The next stage is to define specifications for each expression. This means you can see which expressions are passing or failing at a glance.

Action 160: Return to the **Outputs Setup**, double-click the **Slew Rate Spec** column, select **>** and insert **50M** on the field at its right.

Test	Name	Type	Details	EvalType	Plot	Save	Spec
Filter	Filter	Filter	Filter	Filter			Filter
tran	Vout_tran	expr	VT("/OUT")	point	☒	☐	
tran	Vin_tran	expr	VT("/IN")	point	☒	☐	
tran	Vout_dc	expr	VDC("/OUT")	point	☒	☐	
tran	Vin_dc	expr	VDC("/IN")	point	☒	☐	
tran	Slew Rate	expr	abs(slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time"))	point	☒	☐	> 50M
tran	Settling Time	expr	settlingTime(Vout_tran 0.0 t 2e-06 t 1 nil "time")	point	☒	☐	
tran	Offset	expr	(Vout_dc - Vin_dc)	point	☒	☐	
tran	Pdiss	expr	abs((2 * IDC("/VDD_source/PLUS")))	point	☒	☐	
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))	point	☒	☐	
ac	UGB	expr	cross(Av_ac 0 1 "falling" nil nil nil)	point	☒	☐	
ac	Av0	expr	value(Av_ac 0 ?scale '(roundDown))	point	☒	☐	

This means you want the slew rate value to be higher than 50MV/s.

Action 161: Add specifications for other measurements, as shown below.

Test	Name	Type	Details	EvalType	Plot	Save	Spec
Filter	Filter	Filter	Filter	Filter			Filter
tran	Vout_tran	expr	VT("/OUT")	point	☒	☐	
tran	Vin_tran	expr	VT("/IN")	point	☒	☐	
tran	Vout_dc	expr	VDC("/OUT")	point	☒	☐	
tran	Vin_dc	expr	VDC("/IN")	point	☒	☐	
tran	Slew Rate	expr	abs(slewRate(Vout_tran 1.3 nil 0.7 nil 20 80 nil "time"))	point	☒	☐	> 50M
tran	Settling Time	expr	settlingTime(Vout_tran 0.0 t 2e-06 t 1 nil "time")	point	☒	☐	< 50n
tran	Offset	expr	(Vout_dc - Vin_dc)	point	☒	☐	range -10m 10m
tran	Pdiss	expr	abs((2 * IDC("/VDD_source/PLUS")))	point	☒	☐	< 200u
ac	Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))	point	☒	☐	
ac	UGB	expr	cross(Av_ac 0 1 "falling" nil nil nil)	point	☒	☐	> 50M
ac	Av0	expr	value(Av_ac 0 ?scale '(roundDown))	point	☒	☐	> 60

Action 162: Return to the **Results** tab and press  to reevaluate. The results are shown below.

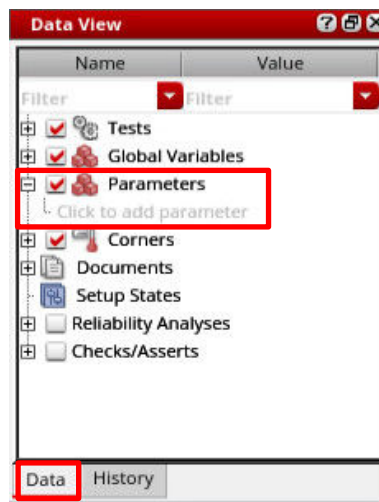
Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
tran	Vout_tran				
tran	Vin_tran				
tran	Vout_dc	998.8m			
tran	Vin_dc	1			
tran	Slew Rate	50.82M	> 50M		pass
tran	Settling Time	22.14n	< 50n		pass
tran	Offset	-1.185m	range -10m 10m		pass
tran	Pdiss	160.7u	< 200u		pass
ac	Av_ac				
ac	UGB	81.01M	> 50M		pass
ac	Av0	51	> 60		fail

All expressions meet the specifications except **Av0**. The red colorization is an indication of the spec failure.

Lab 10: Design Tuning Using Parameters

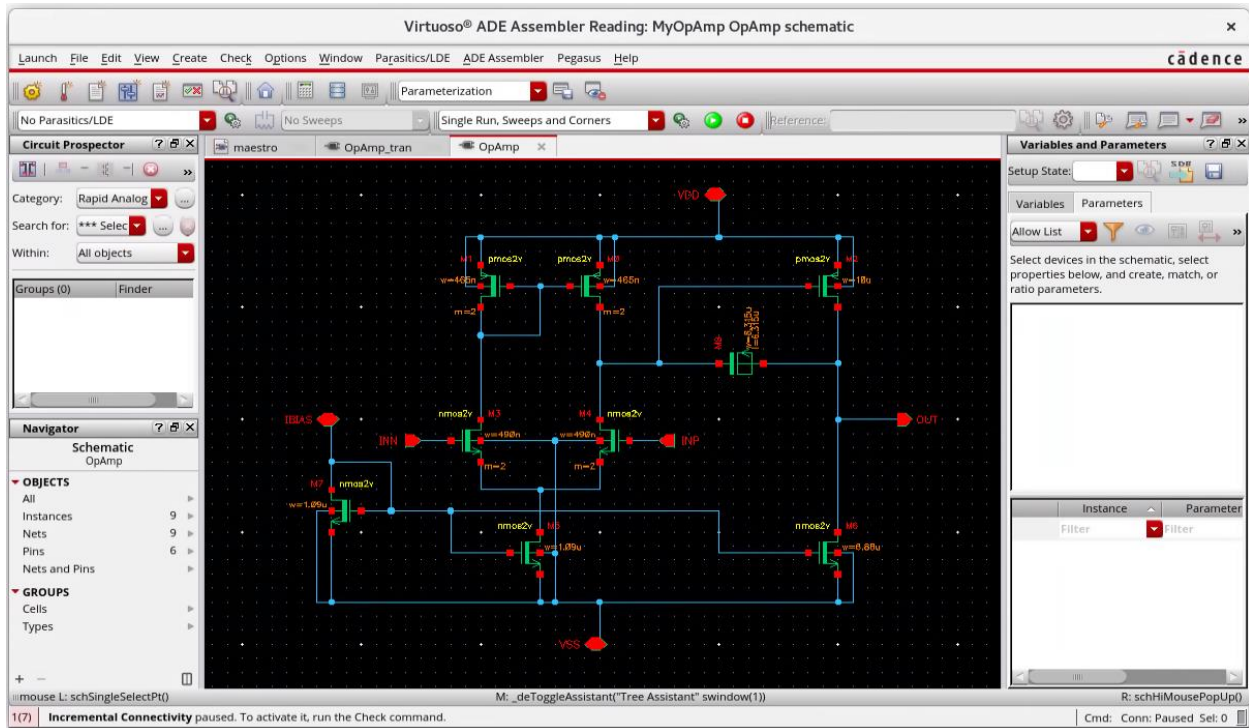
In this section, you will tune the design to pass the specifications. To tune the circuit, adjust the length and width of some transistors. Make the lengths and widths parameters, which will enable you to experiment with these device values without editing the schematic.

Action 163: On the **Data View** assistant, expand the **Parameters** tree and click on the **Click to add parameter** text.



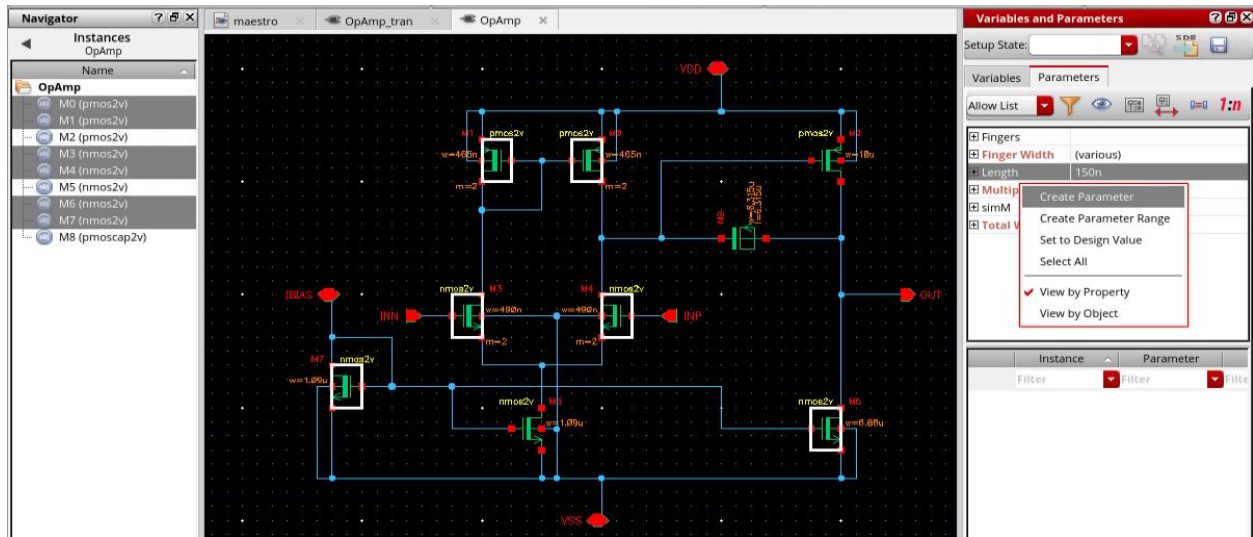
This will open the op-amp schematic and invoke two new assistants: **Circuit Prospector** and **Variables and Parameters**.

Action 164: Go to **Window > Assistants > Navigator** to reinvoke the **Navigator** assistant.

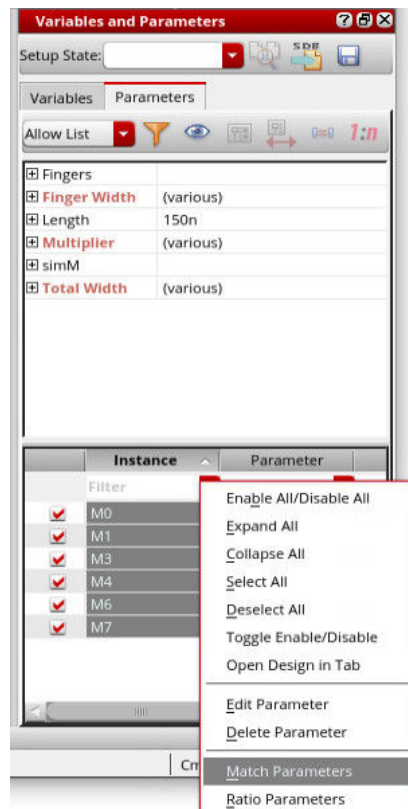


Action 165: Close the **Circuit Prospector** assistant. Select all instances on the **Navigator** except for M2, M5, and M8 (the pmos capacitor). In the **Variables and Parameters** assistant, right-click on the **Length** value of the transistors and select **Create Parameter**.

Basics of Analog Flow: RAK

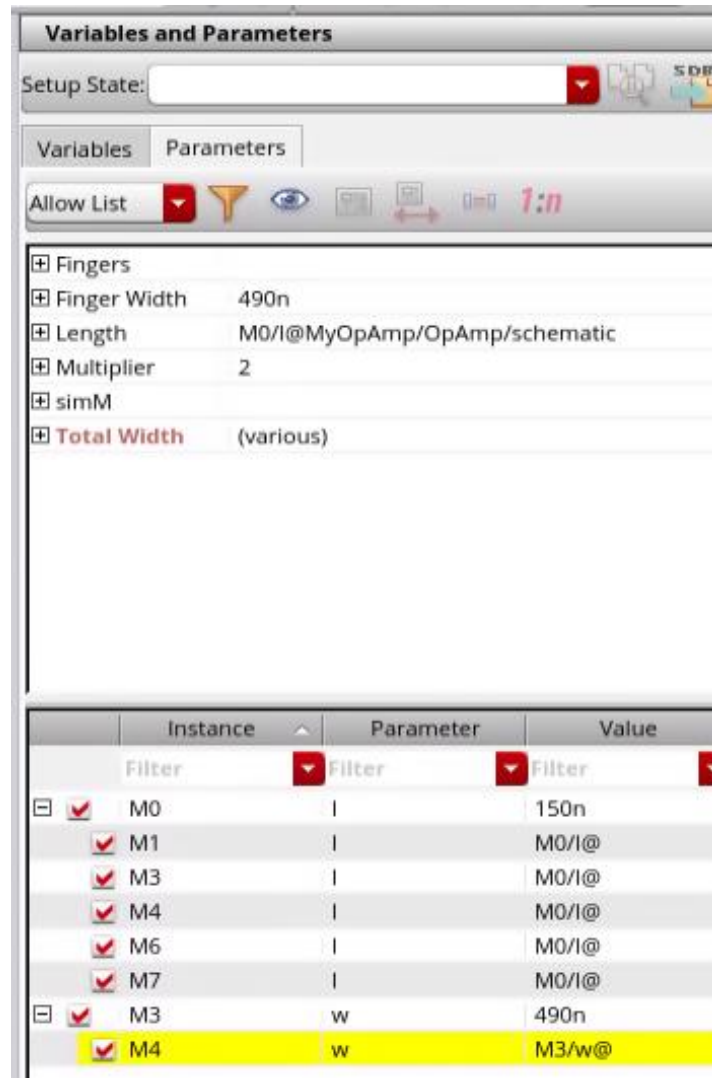


Action 166: Select all six entries in the lower table of the assistant, right-click and choose **Match Parameters**.



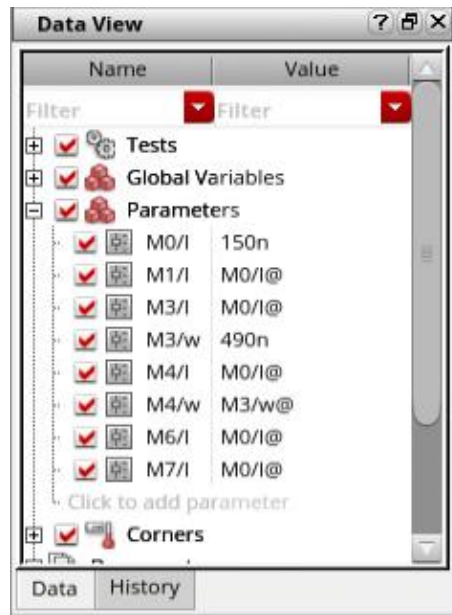
This will cause all lengths to vary as you tune the design. Notice that the parameters are organized in a tree structure with the first parameter as the root.

Action 167: Select only M3 and M4 (the differential pair) and use similar steps to create a parameter for **Total Width**. Select both in the lower table, right-click, and choose **Match** so that they track together.

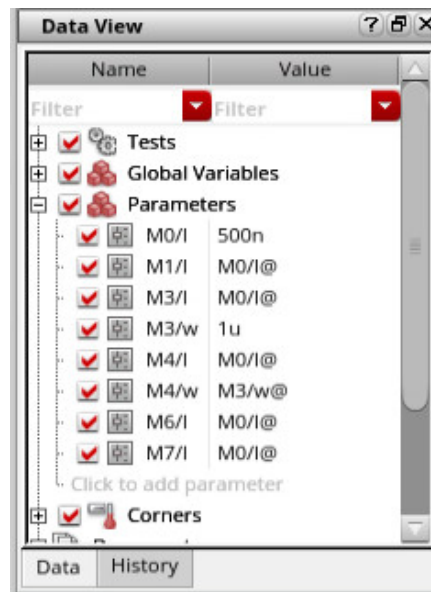


Action 168: Close the schematic view and return to the **maestro** tab.

Action 169: On the **maestro** tab, look at the **Parameters** section on the **Data View** assistant and notice the newly created parameters. From here, you can change these values, avoiding edits to the schematic.



Action 170: Set **M0/I** to **500n** and **M3/w** to **1u**. Here, it is assumed that M0 and M3 are the masters of your two parameter sets.



These parameters will be used for both testbenches. Focus first on the 'ac' results tuning.

Action 171: Descend into the **ac** test by clicking on the down arrow '↓' next to the testbench name.

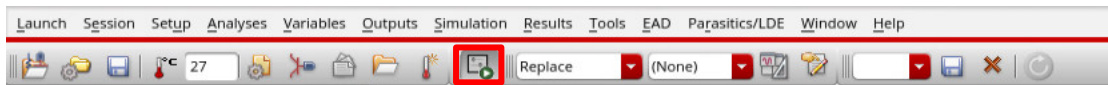
Basics of Analog Flow: RAK

Action 172: Rerun the simulation for this single test using the '▶' icon on the right-side toolbar.

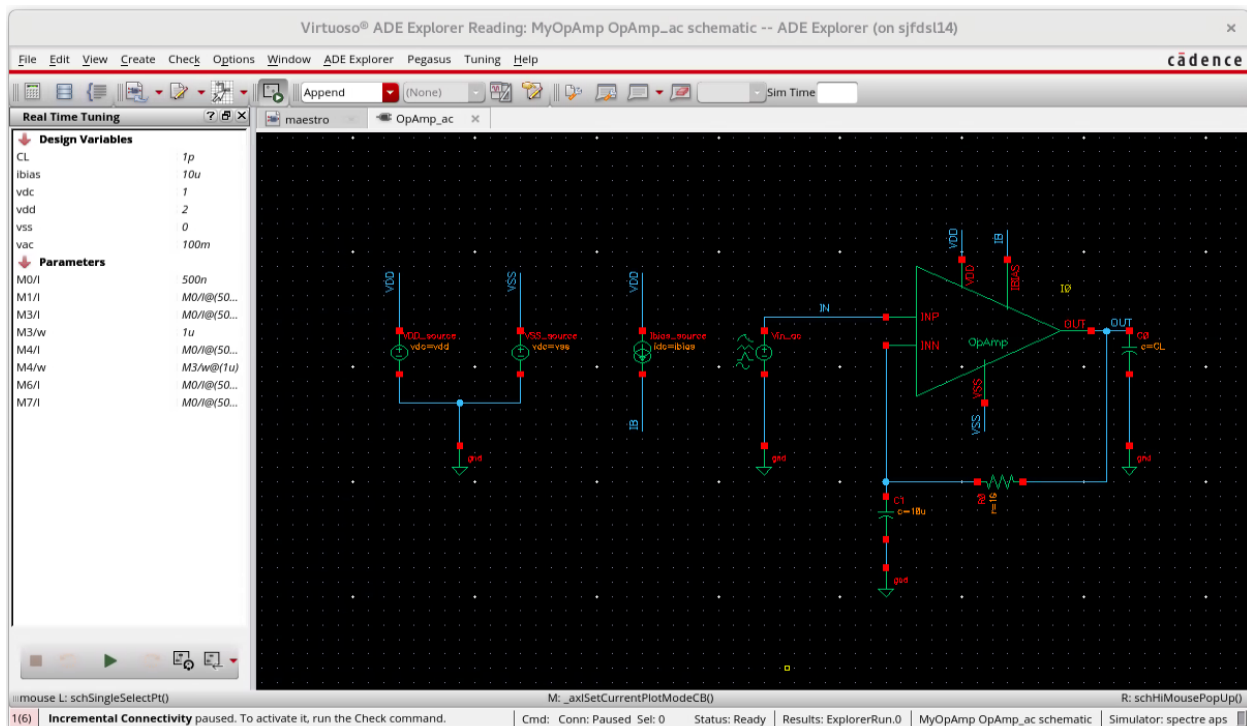
Name	Type	Details	Value	Plot	Save	Spec
Av_ac	expr	dB20((VF("/OUT") / VF("/IN")))		☒	☐	
UGB	expr	cross(Av_ac 0 1 "falling" nil nil nil)	100.1M	☒	☐	> 50M
Av0	expr	value(Av_ac 0 ?scale '(roundDown))	58.5	☒	☐	> 60

As you can see, by increasing **L** and **Wdiff**, you get much closer to the specifications for **Av0**. The yellow colorization indicates that you are within $\pm 10\%$ of the spec.

Action 173: Dismiss the **ViVA** window. Click on the **Real Time Tuning** icon.

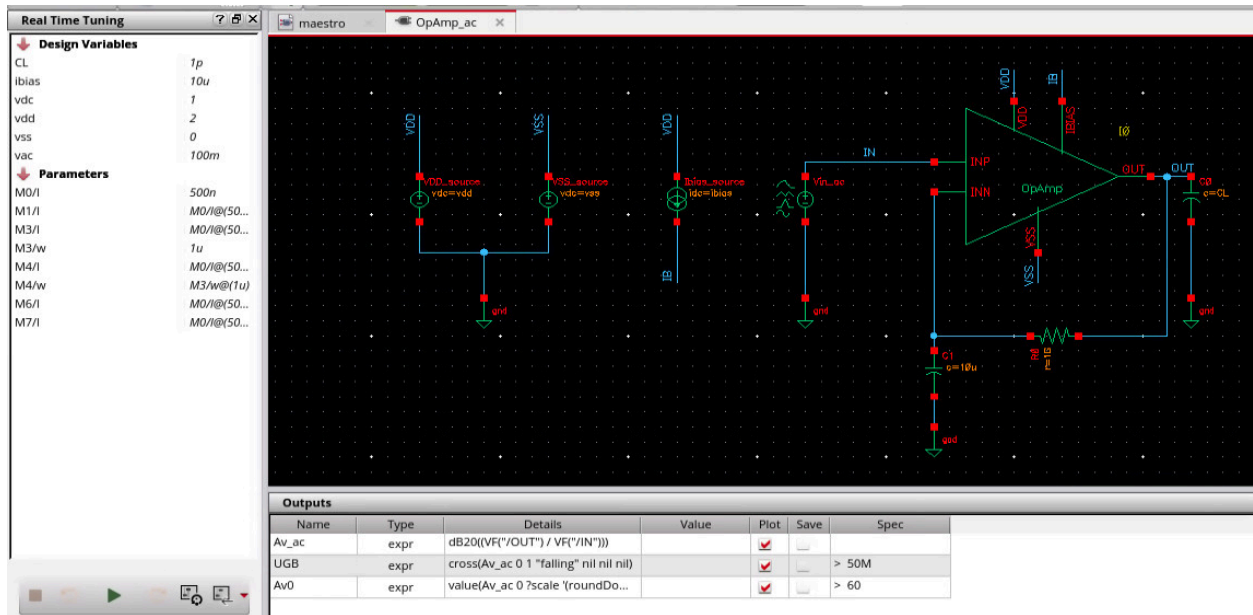


The schematic testbench appears with the **Real Time Tuning** assistant docked on the left.



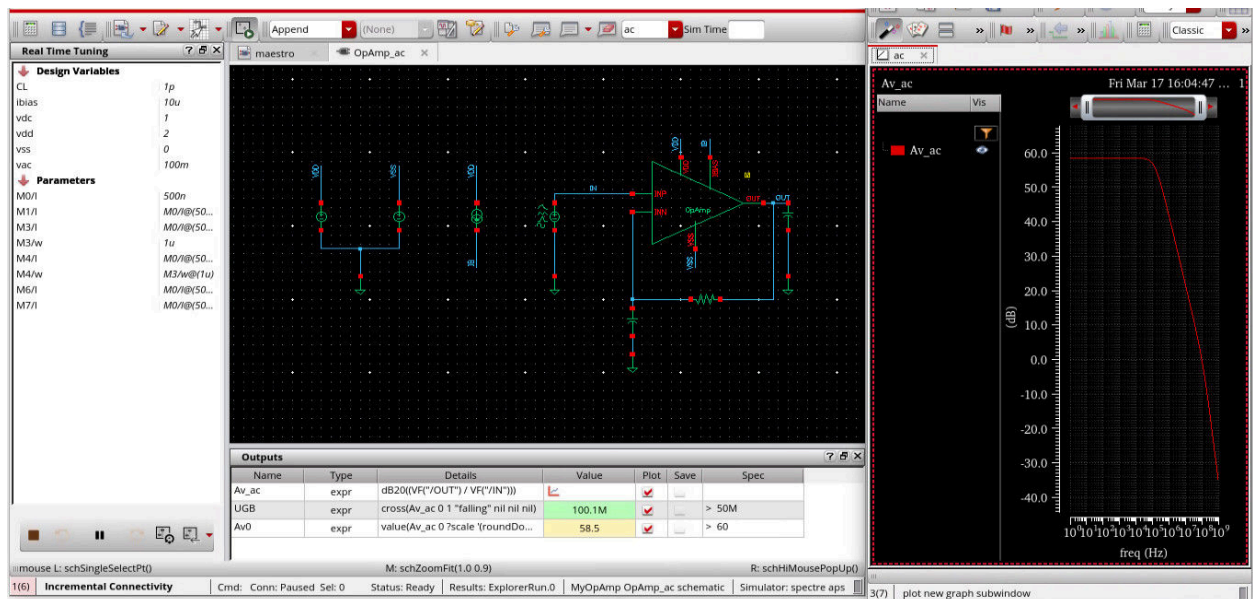
Action 174: Select **Window > Assistants > Outputs** to dock the expressions to the bottom of the window.

Basics of Analog Flow: RAK



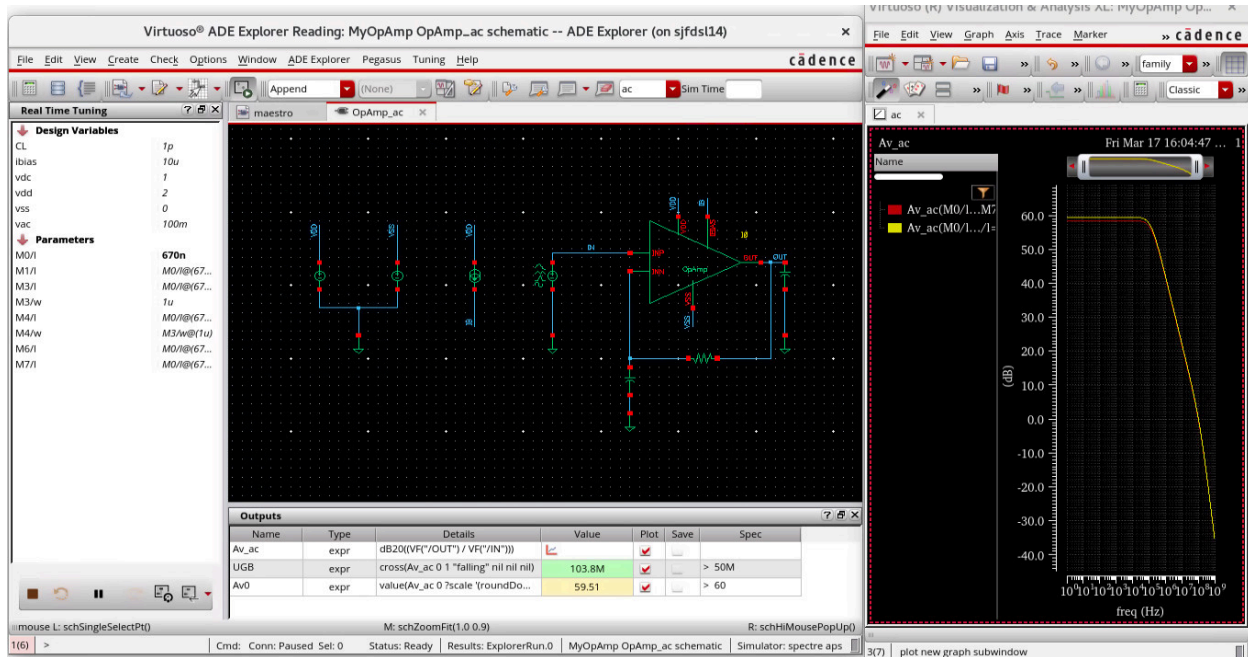
Action 175: Click on the **Run** icon '▶' under the **Tuning** panel to initialize the tuner.

Your initial results should look like as shown below. Arrange the schematic and the waveform so you can see both as you tune.

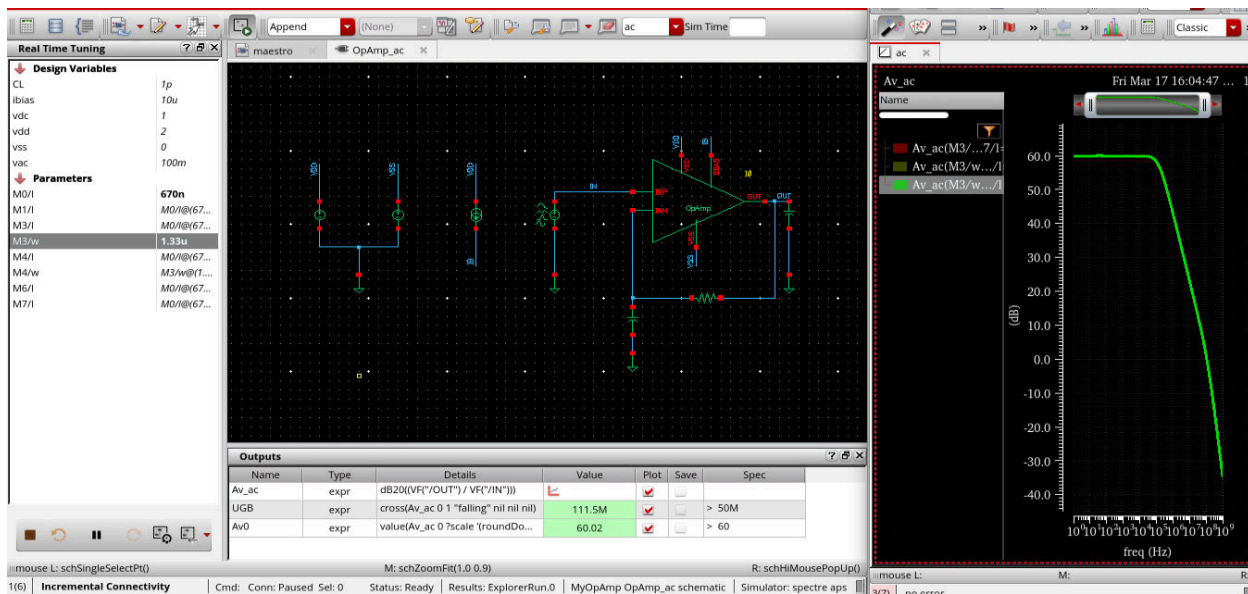


Action 176: Hover your mouse over the **M0/I** entry in the tuner and increase its value to **670n**. Note a change in the scalar expression and the plotted additional signal.

Basics of Analog Flow: RAK



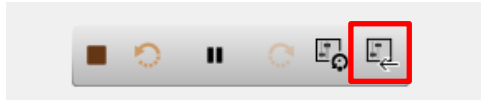
Action 177: Tune M3/w to 1.33u.



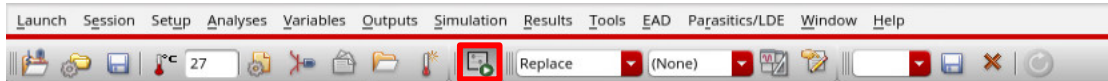
The specifications are now met.

Action 178: As you need to account for parasitic effects on the simulation, you must further tune the design. Increase M0/I to 750n and M3/w to 1.5u. This will increase UGB to about 116MHz and Av0 to about 60.5.

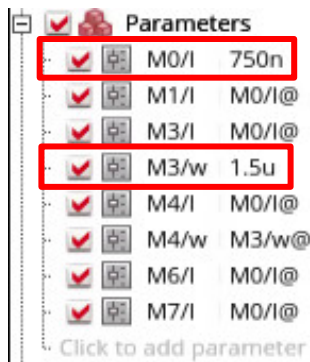
Action 179: Click the **Save back to ADE Explorer** icon in the lower portion of the tuning assistant to save your tuned values to the ADE session.



Action 180: Exit **RTT** by selecting the same icon that you used to invoke it:

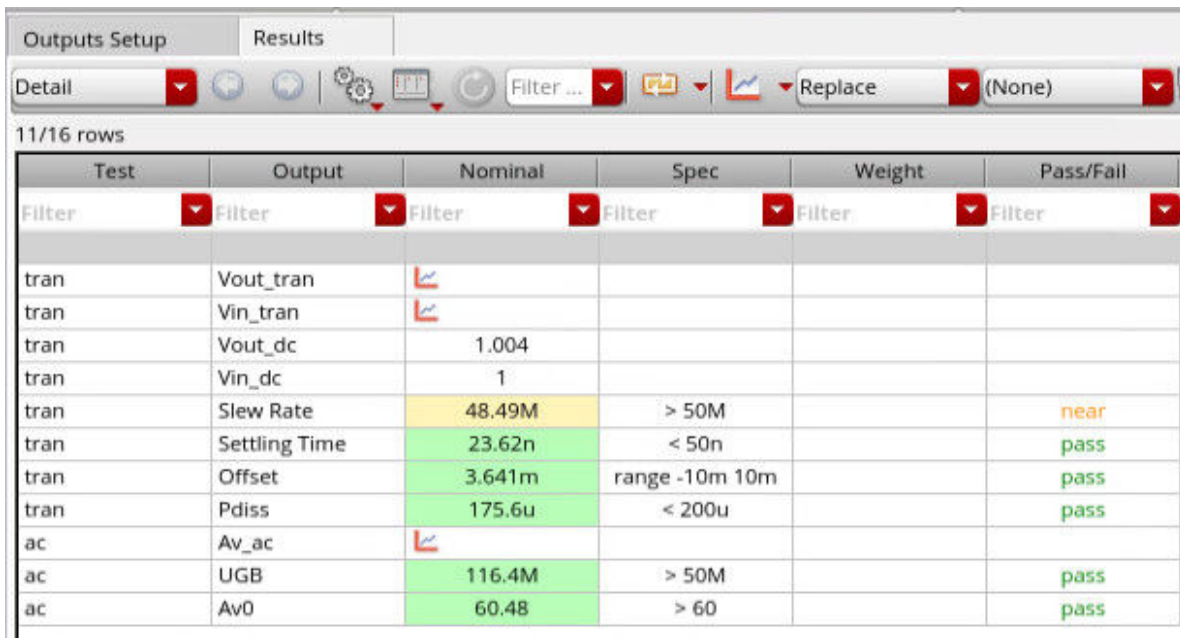


Action 181: Close **ViVA** and the schematic view and return to the **maestro** tab. Expand the **Parameters** section and notice the updated values:



Action 182: Ascend to **ADE Assembler** by clicking on the up arrow '↑' next to the **ac** testbench name. Rerun the simulation to see how tuning impacts the transient results.

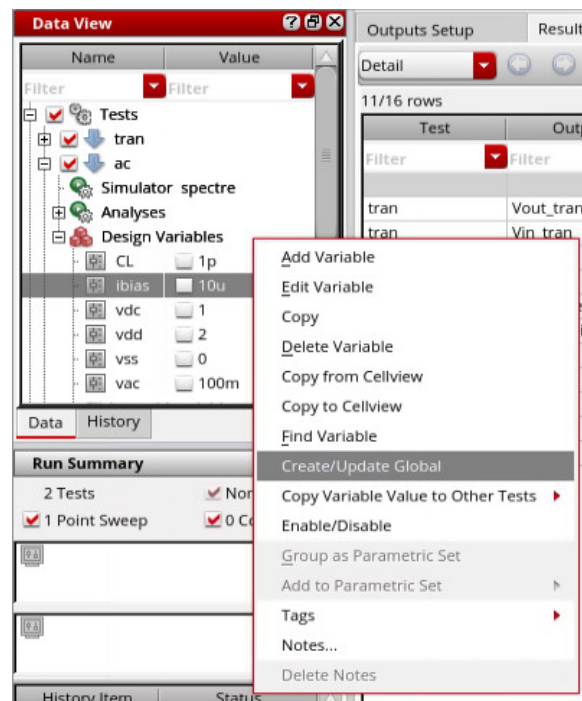
Basics of Analog Flow: RAK



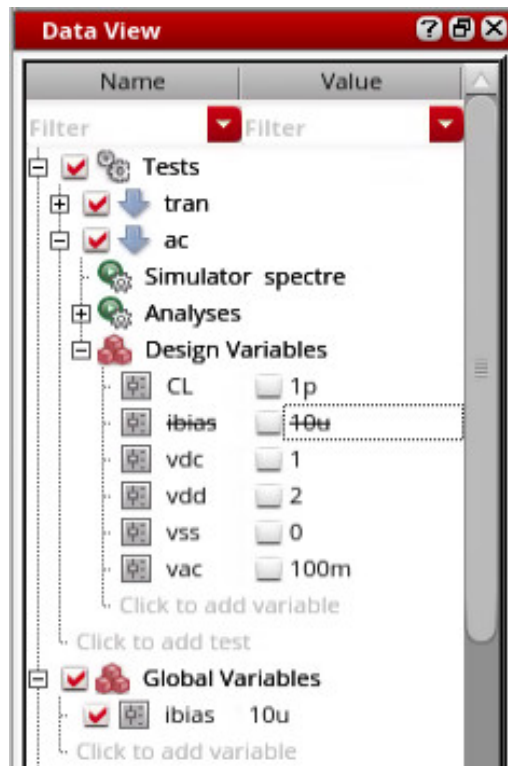
Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
tran	Vout_tran				
tran	Vin_tran				
tran	Vout_dc	1.004			
tran	Vin_dc	1			
tran	Slew Rate	48.49M	> 50M		near
tran	Settling Time	23.62n	< 50n		pass
tran	Offset	3.641m	range -10m 10m		pass
tran	Pdiss	175.6u	< 200u		pass
ac	Av_ac				
ac	UGB	116.4M	> 50M		pass
ac	Av0	60.48	> 60		pass

As you can see, **Slew Rate** has fallen slightly out of spec. To fix this, increase the bias current, **ibias**. As you need to change this value for both tests, make **ibias** a global variable.

Action 183: Expand the **Design Variables** field in the **ac** test, select **ibias**, right-click and select **Create/Update Global**.



Expand the **Global Variables** field and notice that **ibias** is added. If the variable is enabled there, you will see it displayed with a strikethrough in the **Design Variables** section for each test. This indicates that the global variable will be used rather than a local variable.



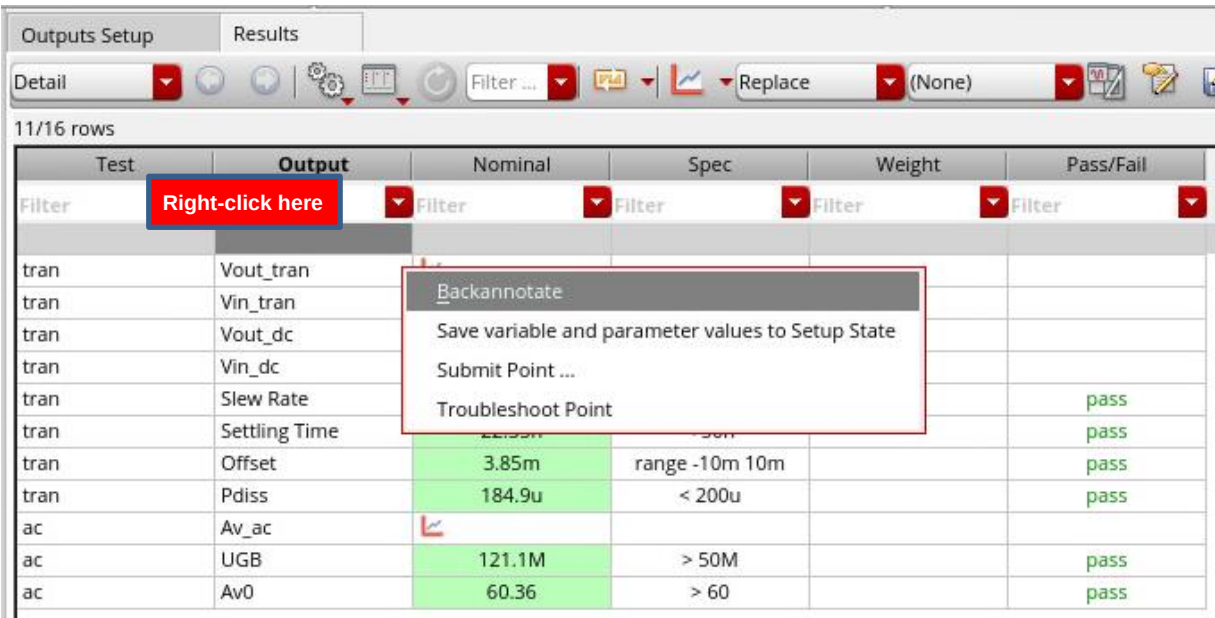
Action 184: Increase the **ibias** global value to 10.5u and rerun the simulation. Your results should look as follows:

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
tran	Vout_tran				
tran	Vin_tran				
tran	Vout_dc	1.004			
tran	Vin_dc	1			
tran	Slew Rate	51.02M	> 50M		pass
tran	Settling Time	22.35n	< 50n		pass
tran	Offset	3.85m	range -10m 10m		pass
tran	Pdiss	184.9u	< 200u		pass
ac	Av_ac				
ac	UGB	121.1M	> 50M		pass
ac	Av0	60.36	> 60		pass

Lab 11: Backannotation of Tuned Parameters

In this section, you will replace the width and length values of your op-amp with the values produced by tuning.

Action 185: In the gray row (just below the **Filter** widgets in the **Results** table), right-click and select **Backannotate**.



Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter		Filter	Filter	Filter	Filter
tran	Vout_tran				
tran	Vin_tran				
tran	Vout_dc				
tran	Vin_dc				
tran	Slew Rate				pass
tran	Settling Time				pass
tran	Offset	3.85m	range -10m 10m		pass
tran	Pdiss	184.9u	< 200u		pass
ac	Av_ac				
ac	UGB	121.1M	> 50M		pass
ac	Av0	60.36	> 60		pass

Action 186: In the **Warning** window that appears, select **Only device parameters** and press **OK**.

Action 187: In the **OpAmp_tran** tab, descend into the OpAmp block and verify that M0, M1, M3, M4, M6, and M7 have **I=750n**. Verify that M3 and M4 have **w=1.5u**.

Module 5: Layout

In this section, you will build a layout for the op-amp. You will use the capabilities in Virtuoso to place instances, match critical devices, and route the design.

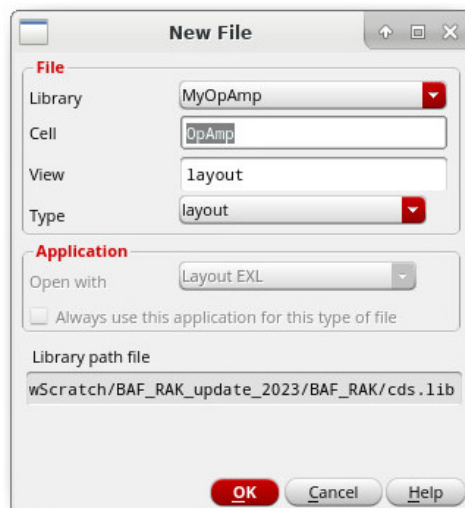
In the case of a large design, it is advisable to create and use constraints to facilitate the placement and routing of devices, making this process more automatic. Since this is a small design, you will not use constraints.

Lab 12: Layout XL

Action 189: From the MyOpAmp/OpAmp/schematic, select **Launch > Layout XL**. Specify **Create New** in the first pop-up window and press **OK**.

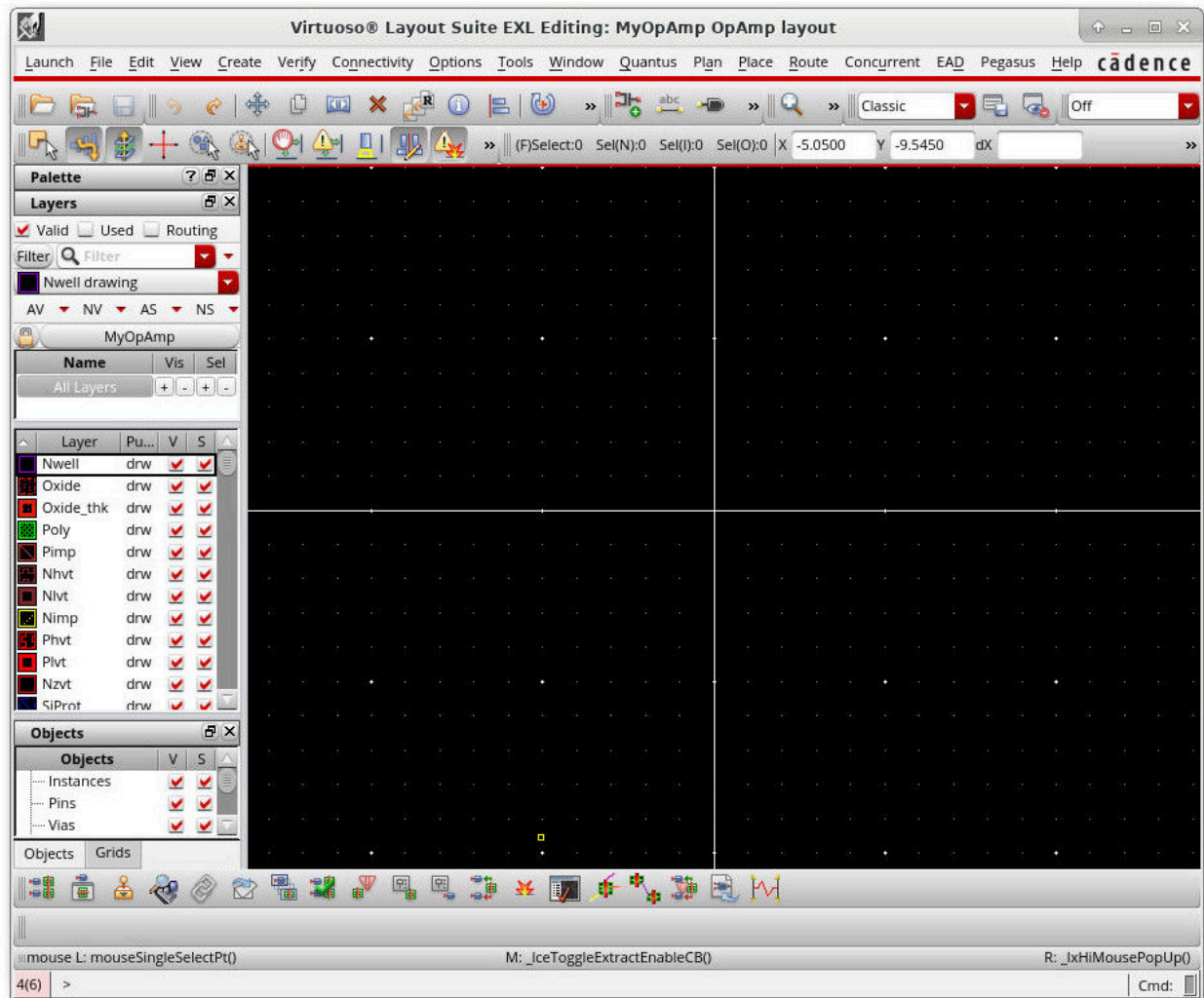


Action 190: In the **New File** window, accept the default values and select **OK**.



The design will open in **Layout XL**.

Basics of Analog Flow: RAK

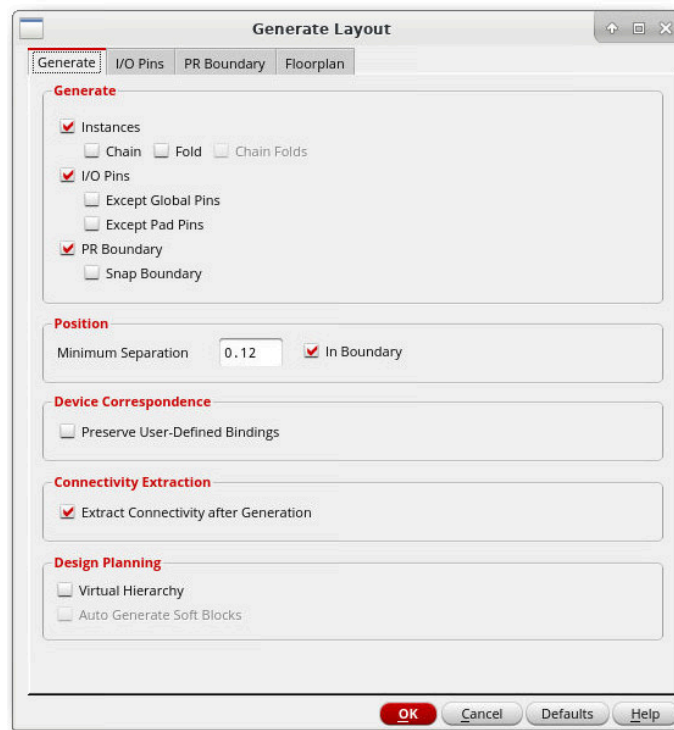


Focus first on generating the devices.

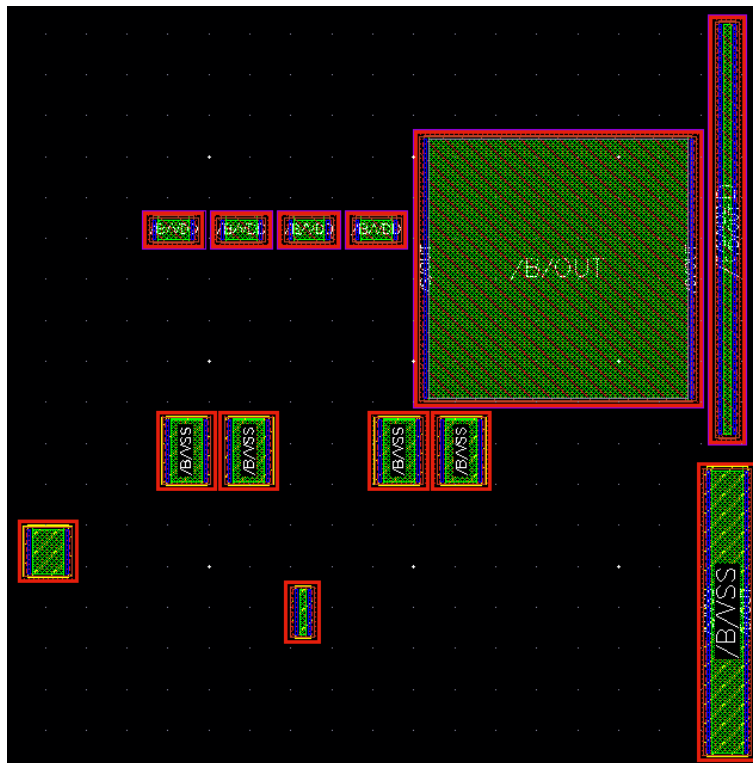
Action 191: Go to **Connectivity > Generate > All From Source** OR press the  icon on the left of the toolbar at the bottom.

Action 192: Ensure the Generate Layout form looks like the image below and click **OK**.

Basics of Analog Flow: RAK



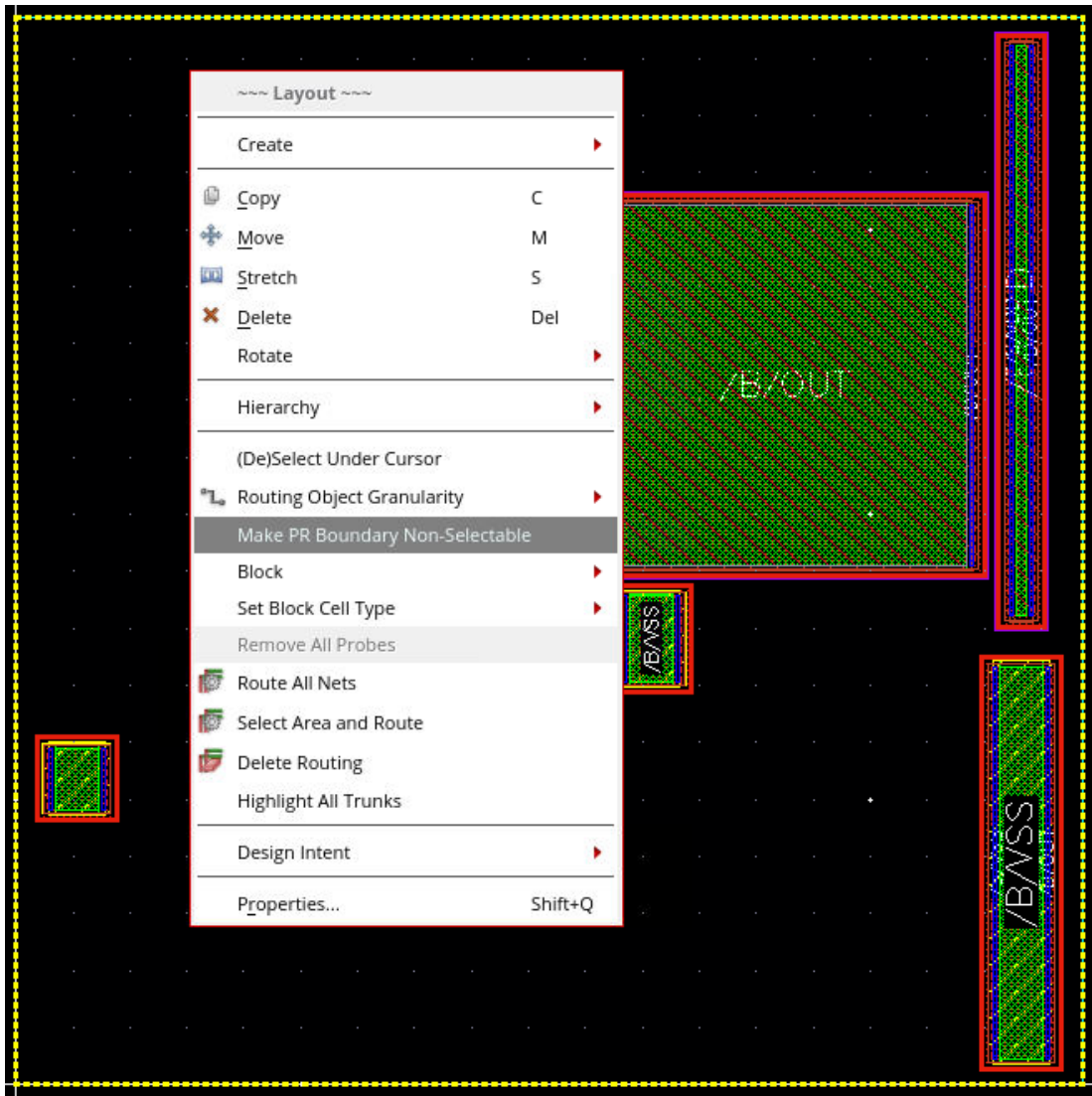
Action 193: Select **Shift+f** to display all layers. Your layout should look as follows:



The **Palette** docked on the left of the layout window shows all the layers defined for the process.

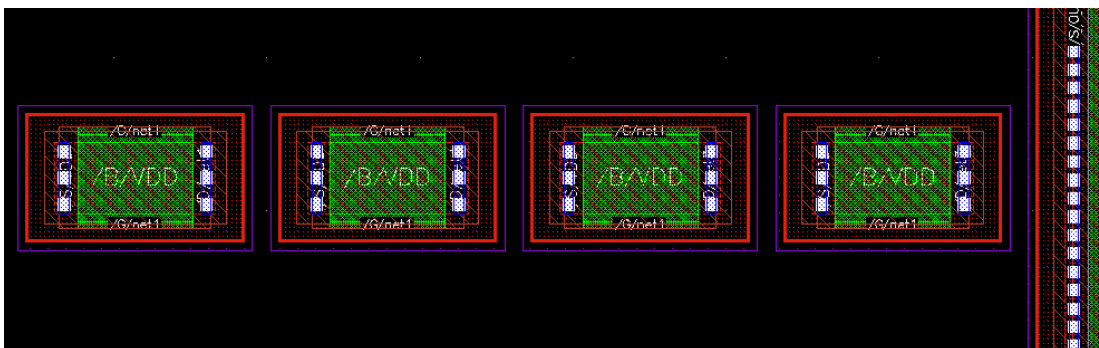


Action 194: Arrange your windows so that the schematic and the layout views are arranged side by side. Select the **M1** instance in the schematic and notice that the corresponding components are selected in the layout.

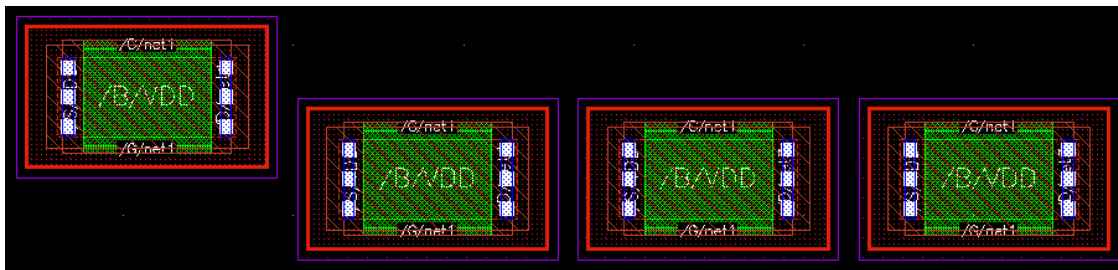


Lab 13: Transistor Abutment

Action 197: In the layout, zoom in on the four components that make up M0 and M1.

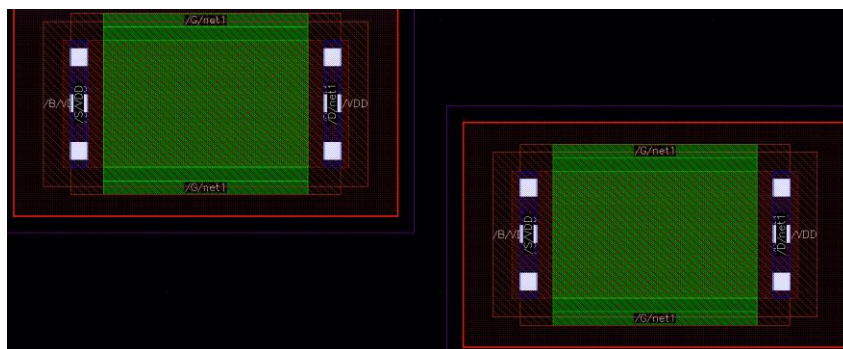


Action 198: Select the leftmost transistor and drag it up a little.



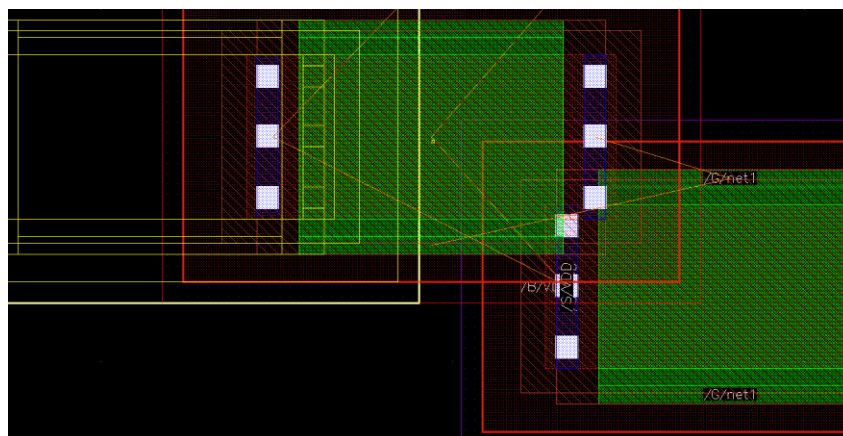
You can also move objects using the **M** bindkey and selecting the device to move.

Action 199: Zoom in on the space between the two transistors on the left.

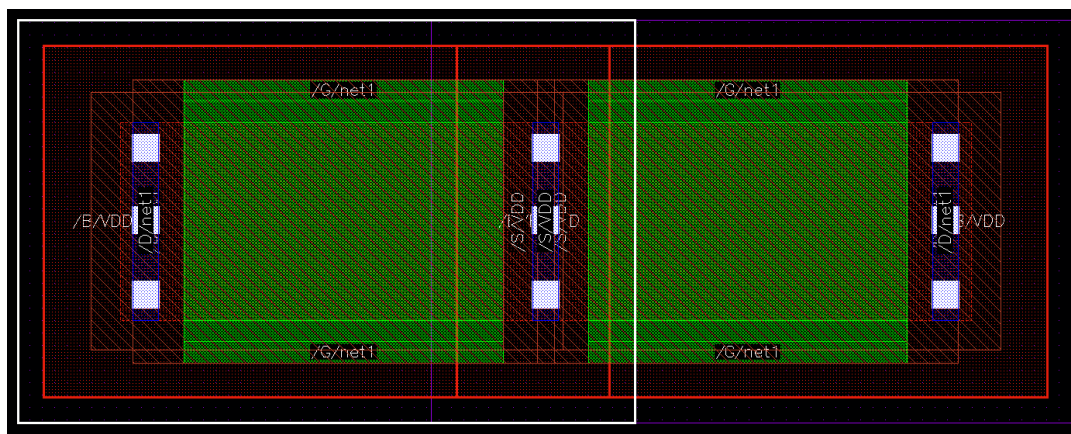


The drain (**D**) of the left member of this pair is next to the source (**S**) of the right member, and so, without the abutment functionality, you would need to manually flip one of them to make the connection correct. You will now see how abutment does that for you automatically.

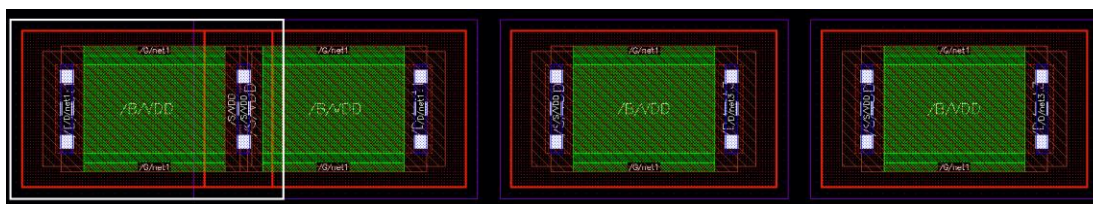
Action 200: Drag the left member M1 to the right and down until its drain just touches the source of the other member, as shown below.



When the move completes, the source connection for the two transistors will be abutted.



Action 201: Zoom out to see the four components.



There is also an automatic abutment option available when you generate the instances. You will look at that next.

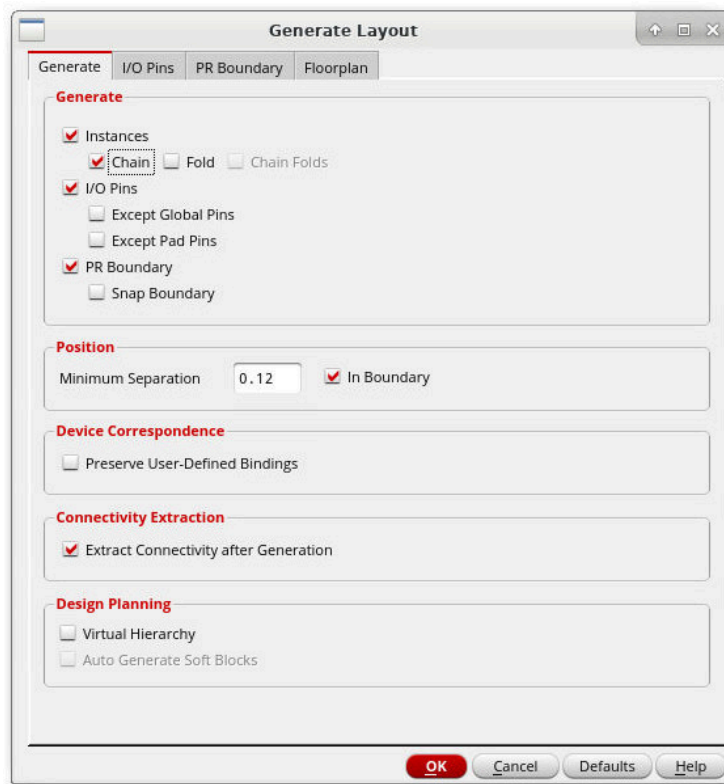
Action 202: Select the **Connectivity > Generate > All From Source** command OR press the  icon on the far left in the bottom toolbar.

Press **Yes** on the following prompt:

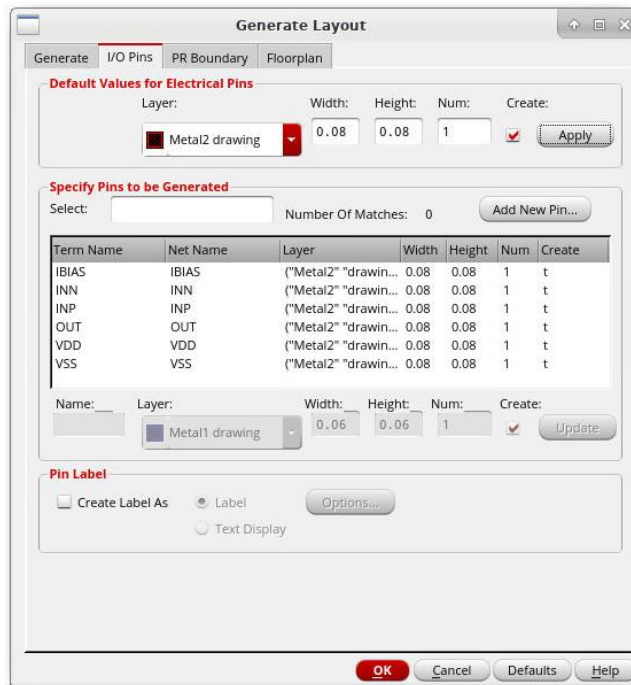


Action 203: On the **Generate Layout** form, select **Chain** under **Instances**.

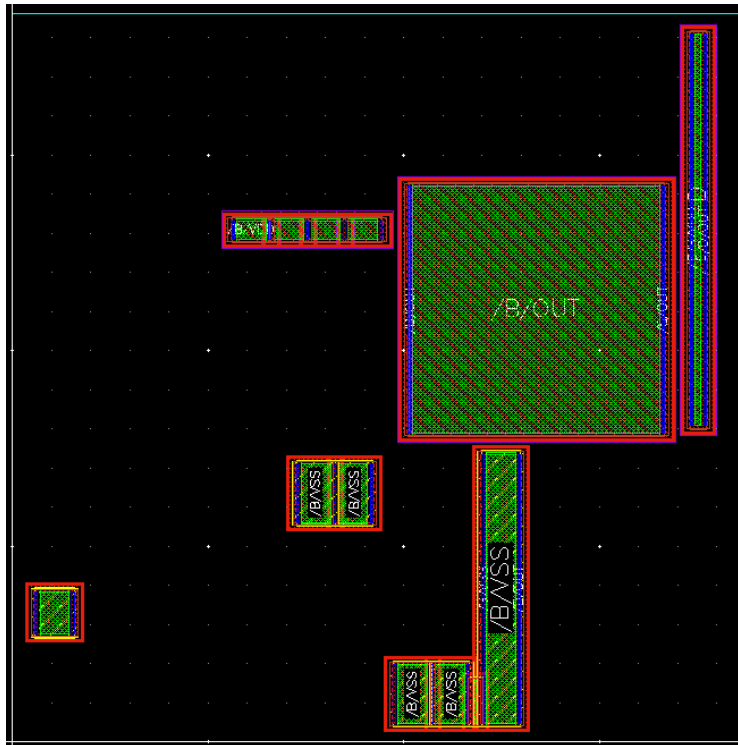
Basics of Analog Flow: RAK



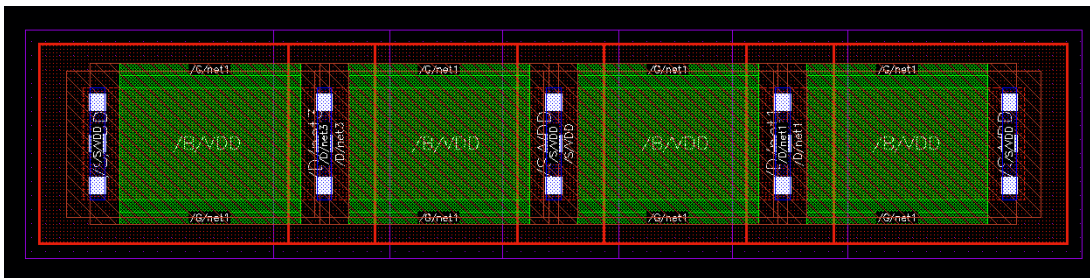
Action 204: On the **I/O Pins** tab, set **Layer** to **Metal2 drawing** and click the **Apply** button on the right. This will change the **Layer** for all pins in the lower table.



Action 205: Click **OK** on the form. Zoom in on M0 and M1 again.



You will see that M1 and M0 are abutted, along with many other transistors in the design.

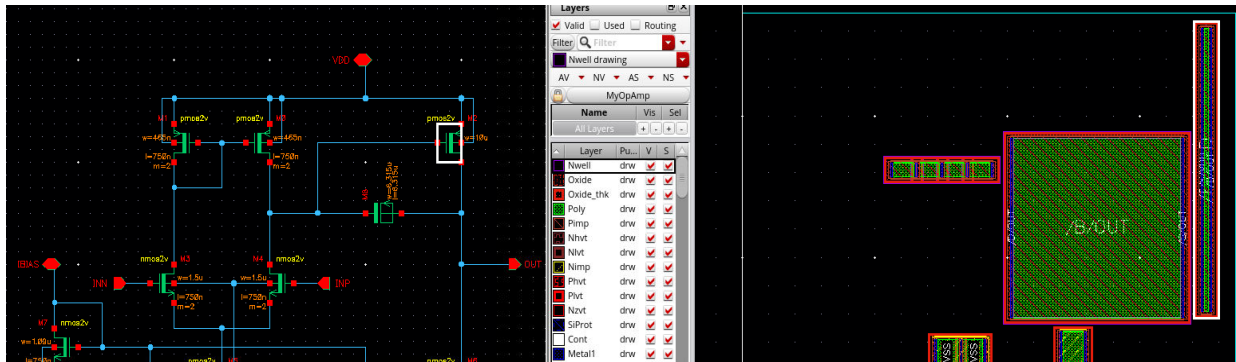


This automatic abutment can save a lot of time when the design is large. To remove an abutment, drag the transistors apart from one another.

Next, you will look at the alignment capability.

Action 206: Select **M2** on the schematic and see that it is also selected in the layout. You may need to move the capacitor so that it is next to the M2 component, as shown below.

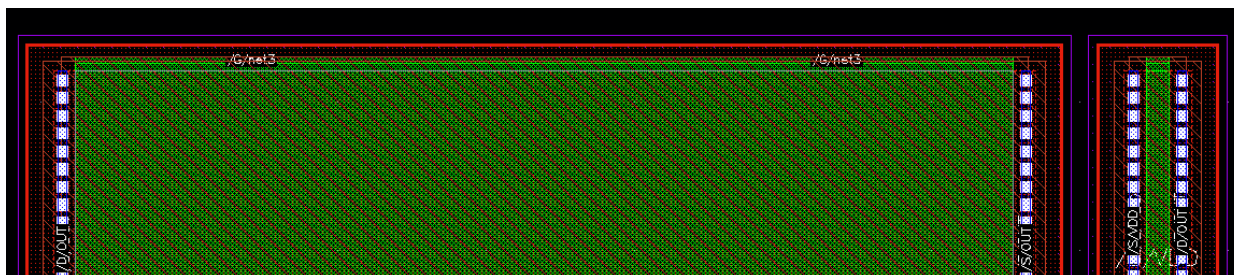
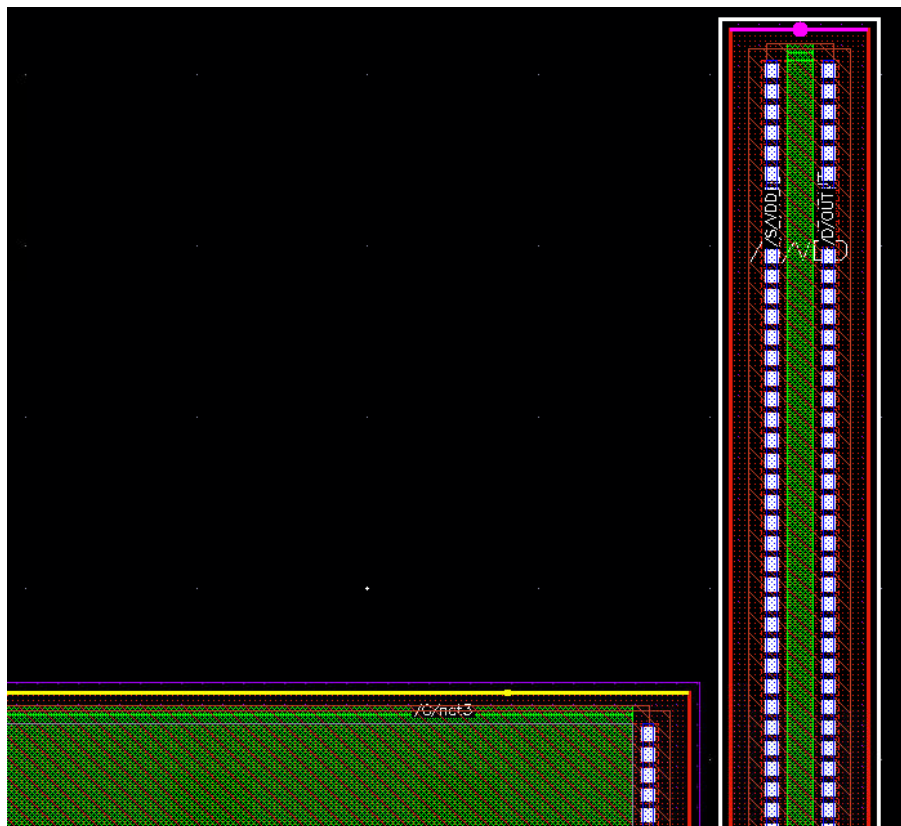
Basics of Analog Flow: RAK



Action 207: Press **A** to invoke the **Align** command and select the top border of M2 on the layout.



Action 208: Now, select the top border of the MOSFET capacitor.

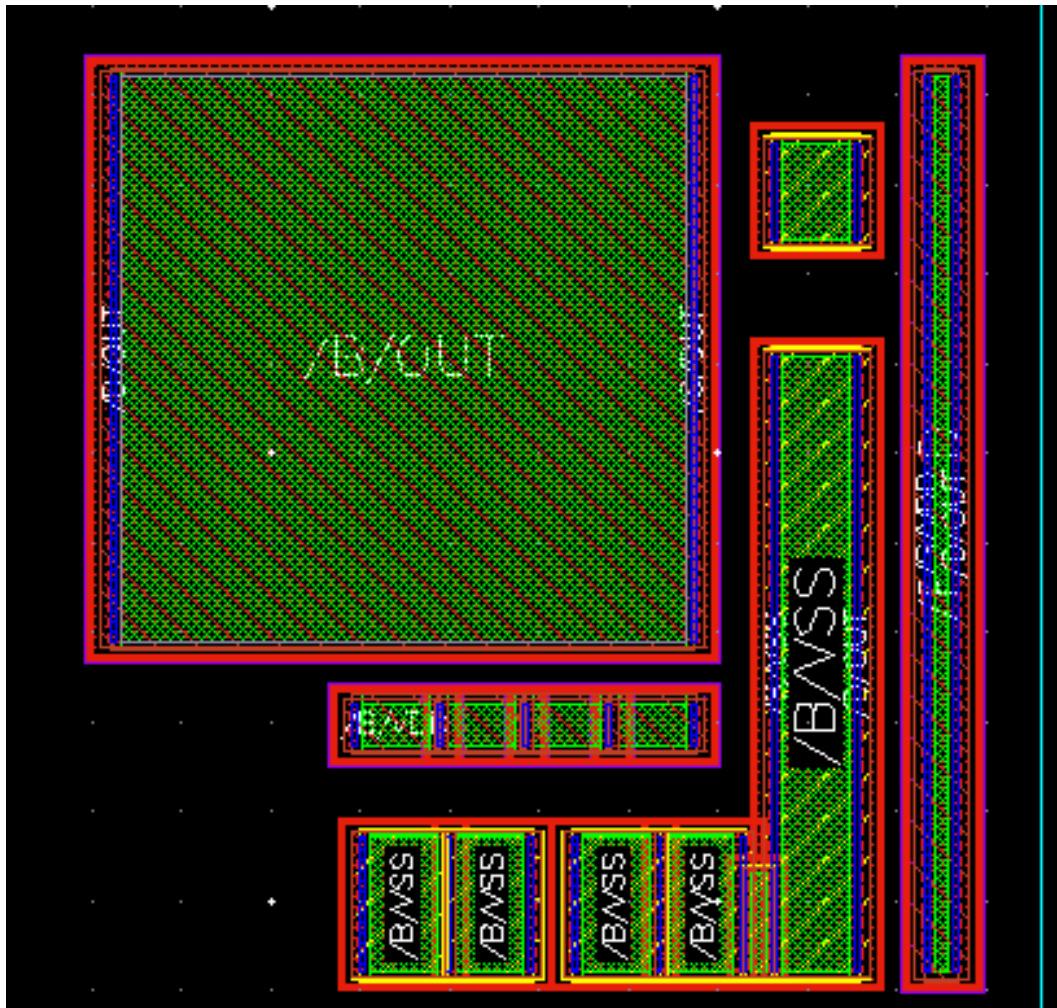


The top edge of M2 will align with the top edge of the capacitor. Again, this is an editing capability that will streamline your work.

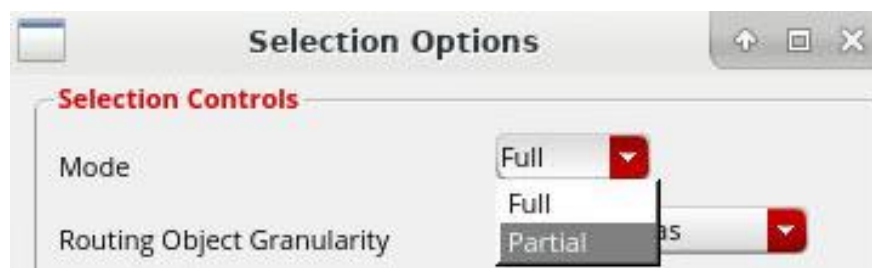
Lab 14: Placement

In this section, you will use some of the manual editing commands in Virtuoso to optimize the area in your layout before routing.

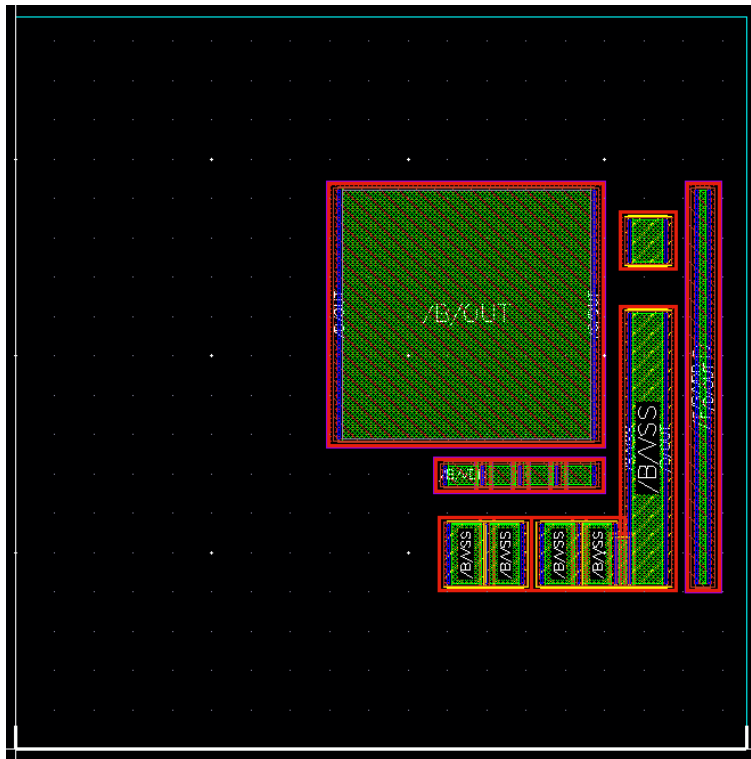
Action 209: Using the **Move** command (the **m** bindkey), place the components as shown below. For abutted devices, select and move them together. You can use the **n** in-command bindkey to change the snap mode during the move.



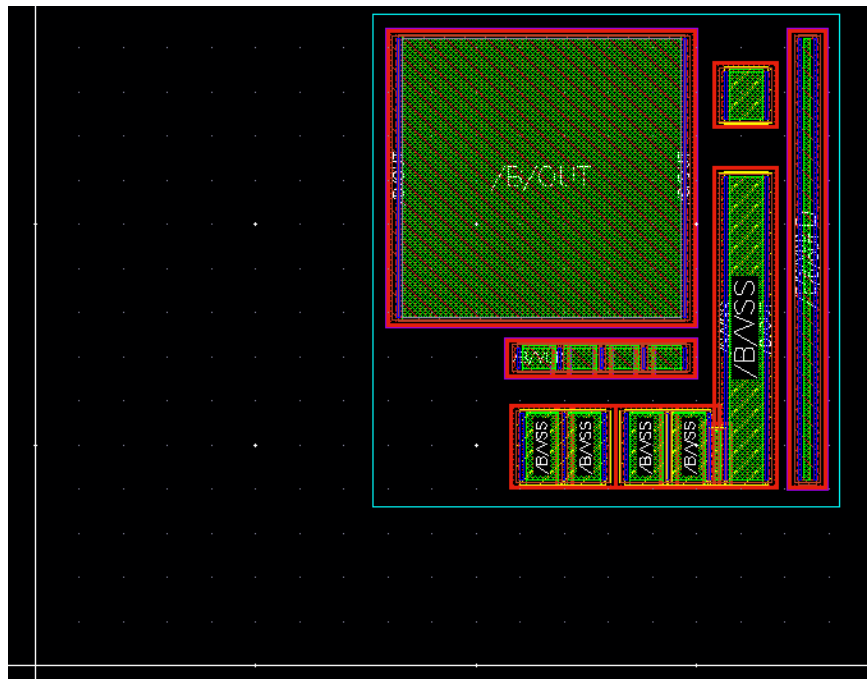
Action 210: Select **Options > Selection** and change the **Mode** option from **Full** to **Partial**. Then, click **OK**.



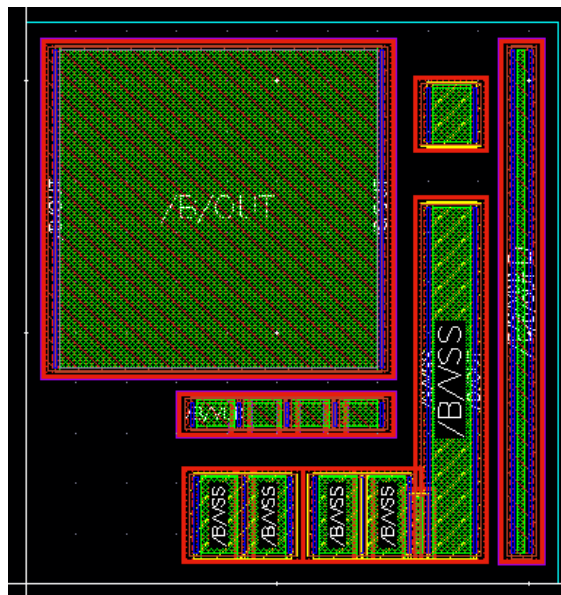
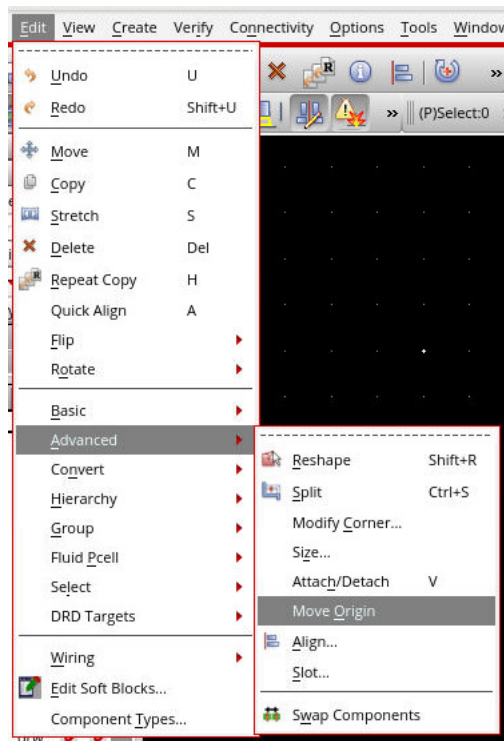
Action 211: Right-click the empty space inside the **PR Boundary** and select **Make PR Boundary Selectable**. Select one of the boundary edges.



Action 212: Then, press **s** to invoke the **Stretch** command. Select and drag the **PR Boundary** to just enclose the components. Repeat with the other edges of the boundary as needed.

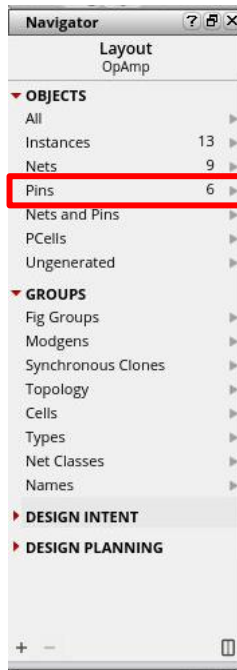


Action 213: To centralize this layout view, select **Edit > Advanced > Move Origin**. Then, select the bottom-left corner of the layout **PR Boundary** rectangle.



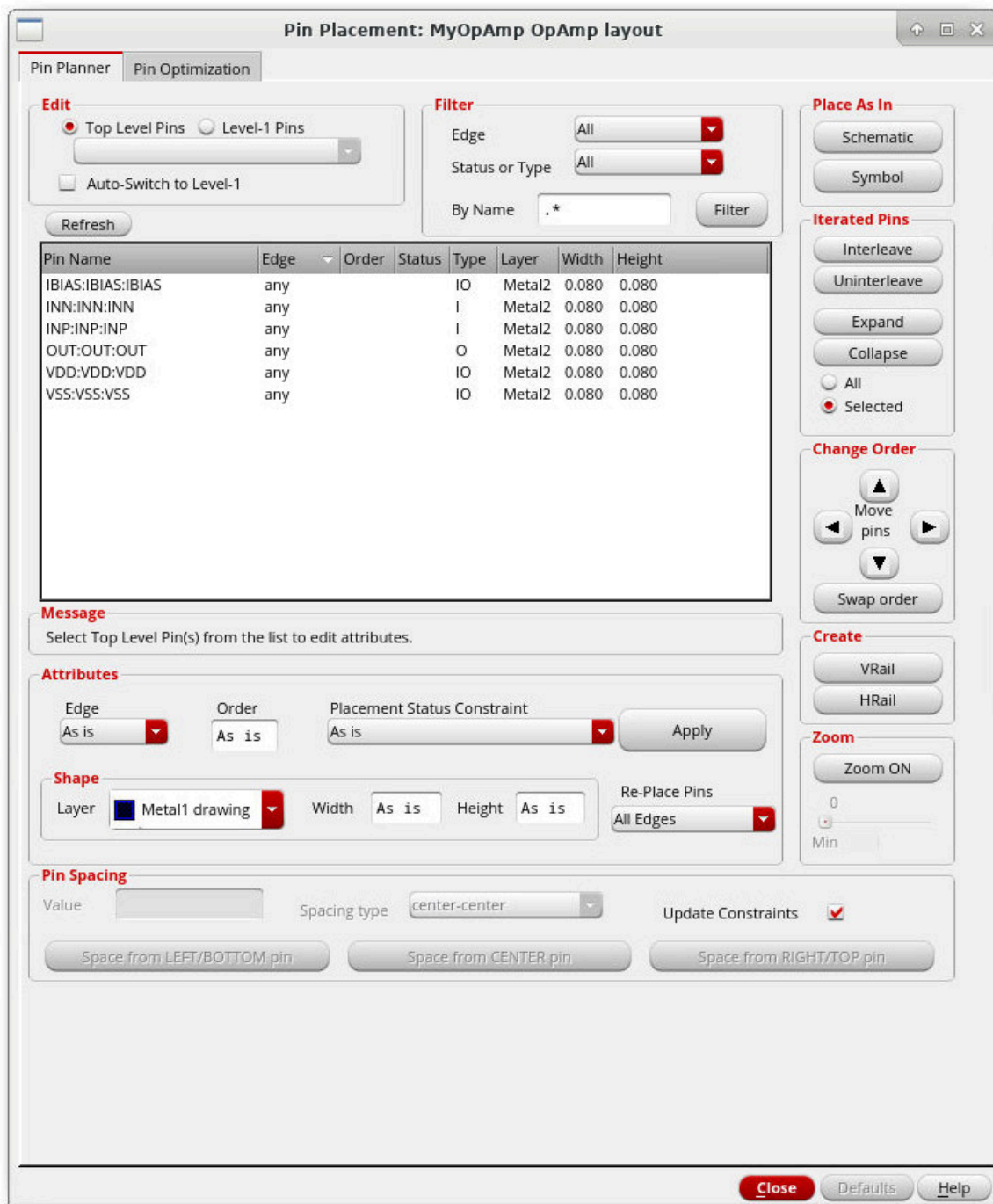
Layout design is now better centralized in the design window.

Action 214: Enable **Navigator** in the layout if it is not already displayed. Select the six pins listed in the Navigator assistant to locate them in the **Layout** view. You may need to use the left-facing arrow in the Navigator to display the Type filter and select **Pins**.



The pins are still placed at their original generated location. To improve their placement, you will use the **Pin Placement** tool.

Action 215: Go to **Place > Pin Placement**. The following form will appear:



The six pins in the design are listed on this form.

Action 216: If the pins are not already selected, use **Shift+LMB** to select all. Set the **Width** and **Height** to **0.1** and select **Apply**.

Pin Name	Edge	Order	Status	Type	Layer	Width	Height
IBIAS:IBIAS:IBIAS	any			IO	Metal2	0.080	0.080
INN:INN:INN	any			I	Metal2	0.080	0.080
INP:INP:INP	any			I	Metal2	0.080	0.080
OUT:OUT:OUT	any			O	Metal2	0.080	0.080
VDD:VDD:VDD	any			IO	Metal2	0.080	0.080
VSS:VSS:VSS	any			IO	Metal2	0.080	0.080

Message
Enter value to change height of selected pins.

Attributes

Edge: Any | Order: | Placement Status Constraint: No Constraint (Floating) | **Apply**

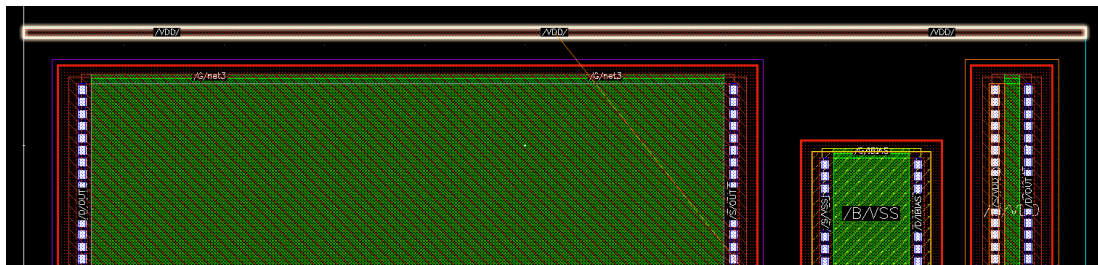
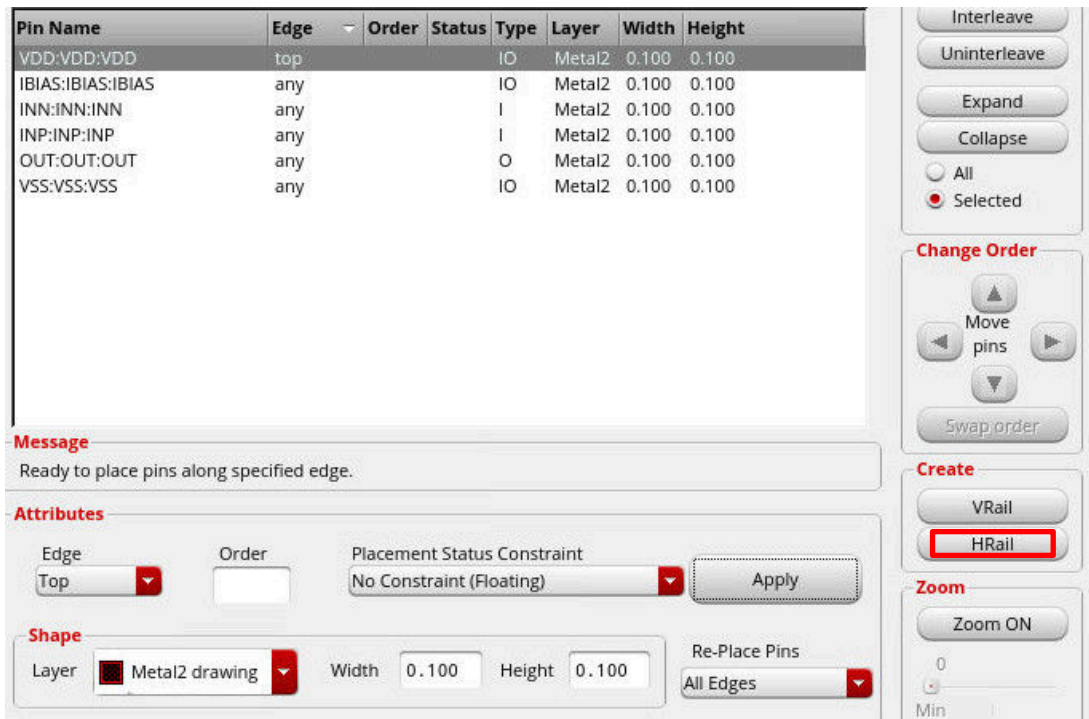
Shape

Layer: Metal2 drawing | Width: 0.100 | Height: 0.100 | Re-Place Pins: All Edges

The dimensions are changed for all pins. Now, set the positions for each pin.

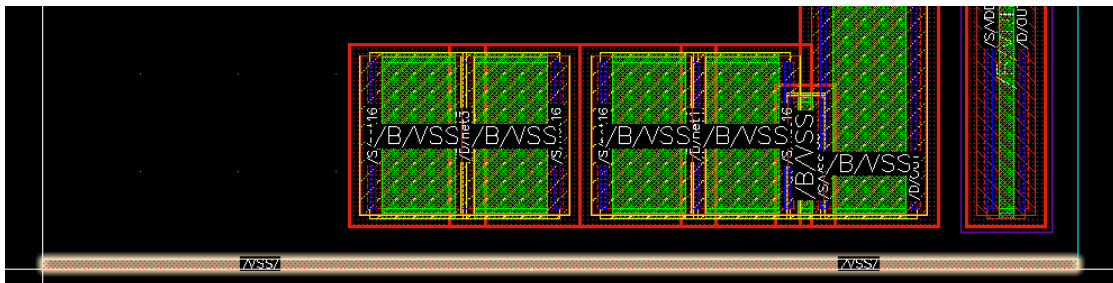
Action 217: Select only the **VDD** pin in the dialog box. Set **Edge** as **Top** and click **Apply**.

Action 218: With **VDD** still selected, select **HRail** in the **Create** section.



This creates a horizontal rail for **VDD** at the top border of the **PR Boundary**.

Action 219: Repeat the previous two Actions for **VSS**, but with **Edge** selected as **Bottom**.



Action 220: Define the following placement for the remaining pins:

- **IBIAS, INN, and INP** on the left

- **OUT** on the right

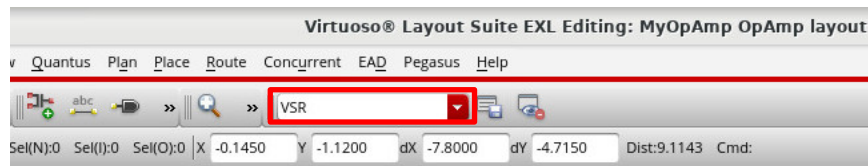
After finishing the above, the form should look like as follows:

Pin Name	Edge	Order	Status	Type	Layer	Width	Height
IBIAS:IBIAS:IBIAS	left			IO	Metal2	0.100	0.100
INN:INN:INN	left			I	Metal2	0.100	0.100
INP:INP:INP	left			I	Metal2	0.100	0.100
VDD:VDD:VDD	top		F	IO	Metal2	10.5...	0.100
OUT:OUT:OUT	right			O	Metal2	0.100	0.100
VSS:VSS:VSS	bottom		F	IO	Metal2	10.5...	0.100

Action 221: Close the form and save the layout view.

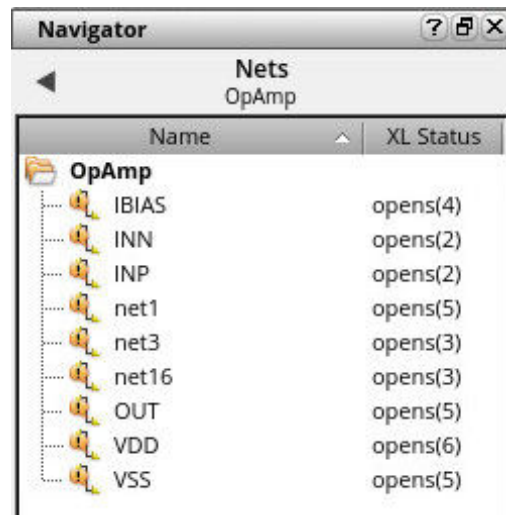
Lab 15: Routing

Action 222: In the layout, change the workspace from **Classic** to **VSR**. This will allow you to see both the **Navigator** and **Palette** assistants, which can be helpful for routing.



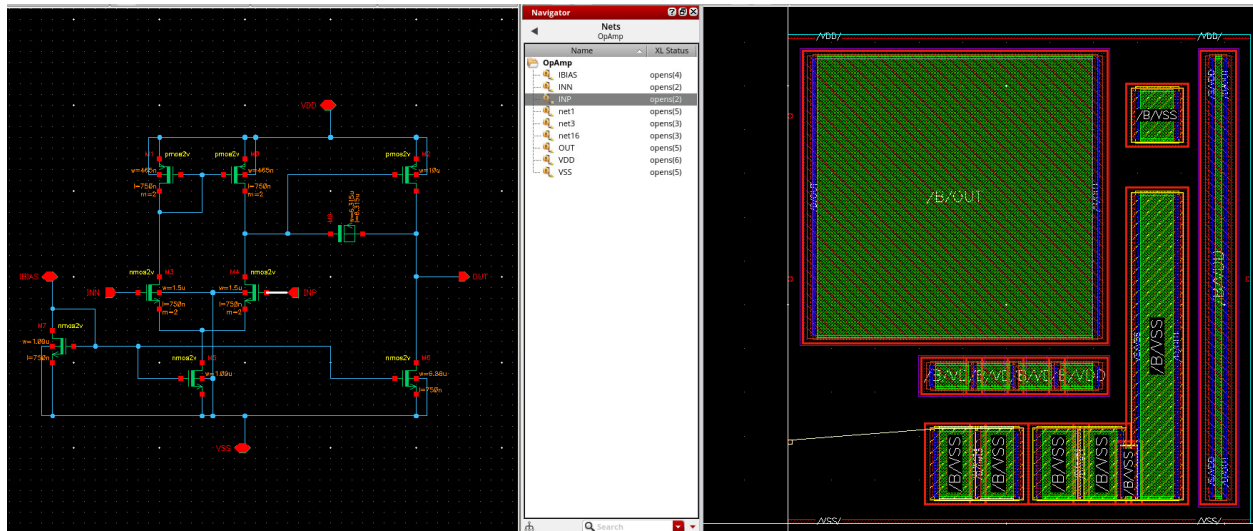
The **Navigator** and the **Palette** assistants are docked and tabbed on the left of the layout canvas.

Action 223: On the **Navigator**, display the **Nets**. Make the assistant wide enough to display the **XL Status** column.



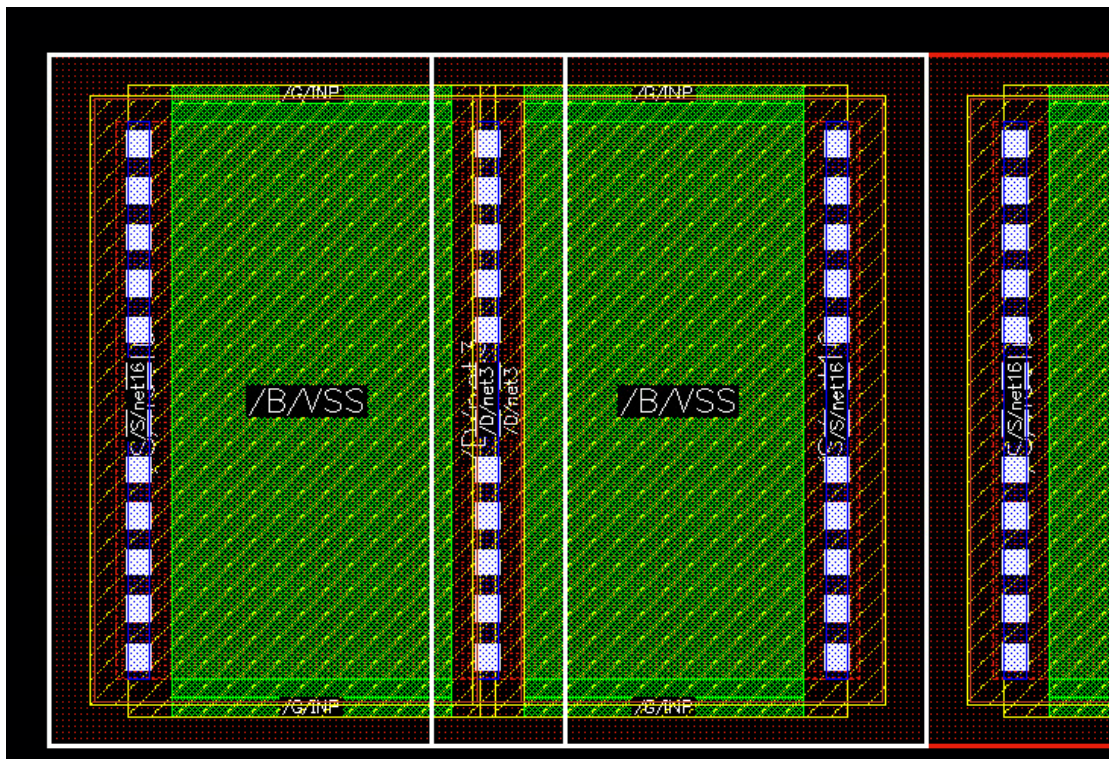
The **XL Status** column shows the number of **opens** for each net.

Action 224: Select the net **INP**.



This highlights the net on the schematic and shows the location where it should be connected in the layout. One easy way to route this net is by using the autoroute feature. First, you need to define gate connections for these transistors.

Action 225: Locate the **M4** instance in the layout. Select the two fingers and zoom in.

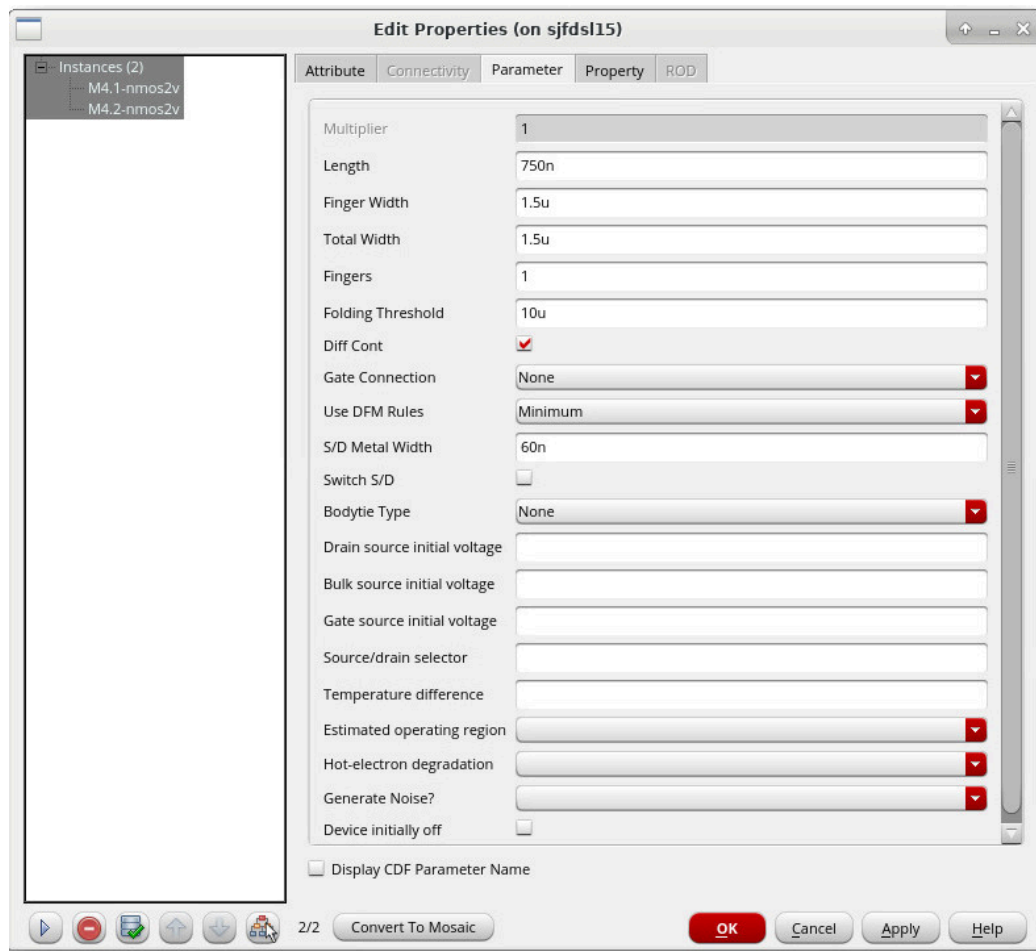


Action 226: Press **q** to invoke **Edit Instance Properties**.

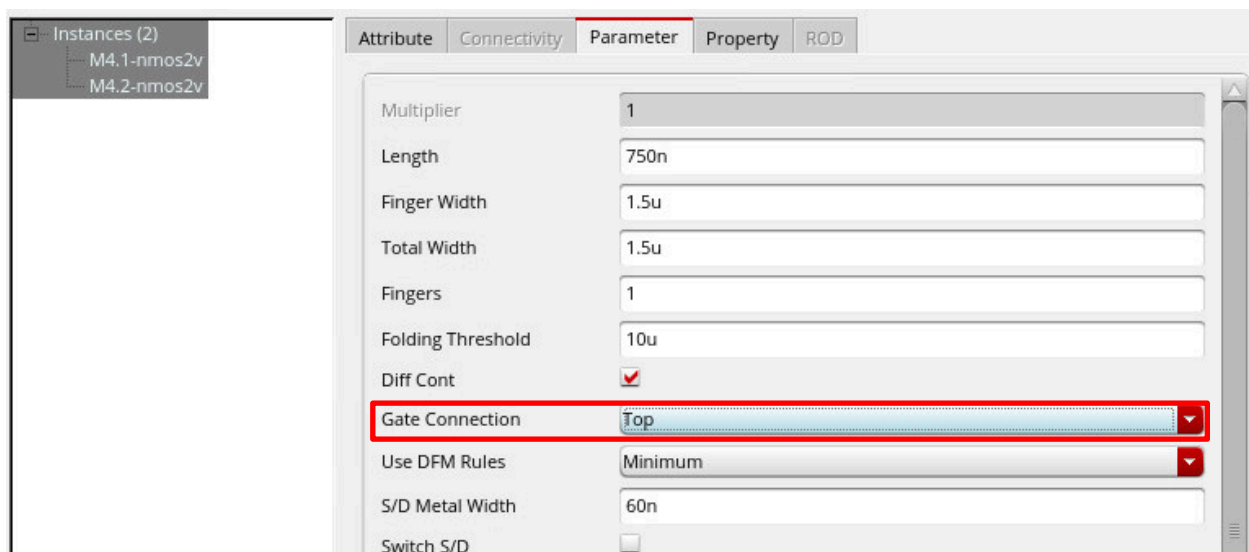


Action 227: Use **Shift** to select both instances in the list on the left. Select the **Parameter** tab.

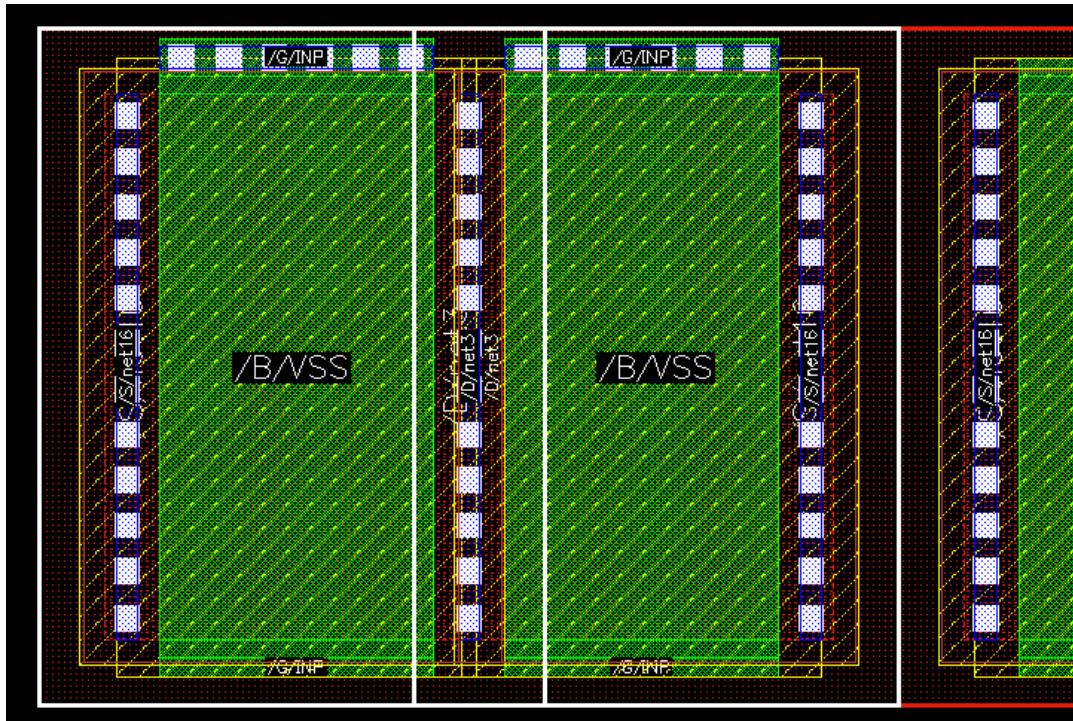
Basics of Analog Flow: RAK



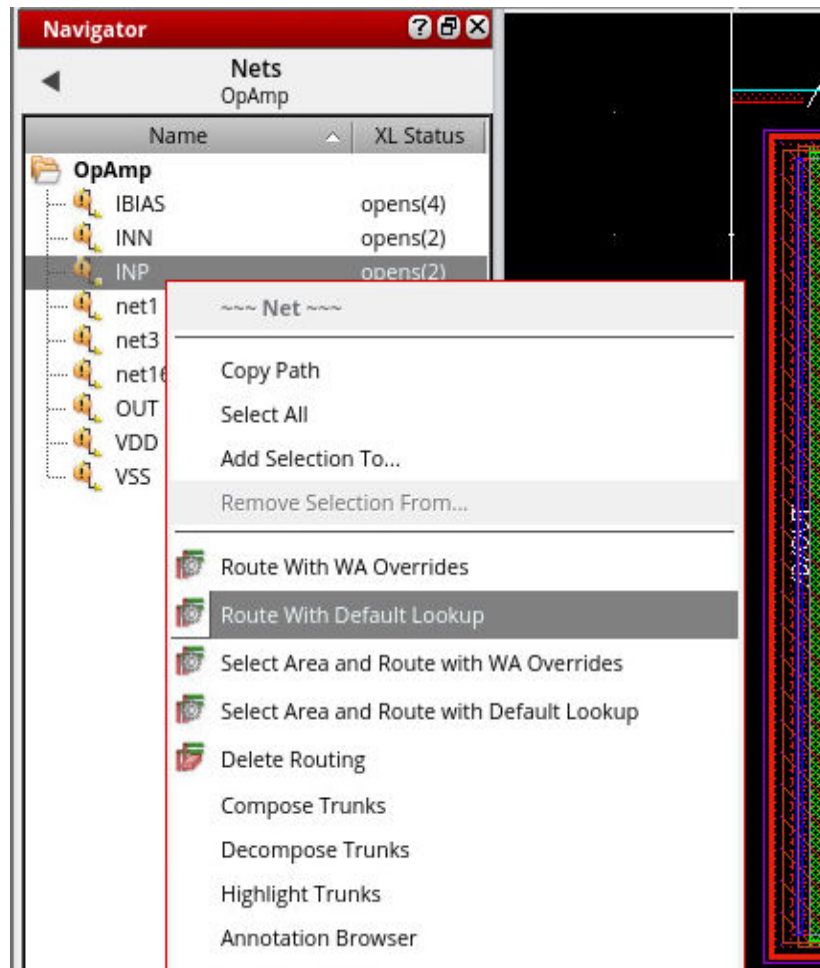
Action 228: Set **Gate Connection** to **Top** and press **OK**.



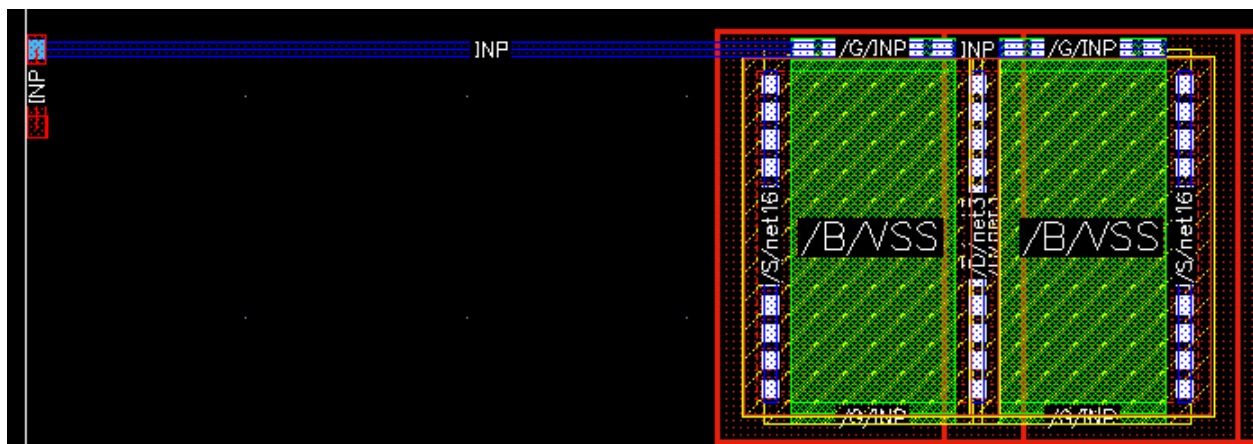
Gate connections will appear, as shown below.



Action 229: Press **f** to zoom out. Go back to the **Navigator** assistant and display the **Nets**. Locate **INP**, right-click, and select **Route with Default Lookup**. Then, unselect the created net.

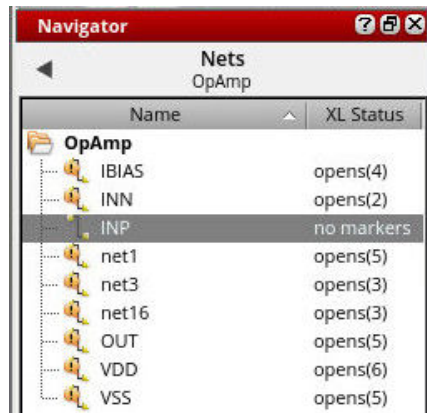


Your layout should look like the following image:



This routing command used Metal1 for horizontal routes and Metal2 for vertical. The appropriate 'via' is created where they intersect.

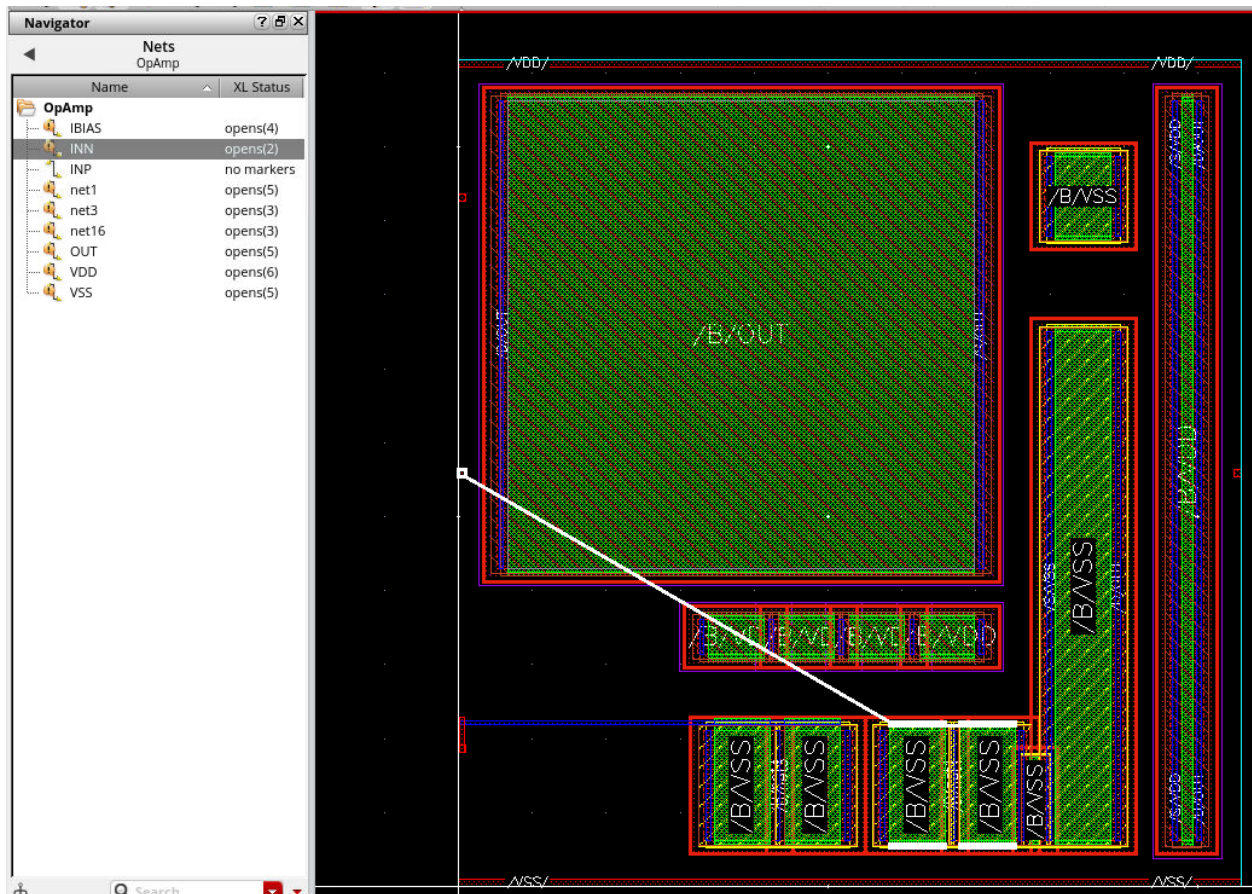
On the **Navigator** assistant, **XL Status** will change to reflect the complete route:



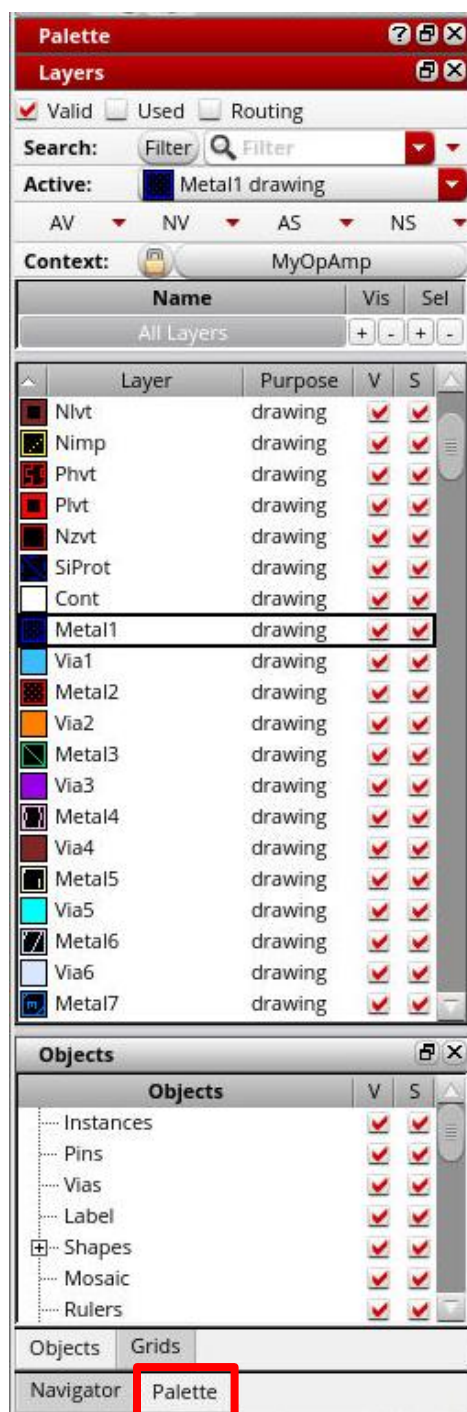
Name	XL Status
OpAmp	
IBIAS	opens(4)
INN	opens(2)
INP	no markers
net1	opens(5)
net3	opens(3)
net16	opens(3)
OUT	opens(5)
VDD	opens(6)
VSS	opens(5)

Next, you will use the manual routing commands.

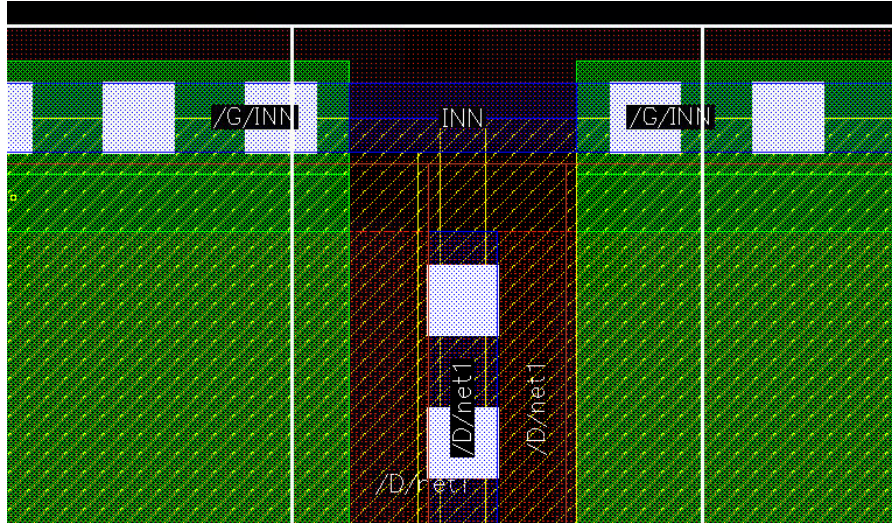
Action 230: Press **f** on the layout to see the full design. Select **INN** in the **Navigator** assistant.



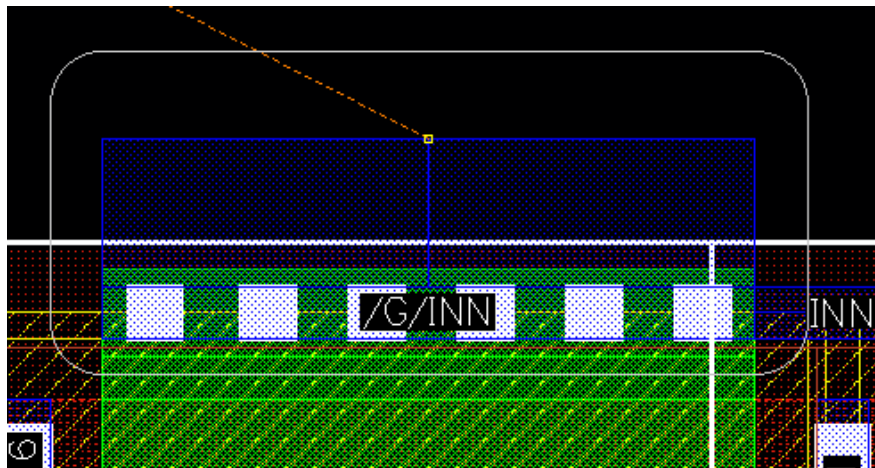
Action 231: Locate M3 and set the gate connection to **Top** as was just done for M4. Then, go to the **Palette** assistant and select the **Metal1** drawing layer.



Action 232: Press **P** to start the **Create Wire** command. Select the right edge of the leftmost M3 member gate and then the left edge of the rightmost M3 transistor gate, wiring their gates together.

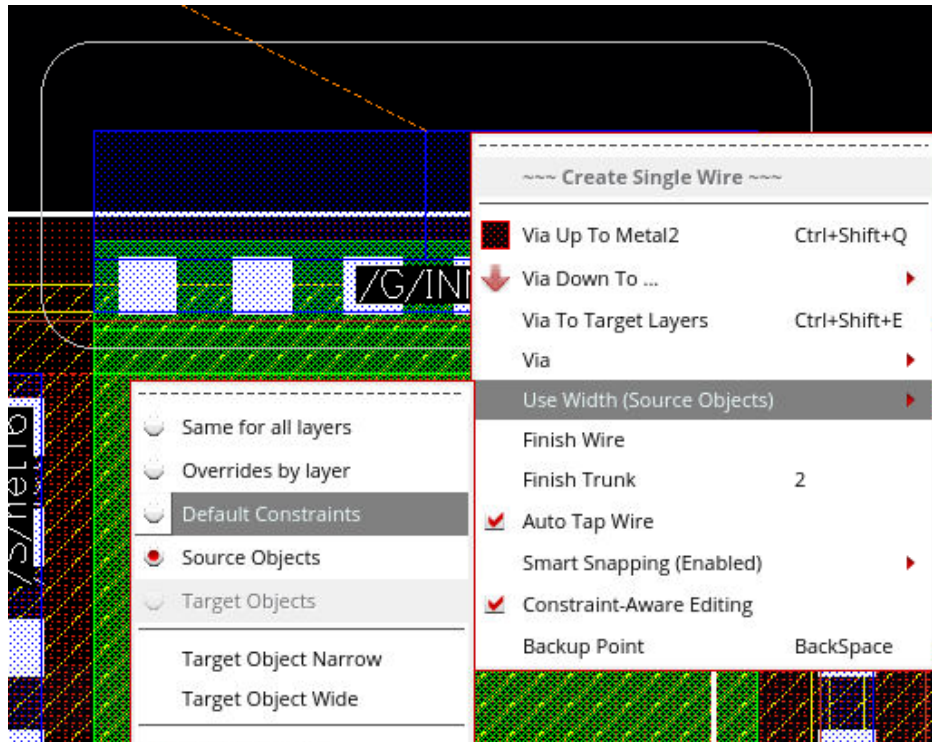


Action 233: Still in the 'wiring' command, zoom in on the M3.1 gate (this is the component on the left) and select the top edge to start a new wire, as shown below.

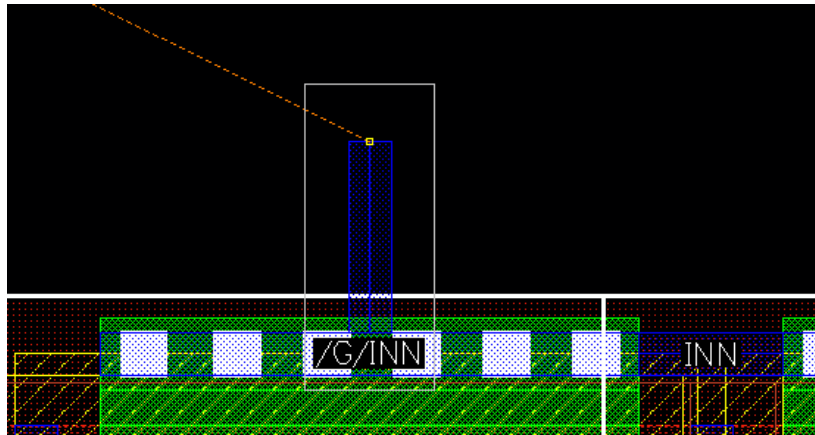


You will see that this will start drawing the wire of the same width as the gate. As you do not want this wire to be that thick, you need to change it from within the command.

Action 234: As you are drawing the wire, right-click and select **Use Width > Default Constraints**.

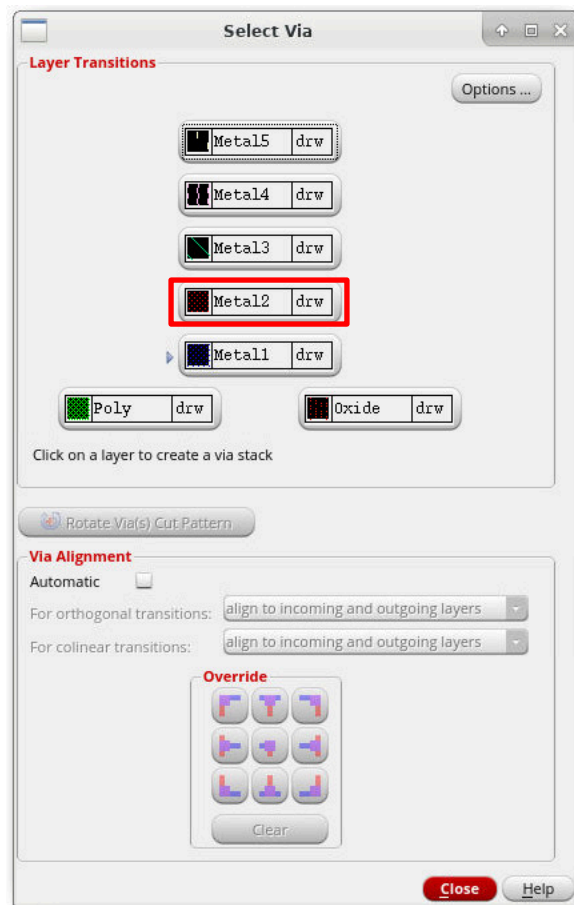


This will change the wire width, as shown below.

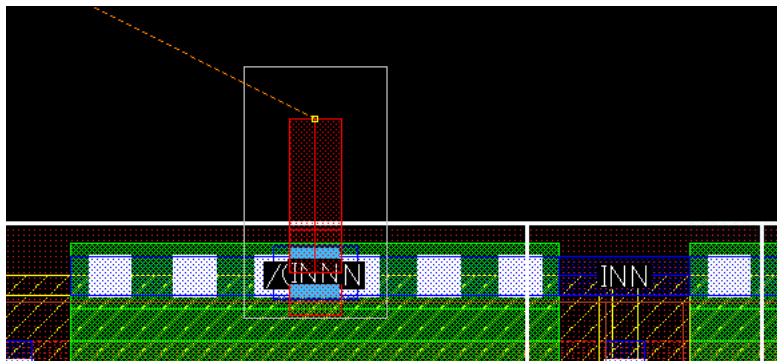


Switch to **Metal2** for this vertical route.

Action 235: Return the cursor to the gate (where the route starts) and press the **Space bar** to open the **Select Via** form. Select **Metal2**. Then, close the form.

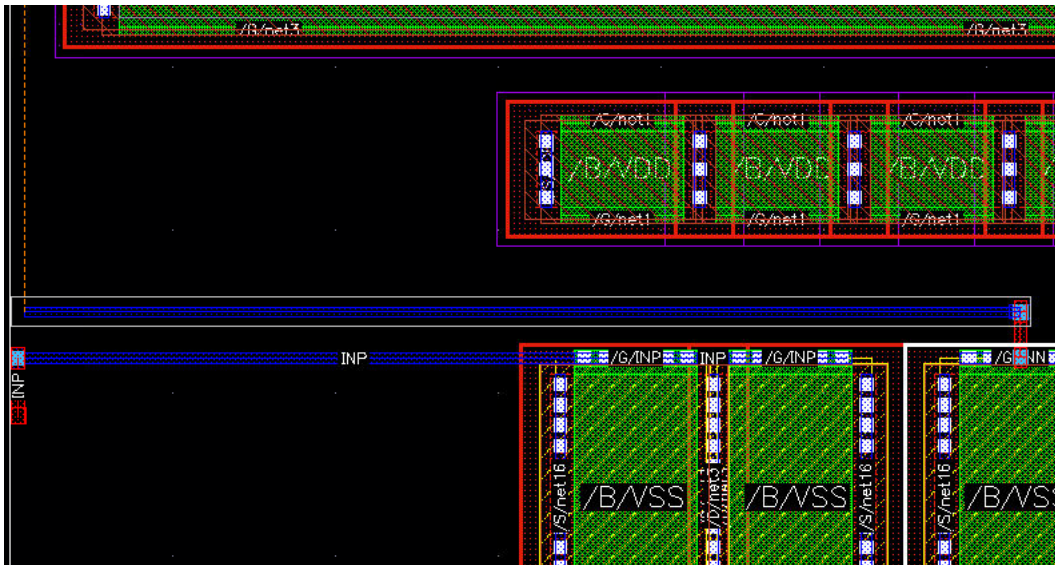


Action 236: Click to place the via and start the **Metal2** wire.

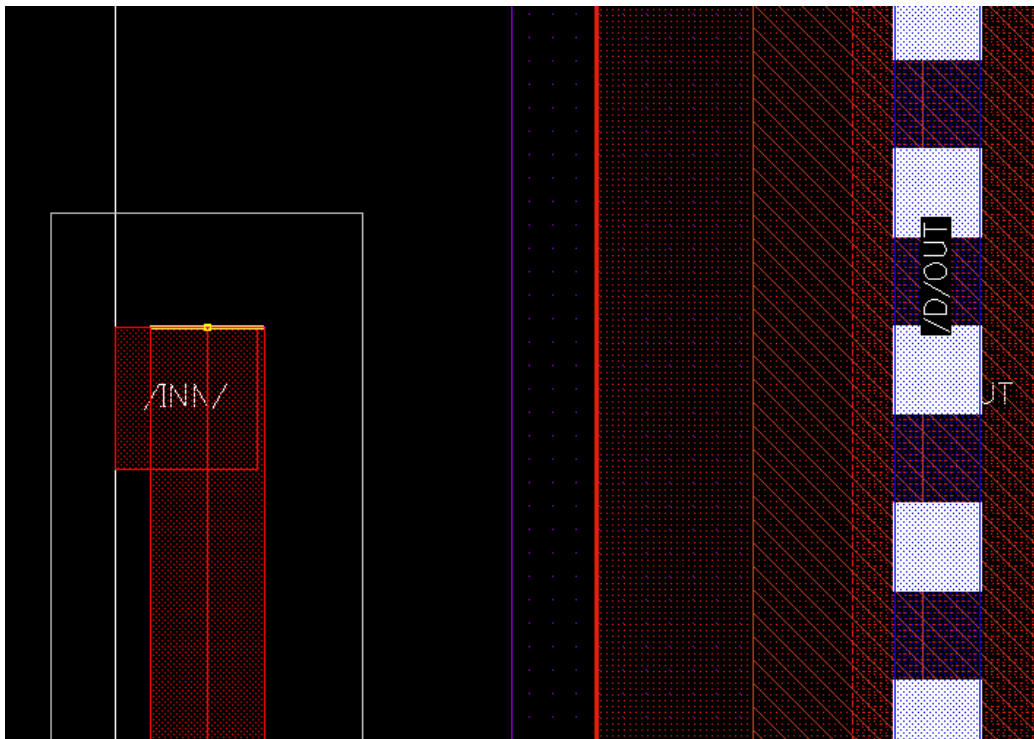


Action 237: Press **Space bar** again and select **Metal1**. Then, close the form.

Action 238: Zoom out and continue the wire on the left border of the layout.

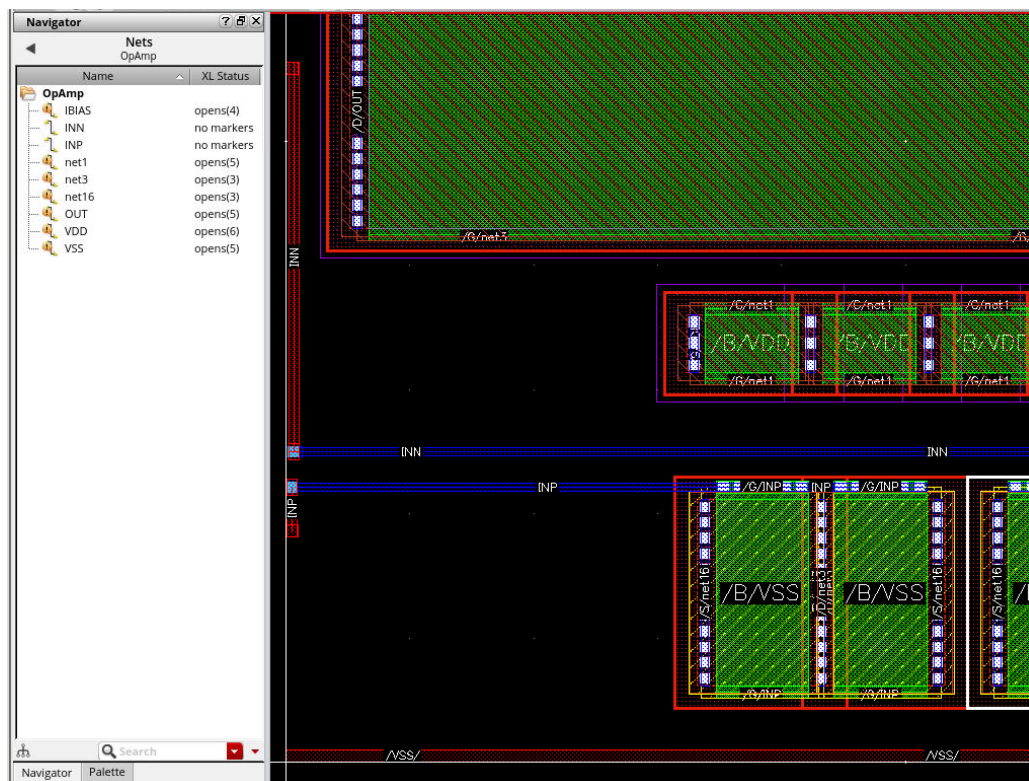


Action 239: The **INN** pin is directly above the wire (if needed, you can move it away from any components). Continue the route from the horizontal segment by dropping another via at the endpoint (use the **Space** bindkey again) and selecting **Metal2**. Continue the route to the **INN** pin. Double-click or press **Enter** to end the connection, and then press **Esc**.





Action 240: Return to the **Navigator** assistant.



As you can see, **INN** is completely routed, with no opens reported.

Action 241: Save the layout and proceed to the next lab.

Lab 16: DRC Tests

You have made several adjustments to the layout so far, but you still do not know if all the physical rules are being followed. You will use Design Rule Checking (DRC) in this lab to check this issue. It is a good practice to do DRC checks after any significant changes to the design before moving further to avoid propagating mistakes.

If you strictly followed the instructions given in the other labs of this module, DRC tests should have similar results to the ones shown in this lab. If not, some differences might appear. In that case, it might be better to copy everything from the `solutions/Module5/Lab15` directory and paste it on the `BAF_RAK` directory, replacing repeated files.

Action 242: Go to **Pegasus > Run DRC**. In the **DRC Run Submission Form**, enter **DRCrun** into the **Run Directory** field and proceed to the **Rules** tab (tab controls are on the left side of the window).

Pegasus 22.20-64b DRC Run Submission Form

File Options Toolbars Help cadence

Run Data

Run Directory DRCrun

Processing

Run Pegasus Single CPU

Run Pegasus on Local host

Config files

Add... View... Edit... Remove...

DRC Options

License Timeout, sec

Core License Name

PegasusUI Log File DRCrun/pegasuidrc.log

Pegasus Log Directory LOGS

Temporary Directory List

☐ Interactive engine

Start RV ☒

Cancel Apply Submit

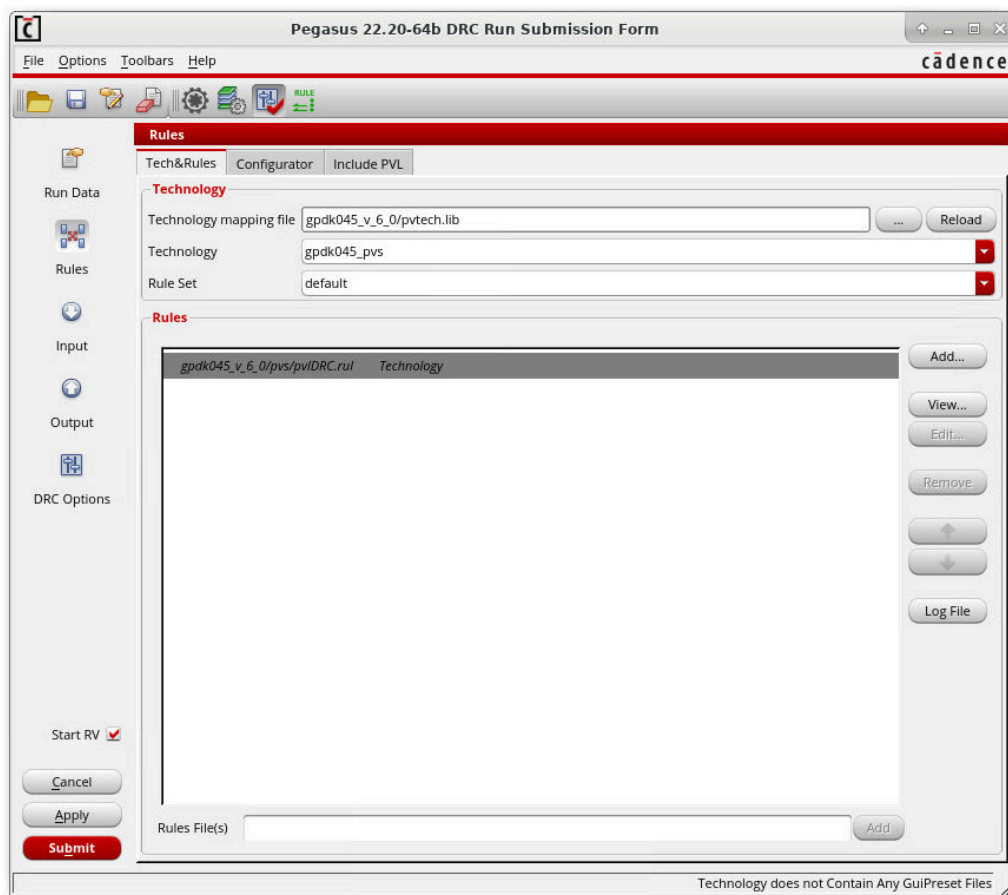
No Presets File was Loaded

Action 243: In the **Rules** tab, press the '...' button on the **Technology** mapping file row and set the path to **gpdk045_v_6_0/pvtech.lib**.

The `pvtech.lib` file defines where the required setup information is stored for design rule checking. The structure is similar to the `cds.lib` file, mapping a technology name to the location where the rules are saved.

Action 244: Choose the **gpdk045_pvs** technology from the drop-down list.

The **DRC Run Submission Form** should look as shown below.

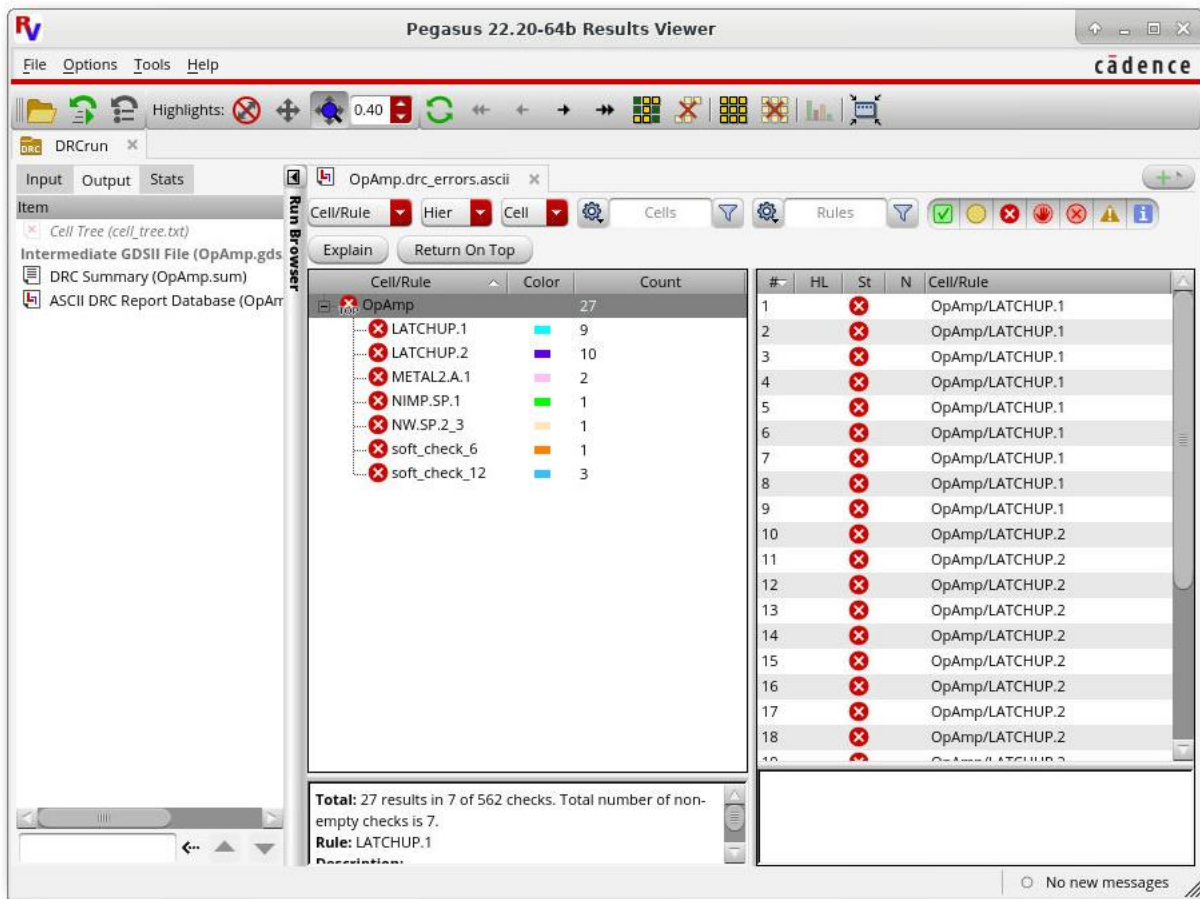


If there are more rules listed in the **Rules** pane, delete them.

Action 245: Leave the rest of the setup as-is and press **Submit**.

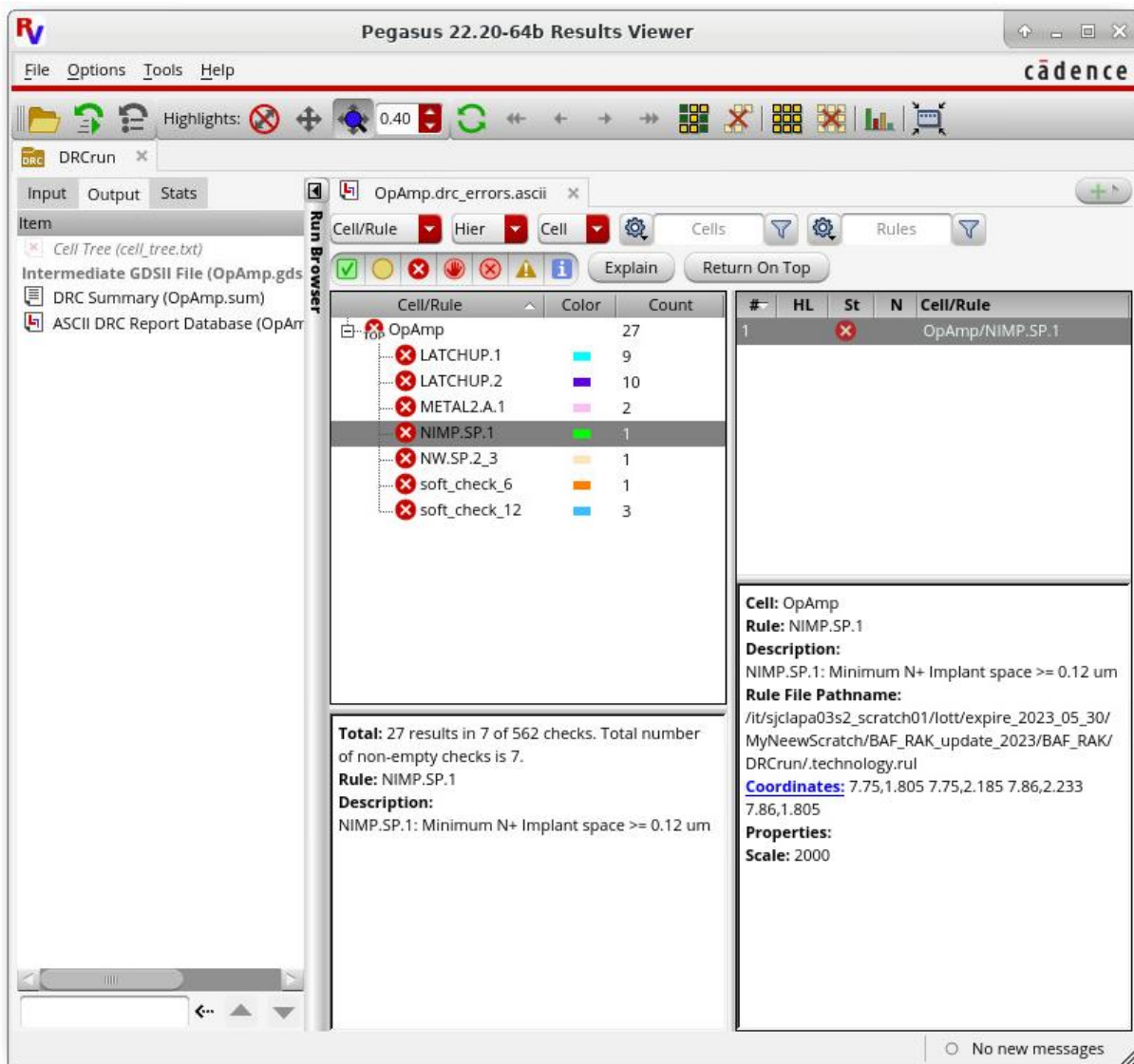
The run will start, and a log file will appear. The following results should appear after the run is finished:

Basics of Analog Flow: RAK



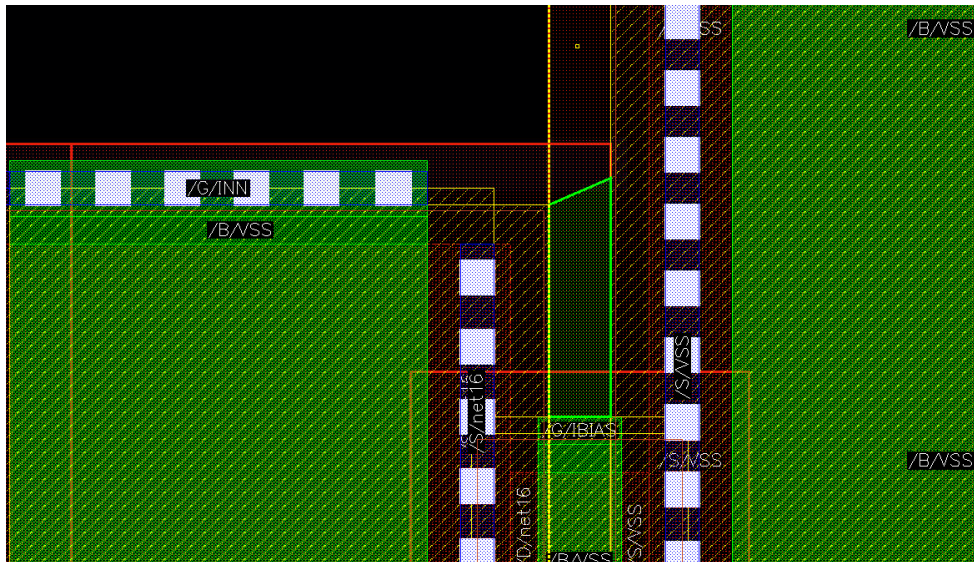
As you can see, **27** DRC errors are found. Your number may vary depending on how closely your design matches this reference.

Action 246: Click on the **NIMP.SP.1** error. Select the **OpAmp/NIMP.SP.1** entry in the right column.



Look in the bottom-right section of the Viewer to see details of the selected error. This is a violation of the **N+ Implant** spacing.

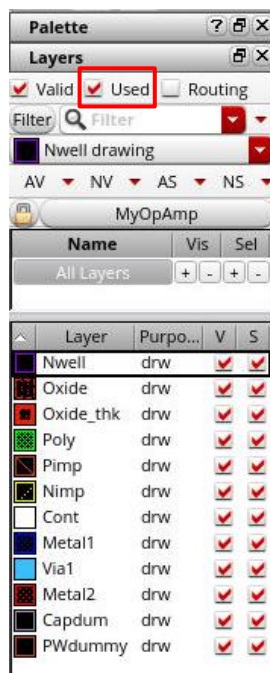
Action 247: Select the **Coordinates** link in the error details. This will zoom into the location of this error in the layout window.



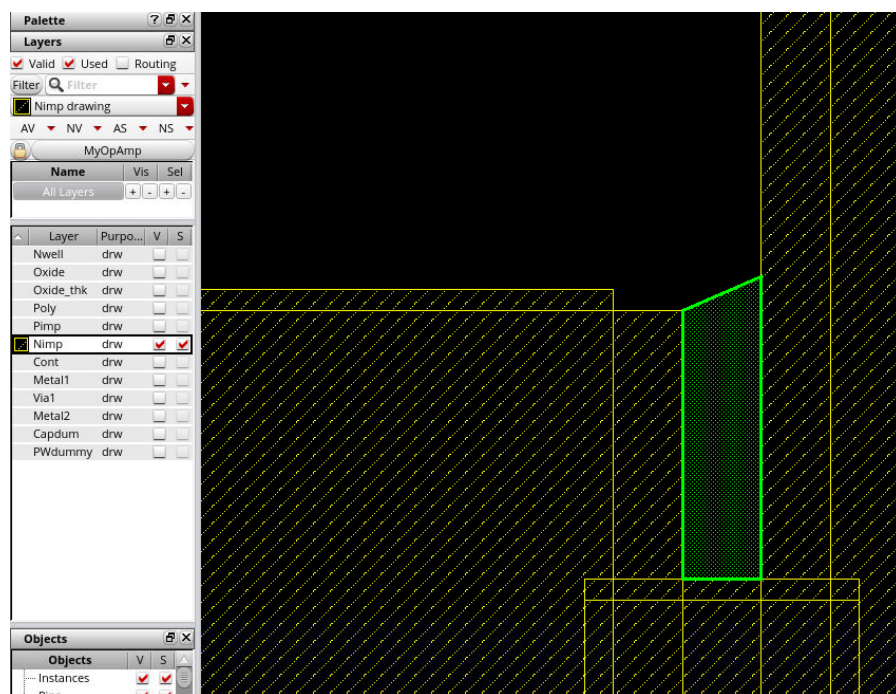
The location of the error is zoomed in and highlighted with the other objects dimmed.

The error shows that the **N+ Implant** layers of the transistors are too closely spaced. You can solve this by joining the N+ Implant layers since they all should connect to the same body.

Action 248: On the **Palette** assistant, select the **Used** checkbox to see only the layers used in the design.



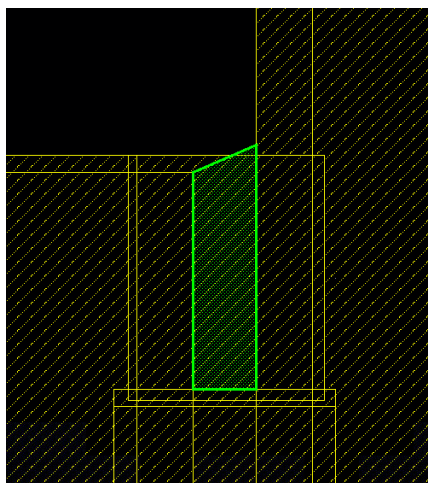
Action 249: Select **Nimp** and click on **NV** (None Visible) to turn off the visibility for all other layers.



The **V** and **S** checkboxes on the right of each layer set each layer as 'visible' and/or 'selectable'.

To solve the spacing error, draw a rectangle on the **Nimp** layer.

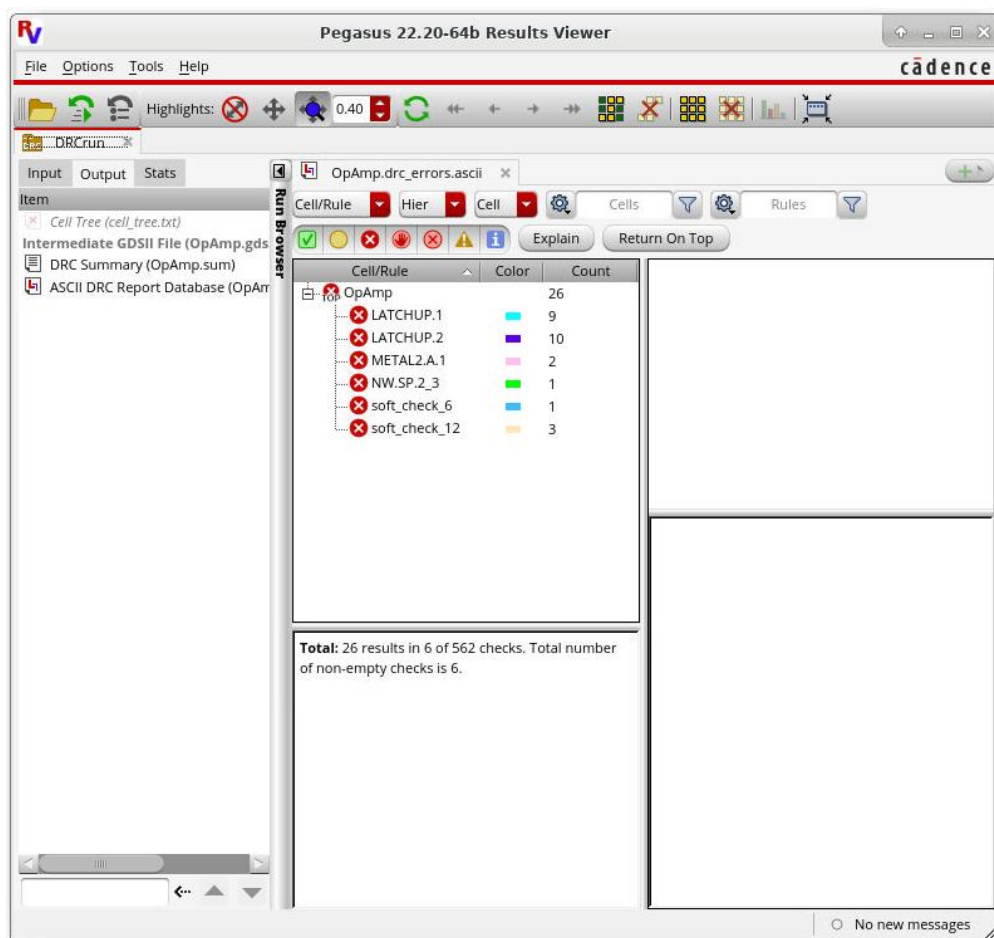
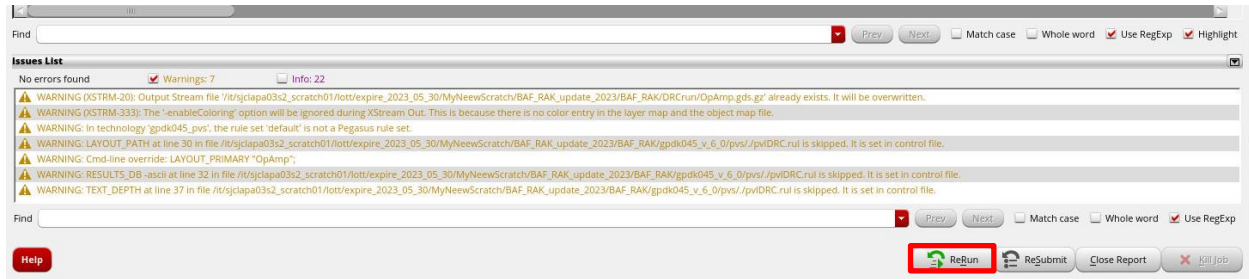
Action 250: With the **Nimp** layer selected, press **r** and draw a rectangle that encloses the space between the implants, as shown below. Then, press **Esc**.



Basics of Analog Flow: RAK

There is no problem if **Nimp** layers overlap each other.

Action 251: Save the view, go back to the **DRC Reports** window, and press **ReRun**.



The **NIMP.SP.1** error should disappear. If it does not, repeat the above process to address the additional spacing violations.

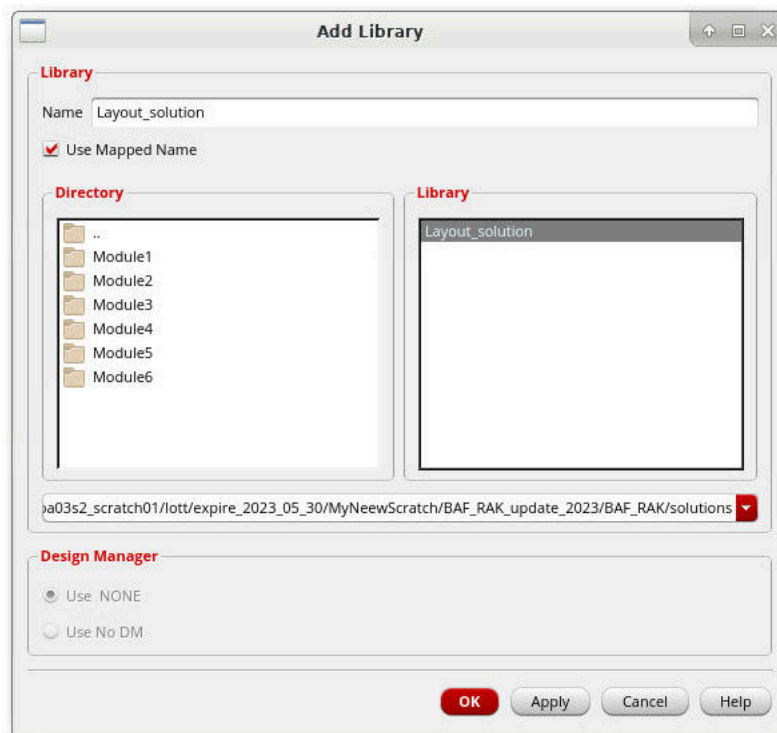
Action 252: Close **all** windows except for the CIW and Library Manager and proceed to the next Lab. You can discard changes if prompted.

Lab 17: LVS (Layout Versus Schematic)

Before running LVS, copy a completed version of the layout into your work area.

Action 253: In the CIW, select **Tools > Library Path Editor** and go to **Edit > Add Library**.

Action 254: In the **Add Library** form, double-click on the **solutions** directory and then select **Layout_solution** in the **Library** field. Press **OK**.

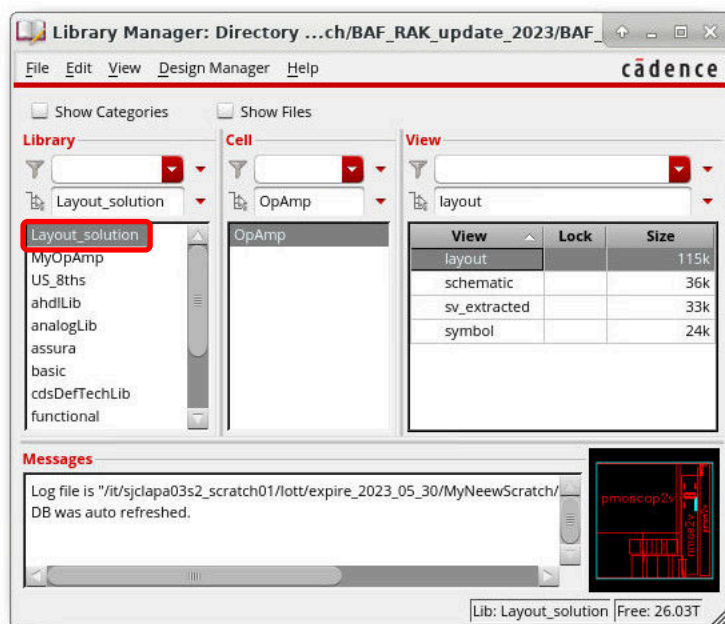


The respective library will be added to your session.

Action 255: Exit the **Library Path Editor** window and save changes when prompted.

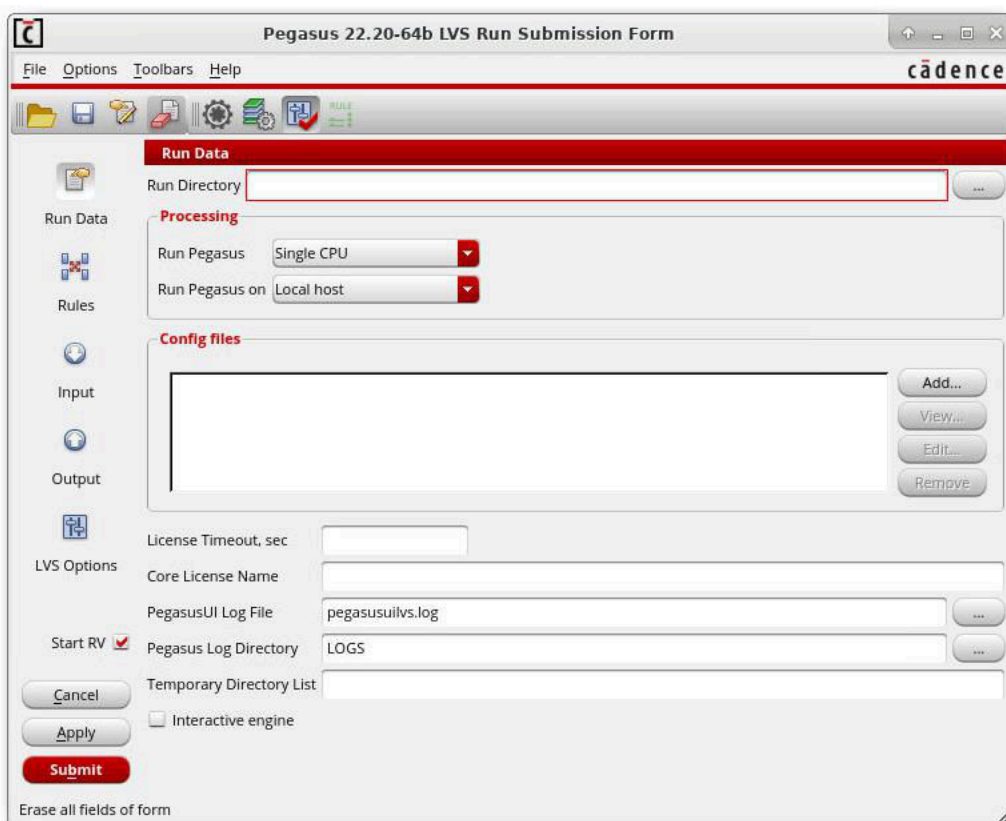
Action 256: Check the **Library Manager** to make sure that the **Layout_solution** library is listed.

Basics of Analog Flow: RAK



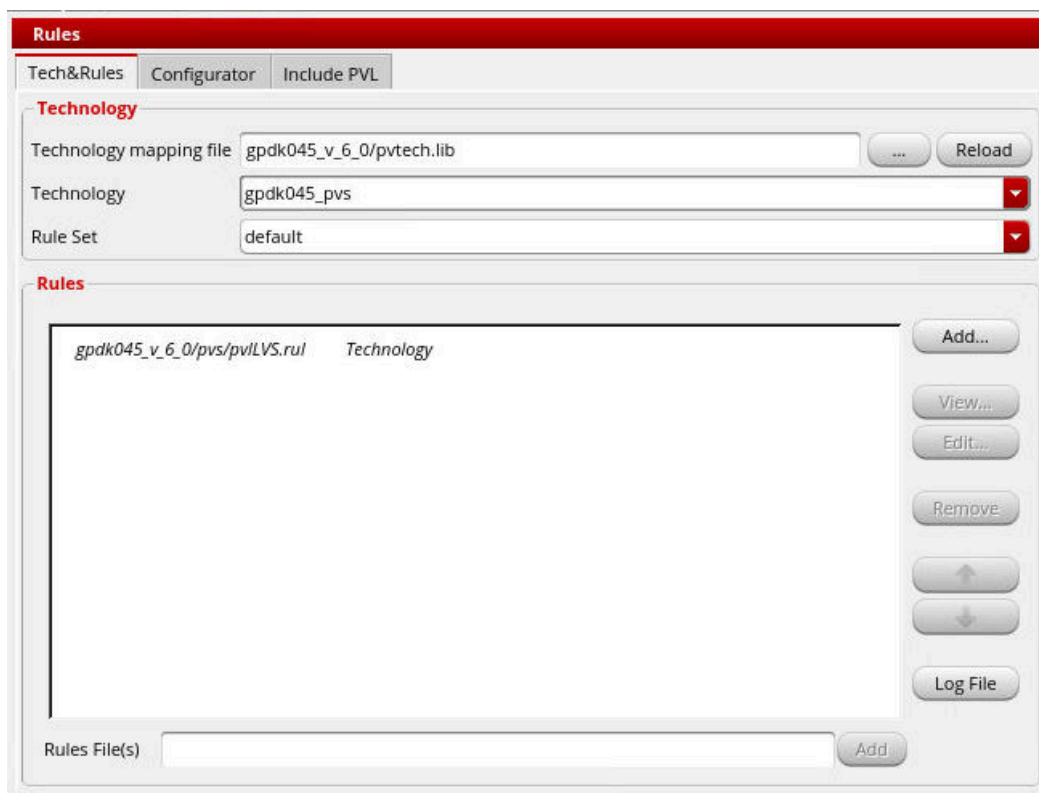
Action 257: Open the **Layout_solution/OpAmp/layout** design.

Action 258: Go to **Pegasus > Run LVS**. The following form opens:

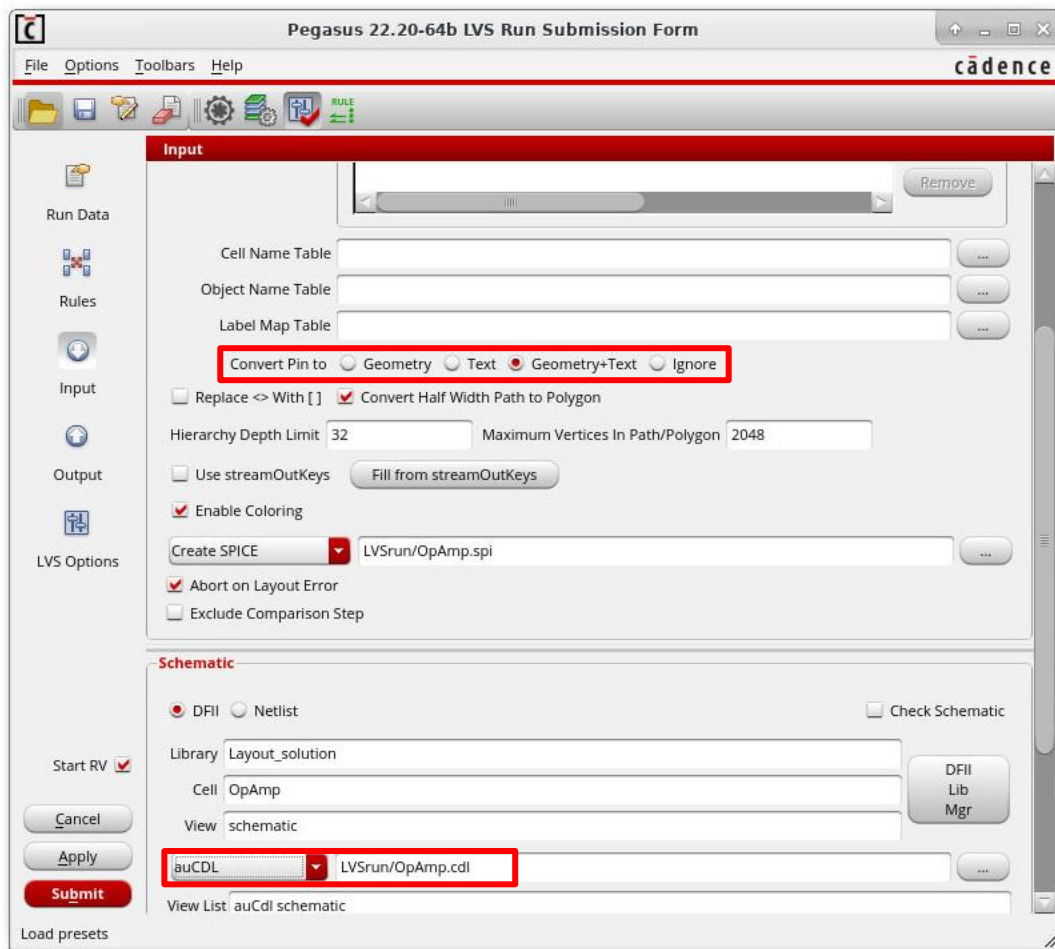


Action 259: Specify **Run Directory** as **LVSrun**.

Action 260: On the **Rules** tab, set **Technology mapping file** as **gpdk045_v_6_0/pvtech.lib** and the technology as **gpdk045_pvs**. Verify that there is a single entry in the **Rules** pane.



Action 261: On the **Input** tab, change the **Convert Pin to** option to **Geometry+Text** and change the **Create CDL** option to **auCDL**, as shown below.



This will make sure that all pins are correctly identified, even if they are not assigned using explicit labels.

Action 262: On the **Output** tab, enable the **Create Quantus Input Data** checkbox, as shown below.

Pegasus 22.20-64b LVS Run Submission Form

File Options Toolbars Help

cadence

Output

H-Cell Settings

☐ Automatch

☐ HCell

☐ GenHierCells

☒ Run ERC Checks

ERC Report

Name: OpAmp.sum

Limit: 1000

☒ Replace File ☐ Append To File ☐ Statistics by Cell

Output Format: ASCII LVSrun/OpAmp.erc_errors.ascii

Output Errors Hierarchically: Rules Definition ☐ Output Complete Pathchk Results

☐ Use Waivers for ERC Checks

LVS Report

Name: OpAmp.rep

Options: -none

Limit: 50 Mismatched Nets Limit: 100

Additional Output

☒ Create Quantus Input Data

Data Dir: svdb

Start RV ☒

Cancel Apply Submit

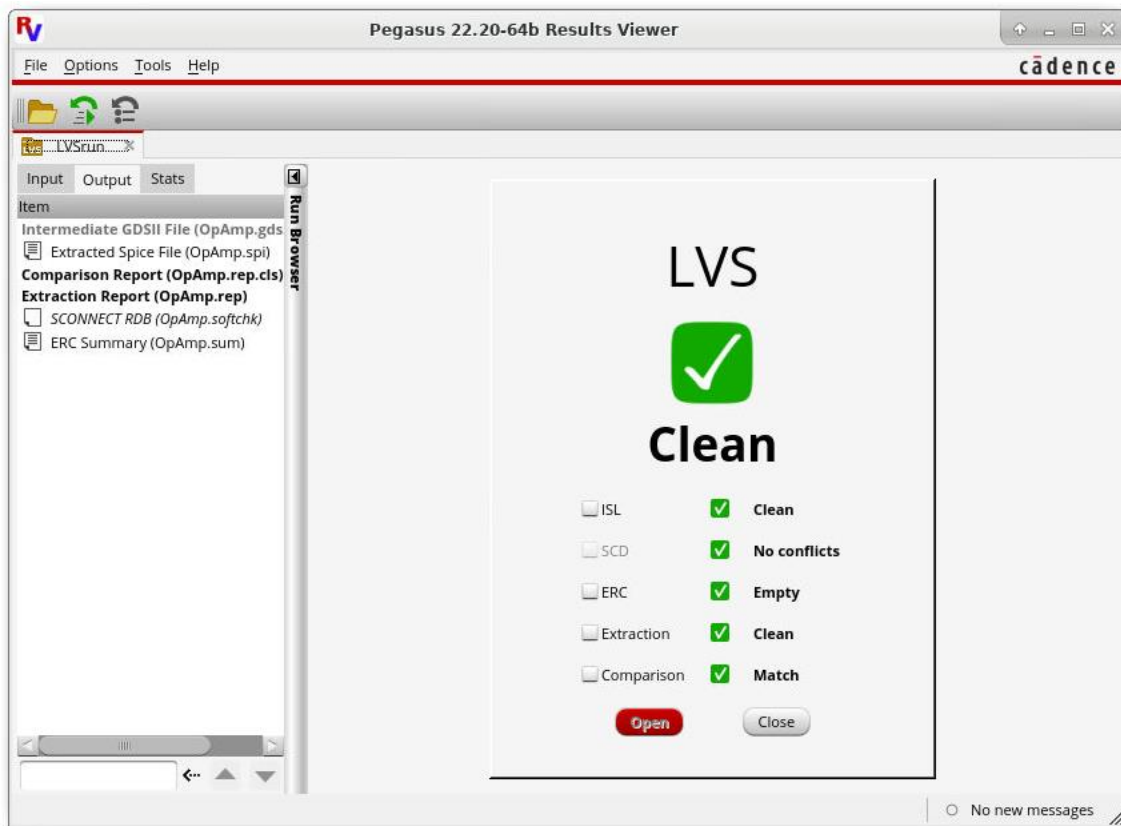
Keep Layers: None

Technology does not Contain Any GuiPreset Files

You will use it for the parasitic extraction in the next module.

Action 263: Leave the rest as default and click **Apply**.

After the LVS run finishes, the following window should appear:



This means that your LVS is clean, and no errors are found in the comparison between your layout view and your schematic view.

Action 264: Close the **Results Viewer** window and close the Pegasus form.

Action 265: Leave the layout view open and proceed to the next module.

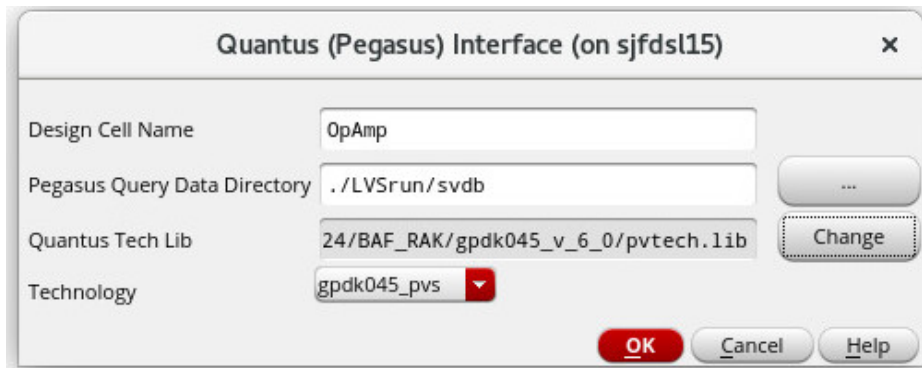
Module 6: Post-Layout Simulation

To be able to simulate with parasitics, you first need to extract parasitics from your layout. For this RAK, you will use **Quantus** to create two parasitic formats: a Detailed Standard Parasitic Format (DSPF) file and Smart View.

DSPF is the most common method of representing parasitics and has been used for many years. Smart View is a new extracted view that enables you to simulate and analyze post-layout design seamlessly in ADE without managing external files or additional translation steps. Smart View has a reduced data size and improved performance over the original Quantus extracted view. The other advantage of Smart View is that you can visualize parasitics on the layout. If your Quantus version does not yet include the Smart View feature, you can skip Lab 19.

Lab 18: Using Quantus to Create a DSPF File

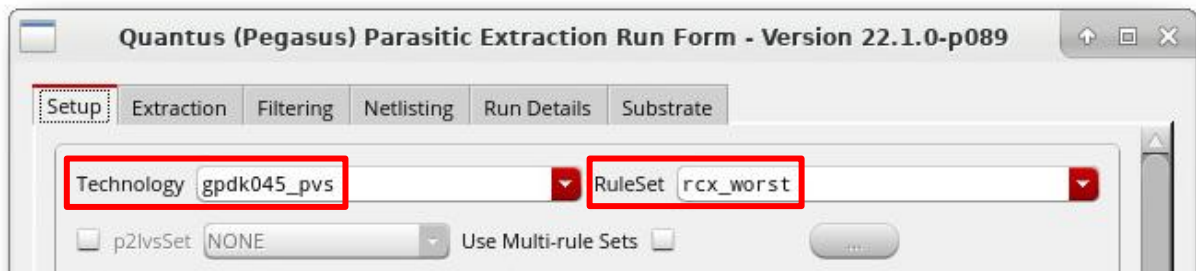
Action 266: On **Layout_solution/OpAmp/layout** design, go to **Quantus > Run Pegasus - Quantus** and fill in the initial form, as shown below.



The image shows the 'Quantus (Pegasus) Interface (on sjfdsl15)' dialog box. It contains the following fields and controls:

- Design Cell Name:** OpAmp
- Pegasus Query Data Directory:** ./LVSRun/svdb
- Quantus Tech Lib:** 24/BAF_RAK/gpdk045_v_6_0/pvtech.lib
- Technology:** gpdk045_pvs (selected from a dropdown menu)
- Buttons:** OK, Cancel, Help, and a 'Change' button next to the Tech Lib field.

Action 267: Click **OK** on the form. On the **Quantus (Pegasus) Parasitic Extraction Run Form**, fill in the **Setup** tab, as shown below.



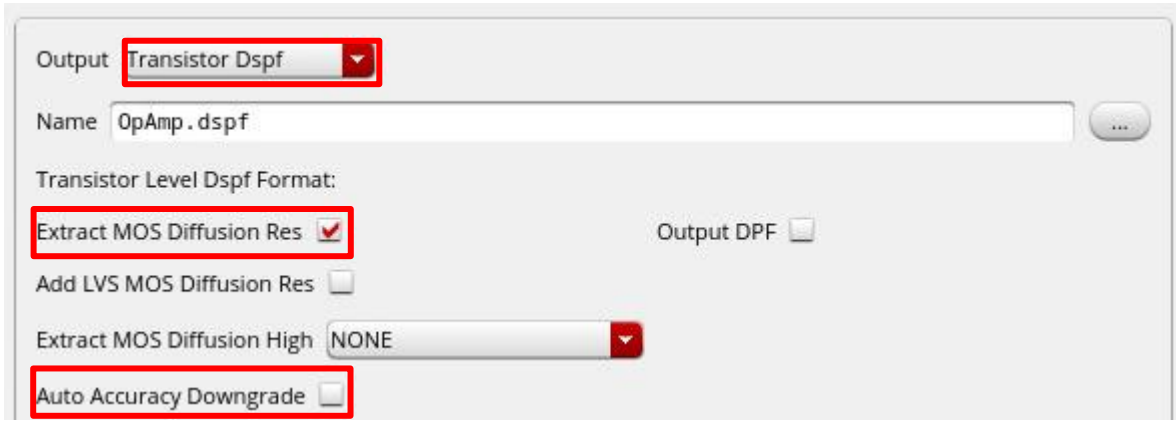
The image shows the 'Quantus (Pegasus) Parasitic Extraction Run Form - Version 22.1.0-p089' window. The 'Setup' tab is selected. The following fields are highlighted with red boxes:

- Technology:** gpdk045_pvs
- RuleSet:** rcx_worst

Other visible fields include:

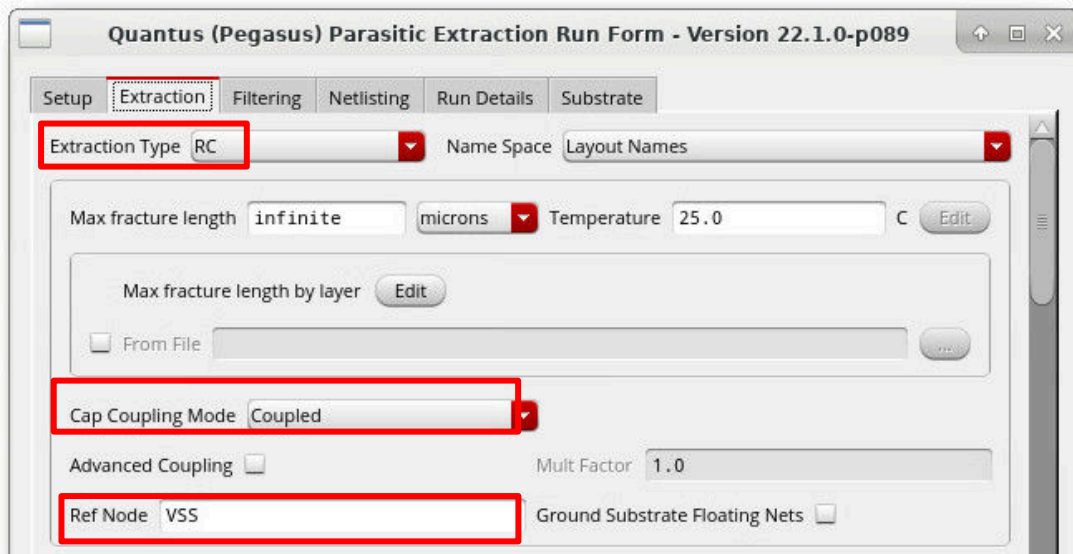
- Extraction:** (empty)
- Filtering:** (empty)
- Netlisting:** (empty)
- Run Details:** (empty)
- Substrate:** (empty)
- p2lvsSet:** NONE
- Use Multi-rule Sets:** (checkbox, unchecked)

Action 268: Scroll down on this tab and change **Output** to **Transistor DSPF**. Enable **Extract MOS Diffusion Res** and make sure that the **Auto Accuracy Downgrade** field is disabled.



The screenshot shows the 'Output' tab of the Parasitic Extraction Run Form. The 'Output' dropdown is set to 'Transistor Dspf'. The 'Name' field is 'OpAmp.dspf'. Under 'Transistor Level Dspf Format', 'Extract MOS Diffusion Res' is checked, 'Add LVS MOS Diffusion Res' is unchecked, and 'Extract MOS Diffusion High' is set to 'NONE'. The 'Auto Accuracy Downgrade' checkbox is unchecked. There is also an 'Output DPF' checkbox which is unchecked.

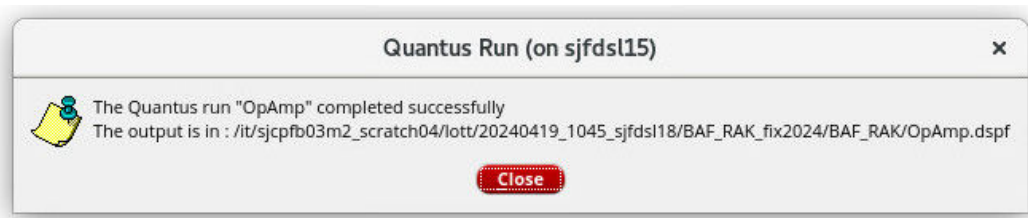
Proceed to the **Extraction** tab and fill it in, as shown below.



The screenshot shows the 'Extraction' tab of the Parasitic Extraction Run Form. The 'Extraction Type' is set to 'RC'. The 'Name Space' is 'Layout Names'. 'Max fracture length' is 'infinite' and 'Temperature' is '25.0 C'. Under 'Max fracture length by layer', there is an 'Edit' button. The 'From File' checkbox is unchecked. 'Cap Coupling Mode' is set to 'Coupled'. 'Advanced Coupling' is unchecked. 'Ref Node' is set to 'VSS'. 'Mult Factor' is '1.0'. The 'Ground Substrate Floating Nets' checkbox is unchecked.

Action 269: Click **Apply** to start the extraction.

When the **Quantus Run** is completed, the following window should open:



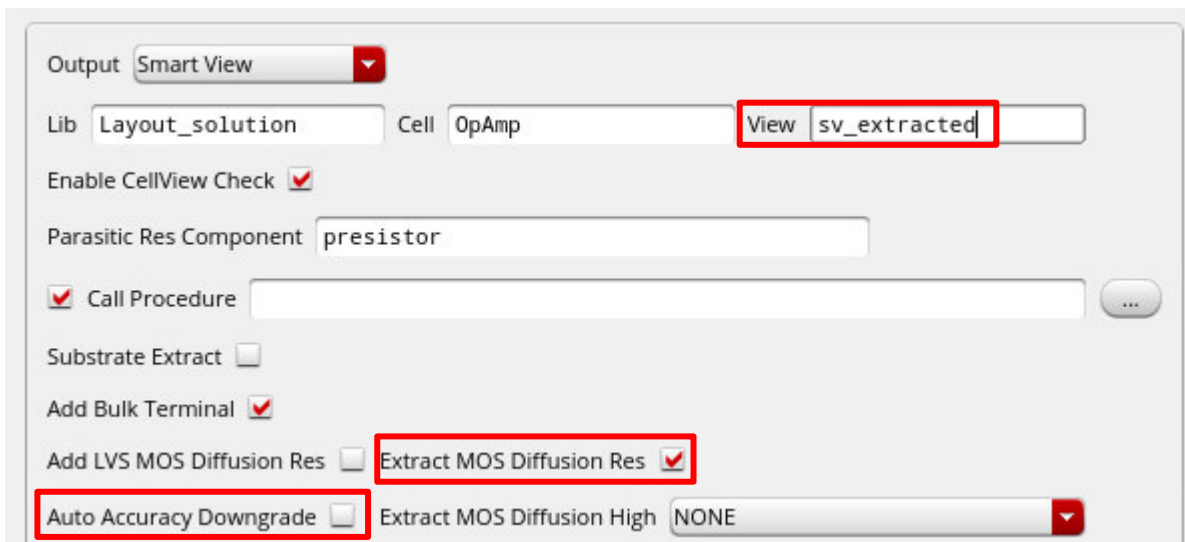
If you check that path, you should be able to see the DSPF file generated.

Action 270: Click on **Close**. Leave everything else including the main Quantus window open and proceed to the next lab.

Lab 19: Using Quantus to Create a Smart View

Action 271: Go back to the **Setup** tab on **Quantus (PVS) Parasitic Extraction Run Form**.

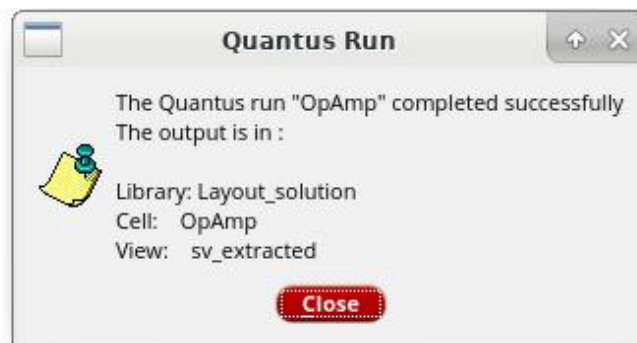
Action 272: Change **Output** to **Smart View**. Then, change the **View** name to **sv_extracted**. Enable **Extract MOS Diffusion Res** and make sure that the **Auto Accuracy Downgrade** field is disabled. Leave everything else set to their default values.



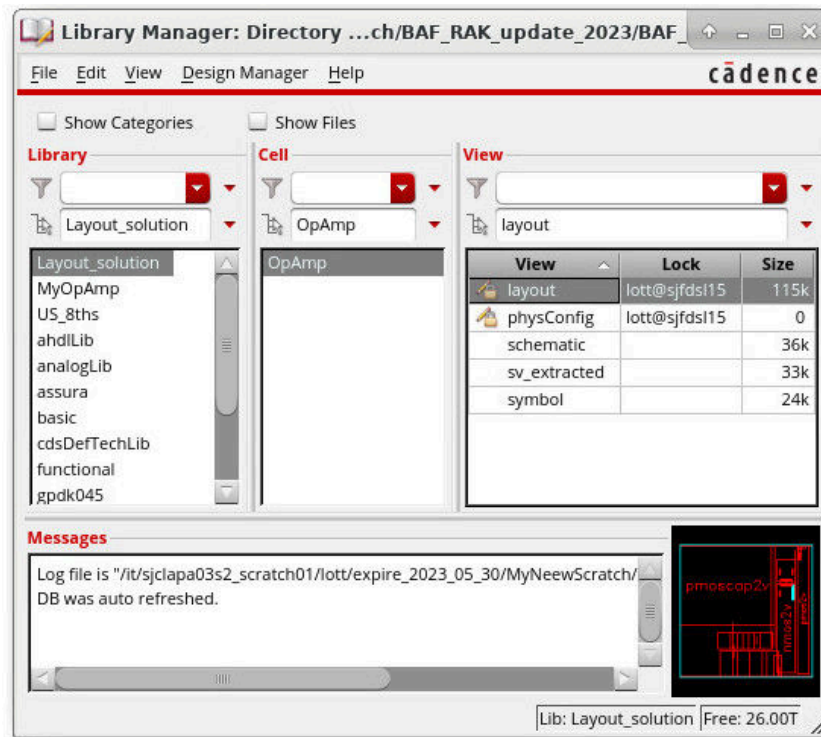
The screenshot shows the 'Setup' tab of the 'Quantus (PVS) Parasitic Extraction Run Form'. The 'Output' dropdown is set to 'Smart View'. The 'Lib' field is 'Layout_solution', 'Cell' is 'OpAmp', and 'View' is 'sv_extracted'. The 'Enable CellView Check' checkbox is checked. The 'Parasitic Res Component' field is 'presistor'. The 'Call Procedure' checkbox is checked. The 'Substrate Extract' checkbox is unchecked. The 'Add Bulk Terminal' checkbox is checked. The 'Add LVS MOS Diffusion Res' checkbox is unchecked. The 'Extract MOS Diffusion Res' checkbox is checked. The 'Auto Accuracy Downgrade' checkbox is unchecked. The 'Extract MOS Diffusion High' dropdown is set to 'NONE'.

Action 273: Click **Apply** again.

When the **Quantus Run** is completed, the following window should open:



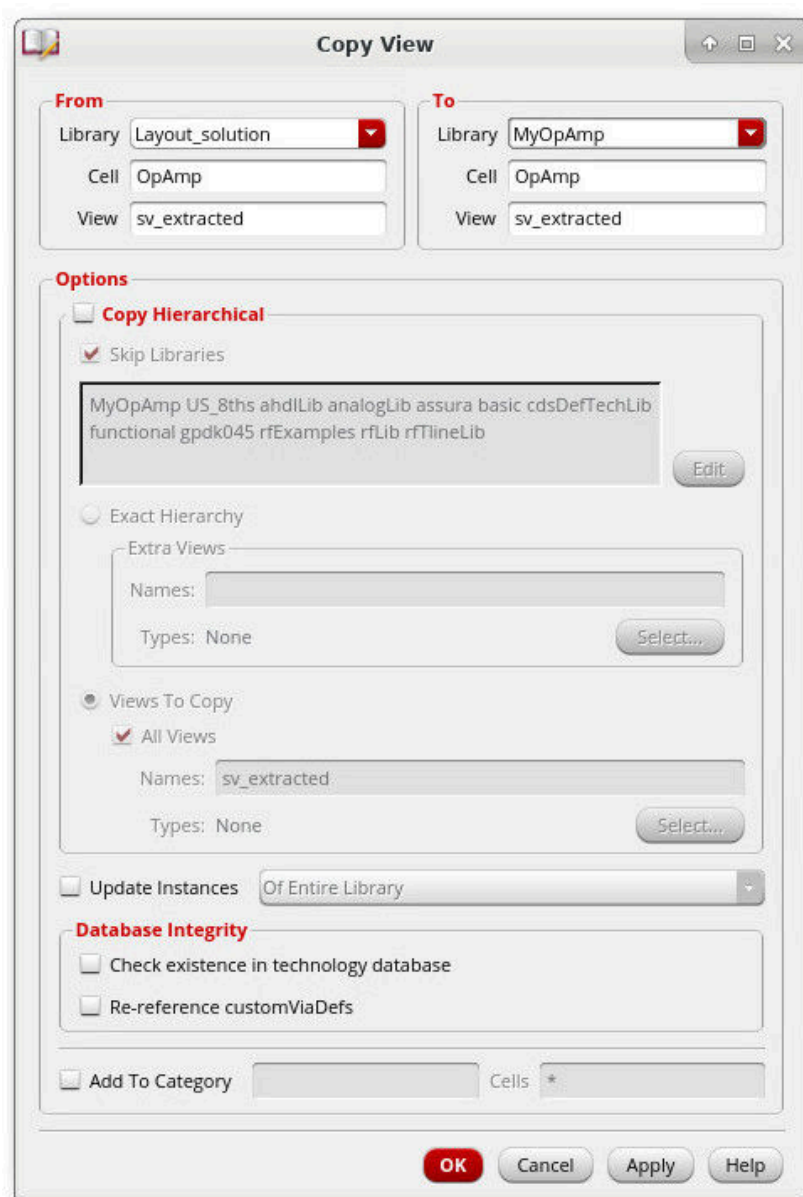
Action 274: Click **Close** and exit **Quantus**. Check the **Library Manager** for the **sv_extracted** view.



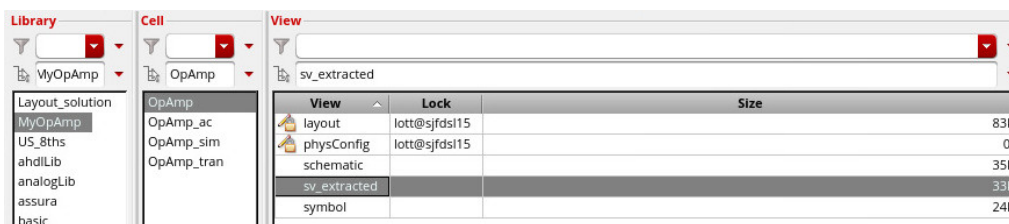
Action 275: Close the schematic and layout views. The CIW and Library Manager should still be open.

Let us now place this view on the MyOpAmp library.

Action 276: Right-click the newly created **sv_extracted** view under **Layout_solution > OpAmp** and select **Copy**. In the **To** section of the **Copy View** form, switch the **Library** from **Layout_solution** to **MyOpAmp**. Your form should look as follows:



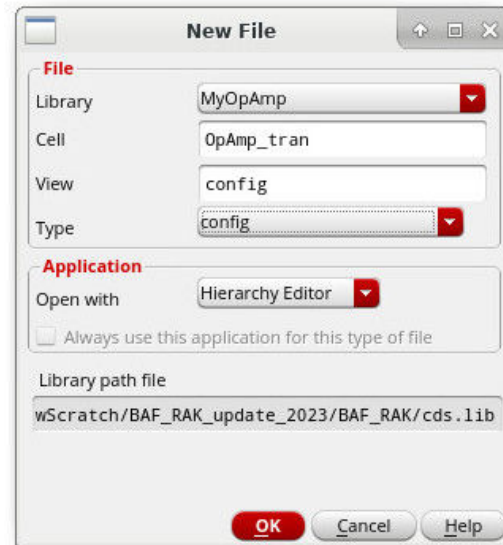
Action 277: Click **OK** on the dialog box. In the **Copy Problems** dialog box, select **OK**. This will create a copy of the extracted view in your working library.



Lab 20: Creating a Config View Using Virtuoso Hierarchy Editor

In this lab, you will create a 'config' view so that you can use the extracted views for a post-layout simulation.

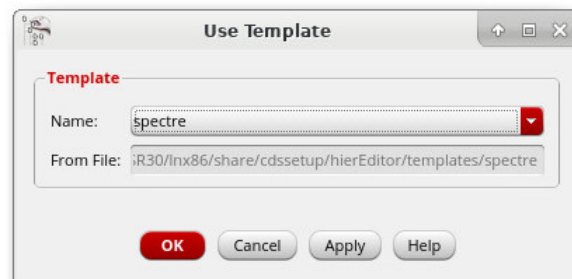
Action 278: In the **Library Manager**, select the **MyOpAmp** library and the **OpAmp_tran** cell. Select **File > New > Cell View** or use the **Ctrl+n** bindkey. Fill in the form (as shown below) and click **OK**.



Virtuoso Hierarchy Editor will open together with the **New Configuration** window.

Action 279: In the **View** field on the **Top Cell** section, select **schematic**.

Action 280: Press the **Use Template** button at the bottom of the dialog box and select **spectre** as the **Name**. Click **OK** on the form.



This populates the **View** and **Stop** lists with view names that are used in Spectre netlisting. These lists will dictate the default traversal behavior as the netlister descends through the design.

The **New Configuration** form should look as follows:

The screenshot shows the 'New Configuration' dialog box with the following fields and options:

- Top Cell**
 - Library: MyOpAmp
 - Cell: OpAmp_tran
 - View: schematic
- Global Bindings**
 - Library List: myLib
 - View List: :tre cmos_sch cmos.sch schematic verilogahdl pspice dspf
 - Stop List: spectre
 - Constraint List: (empty)
- Description**

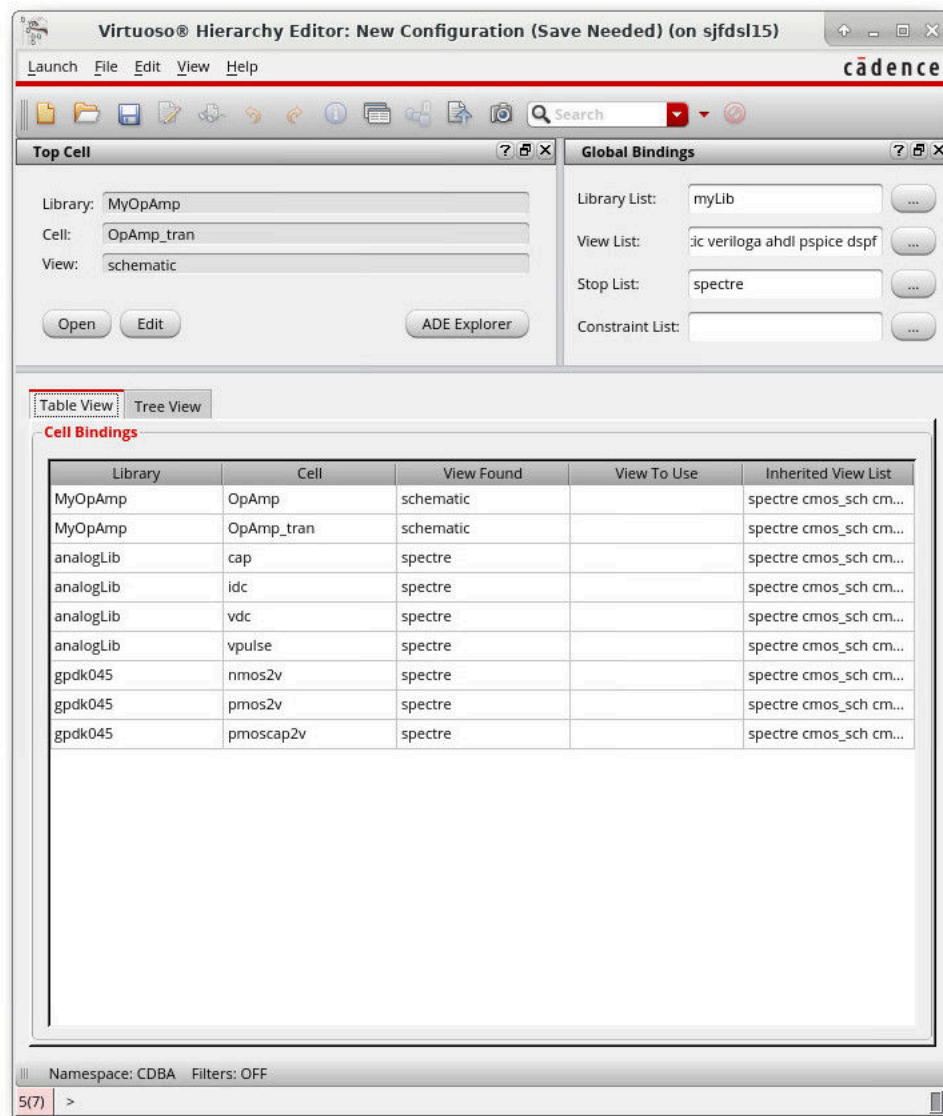
Default template for spectre
Note:
Please remember to replace Top Cell Library, Cell, and View fields with the actual names used by your design.

At the bottom, there are four buttons: **OK**, **Cancel**, **Use Template**, and **Help**.

Note that the **Library List** is used only in text-driven flows, often found in a mixed-signal design. For now, leave the list as-is.

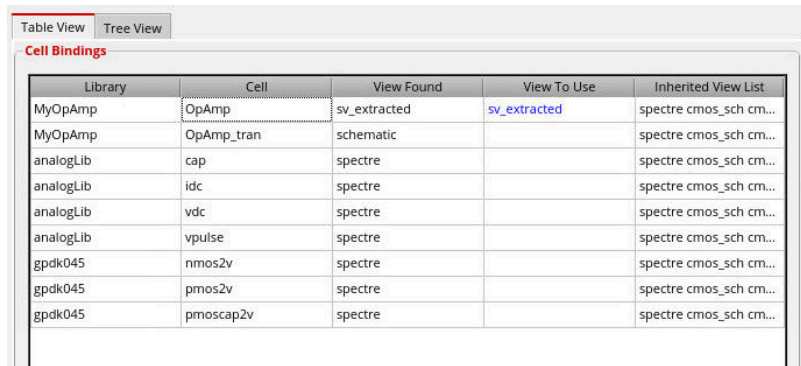
Action 281: Click **OK** on the **New Configuration** dialog box.

The resulting config view should look as follows:



This editor shows every block/component used in the design. The **View Found** column shows the views that will be used during hierarchy traversal and netlisting.

Action 282: Right-click on the **OpAmp Cell** row and select **Set Cell View > sv_extracted**.



Library	Cell	View Found	View To Use	Inherited View List
MyOpAmp	OpAmp	sv_extracted	sv_extracted	spectre cmos_sch cm...
MyOpAmp	OpAmp_tran	schematic		spectre cmos_sch cm...
analogLib	cap	spectre		spectre cmos_sch cm...
analogLib	idc	spectre		spectre cmos_sch cm...
analogLib	vdc	spectre		spectre cmos_sch cm...
analogLib	vpulse	spectre		spectre cmos_sch cm...
gpd045	nmos2v	spectre		spectre cmos_sch cm...
gpd045	pmos2v	spectre		spectre cmos_sch cm...
gpd045	pmoscap2v	spectre		spectre cmos_sch cm...

Action 283: Click on the toolbar icon  to update changes. Press **OK** in the window that pops up.

Next, create a similar view for the **ac** test.

Action 284: In the **Hierarchy Editor**, go to **File > Save As** and change the cell name to **OpAmp_ac**, as shown below. Write **config** in the **View** field. Then, press **OK**.



Now that you made a copy of the config view for the **ac** test, you just need to change the top cell since that is the only difference between both views.

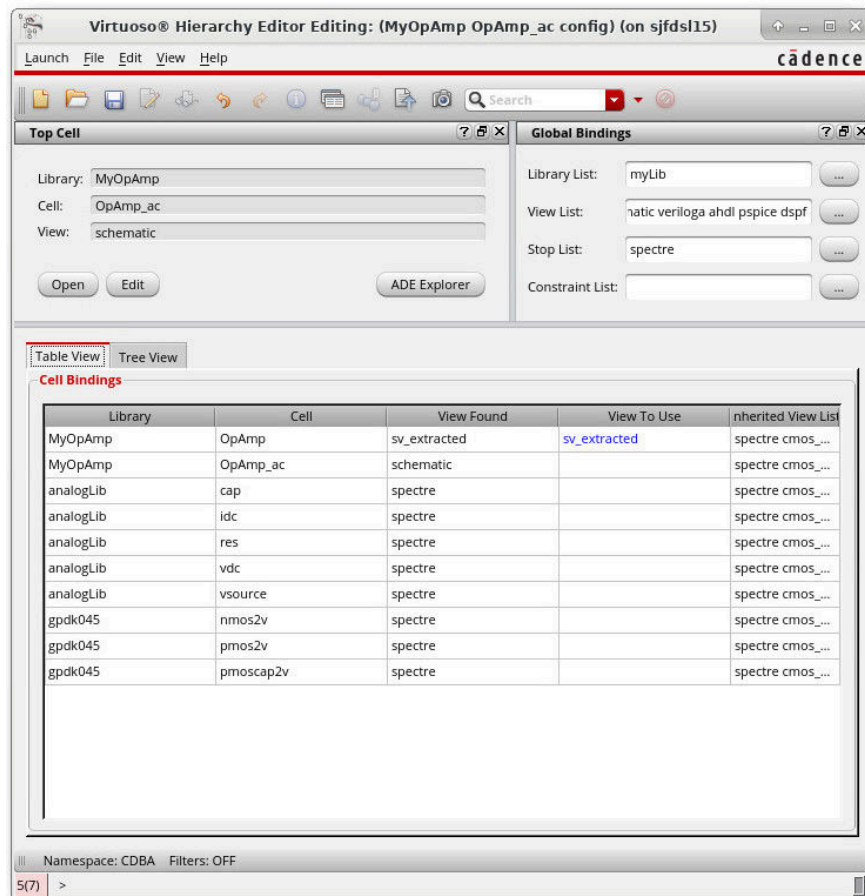
Action 285: In the **OpAmp_ac** config view, select **Edit** in the **Top Cell** section and change the cell name to **OpAmp_ac**, as shown below. Then, press **OK**.



Basics of Analog Flow: RAK



Action 286: Update this config view by pressing the '📄' icon. Press **OK** on the window that pops up.

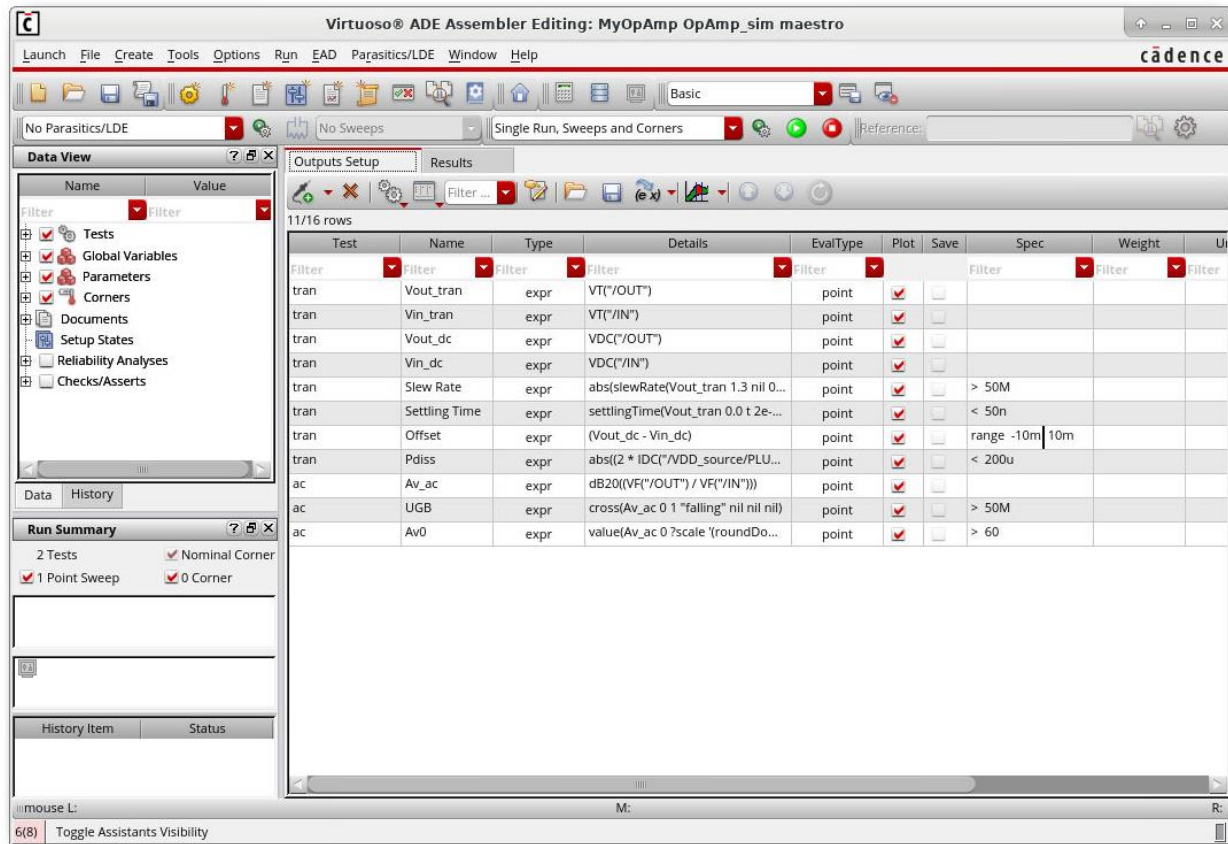


Now, you have successfully created two config views.

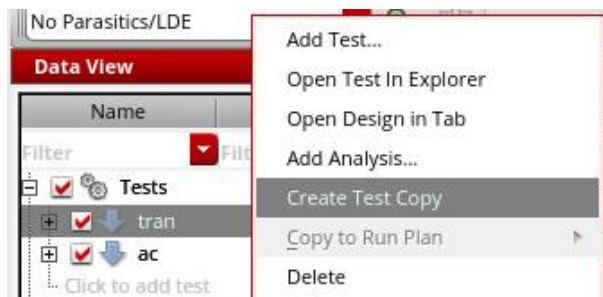
Action 287: Close the config view and proceed to the next lab.

Lab 21: Post-Layout Simulations on ADE Assembler

Action 288: Open the **maestro** view using **MyOpAmp > OpAmp_sim > maestro**. The **ADE Assembler** window will open.

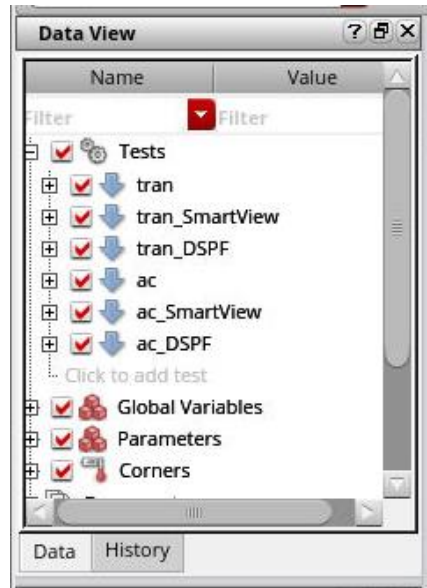


Action 289: Expand **Tests** on the **Data View** assistant, right-click on the **tran** test, and select **Create Test Copy** twice to make two copies of the test.



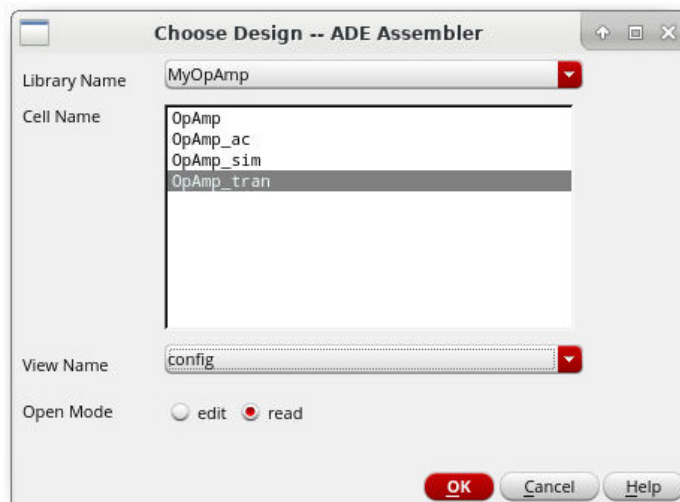
Action 290: Change the name of the two new tests to **tran_SmartView** and **tran_DSPF**.

Action 291: Repeat this same process for the **ac** test so that your maestro view test list appears as shown below. Drag and rearrange the order of the tests so that all **tran** and **ac** tests appear together.

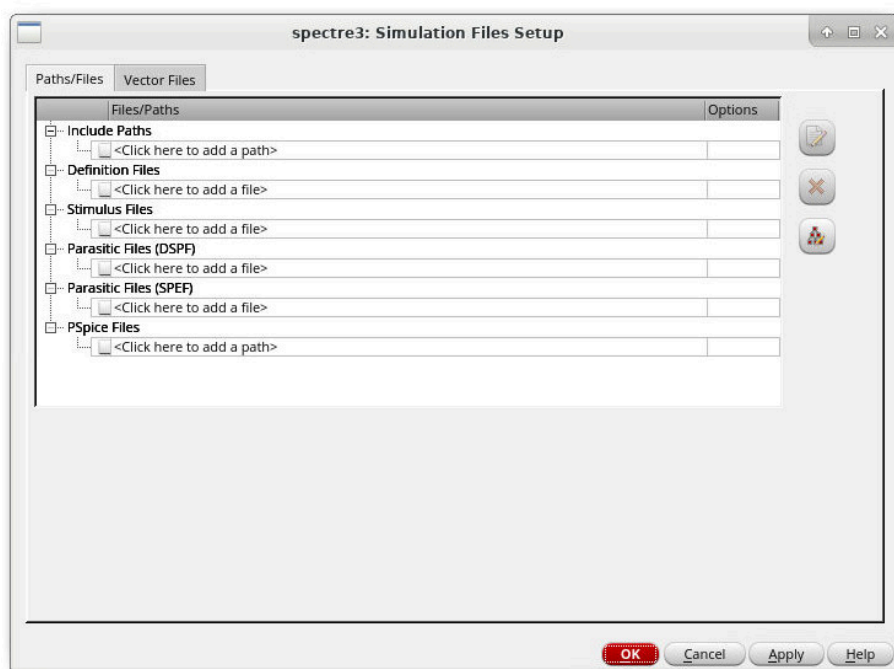


Action 292: Right-click on the **tran_SmartView** test and select **Design**.

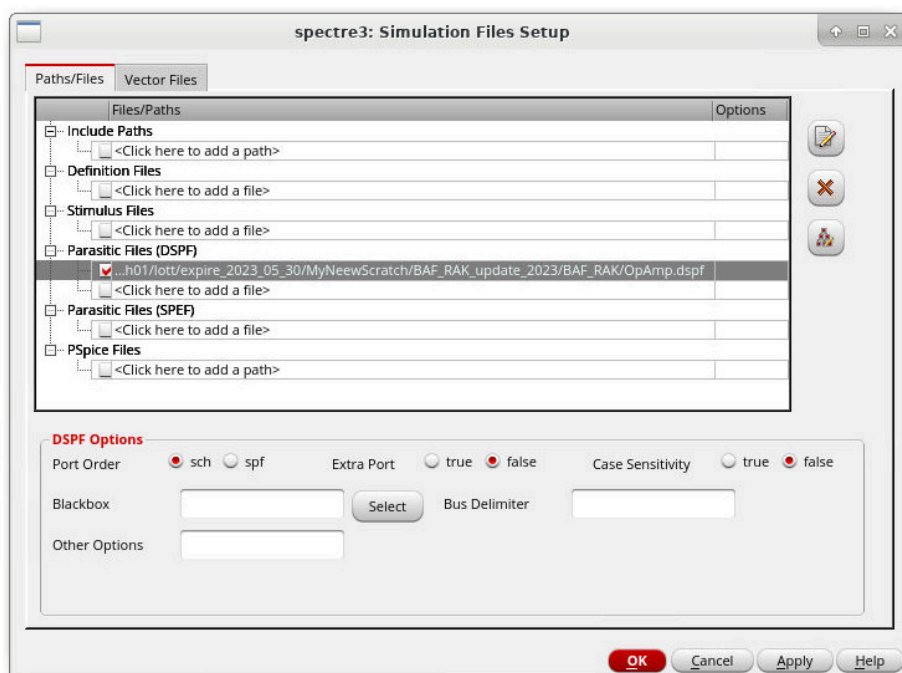
Action 293: In the **Choose Design** dialog box, verify that **OpAmp_tran** is selected, and select **config** as the view. Click **OK**.



Action 294: Right-click on the **tran_DSPF** test and select **Simulation Files**. The **Simulation Files Setup** form will open.



Action 295: Add the **OpAmp.dspf** file you generated to the **Parasitic Files (DSPF)** section.



Action 296: Click **OK** on the form.

Action 297: Repeat this process for **ac_SmartView** and **ac_DSPF**, changing the design reference of the **ac_Smartview** test to use the **config** of **OpAmp_ac** and adding the same DSPF file to **Simulation Files** for **ac_DSPF**.

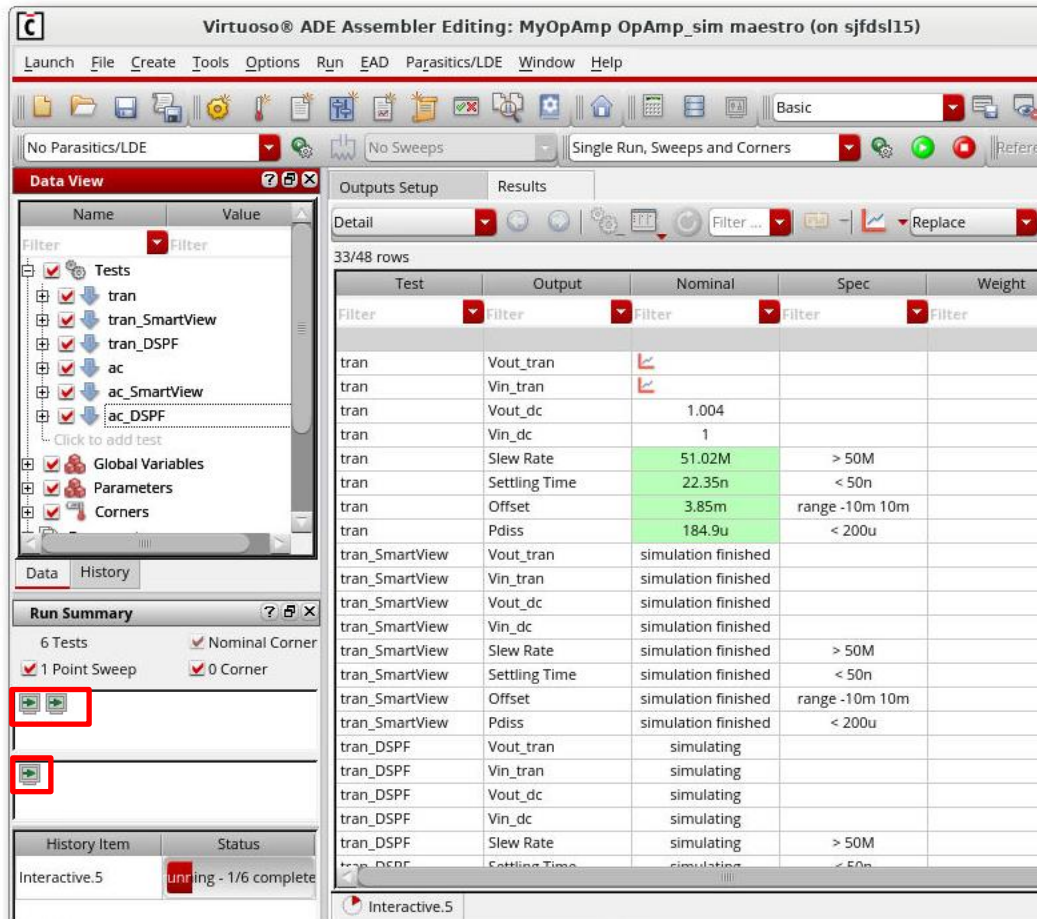
The next step is to run these simulations. Since there are six simulations to be done, configure ADE to allow parallel runs.

Action 298: Go to **Options > Job Setup** and set the **Max. Jobs** field value to **2**, as shown below. Click **OK** on the form.

The screenshot shows the 'Job Policy Setup' dialog box with the 'Simulation' tab selected. The 'Job Policy Name' is 'Maestro Default'. In the 'Setup' section, 'Job Control Mode' is 'LSCS', 'Distribution Method' is 'Local', and 'Max. Jobs' is set to '2' (highlighted with a red box). The 'Timeouts (in Secs.)' section shows 'Start Timeout' as 300, 'Simulation Run Timeout' as empty, and 'Linger Time' as 300. The 'Error reporting' section has 'Show output log on error' unchecked and 'Show errors even if retrying test' checked. At the bottom are buttons for 'OK', 'Cancel', 'Apply', 'Stop All', and 'Help'.

This will launch two Spectre jobs simultaneously.

Action 299: Click the **Run Simulation** icon .



As the simulations run, you will see two terminal icons in the **Run Summary** pane, representing the two parallel simulation jobs. Below that you will see one terminal icon, representing one job for the netlisting stage. If desired, you can also change the maximum number of parallel netlisting jobs in the **Netlisting** tab of the previous form. The tests statuses will change from 'pending' to 'simulating' as the jobs begin.

After all the simulations are finished, numerical results will appear for the measurements and waveform icons for signals.

Basics of Analog Flow: RAK

Outputs Setup

Results

Detail

Filter ...

Replace

(None)

33/48 rows

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
tran	Vout_tran				
tran	Vin_tran				
tran	Vout_dc	1.004			
tran	Vin_dc	1			
tran	Slew Rate	51.02M	> 50M		pass
tran	Settling Time	22.35n	< 50n		pass
tran	Offset	3.85m	range -10m 10m		pass
tran	Pdiss	184.9u	< 200u		pass
tran_SmartView	Vout_tran				
tran_SmartView	Vin_tran				
tran_SmartView	Vout_dc	1.003			
tran_SmartView	Vin_dc	1			
tran_SmartView	Slew Rate	51.83M	> 50M		pass
tran_SmartView	Settling Time	25.49n	< 50n		pass
tran_SmartView	Offset	2.955m	range -10m 10m		pass
tran_SmartView	Pdiss	201.7u	< 200u		near
tran_DSPF	Vout_tran				
tran_DSPF	Vin_tran				
tran_DSPF	Vout_dc	1.003			
tran_DSPF	Vin_dc	1			
tran_DSPF	Slew Rate	51.83M	> 50M		pass
tran_DSPF	Settling Time	25.49n	< 50n		pass
tran_DSPF	Offset	2.955m	range -10m 10m		pass
tran_DSPF	Pdiss	201.7u	< 200u		near
ac	Av_ac				
ac	UGB	121.1M	> 50M		pass
ac	Av0	60.36	> 60		pass
ac_SmartView	Av_ac				
ac_SmartView	UGB	136.2M	> 50M		pass
ac_SmartView	Av0	59.59	> 60		near
ac_DSPF	Av_ac				
ac_DSPF	UGB	136.2M	> 50M		pass
ac_DSPF	Av0	59.59	> 60		near

Most values should be shaded in green, indicating that the design meets specifications in all conditions. Some values are shaded in yellow, which indicates an influence of the parasitic effects on your design.

Additional steps like **transistor matching** and the use of **guard rings** during your layout design might present better post-layout results, but those steps will not be approached here. Notice that the results from the extracted view tests are identical to those from the DSPF file tests. This is expected since the parasitics created by **Quantus** are precisely the same.

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